

RTL Design Sherpa

STREAM Subsystem Micro-Architecture Specification

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1 Stream Index

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2 Document Information

This document describes the STREAM subsystem, a Scatter-gather Transfer Rapid Engine for AXI Memory. STREAM provides descriptor-based DMA functionality with multi-channel support, enabling efficient memory-to-memory data transfers via AXI4 interfaces. The subsystem is designed as a tutorial-focused implementation with intentional simplifications for educational purposes.

2.1 References

2.1.1 Related Documents

Related Documents and Specifications

Source	Title	Version
RTL Design Sherpa	STREAM Product Requirements Document	1.0
RTL Design Sherpa	RAPIDS MAS (Parent Architecture)	0.90
ARM	AMBA AXI and ACE Protocol Specification	IHI0022H
ARM	AMBA APB Protocol Specification	IHI0024C

2.2 Terminology

AXI

Advanced eXtensible Interface. ARM's high-performance, high-frequency protocol for interconnect.

AXI4

Version 4 of the AXI protocol, supporting burst transfers up to 256 beats.

AXIL

AXI4-Lite. A simplified subset of AXI4 for low-throughput register access.

APB

Advanced Peripheral Bus. ARM's simple, low-power bus protocol for peripheral register access.

Beat

A single data transfer within an AXI burst. For STREAM with 512-bit data width, one beat = 64 bytes.

Burst

A group of consecutive data transfers (beats) initiated by a single address phase.

Channel

One of 8 independent DMA channels in STREAM. Each channel has its own descriptor chain and SRAM buffer allocation.

Descriptor

A 256-bit data structure containing transfer parameters: source address, destination address, length, and chain pointer.

Descriptor Chain

A linked list of descriptors where each descriptor points to the next, enabling scatter-gather operations.

DMA

Direct Memory Access. Hardware-controlled data movement between memory locations without CPU intervention.

FUB

Functional Unit Block. A self-contained RTL module with defined interfaces and testbench.

MAC

Macro. An integration-level block that instantiates and connects multiple FUBs.

MonBus

Monitor Bus. A 128-bit packet stream paired with a 64-bit side-band timestamp wire, used for performance monitoring and debug event reporting. Packet field layout is the canonical spec at `docs/markdown/rtl-amba/includes/ocelus_package_spec.md`; the stream `monbus_axil_group` wraps the shared building block at `rtl/amba/ocelus/monbus_axil4_axil4_group.sv` (formerly `rtl/amba/shared/monbus_axil_group.sv`) and emits 3-beat AXI-Lite records (`pkt[63:0]`, `pkt[127:64]`, `ts[63:0]`).

Outstanding Transaction

A transaction that has been issued but not yet completed. V2/V3 performance modes support multiple outstanding transactions.

Scatter-Gather

A DMA technique where non-contiguous memory regions are transferred using a descriptor chain.

SRAM

Static Random Access Memory. Used for internal data buffering between read and write datapaths.

V1/V2/V3

STREAM performance modes: - V1 (Low): Single outstanding transaction, tutorial-focused - V2 (Medium): Command pipelined, best area efficiency - V3 (High): Out-of-order completion, maximum throughput

---

2.3 Revision History

Revisions follow the convention x.y, where x is the major version and y is the minor version. Minor versions indicate incremental updates; major versions indicate significant architectural changes.

STREAM MAS Document Revision History

Rev	Date	Author	Notes
0.90	2025-11-22	seang	Initial STREAM MAS structure and block documentation
0.91	2026-05-14	seang	Sync to RTL changes since 2026-04-17 (commit `be4e5a91`); regenerate artifacts.
0.92	2026-06-05	seang	Sync to RTL state at 2026-06-05 (17 commits since `be4e5a91`). Adds (1) 128-bit monbus packet + 64-bit side-band timestamp documentation in `ch02_blocks/16_monbus_axil_group.md` and `ch03_interfaces/05_monbus_interface_spec.md` — stream wrapper now instantiates the shared building block from `rtl/amba/shared/monbus_axil_group.sv` (since

Rev	Date	Author	Notes
			moved/renamed to `rtl/amba/ocelus/monbus_axil4_axil4_group.sv`), `cfg_ts_append_*` removed, m_axil records locked to 3 beats; (2) channel-observation mux documentation in `ch02_blocks/14_apb_config.md` plus `ch04_registers/register_map.md` (OBS_CTRL/FLAGS/DATA0/DATA1 register set, fed by scheduler error stickies + timeout exposed in `ch02_blocks/04_scheduler.md`); (3) BOTH-end descriptor-path monitoring in `ch02_blocks/05_descriptor_engine.md` (distinct fetch-side / consume-side monitor wrappers); (4) Bug-fix notes for `ch02_blocks/06_axi_read_engine.md` + `12_axi_write_engine.md` (registered `w_arb_request`, commit `4e8f9e02`), `11_stream_drain_c`

Rev	Date	Author	Notes
			`trl.md` + `12_axi_write_engine.md` (stale-view race + post-flop wvalid gate, commit `a82627af`), `08_sram_controller.md` (reset on registered avail outputs, commit `b619eee9`).
0.93	2026-07-02	seang	Monitor/perf registers relocated into a separate `stream_mon_regs` regfile at 0x1000 (single APB slave, 13-bit/8 KB decode) in `ch04_registers/register_map.md` + `ch03_interfaces/03_apb_interface_spec.md`; added `SCHED_TIMEOUT_LIMIT` @ 0x208 and documented the recoverable scheduler write-progress timeout + commit-gated completion + `scheduler_idle` now excluding CH_ERROR (`ch02_blocks/04_scheduler.md`); documented

Rev	Date	Author	Notes
			functional descriptor prefetch throttle + deferred chaining (`ch02_blocks/05_descriptor_engine.md`).
0.94	2026-07-07	seang	Documented the kick-burst fast-path programming model (`ch05_programming/02_single_transfer.md` + `04_multi_channel.md`): program per-channel address registers, then one “go”-bit write fires N channel kicks back-to-back on a single clock cycle via the top-level `i_kick_burst_*` ports, versus the serial APB `CHn_CTRL` kick. Reference implementation = the NexysA7 char harness `CH_KICK_ADDR` shadow registers + `KICK_GO` bitmask CSR (host `harness_kick.py::batch_kick`).
0.96	2026-07-17	seang	Documented

Rev	Date	Author	Notes
			always-on datapath performance meters: `USE_AXI_MONITORS` now gates only the heavy `axi4_master_*_mon` CAM/reporter cones + latency histograms; the in-core `axi_bus_meter` buckets (RDMON/WRMON_PERF prod/bp/starv/idle) and the beat/byte/burst/window counters are always-on and the perf-window `RUN` bits pass through, so a monitors-off production build still measures datapath utilisation. Updated the `USE_AXI_MONITORS = 0` note (`ch02_blocks/01_stream_core.md`) and the MON-block gating clause (`ch04_registers/register_map.md`). Also relocated stream + rapids

Rev	Date	Author	Notes
			under `projects/components/dma-ip/` (path references throughout).
0.95	2026-07-09	seang	Documented read-ahead descriptor prefetch in `ch02_blocks/04_scheduler.md`: the `r_rd_ahead` flag (read flops load the descriptor-FIFO head WITHOUT draining it; write flops load it and drain the FIFO to catch up), the in-place CH_XFER_DATA advance that collapses the per-descriptor bubble to zero, and the registered launch/advance commit-accumulator addend that keeps the descriptor-FIFO output off the accumulator critical path for 100 MHz closure. New runtime field `SCHED_CONFIG.RD_PREFETCH_EN` @

Rev	Date	Author	Notes
			0x204[5] (default on) added to `ch04_registers/register_map.md` + `ch06_configuration/config_reference.md`.
0.97	2026-09-21	seang	No specification change. All 34 committed figures were re-encoded to a 64-colour palette (11.6 MB -> 3.6 MB) with every pixel dimension unchanged; the `regenerate_all_*.sh` scripts in `assets/mermaid/` and `assets/wavedrom/` now render through to `.png` and palette-encode, replacing versions that stopped at `.svg`.
0.98	2026-10-17	seang	Shared descriptor engine: `descriptor_engine_shared` (`SHARED_ENGINES = 1`, the default) in `02_scheduler_group_array.md` (new Figure 2.2.1), `05_descriptor_engine.md`,

Rev	Date	Author	Notes
			01_stream_core.md and 01_architecture.md, with measured area. 16-channel build: stream_top_ch16, the derived 16-channel register map (ch04_registers/register_map.md, new section), REG_NCH in 14_apb_config.md. RTL fixes documented: write-engine W-phase FIFO room gate (12_axi_write_engine.md), descriptor_ready requires channel enable (04_scheduler.md), perf-profiler channel ID widened for 16 channels (15_perf_profiler.md). Diagrams 01_stream_core, 10_axi_write_engine and 14_stream_config_block updated.
0.99	2026-10-17	seang	02_scheduler_group_array.md: post-

Rev	Date	Author	Notes
			route area of the per-channel and shared descriptor engines on the Genesys 2 board build (4,974 / 4,120 -> 3,825 / 1,725 LUTs / flip-flops, 1,840 LUTs of it the per-channel FIFOs), with a pointer to the HAS before-and-after.
0.100	2026-10-20	seang	Block diagrams redrawn to the repo element key (v3): class fills, double-bordered registered blocks, green return paths and a generated key under each figure. No technical content changed.
0.101	2026-10-20	seang	Section (h2) headings now open a new page, the pipeline default. No technical content changed.

Last Updated: 2026-10-17

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3 STREAM Architecture Overview

Component: STREAM (Scatter-gather Transfer Rapid Engine for AXI Memory) **Version:** 0.90
Status: Pre-release

3.1 Introduction

STREAM is a high-performance, multi-channel descriptor-based DMA engine designed for memory-to-memory scatter-gather transfers. It provides an educational yet production-capable architecture demonstrating key DMA concepts while maintaining professional design practices.

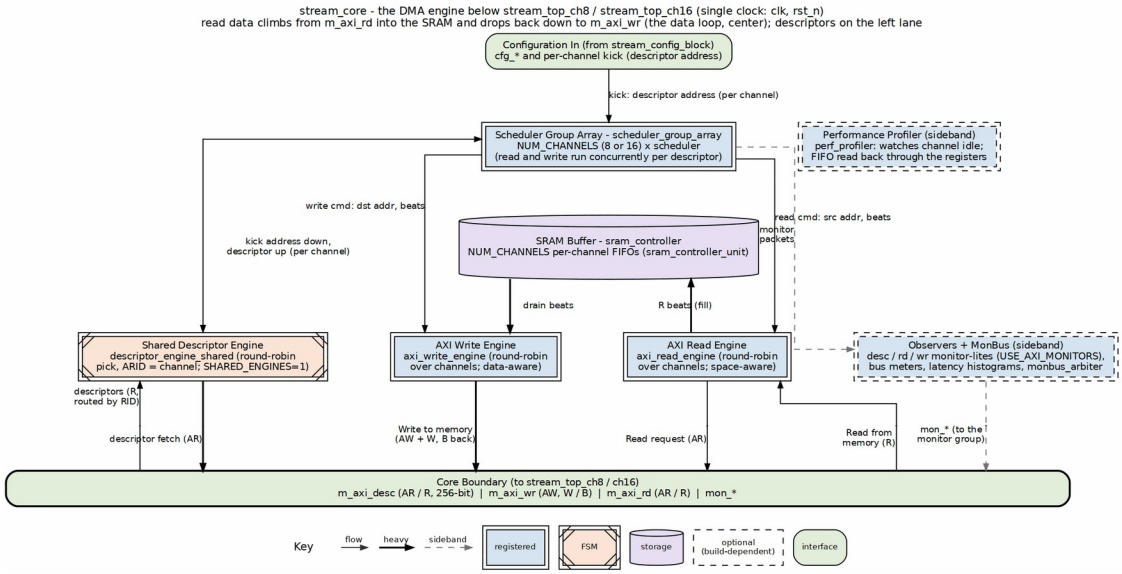
3.1.1 Design Philosophy

1. **Tutorial Focus** - Intentional simplifications for learning
2. **Production Quality** - Industry-standard interfaces and verification
3. **Scalability** - 8 or 16 independent channels (`stream_top_ch8` / `stream_top_ch16`) with shared resource arbitration and, by default, one shared descriptor engine
4. **Modularity** - Clear separation of concerns across functional blocks

3.2 System Architecture

3.2.1 High-Level Block Diagram

3.2.2 Figure 1.1.1: STREAM Architecture Overview



Diagram

Source: 01_stream_core.dot

3.3 Key Architectural Concepts

3.3.1 1. Multi-Channel Design

8 Independent Channels: - Each channel operates autonomously - Separate descriptor chains per channel - Independent FSM state per channel - Concurrent transfers across all channels

Resource Sharing: - Descriptor AXI master (shared via round-robin arbiter) - Data read AXI master (shared via round-robin, space-aware) - Data write AXI master (shared via round-robin, data-aware) - SRAM buffer (per-channel FIFOs, no arbitration)

3.3.2 2. Descriptor-Based Operation

Descriptor Format (256-bit):

```
[255:192] Reserved
[191:160] Next Descriptor Pointer
[159:128] Length (in beats)
[127:64]  Destination Address
[63:0]    Source Address
```

Descriptor Chain: - Single APB write kicks off chain - Automatic chaining via next_descriptor_ptr - Explicit termination (ptr = 0 or last flag)

3.3.3 3. Concurrent Read/Write

CRITICAL Design Pattern:

STREAM uses concurrent read and write operations to handle transfers larger than the SRAM buffer:

```
Example: 100MB transfer with 64KB SRAM buffer

Sequential (WRONG):
  Read 100MB → DEADLOCK at 64KB (SRAM full)

Concurrent (CORRECT):
  Read fills SRAM → SRAM full → Read pauses
  Write drains SRAM → SRAM space freed → Read resumes
  Both continue until 100MB complete
```

Implementation: - Scheduler runs read and write FSMs concurrently in XFER_DATA state - Independent beat counters for read vs write - Transfer completes when BOTH counters reach zero

3.3.4 4. Space-Aware Flow Control

Allocation Controller (Read Path): - Reserves SRAM space BEFORE issuing AXI AR - Prevents overflow due to AR/R latency gap - 2x space margin (accounts for in-flight allocations)

Drain Controller (Write Path): - Reserves SRAM data BEFORE issuing AXI AW - Prevents underflow due to AW/W latency gap - Includes latency bridge occupancy in count

3.3.5 5. Round-Robin Arbitration

Descriptor Fetch: - Round-robin arbitration (fair access)

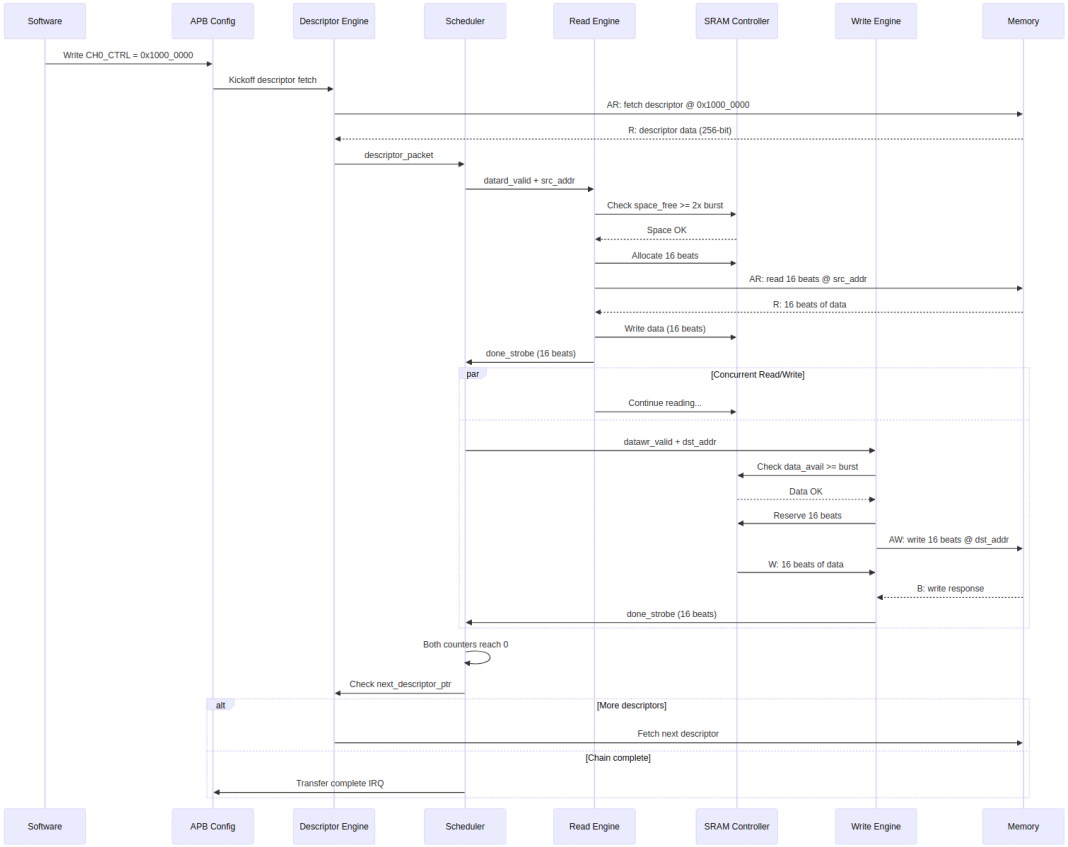
Data Read/Write: - Round-robin arbitration inside each engine (`arbiter_round_robin`) - A channel joins the read pick only with SRAM space for a burst, and the write pick only with SRAM data for a burst - No priority logic: the descriptor priority field is carried but not used for arbitration, and there is no anti-starvation timer

---

3.4 Data Flow Example

3.4.1 Single Channel Transfer

3.4.2 Figure 1.1.2: Single Channel Transfer Sequence



Diagram

3.5 Component Hierarchy

3.5.1 Top-Level Integration (stream_top_ch8.sv / stream_top_ch16.sv)

`stream_top_ch16.sv` is `stream_top_ch8.sv` with `NUM_CHANNELS = 16`, generated by `bin/gen_stream_regs_16ch.py`. The two differ only in the register block they instantiate (`stream_regs` or `stream_regs_16ch`), chosen by the filelist through the `regsel_{8,16}ch` include directory.

```
stream_top_ch8 / stream_top_ch16 (Top-Level Wrapper)
├── APB Interface Path
│   ├── apb4_slave_cdc        (CDC_ENABLE=1: APB clock domain
crossing)
│   └── peakrdl_to_cmdrsp     (CMD/RSP → PeakRDL passthrough; ALL
accesses.
│       │                     No router: the kick is register state at
│       │                     0x000-0x03F plus 0x128 KICK_ENABLE)
│       └── stream_regs       (PeakRDL-generated register file;
│                             stream_regs_16ch in the 16-channel top)
├── stream_config_block      (Register → config signal mapping)
├── stream_core              (Main DMA engine - see below)
└── monbus_axil_group        (USE_AXI_MONITORS=1: MonBus → AXI-Lite)
    ├── Error FIFO (s_axil_err_* slave read)
    └── Master Writer (m_axil_mon_* master write)
```

3.5.2 Core DMA Engine (stream_core.sv)

3.5.3 Figure 1.1.3: Core DMA Engine Hierarchy

```
stream_core
├── scheduler_group_array         (SHARED_ENGINES = 1, the default)
│   ├── scheduler [0..NUM_CHANNELS-1]
│   ├── descriptor_engine_shared  (one fetch engine for every channel)
│   ├── axi4_master_rd_monlite    (descriptor port monitor)
│   └── monbus_arbiter            (NUM_CHANNELS + 2 sources)
│   -- or, SHARED_ENGINES = 0 --
│   ├── scheduler_group [0..NUM_CHANNELS-1]
│   │   ├── descriptor_engine
│   │   ├── scheduler
│   │   └── monbus_arbiter
│   ├── arbiter_round_robin       (descriptor AR, NUM_CHANNELS:1)
│   ├── axi4_master_rd_monlite    (descriptor port monitor)
│   └── monbus_arbiter            (NUM_CHANNELS + 1 sources)
├── monbus_arbiter                (core MonBus merge)
├── axi_read_engine
│   └── arbiter_round_robin       (space-aware channel pick)
├── axi_write_engine
│   ├── arbiter_round_robin       (data-aware channel pick)
│   └── gaxi_fifo_sync            (W-phase queue, per-channel B-phase
```

```
queues)
├── sram_controller
│   └── sram_controller_unit [0..NUM_CHANNELS-1]
│       ├── stream_alloc_ctrl
│       ├── gaxi_fifo_sync
│       ├── stream_drain_ctrl
│       └── stream_latency_bridge
├── perf_profiler
├── axi4_master_rd_monlite / axi4_master_wr_monlite  (data masters;
monitor
│                                                     logic on only
when
│                                                     USE_AXI_MONITORS
= 1)
├── axi_bus_meter x 2             (always built)
└── axi_perf_latency_hist x 2     (USE_AXI_MONITORS = 1 only)
```

3.6 Interface Summary

3.6.1 External Interfaces (stream_top_ch8.sv / stream_top_ch16.sv)

STREAM External Interfaces

Interface	Type	Width	Purpose
APB	Slave	32-bit	Configuration, control, status (with optional CDC)
Descriptor AXI	Master	256-bit	Fetch descriptors from memory
Data Read AXI	Master	512-bit	Read source data
Data Write AXI	Master	512-bit	Write destination data
Error FIFO AXIL	Slave	32-bit	Read monitor error/interrupt FIFO
Monitor Write AXIL	Master	32-bit	Write monitor data to memory
IRQ	Output	1-bit	Interrupt (error FIFO not empty)

3.6.2 Internal MonBus Interface (stream_core.sv)

STREAM Internal MonBus Interface

Interface	Type	Width	Purpose
MonBus	Output	64-bit	Debug/trace event stream to monbus_axil_group

3.6.3 Configuration Registers (APB)

APB Address Map:

```
0x000-0x03F: Channel 0-7 descriptor addresses (staged); 0x128:
KICK_ENABLE (launch)
0x040-0x07F: Channel 8-15 descriptor addresses (16-channel map;
reserved in the 8-channel map)
0x080-0x0FF: Reserved
0x100-0x3FF: PeakRDL configuration registers
```

The 16-channel map is derived from the 8-channel one and differs only where a channel count shows: per-channel fields widen from `[7:0]` to `[15:0]`, and `CH_STATE` moves from 0x150 to 0x180. See the Register Map for the full list.

Key Register Groups: - 0x100: Global Control (enable, reset, version) - 0x120: Per-Channel Control (enable, reset) - 0x140: Per-Channel Status (idle, state, completion) - 0x180: Monitor FIFO Status - 0x200: Scheduler Configuration - 0x220: Descriptor Engine Configuration - 0x10C0-0x111F: AXI Monitor Configuration (DAXMON, RDMON, WRMON) – in the separate `stream_mon_regs` regfile at 0x1000+, not the base regfile - 0x2A0: AXI Transfer Configuration - 0x2B0: Performance Profiler Configuration - 0x2D0-0x2D8: Performance Profiler Data and Status (capture FIFO readback)

See: Register Map for complete documentation.

3.7 Resource Utilization

3.7.1 Measured Area (stream_core)

Out-of-context Vivado synthesis of `stream_core` at 256-bit data, with monitors off and a 100 MHz clock:

STREAM Core Measured Area

Build	LUTs	Flops	Worst slack
8 channels, per-channel descriptor engines (`SHARED_ENGINES = 0`)	25.8k	14.9k	+3.24 ns
8 channels, shared descriptor engine (default)	20.7k	11.7k	+2.50 ns
16 channels, shared descriptor engine	36.8k	19.4k	+1.79 ns

The shared engine saves about 20% of the 8-channel core's LUTs. The per-channel structures that remain are the schedulers, the SRAM controller units and the descriptor FIFOs.

Notes: - SRAM depth configurable (typical: 512 entries × 512 bits × 8 channels = 512KB) - Logic utilization scales with NUM_CHANNELS - Memory utilization scales with SRAM_DEPTH

3.8 Performance Characteristics

3.8.1 Throughput

Theoretical Maximum (512-bit @ 250MHz): - Single channel: 16 GB/s (512 bits × 250 MHz) - 8 or 16 channels concurrent: Limited by memory bandwidth, not DMA

Practical Performance: - Dependent on: - Memory controller latency - Burst sizes (larger = better efficiency) - Transfer alignment - Channel contention

Typical Real-World: - 8-12 GB/s sustained (single channel) - 50-75% of theoretical maximum

3.8.2 Latency

Descriptor Fetch Latency: - AR issue to R data: ~10-50 cycles (memory dependent) - Descriptor parsing: 1 cycle - **Total:** ~11-51 cycles per descriptor

Transfer Initiation: - APB write to first AR: ~5-10 cycles - AR to first R data: ~10-50 cycles - **Total:** ~15-60 cycles from kickoff to first data

Transfer Completion: - Last W beat to B response: ~5-20 cycles - B response to IRQ: 1 cycle - **Total:** ~6-21 cycles from last data to interrupt

---

3.9 Design Decisions

3.9.1 Why These Simplifications?

1. **Aligned Addresses Only:**
   - Eliminates complex alignment fixup logic
   - Clear data path with no byte shifting
   - Focus: Core DMA operation, not edge cases
2. **Length in Beats:**
   - Direct mapping to AXI burst length
   - No unit conversion overhead
   - Focus: AXI protocol understanding
3. **No Credit Management:**
   - Simpler resource arbitration
   - Transaction limits via configuration
   - Focus: Arbitration basics, not complex flow control
4. **No Circular Buffers:**
   - Explicit chain termination
   - Clear end-of-transfer detection
   - Focus: Descriptor chaining, not circular logic

3.9.2 Production Enhancements (Future)

If extending STREAM for production use, consider: - [ ] Alignment fixup (byte-level granularity) - [ ] Length in bytes/chunks - [ ] Credit-based flow control - [ ] Circular buffer support - [ ] Advanced error recovery - [ ] Power management - [ ] Multiple SRAM segments

---

3.10 Testing Strategy

3.10.1 Verification Layers

1. Unit Tests (FUB Level): - Descriptor engine: Fetch, parse, chain - Scheduler: FSM states, concurrent read/write - AXI engines: AR/R and AW/W/B channels - SRAM controller: Allocation, drain, FIFO - Individual tests per module

2. Integration Tests (Multi-Block): - Scheduler + engines: Data flow - Descriptor + scheduler: Descriptor chaining - Full path: APB → Descriptor → Transfer → IRQ

3. System Tests (Full Core): - Single channel end-to-end - Multi-channel concurrent - Error injection and recovery - Performance validation - Long-duration stress tests

3.10.2 Coverage Targets

- **Code Coverage:** >95%
- **Functional Coverage:** >90%
- **Corner Cases:** 100% tested
- **Error Paths:** 100% tested

3.11 Related Documentation

Chapter 2 - Block Specifications: - Scheduler - Core FSM controller - Descriptor Engine - Descriptor fetch/parse - AXI Read Engine - Source data read - AXI Write Engine - Destination data write - SRAM Controller - Buffering and flow control - Scheduler Group - Per-channel wrapper - Scheduler Group Array - Channel array and descriptor fetch (shared or per channel)

Last Updated: 2026-10-17 **Document Version:** 0.92 —

3.12 Revision History

STREAM Architecture Overview Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers
0.92	2026-10-17	seang	16-channel top and register map; shared descriptor engine in the hierarchy; measured area replaces the estimate

Last Updated: 2026-10-17

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4 STREAM Top-Level Port List

Module: `stream_core.sv` **Location:** `projects/components/dma-ip/stream/rtl/macro/`
Last Updated: 2025-12-01

4.1 Overview

This document provides a complete reference for all top-level ports of the STREAM Core module, organized by interface type. All port names, directions, widths, and descriptions are extracted directly from the RTL implementation.

Note: This documents `stream_core.sv` which is instantiated inside `stream_top_ch8.sv` (and `stream_top_ch16.sv`, with `NUM_CHANNELS = 16`). The top-level wrapper adds: - APB slave interface (replaces apb_valid/ready/addr with standard APB protocol) - AXI-Lite interfaces for monitor bus (s_axil_err_, *m_axil_mon_*) - Optional APB CDC crossing (CDC_ENABLE parameter)

For top-level integration, see the Architecture Overview.

Quick Navigation: - Clock and Reset - APB Programming Interface - Configuration Interface - Status Interface - Performance Profiler Interface - AXI4 Master - Descriptor Fetch - AXI4 Master - Data Read - AXI4 Master - Data Write - Status/Debug Outputs - Unified Monitor Bus

---

4.2 Clock and Reset

Clock and Reset Signals

Signal	Direction	Width	Description
`clk`	input	1	System clock (100-500 MHz typical)
`rst_n`	input	1	Active-low asynchronous reset

Notes: - All STREAM logic operates in the `clk` domain - Reset is asynchronous assert, synchronous deassert - All registers use reset macros from `reset_defs.svh`

4.3 APB Programming Interface

Per-channel descriptor kick-off interface. Each channel has independent valid/ready/address signals for descriptor chain start.

APB Programming Interface Signals

Signal	Direction	Width	Description
apb_valid[ch]	input	NUM_CHANNELS	Channel descriptor address valid
apb_ready[ch]	output	NUM_CHANNELS	Channel ready to accept descriptor address
apb_addr[ch]	input	NUM_CHANNELS × ADDR_WIDTH	Descriptor address per channel (64-bit default)

Usage Pattern:

```
// Kick off channel 0 with descriptor at 0x1000_0000
apb_valid[0] = 1'b1;
apb_addr[0] = 64'h0000_0000_1000_0000;
// Wait for handshake
wait (apb_ready[0]);
apb_valid[0] = 1'b0;
```

Notes: - Standard valid/ready handshake protocol - Descriptor address must be aligned to descriptor size (32 bytes) - Channel starts operation immediately after handshake - Default: NUM_CHANNELS = 8, ADDR_WIDTH = 64 (the 16-channel top sets NUM_CHANNELS = 16)

4.4 Configuration Interface

4.4.1 Per-Channel Configuration

Per-Channel Configuration Signals

Signal	Direction	Width	Description
`cfg_channel_enable[ch]`	input	NUM_CHANNELS	Enable channel (0=disabled, 1=enabled)
`cfg_channel_reset[ch]`	input	NUM_CHANNELS	Soft reset channel (FSM → IDLE state)

Notes: - Channel must be enabled before accepting descriptors - Soft reset clears channel FSM but preserves config

4.4.2 Global Scheduler Configuration

Global Scheduler Configuration Signals

Signal	Direction	Width	Description
`cfg_sched_enable`	input	1	Global scheduler enable
`cfg_sched_timeout_cycles`	input	16	Timeout threshold (clock cycles)
`cfg_sched_timeout_enable`	input	1	Enable timeout detection
`cfg_sched_err_enable`	input	1	Enable error event reporting
`cfg_sched_compl_enable`	input	1	Enable completion event reporting
`cfg_sched_perf_enable`	input	1	Enable performance event reporting

Notes: - `cfg_sched_enable` is master enable for all schedulers - Timeout measured from descriptor fetch to completion - Event enables control MonBus traffic

4.4.3 Descriptor Engine Configuration

Descriptor Engine Configuration Signals

Signal	Direction	Width	Description
`cfg_desceng_enable`	input	1	Enable descriptor engine

Signal	Direction	Width	Description
cfg_desceng_prefetch	input	1	Enable descriptor prefetch
cfg_desceng_fifo_thresh	input	4	FIFO threshold for prefetch trigger
cfg_desceng_addr0_base	input	ADDR_WIDTH	Address range 0 base (protection)
cfg_desceng_addr0_limit	input	ADDR_WIDTH	Address range 0 limit (protection)
cfg_desceng_addr1_base	input	ADDR_WIDTH	Address range 1 base (protection)
cfg_desceng_addr1_limit	input	ADDR_WIDTH	Address range 1 limit (protection)

Notes: - Descriptor engine shared across all channels - Prefetch improves latency for chained descriptors - Address ranges provide optional descriptor memory protection

4.4.4 AXI Monitor Configuration

Three identical sets of monitor config signals for descriptor, read, and write AXI masters:

AXI Monitor Signal Prefixes

Signal Prefix	Applies To	Description
cfg_desc_mon_*	Descriptor AXI	Descriptor fetch monitoring
cfg_rdeng_mon_*	Read AXI	Data read monitoring
cfg_wreng_mon_*	Write AXI	Data write monitoring

Each monitor has the following configuration signals:

AXI Monitor Configuration Signal Suffixes

Signal Suffix	Width	Description
_enable	1	Enable monitor
_err_enable	1	Enable error packet reporting
_perf_enable	1	Enable performance

Signal Suffix	Width	Description
		packet reporting
_timeout_enable	1	Enable timeout detection
_timeout_cycles	32	Timeout threshold (cycles)
_latency_thresh	32	Latency threshold for events
_pkt_mask	16	Packet type mask (filter events)
_err_select	4	Error type selector
_err_mask	8	Error event mask
_timeout_mask	8	Timeout event mask
_compl_mask	8	Completion event mask
_thresh_mask	8	Threshold event mask
_perf_mask	8	Performance event mask
_addr_mask	8	Address event mask
_debug_mask	8	Debug event mask

Example Signals:

```
cfg_desc_mon_enable          // Descriptor monitor enable
cfg_desc_mon_timeout_cycles  // Descriptor timeout threshold
cfg_rdeng_mon_err_enable     // Read engine error reporting enable
cfg_wreng_mon_perf_enable    // Write engine perf reporting enable
```

Notes: - Monitors generate MonBus packets for events - Masks control which event types are reported - Timeout measured from AR/AW valid to last R/B response

4.4.5 AXI Transfer Configuration

AXI Transfer Configuration Signals

Signal	Direction	Width	Description
cfg_axi_rd_xfer_beats	input	8	Read burst size (beats, 1-256)

Signal	Direction	Width	Description
cfg_axi_wr_xfer_beats	input	8	Write burst size (beats, 1-256)

Notes: - Configures AXI burst length for read/write engines - Larger bursts improve bandwidth but increase latency - Typical: 16 beats for DDR4, 8 beats for low-latency SRAM

4.4.6 Performance Profiler Configuration

Performance Profiler Configuration Signals

Signal	Direction	Width	Description
cfg_perf_enable	input	1	Enable profiler
cfg_perf_mode	input	1	Profiler mode (0=timestamp, 1=elapsed)
cfg_perf_clear	input	1	Clear profiler counters and FIFO

Notes: - Timestamp mode: Records start/end timestamps (software calculates elapsed) - Elapsed mode: Hardware calculates elapsed time directly - Clear is a single-cycle pulse

---

4.5 Status Interface

4.5.1 Per-Channel Status

Per-Channel Status Signals

Signal	Direction	Width	Description
descriptor_engine_idle[ch]	output	NUM_CHANNELS	Descriptor engine idle (no pending fetch)
scheduler_idle[ch]	output	NUM_CHANNELS	Scheduler in IDLE state (waiting for descriptor)
scheduler_state[ch]	output	NUM_CHANNELS × 7	Scheduler FSM state (ONE-HOT encoding)
sched_error[ch]	output	NUM_CHANNELS	Scheduler error flag (sticky, cleared by reset)
axi_rd_all_complete[ch]	output	NUM_CHANNELS	All read transactions complete for channel
axi_wr_all_complete[ch]	output	NUM_CHANNELS	All write transactions complete for channel

Scheduler State Encoding (ONE-HOT):

```
7'b0000001  // CH_IDLE - Waiting for descriptor
7'b0000010  // CH_FETCH_DESC - Fetching descriptor
7'b0000100  // CH_XFER_DATA - Transfer in progress
7'b0001000  // CH_COMPLETE - Transfer complete
7'b0010000  // CH_NEXT_DESC - Chaining to next descriptor
7'b0100000  // CH_ERROR - Error occurred
7'b1000000  // (Reserved)
```

Transfer Complete Condition:

```
channel_complete = scheduler_idle[ch] &&
                   axi_rd_all_complete[ch] &&
                   axi_wr_all_complete[ch];
```

4.6 Performance Profiler Interface

Performance Profiler Interface Signals

Signal	Direction	Width	Description
perf_fifo_empty	output	1	Profiler FIFO empty flag
perf_fifo_full	output	1	Profiler FIFO full flag
perf_fifo_count	output	16	Profiler FIFO occupancy (0-256)
perf_fifo_rd	input	1	Read profiler entry (pop FIFO)
perf_fifo_data_low	output	32	Profiler data [31:0] (timestamp or elapsed)
perf_fifo_data_high	output	32	Profiler data [63:32] (metadata)

FIFO Entry Format:

Low Word [31:0]: - Timestamp mode: Timestamp in clock cycles - Elapsed mode: Elapsed time in clock cycles

High Word [31:0]: - 8 channels: bit [3] = event type (0=start, 1=end), bits [2:0] = channel ID (0-7), bits [31:4] = 0 - 16 channels: bit [4] = event type, bits [3:0] = channel ID (0-15), bits [31:5] = 0

Read Sequence:

```
// Check FIFO not empty
if (!perf_fifo_empty) begin
    // Read 36-bit entry (two registers)
    perf_fifo_rd = 1'b1;  // Pulse to pop FIFO
    @(posedge clk);
    perf_fifo_rd = 1'b0;

    // Sample data on next cycle
    timestamp = perf_fifo_data_low;
    metadata = perf_fifo_data_high;
    channel_id = metadata[CHID_W-1:0];   // CHID_W = 3 (8 ch) or 4 (16
ch)
    event_type = metadata[CHID_W];
end
```

4.7 AXI4 Master - Descriptor Fetch (256-bit)

Fixed 256-bit width AXI4 master for descriptor fetch from memory.

4.7.1 AR Channel (Read Address)

Descriptor AXI Master AR Channel Signals

Signal	Direction	Width	Description
m_axi_desc_arid	output	AXI_ID_WIDTH	Transaction ID
m_axi_desc_araddr	output	ADDR_WIDTH	Read address
m_axi_desc_arlen	output	8	Burst length - 1 (AXI encoding)
m_axi_desc_arsize	output	3	Burst size = log2(bytes per beat)
m_axi_desc_arburst	output	2	Burst type (01=INCR)
m_axi_desc_arlock	output	1	Lock type (0=normal)
m_axi_desc_arcache	output	4	Cache attributes
m_axi_desc_arprot	output	3	Protection attributes
m_axi_desc_arqos	output	4	QoS value
m_axi_desc_arregion	output	4	Region identifier
m_axi_desc_aruser	output	CHAN_WIDTH	User signal (channel ID)
m_axi_desc_arvalid	output	1	Address valid
m_axi_desc_arready	input	1	Address ready

4.7.2 R Channel (Read Data)

R Channel

Signal	Direction	Width	Description
m_axi_desc_rid	input	AXI_ID_WIDTH	Transaction ID
m_axi_desc_rdata	input	256	Read data (FIXED 256-bit descriptor)
m_axi_desc_rresp	input	2	Response (00=OKAY, 01=EXOKAY, 10=SLVERR, 11=DECERR)
m_axi_desc_rlast	input	1	Last beat of burst
m_axi_desc_ruser	input	CHAN_WIDTH	User signal (channel ID)
m_axi_desc_rvalid	input	1	Read data valid
m_axi_desc_rready	output	1	Read data ready

Notes: - Data width is FIXED at 256 bits (descriptor size) - Typical burst: 1 beat (arlen=0) for single descriptor fetch - arsize = 3'b101 (32 bytes = 2^5) - ID encodes channel number for response routing

---

4.8 AXI4 Master - Data Read (Parameterizable Width)

Parameterizable width AXI4 master for reading source data from memory. Default 512-bit.

4.8.1 AR Channel (Read Address)

AR Channel

Signal	Direction	Width	Description
m_axi_rd_arid	output	AXI_ID_WIDTH	Transaction ID
m_axi_rd_araddr	output	ADDR_WIDTH	Read address
m_axi_rd_arlen	output	8	Burst length - 1
m_axi_rd_arsize	output	3	Burst size = log2(bytes per beat)
m_axi_rd_arburst	output	2	Burst type (01=INCR)
m_axi_rd_arlock	output	1	Lock type (0=normal)
m_axi_rd_arcache	output	4	Cache attributes
m_axi_rd_arprot	output	3	Protection attributes
m_axi_rd_arqos	output	4	QoS value
m_axi_rd_arregion	output	4	Region identifier
m_axi_rd_aruser	output	CHAN_WIDTH	User signal (channel ID)
m_axi_rd_arvalid	output	1	Address valid
m_axi_rd_arready	input	1	Address ready

4.8.2 R Channel (Read Data)

R Channel

Signal	Direction	Width	Description
m_axi_rd_rid	input	AXI_ID_WIDTH	Transaction ID
m_axi_rd_rdata	input	DATA_WIDTH	Read data (default 512-bit)
m_axi_rd_rresp	input	2	Response
m_axi_rd_rlast	input	1	Last beat of burst
m_axi_rd_ruser	input	CHAN_WIDTH	User signal (channel ID)
m_axi_rd_rvalid	input	1	Read data valid
m_axi_rd_rready	output	1	Read data ready

Notes: - Data width configurable via DATA_WIDTH parameter (default 512) - Burst length configured via `cfg_axi_rd_xfer_beats` (default 16) - arsize = log2(DATA_WIDTH/8) automatically calculated - AXI skid buffers on external interface for timing closure

---

4.9 AXI4 Master - Data Write (Parameterizable Width)

Parameterizable width AXI4 master for writing destination data to memory. Default 512-bit.

4.9.1 AW Channel (Write Address)

AW Channel

Signal	Direction	Width	Description
m_axi_wr_awid	output	AXI_ID_WIDTH	Transaction ID
m_axi_wr_awaddr	output	ADDR_WIDTH	Write address
m_axi_wr_awlen	output	8	Burst length - 1
m_axi_wr_awsize	output	3	Burst size = log2(bytes per beat)
m_axi_wr_awburst	output	2	Burst type (01=INCR)
m_axi_wr_awlock	output	1	Lock type (0=normal)
m_axi_wr_awcache	output	4	Cache attributes
m_axi_wr_awprot	output	3	Protection attributes
m_axi_wr_awqos	output	4	QoS value
m_axi_wr_awregion	output	4	Region identifier
m_axi_wr_awuser	output	CHAN_WIDTH	User signal (channel ID)
m_axi_wr_awvalid	output	1	Address valid
m_axi_wr_awready	input	1	Address ready

4.9.2 W Channel (Write Data)

W Channel

Signal	Direction	Width	Description
m_axi_wr_wdata	output	DATA_WIDTH	Write data (default 512-bit)
m_axi_wr_wstrb	output	DATA_WIDTH/8	Write strobes (byte enables, all 1s)
m_axi_wr_wlast	output	1	Last beat of burst
m_axi_wr_wuser	output	CHAN_WIDTH	User signal (channel ID)
m_axi_wr_wvalid	output	1	Write data valid
m_axi_wr_wready	input	1	Write data ready

4.9.3 B Channel (Write Response)

B Channel

Signal	Direction	Width	Description
m_axi_wr_bid	input	AXI_ID_WIDTH	Transaction ID
m_axi_wr_bresp	input	2	Response (00=OKAY, 01=EXOKAY, 10=SLVERR, 11=DECERR)
m_axi_wr_buser	input	CHAN_WIDTH	User signal (channel ID)
m_axi_wr_bvalid	input	1	Response valid
m_axi_wr_bready	output	1	Response ready

Notes: - Data width configurable via DATA_WIDTH parameter (default 512) - Burst length configured via `cfg_axi_wr_xfer_beats` (default 16) - awsize = log2(DATA_WIDTH/8) automatically calculated - wstrb typically all 1s (full data width writes) - AXI skid buffers on external interface for timing closure

---

4.10 Status/Debug Outputs

4.10.1 Descriptor AXI Monitor Status

Descriptor AXI Monitor Status

Signal	Direction	Width	Description
cfg_sts_desc_mon_busy	output	1	Monitor busy processing transaction
cfg_sts_desc_mon_active_txns	output	8	Active transaction count (0-255)
cfg_sts_desc_mon_error_count	output	16	Cumulative error count
cfg_sts_desc_mon_txn_count	output	32	Total transaction count
cfg_sts_desc_mon_conflict_error	output	1	ID conflict detected (sticky)

4.10.2 Read Engine AXI Monitor Status

Read Engine AXI Monitor Status

Signal	Direction	Width	Description
cfg_sts_rdeng_skid_busy	output	1	Skid buffer busy (not empty)
cfg_sts_rdeng_mon_active_txns	output	8	Active transaction count
cfg_sts_rdeng_mon_error_count	output	16	Cumulative error count
cfg_sts_rdeng_mon_txn_count	output	32	Total transaction count
cfg_sts_rdeng_mon_conflict_error	output	1	ID conflict detected (sticky)

4.10.3 Write Engine AXI Monitor Status

Write Engine AXI Monitor Status

Signal	Direction	Width	Description
cfg_sts_wreng_skid_busy	output	1	Skid buffer busy (not empty)
cfg_sts_wreng_mon_active_txns	output	8	Active transaction count
cfg_sts_wreng_mon_error_count	output	16	Cumulative error count
cfg_sts_wreng_mon_txn_count	output	32	Total transaction count
cfg_sts_wreng_mon_conflict_error	output	1	ID conflict detected (sticky)

Notes: - Status signals for debug and performance analysis - Transaction counts never roll over (use for profiling) - Error counts increment on any AXI error response - ID conflicts indicate internal RTL bug (should never occur)

4.11 Unified Monitor Bus Interface

Single output interface for all STREAM monitoring events.

Unified Monitor Bus Interface

Signal	Direction	Width	Description
mon_valid	output	1	Monitor packet valid
mon_ready	input	1	Monitor packet ready (from downstream consumer)
mon_packet	output	64	Monitor packet data (64-bit standard format)

MonBus Packet Format (64-bit):

```
[63:56] - Packet type (event code)
[55:48] - Agent ID (source identifier)
[47:40] - Unit ID (1 for STREAM)
[39:32] - Channel ID (0-7)
[31:0]  - Event-specific data
```

MonBus Sources: - Descriptor engines: 8 sources, agent IDs 16-23 (0x10-0x17) - Schedulers: 8 sources, agent IDs 48-55 (0x30-0x37) - Descriptor AXI monitor: agent ID 8 (0x08) - Read AXI monitor: configurable agent ID - Write AXI monitor: configurable agent ID

Downstream Integration:

```
// Connect to MonBus FIFO for buffering
gaxi_fifo_sync #(
    .DATA_WIDTH(64),
    .DEPTH(256)
) u_gaxi_fifo_sync (
    .axi_aclk               (axi_aclk),
    .axi_aresetn            (axi_aresetn),
    .wr_valid               (wr_valid),
    .wr_ready               (wr_ready),
    .wr_data                (wr_data),
    .rd_ready               (rd_ready),
    .count                  (count),
    .rd_valid               (rd_valid),
    .rd_data                (rd_data)
);
```

Notes: - Standard AMBA monitor bus protocol - Always buffer with FIFO to prevent backpressure to STREAM - Packet format documented in `rtl/amba/includes/ocelus_pkg.sv` - Event codes defined in STREAM package

4.12 Port Count Summary

Port Count Summary

Interface Type	Input Ports	Output Ports	Bidirectional
Clock/Reset	2	0	0
APB Programming	NUM_CHANNELS × (1 + ADDR_WIDTH)	NUM_CHANNELS	0
Configuration	~150	0	0
Status	1 (perf_fifo_rd)	NUM_CHANNELS × 10 + 30	0
Perf Profiler	1	4	0
AXI Desc Master	7	14	0
AXI Read Master	7	14	0
AXI Write Master	10	19	0
MonBus	1	2	0

Approximate Total: ~350 ports (varies with NUM_CHANNELS and ADDR_WIDTH/DATA_WIDTH)

4.13 Default Parameter Values

Default Parameter Values

Parameter	Default	Description
NUM_CHANNELS	8	Number of DMA channels
CHAN_WIDTH	3	Channel ID width (log2(NUM_CHANNELS))
ADDR_WIDTH	64	Address bus width
DATA_WIDTH	512	Data bus width
AXI_ID_WIDTH	8	AXI transaction ID width
FIFO_DEPTH	512	Per-channel FIFO depth
AR_MAX_OUTSTANDING	8	Max concurrent read requests
AW_MAX_OUTSTANDING	8	Max concurrent write requests

4.14 Related Documentation

- **Stream Core Block Spec** - Detailed block documentation
- **Clocks and Reset** - Timing specifications
- **Architecture** - System architecture overview

Last Updated: 2025-12-01 **Maintained By:** STREAM Architecture Team —

4.15 Revision History

STREAM Top-Level Port List Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers

Last Updated: 2026-01-02

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5 Clocks and Reset Specification

Chapter: 01 **Version:** 0.90 **Last Updated:** 2025-11-22

5.1 Overview

STREAM operates in a single clock domain with a single asynchronous active-low reset. This chapter defines clock requirements, reset behavior, and timing constraints for the STREAM subsystem.

5.2 Clock Domain

5.2.1 Primary Clock: `aclk`

Specification: - **Name:** `aclk` (AXI clock) - **Type:** Synchronous, single-edge (rising edge) - **Frequency:** Configurable (typical: 100 MHz - 500 MHz) - **Duty Cycle:** 50% 5% - **Jitter:** < 100 ps peak-to-peak

Usage: - All STREAM internal logic - All AXI master interfaces - All AXIL interfaces - MonBus output - SRAM

5.2.2 Secondary Clock: `pclk` (APB Clock)

Specification: - **Name:** `pclk` (Peripheral clock) - **Type:** Synchronous, single-edge (rising edge) - **Frequency:** Configurable (typical: 50 MHz - 200 MHz) - **Relation to `aclk`:** May be asynchronous

Usage: - APB configuration interface only

Clock Domain Crossing (CDC): - If `pclk` `aclk`: CDC logic required in `apb_config.sv` - If `pclk` = `aclk`: Direct connection (no CDC)

CDC Implementation: - Use `apb4_slave_cdc` wrapper (like HPET example) - `apb4_slave_cdc` implements **handshake-based CDC** using `cdc_handshake` modules - One `cdc_handshake` for command interface (APB → core) - One `cdc_handshake` for response interface (core → APB) - Full req/ack handshake protocol (NOT async FIFO) - Works across all frequency ratios (slow-to-fast, fast-to-slow, any ratio)

5.3 Reset

5.3.1 Primary Reset: aresetn

Specification: - **Name:** aresetn (AXI reset, active-low) - **Type:** Asynchronous assertion, synchronous deassertion - **Polarity:** Active-low (0 = reset, 1 = normal operation) - **Assertion:** Asynchronous (can occur at any time) - **Deassertion:** Synchronous to aclk rising edge - **Duration:** Minimum 10 aclk cycles

Reset Behavior:

```
// Standard reset pattern for all STREAM modules
always_ff @(posedge aclk or negedge aresetn) begin
    if (!aresetn) begin
        // Asynchronous reset assertion
        r_state <= IDLE;
        r_counter <= '0;
        r_valid <= 1'b0;
        // ... all registers to known state
    end else begin
        // Synchronous operation
        r_state <= w_next_state;
        // ... normal logic
    end
end
```

5.3.2 Secondary Reset: presetn

Specification: - **Name:** presetn (APB reset, active-low) - **Type:** Asynchronous assertion, synchronous deassertion - **Polarity:** Active-low (0 = reset, 1 = normal operation) - **Synchronization:** May be asynchronous to aresetn

Usage: - APB configuration interface only - Typically tied to aresetn if pclk = aclk

---

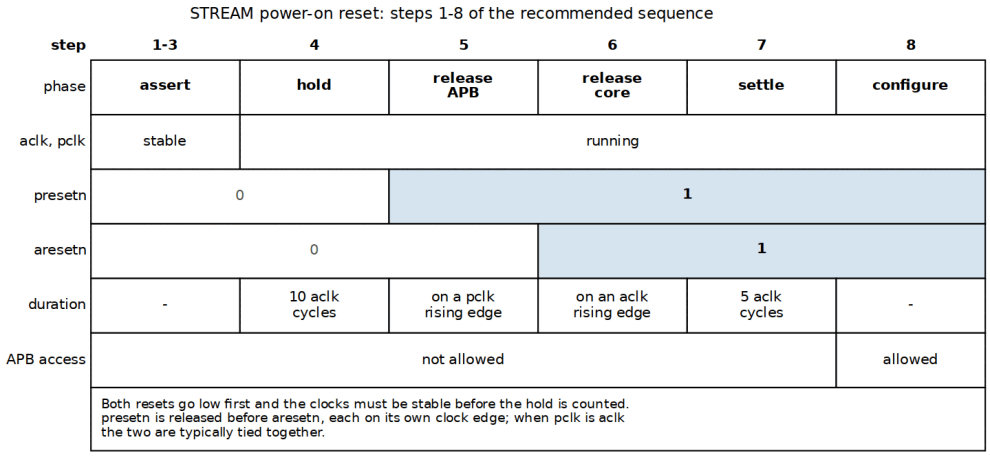
5.4 Reset Sequencing

5.4.1 Power-On Reset

Recommended sequence:

```
1. Assert aresetn (LOW)
2. Assert presetn (LOW)
3. Apply stable clocks (aclk, pclk)
4. Wait   10 aclk cycles
5. Deassert presetn (HIGH) on pclk rising edge
6. Deassert aresetn (HIGH) on aclk rising edge
7. Wait   5 aclk cycles for stabilization
8. Begin APB configuration
```

5.4.2 Figure 1.3.1: STREAM Power-On Reset Sequence



STREAM Power-On Reset Sequence

5.4.3 Functional Reset

Software-initiated reset (per channel):

```
// Reset specific channel via APB
write_apb(ADDR_GLOBAL_CTRL, CHANNEL_0_RESET);  // Auto-clears after 1
cycle

// Hardware response:
// - Channel FSM returns to IDLE
// - Channel registers cleared
// - Outstanding transactions flushed
// - MonBus error packet generated (if mid-transfer)
```

5.4.4 Reset Recovery

After reset deassertion:

Reset Recovery

Cycle	Event
0	`aresetn` deasserted (rising edge)
1-5	Internal state stabilization
6+	Ready for APB configuration
10+	Ready for descriptor transfers

5.5 Clock Requirements by Module

5.5.1 Functional Unit Blocks (FUB)

Functional Unit Blocks

Module	Clock	Reset	Frequency	Notes
descriptor_engine	aclk	aresetn	100-500 MHz	AXI master timing
scheduler	aclk	aresetn	100-500 MHz	Single cycle FSM
axi_read_engine	aclk	aresetn	100-500 MHz	AXI master timing
axi_write_engine	aclk	aresetn	100-500 MHz	AXI master timing
sram_controller_unit	aclk	aresetn	100-500 MHz	Synchronous SRAM buffering (wraps gaxi_fifo_sync)

5.5.2 Integration Blocks (MAC)

Integration Blocks

Module	Clock(s)	Reset(s)	Frequency	Notes
channel_arbiter	aclk	aresetn	100-500 MHz	Single cycle arbitration
apb_config	pclk, (aclk)	presetn, (aresetn)	50-200 MHz (APB)	CDC if async
monbus_axil_group	aclk	aresetn	100-500 MHz	AXIL timing
stream_top	aclk, pclk	aresetn, presetn	Mixed	Top-level

5.6 Timing Constraints

5.6.1 Setup and Hold Times

Internal registers (relative to `aclk`): - Setup time: 0.5 ns (typical) - Hold time: 0.1 ns (typical) - Clock-to-Q: 0.3 ns (typical)

External interfaces: - AXI/AXIL: Per ARM IHI0022E specification - APB: Per ARM IHI0024C specification

5.6.2 Critical Paths

Identified critical paths:

1. **Arbiter -> Scheduler grant:**
   - Latency: 1 cycle
   - Path: Priority encoder -> One-hot grant
2. **AXI read -> SRAM write:**
   - Latency: 1 cycle
   - Path: R data -> SRAM write port
3. **SRAM read -> AXI write:**
   - Latency: 1 cycle
   - Path: SRAM read port -> W data

Maximum frequency estimation: - Typical FPGA (Xilinx 7-series): 250 MHz - High-end FPGA (UltraScale+): 400 MHz - ASIC (28nm): 500 MHz

---

5.7 Clock Domain Crossing (CDC)

5.7.1 APB Configuration CDC

When required: `pclk` `aclk` (asynchronous APB interface)

CDC Implementation:

```
// APB to STREAM domain (pclk -> aclk)
// apb4_slave_cdc is an APB SLAVE on the pclk side and a cmd/rsp pair
on the
// aclk side -- there is no second APB port, so there are no m_p*
signals.
apb4_slave_cdc #(
    .ADDR_WIDTH (32),
    .DATA_WIDTH (32)
    // DEPTH, PROT_WIDTH, STRB_WIDTH, USE_2_PHASE_CDC and USE_JOHNSON
    // select sizing and CDC style; see the module header for their
meaning.
) u_apb_cdc (
    // APB slave side (pclk domain)
    .pclk          (pclk),
    .presetn       (presetn),
    .s_apb_PSEL    (cfg_psel),
    .s_apb_PENABLE (cfg_penable),
    .s_apb_PREADY  (cfg_pready),
    .s_apb_PADDR   (cfg_paddr),
    .s_apb_PWRITE  (cfg_pwrite),
    .s_apb_PWDATA  (cfg_pwdata),
    .s_apb_PSTRB   (cfg_pstrb),
    .s_apb_PPROT   (cfg_pprot),
    .s_apb_PRDATA  (cfg_prdata),
    .s_apb_PSLVERR (cfg_pslverr),

    // STREAM side (aclk domain): command out, response back
    .aclk          (aclk),
    .aresetn       (aresetn),
    .cmd_valid     (cfg_cmd_valid),
    .cmd_ready     (cfg_cmd_ready),
    .cmd_pwrite    (cfg_cmd_pwrite),
    .cmd_paddr     (cfg_cmd_paddr),
    .cmd_pwdata    (cfg_cmd_pwdata),
    .cmd_pstrb     (cfg_cmd_pstrb),
    .cmd_pprot     (cfg_cmd_pprot),
    .rsp_valid     (cfg_rsp_valid),
    .rsp_ready     (cfg_rsp_ready),
    .rsp_prdata    (cfg_rsp_prdata),
```

```
    .rsp_pslverr   (cfg_rsp_pslverr)
);
```

CDC Mechanism: - `apb4_slave_cdc` uses **handshake-based CDC** via `cdc_handshake` modules (NOT async FIFOs) - **Command path** (`pclk` → `aclk`): APB write/read commands cross via req/ack handshake - **Response path** (`aclk` → `pclk`): APB read data crosses back via req/ack handshake - **Internal synchronizers**: Dual-flop synchronizers (2-3 stages) for handshake signals - **ASYNC_REG attribute**: Applied to synchronizer stages for timing tools - **Timing constraints**: Proper constraints required in SDC/XDC for synchronizer paths

Handshake Protocol Benefits: - Works across **all frequency ratios** (slow-to-fast, fast-to-slow, arbitrary ratios) - Guaranteed data integrity (req/ack ensures data stability before sampling) - No FIFO depth management or gray code pointer complexity - Latency: 4-6 APB clock cycles for register access (handshake round-trip)

5.7.2 No CDC Required

Single clock domain: If `pclk` = `aclk` and `presetn` = `aresetn`:

```
// Direct connection (no CDC wrapper)
apb_config #(
    .NUM_CHANNELS(8)
) u_apb_config (
    .pclk(aclk),        // Same clock
    .presetn(aresetn),  // Same reset
    // ... direct APB signals
);
```

5.8 Reset State Initialization

5.8.1 Register Reset Values

All STREAM modules must initialize to known state on reset:

```
// Descriptor Engine
if (!aresetn) begin
    r_desc_fifo_wr_ptr <= '0;
    r_desc_valid <= 1'b0;
    r_desc_error <= 1'b0;
end

// Scheduler
if (!aresetn) begin
    r_current_state <= CH_IDLE;
    r_read_beats_remaining <= '0;
    r_write_beats_remaining <= '0;
    r_timeout_counter <= '0;
end

// AXI Engines
if (!aresetn) begin
    r_burst_counter <= '0;
    m_axi_arvalid <= 1'b0;
    m_axi_awvalid <= 1'b0;
end

// Arbiter
if (!aresetn) begin
    r_last_grant_id <= '0;
    r_grant_valid <= 1'b0;
end
```

5.8.2 SRAM Reset

SRAM contents: Undefined after reset (no initialization required)

SRAM pointers:

```
if (!aresetn) begin
    wr_ptr <= '0;
    rd_ptr <= '0;
end
```

5.9 Clock Gating (Optional)

For power optimization in ASIC implementations:

5.9.1 Per-Channel Clock Gating

```
// Clock gate when channel idle
clock_gate_ctrl u_ch0_clk_gate (
    .clk_in(aclk),
    .cfg_cg_enable(ch0_enable),
    .clk_out(ch0_gated_clk)
);

// Use gated clock for channel logic
scheduler #(.CHANNEL_ID(0)) u_ch0_sched (
    .clk(ch0_gated_clk),  // Gated clock
    .rst_n(aresetn),
    // ...
);
```

Note: Clock gating typically not used in FPGA implementations (tutorial focus).

5.10 Verification Requirements

5.10.1 Clock Checks

Testbench must verify: - [Done] Clock period consistent - [Done] Clock duty cycle 50% tolerance - [Done] No glitches on clock - [Done] Setup/hold times met

5.10.2 Reset Checks

Testbench must verify: - [Done] All registers initialize to known state - [Done] Reset assertion clears FSMs to IDLE - [Done] Reset deassertion synchronous to clock - [Done] Minimum reset duration (10 cycles) enforced - [Done] Operations don't start until stabilization complete

5.10.3 CDC Checks

For APB CDC (if present): - [Done] No metastability violations - [Done] Data integrity across domains - [Done] Proper flag synchronization

---

5.11 Example Reset Testbench

```
# CocoTB testbench pattern
class StreamTB(TBBase):
    async def setup_clocks_and_reset(self):
        """Complete clock and reset initialization"""
        # Start clocks
        await self.start_clock('aclk', freq=10, units='ns')  # 100 MHz
        await self.start_clock('pclk', freq=20, units='ns')  # 50 MHz
(async)

        # Assert reset
        await self.assert_reset()

        # Hold reset for 10 aclk cycles
        await self.wait_clocks('aclk', 10)

        # Deassert reset (synchronous to aclk)
        await self.deassert_reset()

        # Stabilization period
        await self.wait_clocks('aclk', 5)

        # Ready for operation

    async def assert_reset(self):
        """Assert both resets"""
        self.dut.aresetn.value = 0
        self.dut.presetn.value = 0

    async def deassert_reset(self):
        """Deassert both resets synchronously"""
        # Wait for rising edge of aclk
        await RisingEdge(self.dut.aclk)
        self.dut.aresetn.value = 1

        # Wait for rising edge of pclk
        await RisingEdge(self.dut.pclk)
        self.dut.presetn.value = 1
```

5.12 Related Documentation

- **Scheduler FSM:** `fub_02_scheduler.md` - Reset behavior
- **APB Config:** `mac_02_apb_config.md` - CDC implementation

- **Top-Level: `mac_04_stream_top.md` - Clock/reset integration**

5.13 Revision History

Clocks and Reset Specification Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers

Last Updated: 2026-01-02

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6 STREAM Core Specification

Module: `stream_core.sv` **Location:** `projects/components/dma-ip/stream/rtl/macro/`
Status: Implemented **Last Updated:** 2026-05-11

6.1 Overview

The STREAM Core is the top-level integration module that combines all STREAM components into a complete scatter-gather DMA engine. It provides 8 or 16 independent channels (`NUM_CHANNELS`; the shipped tops are `stream_top_ch8` and `stream_top_ch16`) with descriptor-based memory-to-memory transfers.

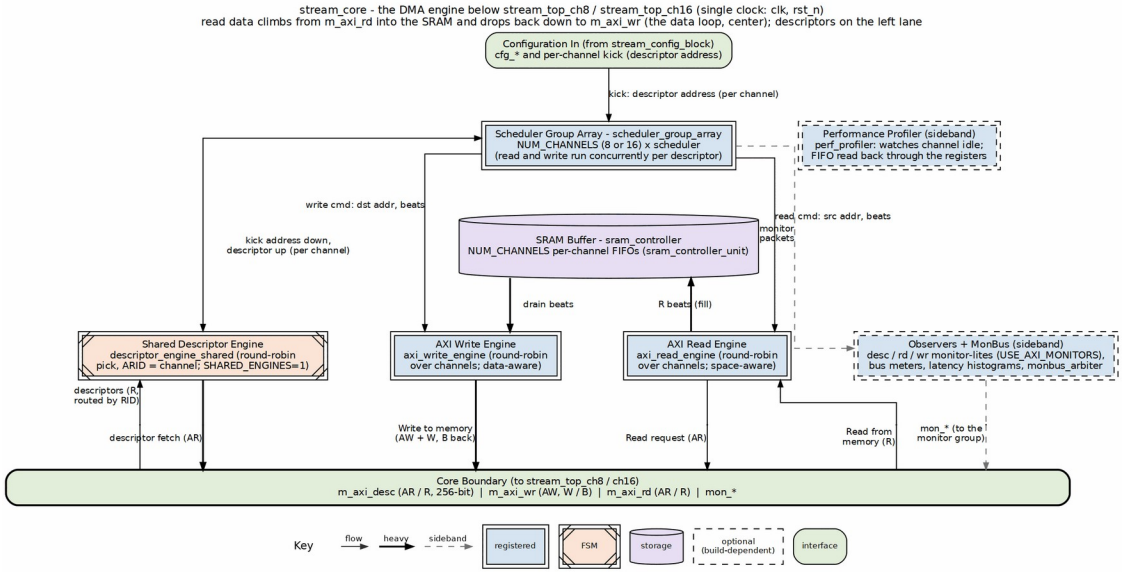
6.1.1 Key Features

- **8 or 16 Independent Channels:** Concurrent descriptor-based transfers
- **Shared Descriptor Engine:** By default (`SHARED_ENGINES = 1`), one `descriptor_engine_shared` fetches for every channel; see Scheduler Group Array
- **Shared AXI Masters:** Three shared AXI4 masters for efficiency
  - Descriptor fetch (256-bit fixed)
  - Data read (parameterizable, default 512-bit)
  - Data write (parameterizable, default 512-bit)
- **Per-Channel Buffering:** Independent FIFO per channel (512 entries default)
- **Performance Monitoring:** Integrated profiler with FIFO readout
- **AXI Skid Buffers:** Timing closure on all external interfaces
- **Unified MonBus:** Single monitor bus output for all events

6.1.2 Block Diagram

The STREAM Core integrates the following major components:

6.1.3 Figure 2.1.1: STREAM Core Block Diagram



STREAM Core Block Diagram

Source: 01_stream_core.dot

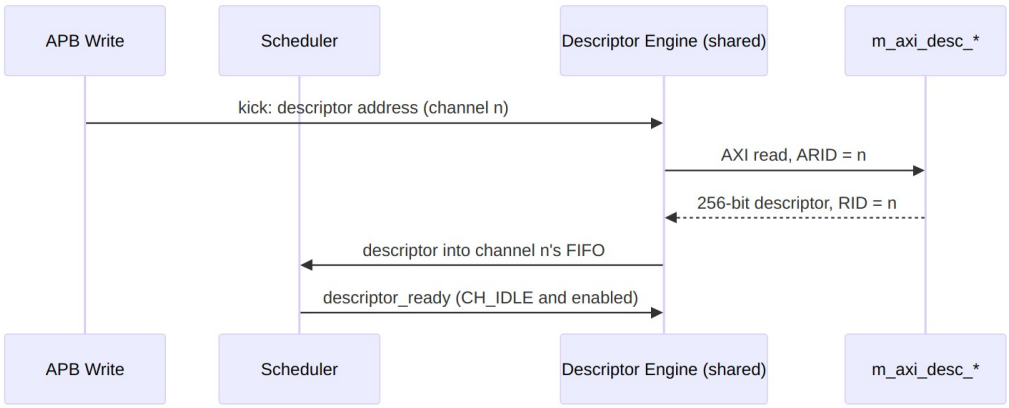
6.2 Architecture

6.2.1 Component Hierarchy

1. **scheduler_group_array** - Top scheduler layer
   - 8 × scheduler_group instances
   - Shared descriptor engine
   - Descriptor fetch arbitration
2. **sram_controller** - Buffering layer
   - 8 × independent FIFOs (gaxi_fifo_sync)
   - Per-channel allocation controllers
   - Per-channel drain controllers
3. **axi_read_engine** - Read datapath
   - Shared AXI master for all channels
   - ID-based routing to SRAM
   - Space allocation flow control
4. **axi_write_engine** - Write datapath
   - Shared AXI master for all channels
   - ID-based routing from SRAM
   - Drain reservation flow control
5. **perf_profiler** - Performance monitoring
   - Transaction counting
   - Bandwidth tracking
   - FIFO readout interface
6. **AXI Skid Buffers** - Timing closure
   - Descriptor AXI (AR/R)
   - Read AXI (AR/R)
   - Write AXI (AW/W/B)

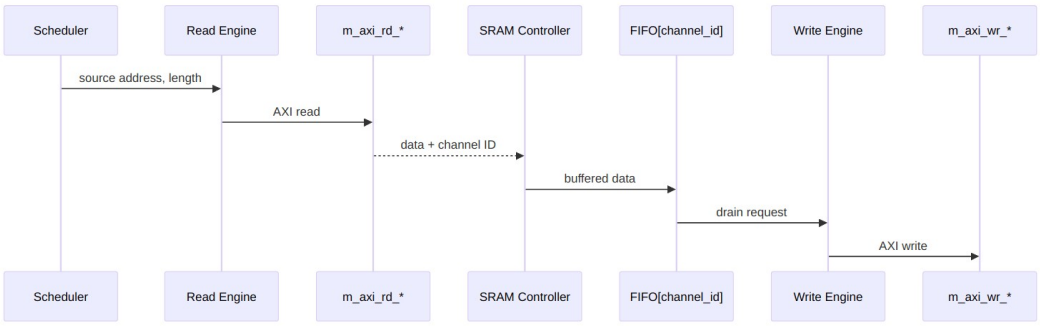
6.2.2 Data Flow

Descriptor Fetch Flow:



.1.2: Descriptor Fetch Flow

Data Transfer Flow:



.1.3: Data Transfer Flow

6.3 Parameters

6.3.1 Primary Configuration

Primary Configuration

Parameter	Type	Default	Description
NUM_CHANNELS	int	8	Number of independent DMA channels
CHAN_WIDTH	int	$clog2(NUM_CHANNELS)	Channel ID width (3 for 8 channels, 4 for 16)
ADDR_WIDTH	int	64	Address bus width
DATA_WIDTH	int	512	Data bus width (must match memory interface)
AXI_ID_WIDTH	int	8	AXI transaction ID width
FIFO_DEPTH	int	512	Per-channel FIFO depth
SHARED_ENGINES	int	1	1: one shared descriptor engine; 0: one descriptor_engine per channel behind an AR arbiter

6.3.2 Monitor Control

Monitor Control

Parameter	Type	Default	Description
USE_AXI_MONITORS	int	1	Enable (1) or disable (0) AXI transaction monitors

Note: USE_AXI_MONITORS gates the monitor-lite logic and the latency histograms. It does **not** gate the bus meters. When USE_AXI_MONITORS = 0: - The monitor-lite logic inside the data-master taps (axi4_master_rd_monlite / axi4_master_wr_monlite, USE_MONITOR = 0) is not built. The taps shrink to bare skid buffers. The descriptor-port axi4_master_rd_monlite in scheduler_group_array is also left unbuilt, and it is off by default even with monitors on (USE_DESC_AXI_MONITOR). The configuration inputs of these monitors are tied off internally. - The per-transaction latency histograms (axi_perf_latency_hist x 2) are not built. - The **bus meters stay built**. The two in-core axi_bus_meter instances (RDMON/WRMON_PERF productive / backpressure / starvation / idle buckets) and the beat, byte, burst and window counters do not depend on USE_AXI_MONITORS, and the perf-window RUN bits pass straight from the registers. A

monitors-off build can still measure datapath utilisation, with no monitor-lite or histogram area. - The per-channel completion/error MonBus emitters are controlled by GEN_MON, not by USE_AXI_MONITORS. - The meters' registers sit in the MON window (0x1000). At top level, USE_MON_REGS defaults to 1, independent of USE_AXI_MONITORS, so a default monitors-off build answers the window and the meters can be armed and read. The monitor configuration registers in the same window (DAXMON / RDMON / WRMON enable, masks, timeouts) then read back what was written but configure nothing, because the monitor-lite logic is not built. Setting USE_MON_REGS = 0 straps those outputs off and makes every MON-window access return PSLVERR, which also makes the meters unreachable.

6.3.3 Outstanding Transaction Limits

Outstanding Transaction Limits

Parameter	Default	Description
AR_MAX_OUTSTANDING	8	Maximum concurrent read address requests
AW_MAX_OUTSTANDING	8	Maximum concurrent write address requests

6.3.4 AXI Skid Buffer Depths

AXI Skid Buffer Depths

Parameter	Default	Purpose
SKID_DEPTH_AR	2	AR channel timing closure
SKID_DEPTH_R	4	R channel timing closure
SKID_DEPTH_AW	2	AW channel timing closure
SKID_DEPTH_W	4	W channel timing closure
SKID_DEPTH_B	2	B channel timing closure

Note: Deeper buffers on data channels (R/W) improve throughput.

6.3.5 MonBus Agent IDs

MonBus Agent IDs

Parameter	Default	Description
DESC_MON_BASE_AGENT_ID	16 (0x10)	Descriptor engines base (16-23)
SCHED_MON_BASE_AGENT_ID	48 (0x30)	Schedulers base (48-55)
DESC_AXI_MON_AGENT_ID	8 (0x08)	Descriptor AXI master monitor
MON_UNIT_ID	1 (0x1)	Unit ID for all STREAM events

6.4 Port List

6.4.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low asynchronous reset

6.4.2 APB Programming Interface

Per-channel descriptor kick-off interface:

APB Programming Interface

Signal	Direction	Width	Description
`apb_valid[ch]`	input	NUM_CHANNELS	Channel descriptor address valid
`apb_ready[ch]`	output	NUM_CHANNELS	Channel ready to accept descriptor address
`apb_addr[ch]`	input	NUM_CHANNELS × ADDR_WIDTH	Descriptor address per channel

Usage:

```
// Kick off channel 0 with descriptor at 0x1000_0000
apb_valid[0] = 1'b1;
apb_addr[0] = 64'h0000_0000_1000_0000;
// Wait for handshake
wait (apb_ready[0]);
apb_valid[0] = 1'b0;
```

6.4.3 Configuration Interface

Per-Channel Configuration:

Configuration Interface

Signal	Direction	Width	Description
cfg_channel_enable[ch]	input	NUM_CHANNELS	Enable channel
cfg_channel_reset[ch]	input	NUM_CHANNELS	Soft reset channel (FSM → IDLE)

Global Scheduler Configuration:

Configuration Interface

Signal	Direction	Width	Description
cfg_sched_enable	input	1	Global scheduler enable
cfg_sched_timeout_cycles	input	16	Timeout threshold (cycles)
cfg_sched_timeout_enable	input	1	Enable timeout detection
cfg_sched_err_enable	input	1	Enable error event reporting
cfg_sched_compl_enable	input	1	Enable completion event reporting
cfg_sched_perf_enable	input	1	Enable performance event reporting

Descriptor Engine Configuration:

Configuration Interface

Signal	Direction	Width	Description
cfg_desceng_enable	input	1	Enable descriptor engine
cfg_desceng_prefetch	input	1	Enable descriptor prefetch
cfg_desceng_fifo_thresh	input	4	FIFO threshold for prefetch
cfg_desceng_addr0_base	input	ADDR_WIDTH	Base address limit 0
cfg_desceng_	input	ADDR_WIDT	Limit address limit 0

Signal	Direction	Width	Description
addr0_limit		H	
cfg_desceng_addr1_base	input	ADDR_WIDTH	Base address limit 1
cfg_desceng_addr1_limit	input	ADDR_WIDTH	Limit address limit 1

AXI Monitor Configuration:

Three identical sets of monitor config signals (descriptor, read, write):

Configuration Interface

Signal Prefix	Applies To	Description
cfg_desc_mon_*	Descriptor AXI	Descriptor fetch monitoring
cfg_rdeng_mon_*	Read AXI	Data read monitoring
cfg_wreng_mon_*	Write AXI	Data write monitoring

Each monitor has:

Signal Suffix	Width	Description
_enable	1	Enable monitor
_err_enable	1	Enable error reporting
_perf_enable	1	Enable performance reporting
_timeout_enable	1	Enable timeout detection
_timeout_cycles	32	Timeout threshold
_latency_thresh	32	Latency threshold for events
_pkt_mask	16	Packet type mask
_err_select	4	Error type selector
_err_mask	8	Error event mask

Signal Suffix	Width	Description
_timeout_mask	8	Timeout event mask
_compl_mask	8	Completion event mask
_thresh_mask	8	Threshold event mask
_perf_mask	8	Performance event mask
_addr_mask	8	Address event mask
_debug_mask	8	Debug event mask

AXI Transfer Configuration:

Signal	Direction	Width	Description
cfg_axi_rd_xfer_beats	input	8	Read burst size (beats)
cfg_axi_wr_xfer_beats	input	8	Write burst size (beats)

Performance Profiler Configuration:

Signal	Direction	Width	Description
cfg_perf_enable	input	1	Enable profiler
cfg_perf_mode	input	1	Profiler mode (0=count, 1=latency)
cfg_perf_clear	input	1	Clear profiler counters

6.4.4 Status Interface

System-Level Status:

Status Interface

Signal	Direction	Width	Description
system_idle	output	1	All channels idle (AND of all scheduler_idle)

Per-Channel Status:

Status Interface

Signal	Direction	Width	Description
descriptor_engine_idle[ch]	output	NUM_CHANNELS	Descriptor engine idle
scheduler_idle[ch]	output	NUM_CHANNELS	Scheduler in IDLE state
scheduler_state[ch]	output	NUM_CHANNELS × 7	Scheduler FSM state (ONE-HOT)
sched_error[ch]	output	NUM_CHANNELS	Scheduler error (sticky)
axi_rd_all_complete[ch]	output	NUM_CHANNELS	All read transactions complete
axi_wr_all_complete[ch]	output	NUM_CHANNELS	All write transactions complete

Performance Profiler Status:

Status Interface

Signal	Direction	Width	Description
perf_fifo_empty	output	1	Profiler FIFO empty
perf_fifo_full	output	1	Profiler FIFO full
perf_fifo_count	output	16	Profiler FIFO occupancy
perf_fifo_rd	input	1	Read profiler entry
perf_fifo_data_low	output	32	Profiler data [31:0]
perf_fifo_data_high	output	32	Profiler data [63:32]

6.4.5 AXI4 Master - Descriptor Fetch (256-bit)

AR Channel:

AXI4 Master - Descriptor Fetch

Signal	Direction	Width	Description
m_axi_desc_arid	output	AXI_ID_WIDTH	Transaction ID
m_axi_desc_araddr	output	ADDR_WIDTH	Address
m_axi_desc_arlen	output	8	Burst length - 1
m_axi_desc_arsize	output	3	Burst size (log2 bytes)
m_axi_desc_arburst	output	2	Burst type (INCR)
m_axi_desc_arlock	output	1	Lock type
m_axi_desc_arcache	output	4	Cache attributes
m_axi_desc_arprot	output	3	Protection attributes
m_axi_desc_arqos	output	4	QoS value
m_axi_desc_arregion	output	4	Region identifier
m_axi_desc_aruser	output	CHAN_WIDTH	User signal (channel ID)
m_axi_desc_arvalid	output	1	Address valid
m_axi_desc_arready	input	1	Address ready

R Channel:

AXI4 Master - Descriptor Fetch

Signal	Direction	Width	Description
m_axi_desc_rid	input	AXI_ID_WIDTH	Transaction ID
m_axi_desc_rdata	input	256	Read data (FIXED 256-bit)
m_axi_desc_r	input	2	Response

Signal	Direction	Width	Description
resp			(OKAY/EXOKAY/SLVERR/DECERR)
m_axi_desc_rlast	input	1	Last beat of burst
m_axi_desc_ruser	input	CHAN_WIDTH	User signal (channel ID)
m_axi_desc_rvalid	input	1	Read data valid
m_axi_desc_rready	output	1	Read data ready

6.4.6 AXI4 Master - Data Read (Parameterizable Width)

AR Channel:

AXI4 Master - Data Read

Signal	Direction	Width	Description
m_axi_rd_arid	output	AXI_ID_WIDTH	Transaction ID
m_axi_rd_araddr	output	ADDR_WIDTH	Address
m_axi_rd_arlen	output	8	Burst length - 1
m_axi_rd_arsize	output	3	Burst size (log2 bytes)
m_axi_rd_arburst	output	2	Burst type (INCR)
m_axi_rd_arlock	output	1	Lock type
m_axi_rd_arcache	output	4	Cache attributes
m_axi_rd_arprot	output	3	Protection attributes
m_axi_rd_arqos	output	4	QoS value
m_axi_rd_arregion	output	4	Region identifier
m_axi_rd_aruser	output	CHAN_WIDTH	User signal

Signal	Direction	Width	Description
			(channel ID)
m_axi_rd_arvalid	output	1	Address valid
m_axi_rd_arready	input	1	Address ready

R Channel:

AXI4 Master - Data Read

Signal	Direction	Width	Description
m_axi_rd_rid	input	AXI_ID_WIDTH	Transaction ID
m_axi_rd_rdata	input	DATA_WIDTH	Read data (default 512-bit)
m_axi_rd_rresp	input	2	Response
m_axi_rd_rlast	input	1	Last beat of burst
m_axi_rd_ruser	input	CHAN_WIDTH	User signal (channel ID)
m_axi_rd_rvalid	input	1	Read data valid
m_axi_rd_rready	output	1	Read data ready

6.4.7 AXI4 Master - Data Write (Parameterizable Width)

AW Channel:

AXI4 Master - Data Write

Signal	Direction	Width	Description
m_axi_wr_awid	output	AXI_ID_WIDTH	Transaction ID
m_axi_wr_awaddr	output	ADDR_WIDTH	Address
m_axi_wr_awlen	output	8	Burst length - 1
m_axi_wr_awsize	output	3	Burst size (log2

Signal	Direction	Width	Description
			bytes)
m_axi_wr_awburst	output	2	Burst type (INCR)
m_axi_wr_awlock	output	1	Lock type
m_axi_wr_awcache	output	4	Cache attributes
m_axi_wr_awprot	output	3	Protection attributes
m_axi_wr_awqos	output	4	QoS value
m_axi_wr_awregion	output	4	Region identifier
m_axi_wr_awuser	output	CHAN_WIDTH	User signal (channel ID)
m_axi_wr_awvalid	output	1	Address valid
m_axi_wr_awready	input	1	Address ready

W Channel:

AXI4 Master - Data Write

Signal	Direction	Width	Description
m_axi_wr_wdata	output	DATA_WIDTH	Write data (default 512-bit)
m_axi_wr_wstrb	output	DATA_WIDTH/8	Write strobes (byte enables)
m_axi_wr_wlast	output	1	Last beat of burst
m_axi_wr_wuser	output	CHAN_WIDTH	User signal (channel ID)
m_axi_wr_wvalid	output	1	Write data valid
m_axi_wr_wready	input	1	Write data ready

B Channel:

AXI4 Master - Data Write

Signal	Direction	Width	Description
m_axi_wr_bid	input	AXI_ID_WIDTH	Transaction ID
m_axi_wr_bresp	input	2	Response
m_axi_wr_buser	input	CHAN_WIDTH	User signal (channel ID)
m_axi_wr_bvalid	input	1	Response valid
m_axi_wr_bready	output	1	Response ready

6.4.8 Status/Debug Outputs

Descriptor AXI Monitor:

Status/Debug Outputs

Signal	Direction	Width	Description
cfg_sts_desc_mon_busy	output	1	Monitor busy
cfg_sts_desc_mon_active_txns	output	8	Active transaction count
cfg_sts_desc_mon_error_count	output	16	Error count
cfg_sts_desc_mon_txn_count	output	32	Total transaction count
cfg_sts_desc_mon_conflict_error	output	1	ID conflict detected

Read Engine AXI Monitor:

Status/Debug Outputs

Signal	Direction	Width	Description
cfg_sts_rdeng_skid_busy	output	1	Skid buffer busy

Signal	Direction	Width	Description
cfg_sts_rdeng_mon_active_txns	output	8	Active transaction count
cfg_sts_rdeng_mon_error_count	output	16	Error count
cfg_sts_rdeng_mon_txn_count	output	32	Total transaction count
cfg_sts_rdeng_mon_conflict_error	output	1	ID conflict detected

Write Engine AXI Monitor:

Status/Debug Outputs

Signal	Direction	Width	Description
cfg_sts_wreng_skid_busy	output	1	Skid buffer busy
cfg_sts_wreng_mon_active_txns	output	8	Active transaction count
cfg_sts_wreng_mon_error_count	output	16	Error count
cfg_sts_wreng_mon_txn_count	output	32	Total transaction count
cfg_sts_wreng_mon_conflict_error	output	1	ID conflict detected

6.4.9 Unified Monitor Bus Interface

Unified Monitor Bus Interface

Signal	Direction	Width	Description
mon_valid	output	1	Monitor packet valid
mon_ready	input	1	Monitor packet ready
mon_packet	output	64	Monitor packet

Signal	Direction	Width	Description
			data

MonBus Sources: - Descriptor engines (8 sources, agent IDs 16-23) - Schedulers (8 sources, agent IDs 48-55) - Descriptor AXI monitor (agent ID 8) - Read AXI monitor (configurable) - Write AXI monitor (configurable)

6.5 Operation

6.5.1 Transfer Initialization

Step 1: Configuration

```
1. Configure global settings (timeouts, monitors, transfer sizes)
2. Enable descriptor engine (cfg_desceng_enable = 1)
3. Enable scheduler (cfg_sched_enable = 1)
4. Enable target channel (cfg_channel_enable[ch] = 1)
```

Step 2: Descriptor Kick-off

```
1. Assert apb_valid[ch] = 1
2. Provide descriptor address on apb_addr[ch]
3. Wait for apb_ready[ch] handshake
4. Deassert apb_valid[ch]
```

Step 3: Automatic Transfer

```
1. Descriptor engine fetches descriptor via m_axi_desc_*
2. Scheduler receives descriptor, starts transfer
3. Read engine: memory → SRAM (via m_axi_rd_*)
4. Write engine: SRAM → memory (via m_axi_wr_*)
5. Scheduler reports completion via MonBus
```

6.5.2 Monitoring Transfer Progress

Check Scheduler State:

```
// Monitor scheduler state
case (scheduler_state[ch])
    7'b0000001: // CH_IDLE - waiting for descriptor
    7'b0000010: // CH_FETCH_DESC - fetching descriptor
    7'b0000100: // CH_XFER_DATA - transfer in progress
    7'b0001000: // CH_COMPLETE - transfer complete
    7'b0010000: // CH_NEXT_DESC - chaining to next
    7'b0100000: // CH_ERROR - error occurred
endcase
```

Check Completion:

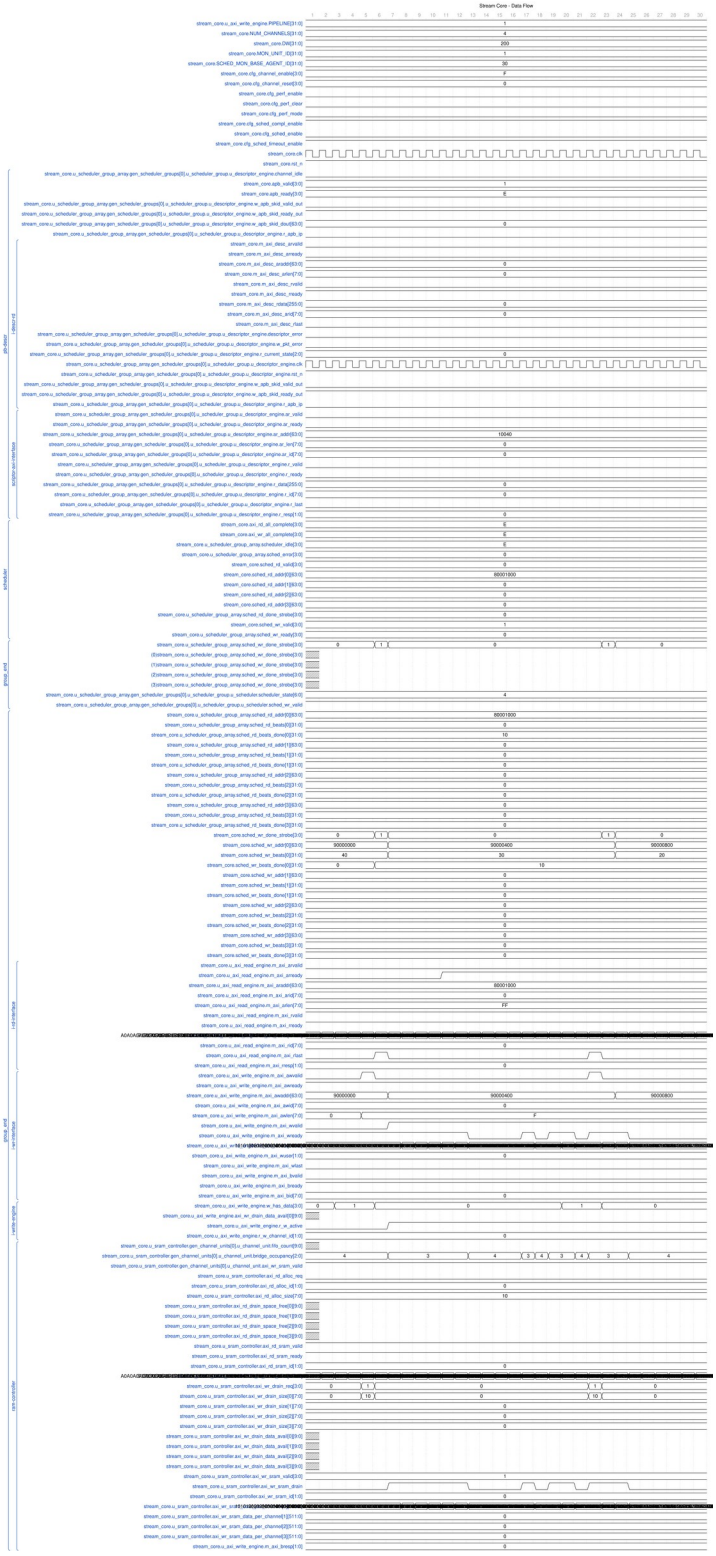
```
// All complete when:
complete = scheduler_idle[ch] &&
           axi_rd_all_complete[ch] &&
           axi_wr_all_complete[ch];
```

6.6 Timing Diagrams

6.6.1 Complete Data Flow Timing

The following diagram shows the complete data flow through the STREAM core, from descriptor kick-off through data transfer completion:

6.6.1.1 Waveform 2.1.1: STREAM Core Data Flow



STREAM Core Data Flow

Source: stream_core_data_flow.json

6.7 Testing

Test Location: `projects/components/dma-ip/stream/dv/tests/top/`

Key Test Scenarios:

1. **Single channel transfer** - Basic end-to-end operation
2. **Multi-channel concurrent** - 2 to 16 channels simultaneously (the 16-channel top runs the high channels and all channels at once)
3. **Descriptor chaining** - 2-5 descriptors linked
4. **FIFO overflow prevention** - Large transfer with small FIFO
5. **Error handling** - AXI errors, timeouts
6. **Performance profiling** - Bandwidth measurements
7. **MonBus event checking** - Verify all events reported

Test Configuration:

```
# Basic configuration
NUM_CHANNELS = 4
DATA_WIDTH = 128
FIFO_DEPTH = 512
cfg_axi_rd_xfer_beats = 16
cfg_axi_wr_xfer_beats = 16
```

6.8 Resource Utilization

Measured (out-of-context Vivado, 256-bit data, monitors off, 100 MHz):

Resource Utilization

Build	LUTs	Flops	Worst slack
8 channels, `SHARED_ENGINES = 0`	25.8k	14.9k	+3.24 ns
8 channels, `SHARED_ENGINES = 1`	20.7k	11.7k	+2.50 ns
16 channels, `SHARED_ENGINES = 1`	36.8k	19.4k	+1.79 ns

Critical Paths: - AXI handshake paths (improved by skid buffers) - SRAM address decode - MonBus arbiter

---

6.9 Integration Example

```
stream_core #(
    .NUM_CHANNELS(8),
    .DATA_WIDTH(512),
    .FIFO_DEPTH(512)
) u_stream (
    .clk                    (system_clk),
    .rst_n                  (system_rst_n),

    // APB kick-off
    .apb_valid              (stream_apb_valid),
    .apb_ready              (stream_apb_ready),
    .apb_addr               (stream_apb_addr),

    // Configuration
    .cfg_channel_enable     (stream_ch_enable),
    .cfg_sched_enable       (1'b1),
    .cfg_axi_rd_xfer_beats  (8'd16),
    .cfg_axi_wr_xfer_beats  (8'd16),
    // ... other config

    // AXI Descriptor Master
    .m_axi_desc_arid        (desc_arid),
    .m_axi_desc_araddr      (desc_araddr),
    // ... full AXI AR/R

    // AXI Read Master
    .m_axi_rd_arid          (rd_arid),
    .m_axi_rd_araddr        (rd_araddr),
    // ... full AXI AR/R

    // AXI Write Master
    .m_axi_wr_awid          (wr_awid),
    .m_axi_wr_awaddr        (wr_awaddr),
    // ... full AXI AW/W/B

    // MonBus
    .mon_valid              (stream_mon_valid),
    .mon_ready              (stream_mon_ready),
    .mon_packet             (stream_mon_packet)
);
```

6.10 Related Documentation

- **Scheduler Group Array:** `02_scheduler_group_array.md` - Multi-channel scheduler integration
- **Scheduler Group:** `03_scheduler_group.md` - Single channel scheduler + descriptor engine
- **Scheduler:** `04_scheduler.md` - Channel state machine and transfer coordination
- **Descriptor Engine:** `05_descriptor_engine.md` - Descriptor fetch and parsing
- **AXI Read Engine:** `06_axi_read_engine.md` - Read datapath
- **Stream Alloc Ctrl:** `07_stream_alloc_ctrl.md` - Space allocation controller
- **SRAM Controller:** `08_sram_controller.md` - Per-channel buffering
- **SRAM Controller Unit:** `09_sram_controller_unit.md` - Single channel SRAM unit
- **Stream Latency Bridge:** `10_stream_latency_bridge.md` - Timing bridge
- **Stream Drain Ctrl:** `11_stream_drain_ctrl.md` - Drain flow controller
- **AXI Write Engine:** `12_axi_write_engine.md` - Write datapath
- **Performance Profiler:** `15_perf_profiler.md` - Performance monitoring

- **MonBus AXI-Lite Group: `16_monbus_axil_group.md` - Monitor bus arbitration**

6.11 Revision History

STREAM Core Specification Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers
0.92	2026-10-17	seang	8 or 16 channels; `SHARED_ENGINES` parameter and shared descriptor fetch flow; measured area

Last Updated: 2026-10-17

7 Scheduler Group Array

Module: `scheduler_group_array.sv` **Location:** `projects/components/dma-ip/stream/rtl/macro/` **Category:** MACRO (Top-level Integration)
Parent: `stream_core.sv` **Status:** Implemented **Last Updated:** 2026-10-17

7.1 Overview

The scheduler_group_array is the scheduler layer of the STREAM engine: one scheduler per channel, the descriptor fetch that feeds them, and the MonBus aggregation for the layer. SHARED_ENGINES selects how descriptor fetch is built:

- **SHARED_ENGINES = 1 (default):** one descriptor_engine_shared serves every channel. It picks among the schedulers round-robin and owns the descriptor AXI master outright.
- **SHARED_ENGINES = 0:** one scheduler_group per channel, each with its own descriptor_engine, and a round-robin arbiter merging their AR requests onto the descriptor AXI master (the original structure).

Both structures present the same ports, so stream_core and everything above it are unchanged. A build has 8 channels (stream_top_ch8) or 16 (stream_top_ch16).

7.1.1 Key Features

- **Independent channels:** each channel has its own scheduler FSM.
- **Descriptor fetch:** one shared engine, or one engine per channel behind a round-robin arbiter. Either way, the channel number is the AXI ID, so responses route back by RID.
- **Per-channel data interfaces:** passed straight to the AXI engines, with no arbitration at this level.
- **Unified MonBus aggregation:** NUM_CHANNELS + 2 sources with shared engines (one per scheduler, the shared engine, the descriptor AXI monitor), NUM_CHANNELS + 1 with per-channel engines (one per scheduler group, the monitor).

7.1.2 Simplified from RAPIDS

This module is intentionally simplified from RAPIDS for tutorial focus: - No control read/write engines - No program engine - No network interfaces (EOS completion) - No alignment bus - Direct data path interfaces (beats-based, no chunks)

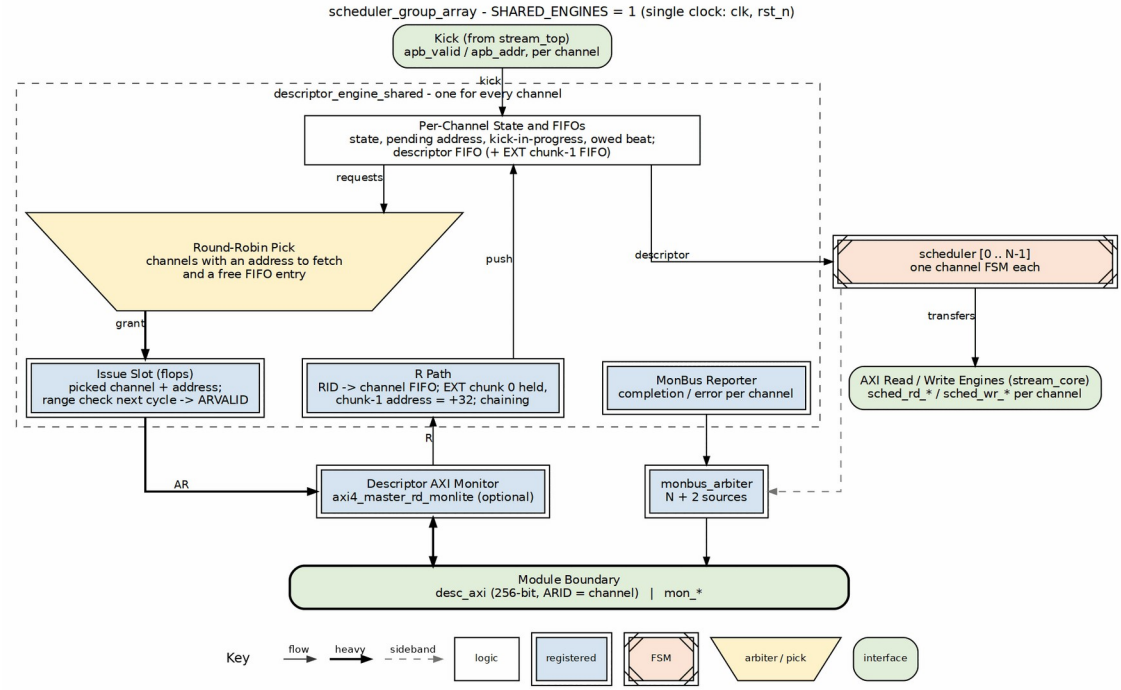
7.2 Architecture

7.2.1 Component Hierarchy

```
scheduler_group_array  (SHARED_ENGINES = 1, default)
   scheduler[0..NUM_CHANNELS-1]   # Channel FSMs
   descriptor_engine_shared       # One fetch engine for every channel
   axi4_master_rd_monlite         # Descriptor AXI monitor
(USE_DESC_AXI_MONITOR)
   monbus_arbiter                 # NUM_CHANNELS + 2 sources

scheduler_group_array  (SHARED_ENGINES = 0)
   scheduler_group[0..NUM_CHANNELS-1]
      scheduler                  # Channel FSM
      descriptor_engine          # Descriptor fetch, per channel
   arbiter_round_robin            # AR channel arbitration
   axi4_master_rd_monlite         # Descriptor AXI monitor
   monbus_arbiter                 # NUM_CHANNELS + 1 sources
```

7.2.2 Figure 2.2.1: Scheduler Group Array, Shared Engine



Scheduler group array with the shared descriptor engine

Source: 02_scheduler_group_array.dot

7.2.3 Shared Descriptor Engine

With `SHARED_ENGINES = 1`, one `descriptor_engine_shared` replaces the per-channel engines. Each scheduler sees exactly the per-channel engine's behaviour: the same kick policy, refusal of a kicked address outside both ranges, the EXT second-chunk fetch, the prefetch throttle, autonomous chaining, the error pulse, channel reset, and the completion and error MonBus packets.

Shared Descriptor Engine

Shared by every channel	Kept per channel
The AR issuer, picked round-robin among channels with an address to fetch	A state code and one pending fetch address
The address-range comparators	The kick-in-progress flag and the owed-beat flag
The R path: each beat routed by RID straight into its channel's FIFO	The descriptor FIFO (and the EXT chunk-1 FIFO)
The MonBus reporter	

- **One fetch in flight per channel,** in order. Different channels' fetches overlap on the bus because each carries its own ARID.
- **Issue slot:** the pick and the address mux end at a register, and the range check runs on the registered address in the next cycle, so the 16-way pick and the 64-bit compares are not one path. The second chunk's +32 is applied on the R side when chunk 0 arrives (one beat per cycle, one shared adder).
- **A beat for a channel that is not waiting** (one reset mid-fetch) is taken and dropped rather than left on the shared R channel; the channel takes no new kick until that owed beat has drained.

Measured out-of-context, `stream_core` at 256 bits with monitors off and a 100 MHz clock:

Area and Timing by Structure

Build	LUTs	Flops	Worst slack
8 channels, per-channel engines	25.8k	14.9k	+3.24 ns
8 channels, shared engine	20.7k	11.7k	+2.50 ns
16 channels,	36.8k	19.4k	+1.79 ns

Build	LUTs	Flops	Worst slack
shared engine			

On the Genesys 2 board build (8 channels, 128 bits, `GEN_MON = 0`, placed and routed) the eight per-channel engines take 4,974 LUTs and 4,120 flip-flops, and the shared engine 3,825 LUTs and 1,725 flip-flops, of which 1,840 LUTs are the per-channel FIFOs in distributed RAM. The saving there is smaller than the out-of-context figure because `GEN_MON = 0` removes the per-channel MonBus reporters from both builds. The full before-and-after is in the HAS Resource Estimates chapter.

7.2.4 Descriptor AXI Arbitration (`SHARED_ENGINES = 0`)

AR Channel Flow:

```
Channel 0..N-1 AR Requests -> Round-Robin Arbiter -> Single AR to
External AXI
                          |
               Embed channel_id in AXI ID lower bits
```

R Channel Demux:

```
External R Response -> Extract channel_id from RID[CHAN_WIDTH-1:0] ->
Route to correct channel
```

Critical: Channel ID is embedded in the lower bits of the AXI ID field, in both structures:

```
// In AR mux
desc_axi_int_arid = {{(AXI_ID_WIDTH-CHAN_WIDTH){1'b0}}, ch[CHAN_WIDTH-
1:0]};

// In R demux
desc_r_channel_id = desc_axi_int_rid[CHAN_WIDTH-1:0];
```

7.2.5 Data Path Interfaces

IMPORTANT: Data read and write interfaces are NOT arbitrated at this level!

- `sched_rd_*` - Per-channel arrays passed directly to `axi_read_engine`
- `sched_wr_*` - Per-channel arrays passed directly to `axi_write_engine`

The AXI engines themselves handle channel arbitration internally.

7.3 Parameters

Parameters

Parameter	Type	Default	Description
NUM_CHANNELS	int	8	Number of DMA channels (8 or 16 in the shipped tops)
CHAN_WIDTH	int	$clog2(NUM_CHANNELS)	Channel ID width (3 bits for 8 channels, 4 for 16)
SHARED_ENGINES	int	1	1: one shared descriptor engine; 0: one per channel
ADDR_WIDTH	int	64	Address bus width
DATA_WIDTH	int	512	Data bus width
AXI_ID_WIDTH	int	8	AXI transaction ID width

7.3.1 Monitor Bus Agent IDs

Monitor Bus Agent IDs

Parameter	Default	Description
DESC_MON_BASE_AGENT_ID	16 (0x10)	Descriptor engine base (agent = base + channel, both structures)
SCHED_MON_BASE_AGENT_ID	48 (0x30)	Scheduler base (agent = base + channel)
DESC_AXI_MON_AGENT_ID	8 (0x08)	Descriptor AXI monitor
MON_UNIT_ID	1	Unit ID for all STREAM events
MON_MAX_TRANSACTIONS	16	Max outstanding transactions for monitor

7.4 Port List

7.4.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low asynchronous reset

7.4.2 APB Programming Interface (Per-Channel)

APB Programming Interface

Signal	Direction	Width	Description
apb_valid[ch]	input	NUM_CHANNELS	Descriptor address valid
apb_ready[ch]	output	NUM_CHANNELS	Ready to accept descriptor
apb_addr[ch]	input	NUM_CHANNELS x ADDR_WIDTH	Descriptor address

7.4.3 Configuration Interface (Per-Channel)

Configuration Interface

Signal	Direction	Width	Description
cfg_channel_enable[ch]	input	NUM_CHANNELS	Enable channel
cfg_channel_reset[ch]	input	NUM_CHANNELS	Soft reset channel

7.4.4 Global Scheduler Configuration

Global Scheduler Configuration

Signal	Direction	Width	Description
cfg_sched_enable	input	1	Master scheduler

Signal	Direction	Width	Description
			enable
cfg_sched_timeout_cycles	input	16	Timeout threshold
cfg_sched_timeout_enable	input	1	Enable timeout detection
cfg_sched_err_enable	input	1	Enable error reporting
cfg_sched_compl_enable	input	1	Enable completion reporting
cfg_sched_perf_enable	input	1	Enable performance monitoring

7.4.5 Descriptor Engine Configuration

Descriptor Engine Configuration

Signal	Direction	Width	Description
cfg_desceng_enable	input	1	Enable descriptor engine
cfg_desceng_prefetch	input	1	Enable descriptor prefetch
cfg_desceng_fifo_thresh	input	4	FIFO threshold for prefetch
cfg_desceng_addr0_base	input	ADDR_WIDTH	Address range 0 base
cfg_desceng_addr0_limit	input	ADDR_WIDTH	Address range 0 limit
cfg_desceng_addr1_base	input	ADDR_WIDTH	Address range 1 base
cfg_desceng_addr1_limit	input	ADDR_WIDTH	Address range 1 limit

7.4.6 Status Interface (Per-Channel)

Status Interface

Signal	Direction	Width	Description
descriptor_engine_idle[ch]	output	NUM_CHANNELS	Descriptor engine idle
scheduler_idle[ch]	output	NUM_CHANNELS	Scheduler idle
scheduler_state[ch]	output	NUM_CHANNELS x 7	Scheduler FSM state (ONE-HOT)
sched_error[ch]	output	NUM_CHANNELS	Scheduler error (sticky)

7.4.7 Shared Descriptor AXI4 Master Read Interface

AR Channel:

Shared Descriptor AXI4 Master Read Interface

Signal	Direction	Width	Description
desc_axi_arvalid	output	1	Address valid
desc_axi_arready	input	1	Address ready
desc_axi_araddr	output	ADDR_WIDTH	Address
desc_axi_arlen	output	8	Burst length - 1
desc_axi_arsize	output	3	Burst size (log2 bytes)
desc_axi_arburst	output	2	Burst type (INCR)
desc_axi_arid	output	AXI_ID_WIDTH	Transaction ID
desc_axi_arlock	output	1	Lock type
desc_axi_arcache	output	4	Cache attributes
desc_axi_arprot	output	3	Protection attributes

Signal	Direction	Width	Description
desc_axi_arqos	output	4	QoS value
desc_axi_arregion	output	4	Region identifier

R Channel (FIXED 256-bit):

Shared Descriptor AXI4 Master Read Interface

Signal	Direction	Width	Description
desc_axi_rvalid	input	1	Read data valid
desc_axi_rready	output	1	Read data ready
desc_axi_rdata	input	256	Read data (FIXED 256-bit descriptor)
desc_axi_rresp	input	2	Response
desc_axi_rlast	input	1	Last beat of burst
desc_axi_rid	input	AXI_ID_WIDTH	Transaction ID

7.4.8 Shared Data Read Interface (Per-Channel Arrays)

Shared Data Read Interface

Signal	Direction	Width	Description
sched_rd_valid[ch]	output	NUM_CHANNELS	Read request valid
sched_rd_addr[ch]	output	NUM_CHANNELS x ADDR_WIDTH	Source address
sched_rd_beats[ch]	output	NUM_CHANNELS x 32	Beats to read
sched_rd_done_strobe[ch]	input	NUM_CHANNELS	Read completion strobe
sched_rd_beats_done[ch]	input	NUM_CHAN	Beats completed

Signal	Direction	Width	Description
		NELS x 32	
sched_rd_error[ch]	input	NUM_CHANNELS	Read error

7.4.9 Shared Data Write Interface (Per-Channel Arrays)

Shared Data Write Interface

Signal	Direction	Width	Description
sched_wr_valid[ch]	output	NUM_CHANNELS	Write request valid
sched_wr_ready[ch]	input	NUM_CHANNELS	Write ready
sched_wr_addr[ch]	output	NUM_CHANNELS x ADDR_WIDTH	Destination address
sched_wr_beats[ch]	output	NUM_CHANNELS x 32	Beats to write
sched_wr_done_strobe[ch]	input	NUM_CHANNELS	Write completion strobe
sched_wr_beats_done[ch]	input	NUM_CHANNELS x 32	Beats completed
sched_wr_error[ch]	input	NUM_CHANNELS	Write error

7.4.10 Unified Monitor Bus Interface

Unified Monitor Bus Interface

Signal	Direction	Width	Description
mon_valid	output	1	Monitor packet valid
mon_ready	input	1	Monitor packet ready
mon_packet	output	64	Monitor packet data

7.5 Operation

7.5.1 Descriptor Fetch Arbitration

1. **Request Phase:** Channels assert `desc_ar_valid[ch]` when needing descriptor fetch
2. **Grant Phase:** Round-robin arbiter selects one channel
3. **AR Issue:** Granted channel's AR signals muxed to external interface
4. **ID Embedding:** Channel ID placed in lower bits of `arid`
5. **Grant ACK:** Arbiter acknowledged when `arready` received
6. **R Routing:** Response routed to correct channel based on `rid[CHAN_WIDTH-1:0]`

7.5.2 Data Path Flow

Data interfaces pass directly through without arbitration:

```
scheduler_group[ch].sched_rd_* -> sched_rd_*[ch] -> axi_read_engine
(handles arbitration)
scheduler_group[ch].sched_wr_* -> sched_wr_*[ch] -> axi_write_engine
(handles arbitration)
```

7.5.3 MonBus Aggregation

9 sources aggregated via `monbus_arbiter`: - Sources 0-7: Per-channel `mon_*` from scheduler_groups - Source 8: Descriptor AXI master monitor

---

7.6 Integration Example

```
scheduler_group_array #(
    .NUM_CHANNELS           (8),
    .ADDR_WIDTH             (64),
    .DATA_WIDTH             (512),
    .AXI_ID_WIDTH           (8)
) u_scheduler_group_array (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // APB kick-off
    .apb_valid              (apb_valid),
    .apb_ready              (apb_ready),
    .apb_addr               (apb_addr),

    // Configuration
    .cfg_channel_enable     (cfg_channel_enable),
    .cfg_channel_reset      (cfg_channel_reset),
    .cfg_sched_enable       (cfg_sched_enable),
    // ... other config

    // Descriptor AXI master
    .desc_axi_arvalid       (desc_axi_arvalid),
    .desc_axi_arready       (desc_axi_arready),
    // ... full AR/R interface

    // Data read interface (to axi_read_engine)
    .sched_rd_valid         (sched_rd_valid),
    .sched_rd_addr          (sched_rd_addr),
    .sched_rd_beats         (sched_rd_beats),
    .sched_rd_done_strobe   (rd_done_strobe),
    .sched_rd_beats_done    (rd_beats_done),

    // Data write interface (to axi_write_engine)
    .sched_wr_valid         (sched_wr_valid),
    .sched_wr_ready         (sched_wr_ready),
    .sched_wr_addr          (sched_wr_addr),
    .sched_wr_beats         (sched_wr_beats),
    .sched_wr_done_strobe   (wr_done_strobe),
    .sched_wr_beats_done    (wr_beats_done),

    // Monitor bus
    .mon_valid              (sga_mon_valid),
    .mon_ready              (sga_mon_ready),
    .mon_packet             (sga_mon_packet)
);
```

7.7 Related Documentation

- **Parent:** `01_stream_core.md` - Top-level integration
- **Child:** `03_scheduler_group.md` - Single channel scheduler + descriptor engine
- **Scheduler:** `04_scheduler.md` - Channel state machine
- **Descriptor Engine:** `05_descriptor_engine.md` - Descriptor fetch and parsing
- **AXI Read Engine:** `06_axi_read_engine.md` - Read datapath (receives sched_rd_*)
- **AXI Write Engine:** `12_axi_write_engine.md` - Write datapath (receives sched_wr_*)

- **MonBus Arbiter: `16_monbus_axil_group.md` - Monitor bus aggregation**

7.8 Revision History

Scheduler Group Array Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers

Last Updated: 2026-01-02

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8 Scheduler Group

Module: scheduler_group.sv **Location:** projects/components/dma-ip/stream/rtl/macro/ **Category:** MACRO (Channel Wrapper) **Parent:** scheduler_group_array.sv (built only when SHARED_ENGINES = 0) **Status:** Implemented **Last Updated:** 2026-10-17

> **Build note:** scheduler_group is used only when SHARED_ENGINES = 0. The default build (SHARED_ENGINES = 1) instantiates each channel's scheduler directly in scheduler_group_array, alongside one descriptor_engine_shared that serves every channel. See Scheduler Group Array and Descriptor Engine.

8.1 Overview

The `scheduler_group` module is a wrapper combining the scheduler and descriptor_engine for a single STREAM channel. It provides a simplified interface compared to RAPIDS by removing unused features while maintaining the core descriptor-based DMA functionality.

8.1.1 Key Features

- **Single Channel Wrapper:** Combines scheduler + descriptor_engine
- **Simplified from RAPIDS:** No program engine, no control engines, no alignment bus
- **MonBus Aggregation:** Combines 2 internal sources (scheduler, descriptor_engine)
- **Direct Data Interfaces:** Per-channel read/write interfaces to shared engines

8.1.2 Simplified from RAPIDS

Removed from RAPIDS scheduler_group: - Program engine - Control read/write engines - Alignment bus - Network interfaces (EOS completion) - Credit management

---

8.2 Architecture

8.2.1 Component Hierarchy

```
scheduler_group
 scheduler                      # Channel state machine
 descriptor_engine              # Descriptor fetch and parsing
 arbiter_round_robin_monbus     # 2-source MonBus arbiter
```

8.2.2 Data Flow

```
APB kick-off -> descriptor_engine -> scheduler
                        |                   |

: Data Flow
                        v                   v
              Descriptor AXI        sched_rd_* / sched_wr_*
              (to arbiter)          (to shared engines)
```

Descriptor Flow: 1. APB kick-off provides descriptor address 2. Descriptor engine fetches via shared AXI (arbitrated externally) 3. Parsed descriptor delivered to scheduler 4. Scheduler coordinates read/write phases

Data Path Flow: 1. Scheduler asserts sched_rd_valid with source address/beats 2. External read engine issues AXI reads, returns completion strobe 3. Scheduler asserts sched_wr_valid with destination address/beats 4. External write engine issues AXI writes, returns completion strobe 5. Scheduler advances to next descriptor or completes

---

8.3 Parameters

Parameters

Parameter	Type	Default	Description
CHANNEL_ID	int	0	Channel identifier (0-7)
NUM_CHANNELS	int	8	Total number of channels
CHAN_WIDTH	int	$clog2(NUM_CHANNELS)	Channel ID width
ADDR_WIDTH	int	64	Address bus width
DATA_WIDTH	int	512	Data bus width
AXI_ID_WIDTH	int	8	AXI transaction ID width

8.3.1 Monitor Bus Agent IDs

Monitor Bus Agent IDs

Parameter	Type	Default	Description
DESC_MON_AGENT_ID	8-bit	0x10+ch	Descriptor engine agent ID
SCHED_MON_AGENT_ID	8-bit	0x30+ch	Scheduler agent ID
MON_UNIT_ID	4-bit	0x1	Unit ID
MON_CHANNEL_ID	6-bit	ch	Channel ID in packets

8.4 Port List

8.4.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low asynchronous reset

8.4.2 APB Programming Interface

APB Programming Interface

Signal	Direction	Width	Description
apb_valid	input	1	Descriptor address valid
apb_ready	output	1	Ready to accept descriptor
apb_addr	input	ADDR_WIDTH	Descriptor address

8.4.3 Configuration Interface

Configuration Interface

Signal	Direction	Width	Description
cfg_channel_enable	input	1	Enable this channel
cfg_channel_reset	input	1	Soft reset this channel

8.4.4 Scheduler Configuration

Scheduler Configuration

Signal	Direction	Width	Description
cfg_sched_timeout_cycles	input	16	Timeout threshold

Signal	Direction	Width	Description
cfg_sched_timeout_enable	input	1	Enable timeout detection
cfg_sched_err_enable	input	1	Enable error reporting
cfg_sched_compl_enable	input	1	Enable completion reporting
cfg_sched_perf_enable	input	1	Enable performance monitoring

8.4.5 Descriptor Engine Configuration

Descriptor Engine Configuration

Signal	Direction	Width	Description
cfg_desceng_prefetch	input	1	Enable descriptor prefetch
cfg_desceng_fifo_thresh	input	4	FIFO threshold for prefetch
cfg_desceng_addr0_base	input	ADDR_WIDTH	Address range 0 base
cfg_desceng_addr0_limit	input	ADDR_WIDTH	Address range 0 limit
cfg_desceng_addr1_base	input	ADDR_WIDTH	Address range 1 base
cfg_desceng_addr1_limit	input	ADDR_WIDTH	Address range 1 limit

8.4.6 Status Interface

Status Interface

Signal	Direction	Width	Description
descriptor_engine_idle	output	1	Descriptor engine idle
scheduler_idle	output	1	Scheduler in

Signal	Direction	Width	Description
			IDLE state
scheduler_state	output	7	Scheduler FSM state (ONE-HOT)
sched_error	output	1	Scheduler error (sticky)

8.4.7 Descriptor AXI Interface (to external arbiter)

AR Channel:

Descriptor AXI Interface

Signal	Direction	Width	Description
desc_ar_valid	output	1	Address valid
desc_ar_ready	input	1	Address ready
desc_ar_addr	output	ADDR_WIDTH	Address
desc_ar_len	output	8	Burst length - 1
desc_ar_size	output	3	Burst size
desc_ar_burst	output	2	Burst type
desc_ar_id	output	AXI_ID_WIDTH	Transaction ID
desc_ar_lock	output	1	Lock type
desc_ar_cache	output	4	Cache attributes
desc_ar_prot	output	3	Protection attributes
desc_ar_qos	output	4	QoS value
desc_ar_region	output	4	Region identifier

R Channel (256-bit fixed):

Descriptor AXI Interface

Signal	Direction	Width	Description
desc_r_valid	input	1	Read data valid

Signal	Direction	Width	Description
desc_r_ready	output	1	Read data ready
desc_r_data	input	256	Read data (256-bit descriptor)
desc_r_resp	input	2	Response
desc_r_last	input	1	Last beat
desc_r_id	input	AXI_ID_WIDTH	Transaction ID

8.4.8 Data Read Interface (to shared read engine)

Data Read Interface

Signal	Direction	Width	Description
sched_rd_valid	output	1	Read request valid
sched_rd_addr	output	ADDR_WIDTH	Source address
sched_rd_beats	output	32	Beats to read
sched_rd_done_strobe	input	1	Read completion strobe
sched_rd_beats_done	input	32	Beats completed
sched_rd_error	input	1	Read error

8.4.9 Data Write Interface (to shared write engine)

Data Write Interface

Signal	Direction	Width	Description
sched_wr_valid	output	1	Write request valid
sched_wr_ready	input	1	Write ready
sched_wr_addr	output	ADDR_WIDTH	Destination address
sched_wr_beats	output	32	Beats to write

Signal	Direction	Width	Description
sched_wr_done_strobe	input	1	Write completion strobe
sched_wr_beats_done	input	32	Beats completed
sched_wr_error	input	1	Write error

8.4.10 Monitor Bus Interface

Monitor Bus Interface

Signal	Direction	Width	Description
mon_valid	output	1	Monitor packet valid
mon_ready	input	1	Monitor packet ready
mon_packet	output	64	Monitor packet data

8.5 Operation

8.5.1 Transfer Sequence

1. **APB Kick-off:** `apb_valid` with descriptor address
2. **Descriptor Fetch:** descriptor_engine fetches via AXI
3. **Descriptor Parse:** 256-bit descriptor parsed into fields
4. **Read Phase:** scheduler asserts `sched_rd_valid`
5. **Read Completion:** engine returns `sched_rd_done_strobe`
6. **Write Phase:** scheduler asserts `sched_wr_valid`
7. **Write Completion:** engine returns `sched_wr_done_strobe`
8. **Chain Check:** if `next_descriptor_ptr != 0`, fetch next
9. **Complete:** scheduler returns to IDLE

8.5.2 MonBus Sources

The scheduler_group aggregates 2 MonBus sources: 1. **Descriptor Engine:** Fetch events, errors 2. **Scheduler:** State transitions, completions, errors

---

8.6 Integration Example

```
scheduler_group #(
    .CHANNEL_ID             (ch),
    .NUM_CHANNELS           (8),
    .ADDR_WIDTH             (64),
    .DATA_WIDTH             (512),
    .AXI_ID_WIDTH           (8),
    .DESC_MON_AGENT_ID      (8'h10 + ch),
    .SCHED_MON_AGENT_ID     (8'h30 + ch)
) u_scheduler_group (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // APB interface
    .apb_valid              (apb_valid[ch]),
    .apb_ready              (apb_ready[ch]),
    .apb_addr               (apb_addr[ch]),

    // Configuration
    .cfg_channel_enable     (cfg_channel_enable[ch]),
    .cfg_channel_reset      (cfg_channel_reset[ch]),
    // ... other config

    // Descriptor AXI (to arbiter)
    .desc_ar_valid          (desc_ar_valid[ch]),
    .desc_ar_ready          (desc_ar_ready[ch]),
    // ... full AR/R interface

    // Data interfaces (direct passthrough)
    .sched_rd_valid         (sched_rd_valid[ch]),
    .sched_rd_addr          (sched_rd_addr[ch]),
    .sched_rd_beats         (sched_rd_beats[ch]),
    .sched_rd_done_strobe   (sched_rd_done_strobe[ch]),
    .sched_rd_beats_done    (sched_rd_beats_done[ch]),

    .sched_wr_valid         (sched_wr_valid[ch]),
    .sched_wr_ready         (sched_wr_ready[ch]),
    .sched_wr_addr          (sched_wr_addr[ch]),
    .sched_wr_beats         (sched_wr_beats[ch]),
    .sched_wr_done_strobe   (sched_wr_done_strobe[ch]),
    .sched_wr_beats_done    (sched_wr_beats_done[ch]),

    // Monitor bus
    .mon_valid              (mon_valid_ch[ch]),
    .mon_ready              (mon_ready_ch[ch]),
```

```
    .mon_packet             (mon_packet_ch[ch])
);
```

8.7 Related Documentation

- **Parent:** `02_scheduler_group_array.md` - Multi-channel array
- **Scheduler:** `04_scheduler.md` - Channel state machine details
- **Descriptor Engine:** `05_descriptor_engine.md` - Descriptor fetch logic
- **AXI Read Engine:** `06_axi_read_engine.md` - Shared read engine

- **AXI Write Engine: `12_axi_write_engine.md` - Shared write engine**

8.8 Revision History

Scheduler Group Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers
0.92	2026-10-17	seang	Built only with `SHARED_ENGINES = 0`; the default build uses `descriptor_engine_shared`

Last Updated: 2026-10-17

9 Scheduler Specification

Module: `scheduler.sv` **Location:** `projects/components/dma-ip/stream/rtl/fub/` **Status:** Implemented **Last Updated:** 2025-11-30

9.1 Overview

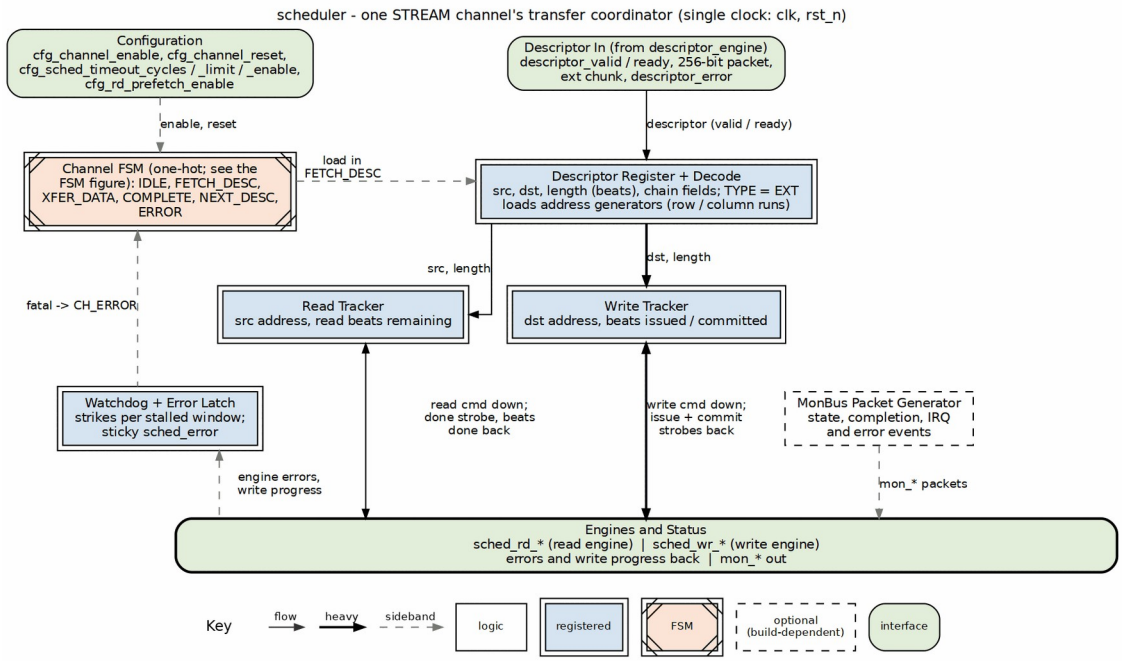
The Scheduler coordinates descriptor-based memory-to-memory DMA transfers for a single channel. It receives descriptors, manages concurrent read/write operations, and handles descriptor chaining.

9.1.1 Key Features

- **Concurrent read/write:** Read and write engines run simultaneously (prevents deadlock)
- **Read-ahead descriptor prefetch:** On a chained legacy descriptor the read side advances to the next descriptor while the write side still drains the current one, eliminating the per-descriptor boundary bubble (runtime `SCHED_CONFIG.RD_PREFETCH_EN`, default on)
- **Beat-based tracking:** Length in data width units (STREAM simplification)
- **Aligned addresses:** No alignment fixup logic (must be pre-aligned)
- **Descriptor chaining:** Follows next_descriptor_ptr for multi-buffer transfers
- **Interrupt generation:** MonBus IRQ event when gen_irq flag set
- **Commit-gated completion:** Channel completes only when all reads are done AND all writes are committed (B responses), not merely issued
- **Recoverable write-progress timeout:** Soft timeout re-arms each window and only escalates to a sticky CH_ERROR after `cfg_sched_timeout_limit` consecutive stalled windows (0 = never escalate)
- **Error handling:** Timeout detection, error aggregation from engines
- **MonBus integration:** State transition and IRQ event reporting

9.1.2 Block Diagram

9.1.3 Figure 2.4.1: Scheduler Block Diagram



Diagram

Source: 02_scheduler.dot

9.2 CRITICAL: Concurrent Read/Write Design

Why Concurrent Operation is Essential:

The scheduler runs read and write engines **CONCURRENTLY** in CH_XFER_DATA state. This prevents deadlock when transfer size exceeds SRAM buffer capacity:

```
Example: 100MB transfer with 2KB SRAM buffer

Sequential operation (WRONG):
1. Read 100MB → DEADLOCK at 2KB (SRAM full, can't complete read)

Concurrent operation (CORRECT):
1. Read starts filling SRAM → SRAM becomes full (2KB)
2. Read pauses (natural backpressure)
3. Write drains SRAM → SRAM has free space
4. Read resumes → Both continue until 100MB complete
```

Implementation: - Both `sched_rd_valid` and `sched_wr_valid` asserted in CH_XFER_DATA - Independent beat counters: `r_read_beats_remaining`, `r_write_beats_remaining` - Exit when **BOTH** counters reach zero

---

9.3 Read-Ahead Descriptor Prefetch

Runtime enable: `SCHED_CONFIG.RD_PREFETCH_EN` (default **on**). Applies to chained **legacy** descriptors only; EXT (row/col-major) descriptors always run lockstep.

Problem it solves. Without prefetch, when descriptor N's reads finish the FSM walks `CH_XFER_DATA -> CH_COMPLETE -> CH_NEXT_DESC -> CH_FETCH_DESC` before the read engine can restart, and the read pipeline (AR -> R) goes cold at every boundary. This is a per-descriptor **bubble** (~13 cycles measured) that caps efficiency on short-descriptor chains.

Mechanism. The descriptor FIFO (in the descriptor engine, already prefetched) is used as a one-entry holding buffer between the read and write sides. A single flag, `r_rd_ahead`, sequences the handoff:

- **SET** when the read flops load the next descriptor from the FIFO **head** *without draining the FIFO* — the read side runs one descriptor ahead, filling SRAM with descriptor N+1 while the write side still drains N.
- **CLEAR** when the write flops load that same descriptor and *drain the FIFO* (the write side catches up; read and write are lockstep again on N+1).

The SRAM sits between the two directions as an in-order FIFO, so racing the reads ahead is safe: descriptor N+1's data simply queues behind N's, and the write side counts beats to know where N ends and N+1 begins. The in-place advance collapses the four-state FSM walk into a single-cycle reload that stays in CH_XFER_DATA, and because the read pipeline never goes cold the AR->R refill latency is hidden.

Result. The per-descriptor bubble drops from ~13 cycles to 0 — chained descriptors stream at full rate; only a one-time fill/drain remains, independent of chain length. On-silicon A/B (16 x 64-beat descriptors, one bitstream): datapath utilisation 95.3% with prefetch on vs 76.1% off; longer descriptors reach 99-100%. See HAS Section 5.1.

Timing note. The write-commit accumulator's launch/advance addend is registered one cycle early (the add lands the following cycle; safe because commit B-responses always lag launches) so the descriptor-FIFO output and the launch/advance decode stay off the accumulator's combinational path — required to close 100 MHz timing with the added read-ahead fanout.

9.4 Parameters

```
parameter int CHANNEL_ID = 0;                    // Channel identifier
parameter int NUM_CHANNELS = 8;                  // Total channels in
system
parameter int CHAN_WIDTH = $clog2(NUM_CHANNELS); // Channel ID width
parameter int ADDR_WIDTH = 64;                   // Address bus width
parameter int DATA_WIDTH = 512;                  // Data bus width
(beats)

// Monitor Bus Parameters
parameter logic [7:0] MON_AGENT_ID = 8'h40;      // STREAM Scheduler
Agent ID
parameter logic [3:0] MON_UNIT_ID = 4'h1;        // Unit identifier
parameter logic [5:0] MON_CHANNEL_ID = 6'h0;     // Base channel ID

// Descriptor Width (FIXED at 256-bit for STREAM)
parameter int DESC_WIDTH = 256;
```

Validation:

```
// Scheduler only supports 256-bit STREAM descriptors
if (DESC_WIDTH != 256)
    $fatal("DESC_WIDTH must be 256 for STREAM scheduler");
```

9.5 Port List

9.5.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low asynchronous reset

9.5.2 Configuration Interface

Configuration Interface

Signal	Direction	Width	Description
cfg_channel_enable	input	1	Enable this channel
cfg_channel_reset	input	1	Channel soft reset (FSM → IDLE)
cfg_sched_timeout_cycles	input	32	Timeout threshold in clock cycles (32-bit, runtime config)
cfg_sched_timeout_enable	input	1	Enable timeout detection
cfg_sched_timeout_limit	input	8	Consecutive timeout windows before fatal escalation (0 = never escalate; from SCHED_TIMEOUT_LIMIT @ 0x208)

9.5.3 Status Interface

Status Interface

Signal	Direction	Width	Description
scheduler_idle	output	1	CH_IDLE, no channel reset pending, descriptor skid empty
scheduler_st	output	7	Current FSM state (one-

Signal	Direction	Width	Description
ate			hot encoding)

9.5.4 Descriptor Engine Interface

Descriptor Engine Interface

Signal	Direction	Width	Description
descriptor_valid	input	1	Descriptor valid from descriptor engine
descriptor_ready	output	1	The descriptor skid has room (see Descriptor Skid Buffer)
descriptor_packet	input	256	256-bit STREAM descriptor (see format below)
descriptor_error	input	1	Error signal from descriptor engine

9.5.5 Data Read Interface

To AXI Read Engine:

Data Read Interface

Signal	Direction	Width	Description
sched_rd_valid	output	1	Channel requests read
sched_rd_addr	output	ADDR_WIDTH	Source address (aligned, static during burst)
sched_rd_beats	output	32	Beats remaining to read

Completion from Read Engine:

Data Read Interface

Signal	Direction	Width	Description
sched_rd_done_strobe	input	1	Read burst completed (1-cycle pulse)
sched_rd_beats_done	input	32	Number of beats completed in burst

Signal	Direction	Width	Description
sched_rd_error	input	1	Read engine error (sticky)

9.5.6 Data Write Interface

To AXI Write Engine:

Data Write Interface

Signal	Direction	Width	Description
sched_wr_valid	output	1	Channel requests write
sched_wr_ready	input	1	Engine ready for channel (completion handshake)
sched_wr_addr	output	ADDR_WIDTH	Destination address (aligned, static during burst)
sched_wr_beats	output	32	Beats remaining to write

Completion from Write Engine:

Data Write Interface

Signal	Direction	Width	Description
sched_wr_done_strobe	input	1	Write burst completed (1-cycle pulse)
sched_wr_beats_done	input	32	Number of beats completed in burst
sched_wr_error	input	1	Write engine error (sticky)

9.5.7 Error Signals

Error Signals

Signal	Direction	Width	Description
sched_error	output	1	Scheduler error output (aggregates rd/wr errors, sticky)

9.5.8 Monitor Bus Interface

Monitor Bus Interface

Signal	Direction	Width	Description
mon_valid	output	1	Monitor packet valid
mon_ready	input	1	Monitor bus ready
mon_packet	output	64	64-bit monitor bus packet

9.6 Interface

9.6.1 Clock and Reset

```
input  logic                        clk;
input  logic                        rst_n;      // Active-low
asynchronous reset
```

9.6.2 Configuration Interface

```
input  logic                        cfg_channel_enable;       //
Enable this channel
input  logic                        cfg_channel_reset;        //
Channel reset (soft reset)
input  logic [31:0]                 cfg_sched_timeout_cycles; //
Timeout threshold (runtime config)
input  logic                        cfg_sched_timeout_enable; //
Enable timeout detection
input  logic [7:0]                  cfg_sched_timeout_limit;  //
Consecutive timeout windows before escalation (0=never)
```

Channel Reset Behavior: - `cfg_channel_reset` forces FSM to CH_IDLE immediately - Clears descriptor_loaded flag - Resets beat counters - Independent of global `rst_n`

Timeout Configuration: - Runtime configurable via `cfg_sched_timeout_cycles` (replaces compile-time TIMEOUT_CYCLES parameter) - Can be disabled/enabled dynamically via `cfg_sched_timeout_enable`

9.6.3 Status Interface

```
output logic                        scheduler_idle;         //
CH_IDLE, no reset, skid empty
output logic [6:0]                  scheduler_state;        // Current
state (ONE-HOT)
```

State Encoding (ONE-HOT): - `[0]` = CH_IDLE - `[1]` = CH_FETCH_DESC - `[2]` = CH_XFER_DATA - `[3]` = CH_COMPLETE - `[4]` = CH_NEXT_DESC - `[5]` = CH_ERROR - `[6]` = Reserved

9.6.4 Descriptor Engine Interface

```
input  logic                        descriptor_valid;
output logic                        descriptor_ready;
input  logic [DESC_WIDTH-1:0]       descriptor_packet;     // 256-bit
STREAM descriptor
input  logic                        descriptor_error;       // Error
from descriptor engine
```

Descriptor Handshake: - `descriptor_ready` is the descriptor skid's room (below); the FSM takes from the skid in CH_IDLE (only while `cfg_channel_enable` is set), CH_NEXT_DESC, or on a write-side advance - Descriptor captured when the skid head is valid and the FSM takes it - Supports descriptor chaining (next_descriptor_ptr)

9.6.5 Descriptor Skid Buffer

Every scheduler takes its descriptors through a two-deep skid buffer (`gaxi_skid_buffer`, DEPTH 2) on `{descriptor_packet, descriptor_ext_packet}`, in front of the descriptor engine's per-channel FIFO, which stays (stream TASK-017). The FSM reads the descriptor from the skid's flops rather than straight off the engine's 256-bit output, one engine for every channel under `SHARED_ENGINES`, and a running channel holds two more descriptors.

- `descriptor_ready` is the skid's `wr_ready`: high while it has room, whatever the FSM state. The FSM's own take (CH_IDLE while `cfg_channel_enable` is set, CH_NEXT_DESC, or a write-side advance) pops the skid.
- A disabled channel's descriptor waits in the skid (or behind it in the engine FIFO) until the channel is enabled; it is not consumed and dropped.
- `scheduler_idle` stays low while the skid holds a descriptor, so the engine and the scheduler never both read idle with work pending.
- Channel reset: the engine masks `descriptor_valid` during its reset, so nothing is written during one, and the skid drains while the reset is active. Each entry carries the reset epoch it was written in, and the epoch flips as a reset starts, so an entry a reset too short to drain it left behind is dropped at the head, never run.

9.6.6 Data Read Interface (to AXI Read Engine)

Request:

```
output logic                        sched_rd_valid;         // Request
read access
output logic [ADDR_WIDTH-1:0]       sched_rd_addr;          // Source
address (static base)
output logic [31:0]                 sched_rd_beats;         // Total
beats to read
```

Completion:

```
input  logic                        sched_rd_done_strobe;   // Read
engine completed beats
input  logic [31:0]                 sched_rd_beats_done;    // Number
of beats completed
```

Error:

```
input  logic                        sched_rd_error;         // Read
engine error
```

Address Management: - Scheduler provides **static base address** in `sched_rd_addr` - Read engine handles address increment internally - Scheduler does NOT update `sched_rd_addr` after each burst

Interface Timing Notes (November 2025 Updates): - Scheduler holds `sched_rd_valid` high while `sched_rd_beats > 0` - Engine reports completion via `sched_rd_done_strobe` (pulsed) - Scheduler decrements `r_read_beats_remaining` by `sched_rd_beats_done` - Process repeats until `r_read_beats_remaining == 0`

9.6.7 Data Write Interface (to AXI Write Engine)

Request:

```
output logic                        sched_wr_valid;         // Request
write access
input  logic                        sched_wr_ready;         // Engine
grants access
output logic [ADDR_WIDTH-1:0]       sched_wr_addr;          //
Destination address (static base)
output logic [31:0]                 sched_wr_beats;         // Total
beats to write
```

Completion:

```
input  logic                        sched_wr_done_strobe;   // Write
engine completed beats
input  logic [31:0]                 sched_wr_beats_done;    // Number
of beats completed
```

Error:

```
input  logic                        sched_wr_error;         // Write
engine error
```

Interface Timing Notes (November 2025 Updates):

sched_wr_ready Timing: - **REGISTERED OUTPUT** from write engine - Not combinatorial from completion signals - Asserts 1 cycle after channel becomes ready to accept new request - Cleared 1 cycle after handshake (valid && ready) - **Do not expect immediate deassertion on handshake** - takes 1 cycle

Example Timing:

```
Cycle  sched_wr_valid  sched_wr_ready  sched_wr_done_strobe  Notes
-----  --------------  --------------  --------------------  -----
  0    0               0               0                     Idle
  1    1               0               0                     Request
(not ready yet)
  2    1               1               0
Handshake!
```

```
  3    1               0               0                     Ready
cleared (registered)
 ...   (W-phase executes)
 50    1               0               1                     B
response, done strobe
 51    1               1               0                     Ready re-
asserts (next cycle)
```

9.6.8 Monitor Bus Interface

```
output logic                        mon_valid;
input  logic                        mon_ready;
output logic [63:0]                 mon_packet;
```

MonBus Events Generated: - State transitions (IDLE → FETCH_DESC, etc.) - IRQ event (when descriptor.gen_irq set) - Error events

---

9.7 Descriptor Format

9.7.1 STREAM Descriptor (256-bit)

```
typedef struct packed {
    logic [47:0]  reserved;             // [255:208] Reserved
    logic [7:0]   desc_priority;        // [207:200] Transfer
priority
    logic [3:0]   channel_id;           // [199:196] Channel ID
(informational)
    logic         error;                // [195] Error flag
    logic         last;                 // [194] Last in chain flag
    logic         gen_irq;              // [193] Generate interrupt
on completion
    logic         valid;                // [192] Valid descriptor
    logic [31:0]  next_descriptor_ptr;  // [191:160] Next descriptor
address (0 = last)
    logic [31:0]  length;               // [159:128] Length in BEATS
    logic [63:0]  dst_addr;             // [127:64] Destination
address (aligned)
    logic [63:0]  src_addr;             // [63:0] Source address
(aligned)
} descriptor_t;
```

Field Constraints: - `src_addr` / `dst_addr`: Must be aligned to DATA_WIDTH (e.g., 64-byte aligned for 512-bit data) - `length`: Transfer size in **BEATS** (not bytes or chunks) - `next_descriptor_ptr`: 0 or address of next descriptor - `valid`: Must be 1 for descriptor to be accepted - `last`: Terminates chain (overrides next_descriptor_ptr) - `gen_irq`: Generates STREAM_EVENT_IRQ via MonBus when transfer completes

Descriptor Bit Positions:

```
DESC_SRC_ADDR:  [63:0]
DESC_DST_ADDR:  [127:64]
DESC_LENGTH:    [159:128]
DESC_NEXT_PTR:  [191:160]
DESC_VALID_BIT: [192]
DESC_GEN_IRQ:   [193]
DESC_LAST:      [194]
```

9.8 FSM Operation

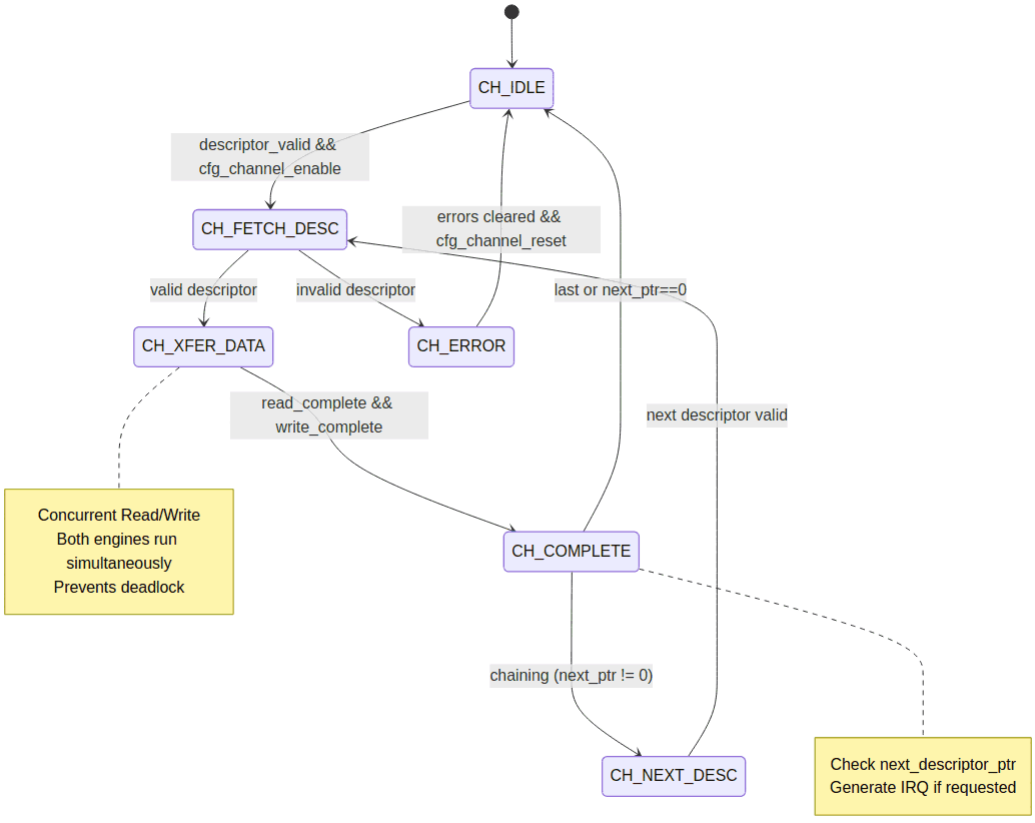
9.8.1 State Machine

States (ONE-HOT encoded):

```
CH_IDLE         - Waiting for descriptor
CH_FETCH_DESC   - Latch and validate descriptor
CH_XFER_DATA    - Concurrent read/write transfer
CH_COMPLETE     - Transfer done, check chaining
CH_NEXT_DESC    - Fetch next chained descriptor
CH_ERROR        - Error condition
```

FSM Flow:

9.8.2 Figure 2.4.2: Scheduler FSM



Diagram

9.8.3 State Transitions

CH_IDLE: - **Wait for:** `descriptor_valid && cfg_channel_enable` - **Action:** Assert `descriptor_ready` (gated by `cfg_channel_enable`) - **Next:** CH_FETCH_DESC (when handshake occurs)

CH_FETCH_DESC: - **Action:** - Latch descriptor fields into `r_descriptor` - Initialize working registers (`r_src_addr`, `r_dst_addr`, `r_*_beats_remaining`) - Validate `descriptor.valid` bit - **Next:** - CH_XFER_DATA (if valid) - CH_ERROR (if invalid)

CH_XFER_DATA: - **Action:** - Assert **BOTH** `sched_rd_valid` and `sched_wr_valid` (concurrent operation!) - Decrement `r_read_beats_remaining` on `sched_rd_done_strobe` - Decrement `r_write_beats_remaining` on `sched_wr_done_strobe` - Monitor timeout counter - **Exit When:** `r_read_beats_remaining == 0 && r_write_beats_remaining == 0` - **Next:** CH_COMPLETE

CH_COMPLETE: - **Action:** - Generate MonBus IRQ event (if `gen_irq` set) - Check `next_descriptor_ptr` and `last` flag - Clear `descriptor_loaded` flag - **Next:** - CH_NEXT_DESC (if chaining) - CH_IDLE (if last or no chain)

CH_NEXT_DESC: - **Wait for:** `descriptor_valid` (descriptor engine fetches next) - **Action:** Assert `descriptor_ready` - **Next:** CH_FETCH_DESC

CH_ERROR: - **STICKY STATE:** Once in error, stays here until channel reset - **Action:** Report error via MonBus - **Recovery:** Requires `cfg_channel_reset` assertion (or global reset) - **Note:** `scheduler_idle` does **NOT** assert in this state — a faulted channel must not report idle. The fault is surfaced via `sched_error`/CHANNEL_ERROR instead.

9.9 Beat Tracking

9.9.1 Independent Counters

Initialization (CH_FETCH_DESC):

```
r_read_beats_remaining  <= r_descriptor.length;
r_write_beats_remaining <= r_descriptor.length;
```

Decrement (CH_XFER_DATA):

```
// Read progress (independent)
if (sched_rd_done_strobe) begin
    r_read_beats_remaining <= (r_read_beats_remaining >=
sched_rd_beats_done) ?
                              (r_read_beats_remaining -
sched_rd_beats_done) : 32'h0;
end

// Write progress (independent)
if (sched_wr_done_strobe) begin
    r_write_beats_remaining <= (r_write_beats_remaining >=
sched_wr_beats_done) ?
                               (r_write_beats_remaining -
sched_wr_beats_done) : 32'h0;
end
```

Saturation: - Counters saturate at 0 (prevent underflow) - Safety check for engine misbehavior

9.9.2 Completion Detection

Write completion is gated on COMMIT, not issue. Read completion tracks the read-beat counter, but write completion tracks a separate `r_write_beats_to_commit` counter that decrements only on write-data **commit** (B responses), not on AW issue. This prevents the channel from signalling done/interrupt while write data is still draining to memory.

Combinational Flags:

```
w_read_complete     = (r_read_beats_remaining == 0);
// Write is complete only when all issued beats have been COMMITTED (B
responses),
// not merely issued (r_write_beats_remaining).
w_write_complete    = (r_write_beats_to_commit == 0);
w_transfer_complete = w_read_complete && w_write_complete;
```

Write-request gating: `sched_wr_valid` is additionally gated on remaining *issue* beats (`r_write_beats_remaining != 0`). Once all beats are issued, the scheduler stops requesting new writes but stays in CH_XFER_DATA waiting for the outstanding commits (B responses) to retire `r_write_beats_to_commit`.

State Exit:

```
// In CH_XFER_DATA:
if (w_transfer_complete) begin  // all reads done AND all writes
committed
    w_next_state = CH_COMPLETE;
end
```

9.9.3 Multiple Requests per Descriptor

For large transfers, scheduler issues multiple requests as engines complete work:

```
Descriptor: length = 256 beats (16KB @ 512-bit data)
Engine burst size: 16 beats (configured via cfg_axi_wr_xfer_beats)

Request sequence:
  Request 1: sched_wr_valid=1, sched_wr_beats=256, ready → handshake
             Engine executes 16-beat burst
             sched_wr_done_strobe=1, sched_wr_beats_done=16
             Scheduler updates: 256 - 16 = 240 beats remaining

  Request 2: sched_wr_valid=1, sched_wr_beats=240, ready → handshake
             Engine executes 16-beat burst
             sched_wr_done_strobe=1, sched_wr_beats_done=16
             Scheduler updates: 240 - 16 = 224 beats remaining

  ... (continues for 16 requests total)

  Request 16: sched_wr_valid=1, sched_wr_beats=16, ready → handshake
              Engine executes 16-beat burst
              sched_wr_done_strobe=1, sched_wr_beats_done=16
              Scheduler updates: 16 - 16 = 0 beats remaining →
COMPLETE
```

Key Point: Scheduler holds `sched_wr_valid` high and keeps reissuing requests as long as `sched_wr_beats > 0`. Engines handle each request independently.

9.10 Address Management

9.10.1 Static Base Address

Scheduler Provides:

```
assign sched_rd_addr = r_src_addr;  // Static base, set in
CH_FETCH_DESC
assign sched_wr_addr = r_dst_addr;  // Static base, set in
CH_FETCH_DESC
```

Scheduler Does NOT: - Increment addresses after each burst - Calculate byte offsets - Handle alignment

Engine Responsibility: - Read engine: `m_axi_araddr = sched_rd_addr + (beats_issued << AXSIZE)` - Write engine: `m_axi_awaddr = sched_wr_addr + (beats_issued << AXSIZE)`

9.11 Timeout Detection (Recoverable)

The write-progress timeout is a **recoverable liveness fault**, not a hard error. A single expired window does NOT wedge the channel into CH_ERROR; the timeout only escalates to a fatal, sticky error after a configurable number of consecutive windows with no write progress.

9.11.1 Timeout Counter

Configuration: - Runtime configurable via `cfg_sched_timeout_cycles` (32-bit) - Can be enabled/disabled via `cfg_sched_timeout_enable` - Escalation limit via `cfg_sched_timeout_limit` (8-bit, from SCHED_TIMEOUT_LIMIT @ 0x208) - Replaces compile-time TIMEOUT_CYCLES parameter

Increment (counts stalled write cycles):

```
// In CH_XFER_DATA when the write request is stalled (valid high,
ready low)
if (cfg_sched_timeout_enable && sched_wr_valid && !sched_wr_ready)
begin
    r_timeout_counter <= r_timeout_counter + 1;
end
```

Window Expiry (re-arming, one pulse per window):

```
// w_timeout_expired is a one-cycle PULSE per elapsed window (not a
latched level).
assign w_timeout_expired = cfg_sched_timeout_enable &&
                           (r_timeout_counter >=
cfg_sched_timeout_cycles);
// On expiry the counter re-arms (clears) so a new window can begin.
```

9.11.2 Consecutive-Window Strike Counter and Escalation

```
// r_timeout_strikes counts consecutive expired windows with NO write
progress.
// Any write progress (done or commit strobe) clears the strike
counter.
if (write_progress)                r_timeout_strikes <= 8'h0;
else if (w_timeout_expired)        r_timeout_strikes <=
r_timeout_strikes + 8'h1;

// Escalate to a fatal (sticky CH_ERROR) fault only after
cfg_sched_timeout_limit
// consecutive windows. cfg_sched_timeout_limit == 0 means never
escalate.
assign w_timeout_escalate = (cfg_sched_timeout_limit != 8'd0) &&
                            (r_timeout_strikes >=
cfg_sched_timeout_limit);
```

Action: - Each expired window emits a MonBus timeout event but leaves the channel running (soft timeout). - Any write progress (a `sched_wr_done_strobe` or a commit strobe) clears `r_timeout_strikes`. - Only `w_timeout_escalate` drives the FSM to the sticky CH_ERROR; recovery then requires `cfg_channel_reset` (or global reset). - With `cfg_sched_timeout_limit == 0`, the timeout never escalates (pure soft-timeout reporting).

9.12 Error Handling

9.12.1 Error Sources

External: - `descriptor_error` - Descriptor engine reports error - `sched_rd_error` - Read engine error (AXI RRESP != OKAY, etc.) - `sched_wr_error` - Write engine error (AXI BRESP != OKAY, etc.)

Internal: - `w_timeout_escalate` - Write-progress timeout persisted for `cfg_sched_timeout_limit` consecutive windows (a single `w_timeout_expired` window is recoverable and does NOT trigger CH_ERROR) - `!r_descriptor.valid` - Invalid descriptor in CH_FETCH_DESC

9.12.2 Sticky Error Flags

```
logic r_read_error_sticky;   // Set on sched_rd_error, cleared in
CH_IDLE
logic r_write_error_sticky;  // Set on sched_wr_error, cleared in
CH_IDLE
logic r_descriptor_error;    // Set on descriptor_error or validation
failure
```

Set Condition:

```
if (sched_rd_error)
    r_read_error_sticky <= 1'b1;

if (sched_wr_error)
    r_write_error_sticky <= 1'b1;
```

Clear Condition:

```
if (r_current_state == CH_IDLE)
    r_*_error_sticky <= 1'b0;
```

9.12.3 Error Recovery

Error Transition:

```
// Any state with a FATAL error condition (note: w_timeout_escalate,
NOT the
// per-window w_timeout_expired pulse, which is a recoverable soft
timeout).
if (descriptor_error || sched_rd_error || sched_wr_error ||
    r_read_error_sticky || r_write_error_sticky || w_timeout_escalate)
begin
    w_next_state = CH_ERROR;
end
```

CH_ERROR is STICKY:

```
CH_ERROR: begin
    // Error state - STICKY, stay here until reset
    // Once in error, only way out is through reset
    w_next_state = CH_ERROR;
end
```

Recovery: - CH_ERROR is a **sticky state** - does NOT auto-recover - Software **must** assert `cfg_channel_reset` (or global `rst_n`) - On channel reset: FSM → CH_IDLE, sticky flags cleared

---

9.13 Interrupt Generation

9.13.1 IRQ via MonBus

Trigger: - Descriptor completes (CH_COMPLETE state) - `r_descriptor.gen_irq` flag set

MonBus Event:

```
// In CH_COMPLETE state with gen_irq set
mon_packet = {
    MON_AGENT_ID,           // [63:56] Agent ID (0x40 = STREAM
Scheduler)
    MON_UNIT_ID,            // [55:52] Unit ID
    MON_CHANNEL_ID,         // [51:46] Channel ID
    STREAM_EVENT_IRQ,       // [45:40] Event code (IRQ)
    descriptor_fields       // [39:0]  Descriptor info
};
```

No Separate IRQ Signal: - IRQ communicated via MonBus only - Software monitors MonBus for IRQ events - Event includes channel ID for routing

---

9.14 Descriptor Chaining

9.14.1 Chain Detection

In CH_COMPLETE:

```
if (r_descriptor.next_descriptor_ptr != 32'h0 && !r_descriptor.last)
begin
    w_next_state = CH_NEXT_DESC;  // Chain to next descriptor
end else begin
    w_next_state = CH_IDLE;       // Complete (last or no chain)
end
```

9.14.2 Chain Termination

Explicit Termination: - `next_descriptor_ptr == 0` → Stop - `last == 1` → Stop (overrides next_descriptor_ptr)

Example Chain:

```
Descriptor 0 @ 0x1000:
  src_addr = 0x2000, dst_addr = 0x3000, length = 64
  next_descriptor_ptr = 0x1040, last = 0
  → Chains to next

Descriptor 1 @ 0x1040:
  src_addr = 0x2100, dst_addr = 0x3100, length = 32
  next_descriptor_ptr = 0x0000, last = 1
  → Last in chain
```

9.15 MonBus Integration

9.15.1 Event Types

State Transitions: - IDLE → FETCH_DESC: Descriptor fetch start - FETCH_DESC → XFER_DATA: Transfer start - XFER_DATA → COMPLETE: Transfer complete - COMPLETE → NEXT_DESC: Chain fetch - Any → ERROR: Error occurred

Special Events: - STREAM_EVENT_IRQ: Interrupt generation (gen_irq flag) - STREAM_EVENT_TIMEOUT: Timeout expired - STREAM_EVENT_ERROR: Error condition

9.15.2 MonBus Packet Format

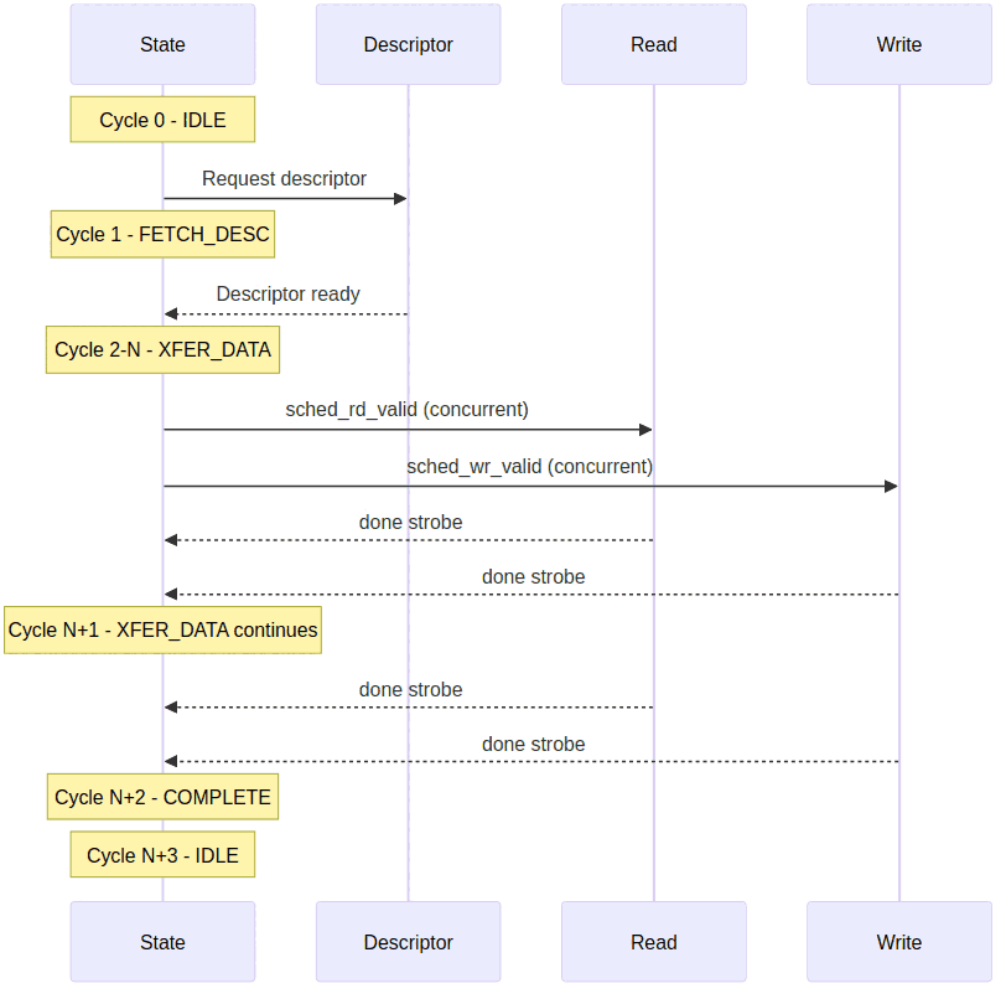
Generic Event:

```
[63:56] - MON_AGENT_ID (0x40)
[55:52] - MON_UNIT_ID
[51:46] - MON_CHANNEL_ID + CHANNEL_ID
[45:40] - Event code
[39:0]  - Event-specific data
```

9.16 Timing Diagrams

9.16.1 Normal Transfer (No Chaining)

9.16.2 Figure 2.4.3: Scheduler Normal Transfer Timing



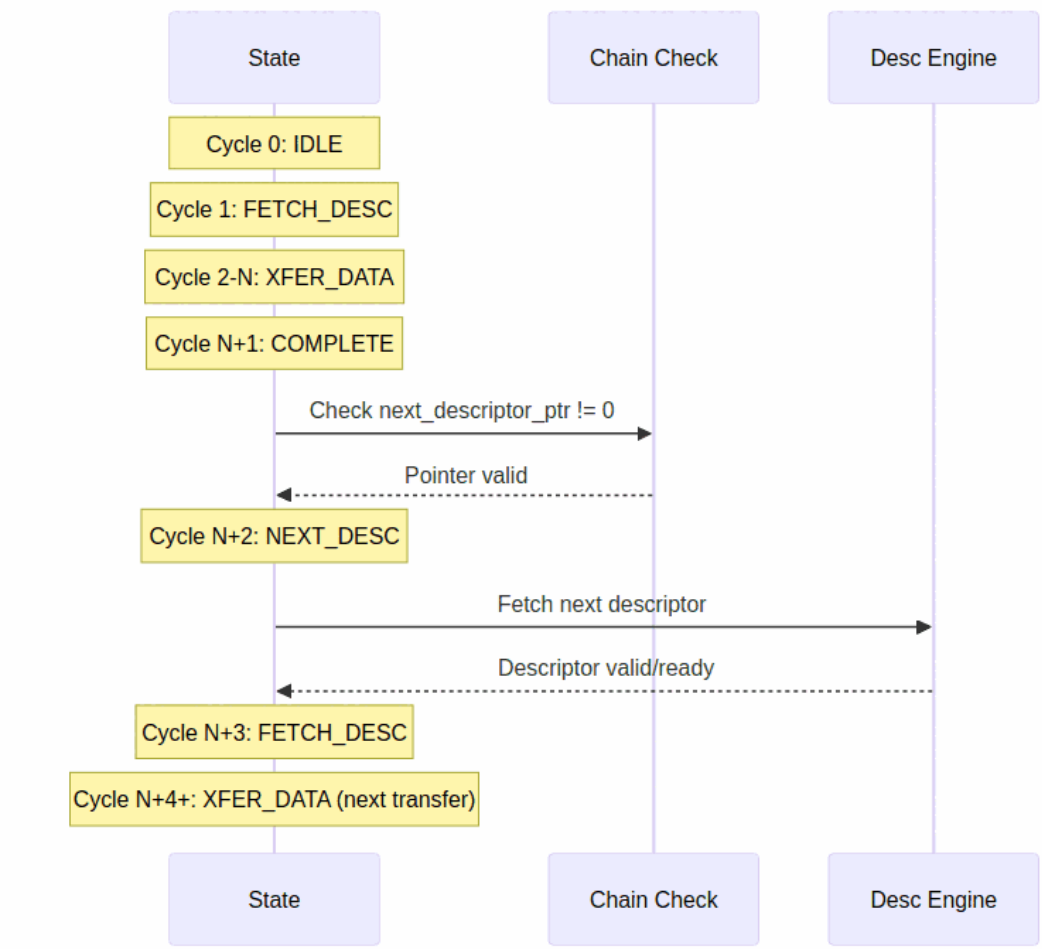
Scheduler Normal Transfer Timing

Source: 04_scheduler_normal_transfer.mmd

Notes: - Both read and write run concurrently in XFER state - Independent done strobes decrement separate counters

9.16.3 Descriptor Chaining

9.16.4 Figure 2.4.4: Scheduler Descriptor Chaining Timing



Scheduler Descriptor Chaining Timing

Source: 04_scheduler_chaining.mmd

9.16.5 Single-Descriptor Transfer (Detailed)

```
Cycle  State        sched_wr_valid  sched_wr_ready  sched_wr_beats
Notes
-----  -----------  --------------  --------------  --------------
-----
  0    IDLE         0               0               0
Waiting
  1    FETCH_DESC   0               0               0
Request descriptor
  2    FETCH_DESC   0               0               0
(fetch latency)
```

```
  3    XFER_DATA    0              0              0
Descriptor received
 ...   (read phase)
 50    XFER_DATA    1              0              256
Assert write request
 51    XFER_DATA    1              1              256
Engine ready, handshake!
 52    XFER_DATA    1              0              256
Ready cleared (registered)
 ...   (engine executes burst)
100    XFER_DATA    1              0              256
done_strobe=1, beats_done=16
101    XFER_DATA    1              1              240
Ready re-asserts, new beats_remaining
102    XFER_DATA    1              0              240
Handshake again
...    (continues until beats_remaining == 0)
```

9.16.6 Chained Descriptors (Detailed)

```
Descriptor chain:
  Desc 0: length=128, next_ptr=0x1000_0100
  Desc 1: length=64,  next_ptr=0

Cycle  State        Descriptor  sched_wr_beats  Notes
-----  -----------  ----------  --------------  -----
  0    IDLE         -           0               Start
  1    FETCH_DESC   Req @0x1000 0               Fetch first
  3    XFER_DATA    Desc 0      128             Transfer desc 0
 ...   (8 bursts × 16 beats)
200    COMPLETE     Desc 0      0               Transfer done
201    NEXT_DESC    Check ptr   0               next_ptr = 0x1000_0100
202    FETCH_DESC   Req @0x1100 0               Fetch second
204    XFER_DATA    Desc 1      64              Transfer desc 1
 ...   (4 bursts × 16 beats)
400    COMPLETE     Desc 1      0               Transfer done
401    NEXT_DESC    Check ptr   0               next_ptr = 0 →
terminate
402    IDLE         -           0               Chain complete
```

9.17 Testing

Test Location: `projects/components/dma-ip/stream/dv/tests/fub/test_scheduler.py`

Key Test Scenarios:

1. **Single descriptor transfer** - Basic operation
2. **Descriptor chaining** - 2-4 descriptors linked
3. **Concurrent read/write** - Verify no deadlock with small SRAM
4. **Large transfer (> SRAM)** - 100MB transfer with 2KB SRAM
5. **IRQ generation** - gen_irq flag set
6. **Error handling** - Descriptor, read, write errors
7. **Timeout detection** - Engine stall scenarios
8. **Channel reset** - cfg_channel_reset during transfer
9. **Runtime timeout config** - Change cfg_sched_timeout_cycles dynamically
10. **sched_rd_ready / sched_wr_ready timing validation** - Verify registered ready behavior

---

9.18 Performance Considerations

9.18.1 Concurrent Operation Benefit

Without Concurrency (Sequential): - Max transfer size = SRAM buffer size - Deadlock when transfer > buffer - Throughput = min(read_bw, write_bw)

With Concurrency: - No transfer size limit - Natural flow control via SRAM full/empty - Throughput = max(read_bw, write_bw) (pipeline overlap)

9.18.2 Example Performance

Configuration: - DATA_WIDTH = 512 bits (64 bytes/beat) - SRAM = 2KB (32 beats) - Transfer = 100MB (1,562,500 beats)

Sequential (hypothetical): - DEADLOCK at 2KB (can't complete read)

Concurrent: - Read fills SRAM (32 beats) - Write drains SRAM concurrently - Both engines sustain ~0.9 beats/cycle - Total time: ~1.7M cycles

---

9.19 Observation and Diagnostics (v0.92+)

9.19.1 OBS_FLAGS Register Mapping

The scheduler exposes two observation flags via the APB configuration block's `OBS_FLAGS` register:

Bit	Signal	Meaning
[0]	`SCH_ERROR_STICKY`	Error sticky flag (descriptor invalid, descriptor address invalid, AXI error, or timeout detected)
[1]	`SCH_TIMEOUT`	Timeout active flag (waiting on AXI master and timeout counter has elapsed)

Both flags are readable via APB `OBS_FLAGS[CHANNEL_SEL]` after selecting the channel via `OBS_CTRL`. The error sticky persists until `cfg_channel_reset` is asserted; the timeout flag clears when AXI activity resumes.

9.19.2 Observation Data Fields (OBS_DATA0 / OBS_DATA1)

Channel-specific diagnostic data is mirrored into the APB registers to enable software monitoring without stopping the scheduler. Contents may include current beat counter state, descriptor pointer, or FSM state depending on the channel's execution phase.

9.20 Related Documentation

- **Descriptor Engine:** `05_descriptor_engine.md` - Descriptor fetch
- **AXI Read Engine:** `06_axi_read_engine.md` - Source data read
- **AXI Write Engine:** `12_axi_write_engine.md` - Destination data write
- **SRAM Controller:** `08_sram_controller.md` - Buffer management
- **Scheduler Group:** `03_scheduler_group.md` - Single-channel integration
- **Scheduler Group Array:** `02_scheduler_group_array.md` - Multi-channel integration

---

9.21 Revision History

Revision History

Date	Version	Changes
2025-10-17	1.0	Initial documentation with old signal names (datard_, datawr_)
2025-11-16	1.5	Enhanced documentation with detailed sections
2025-11-21	2.0	**Merged documentation:**- Updated all signal names (sched_rd_, *sched_wr_*)- Added runtime timeout configuration (cfg_sched_timeout_cycles/enable)- Registered ready signal timing clarification- Added multiple requests per descriptor section- Enhanced beat tracking and error handling details- Updated all code examples and timing diagrams- Added timing examples for chained descriptors- Combined best content from multiple documentation sources
2025-11-30	2.1	**RTL Sync Update:**- CH_ERROR is now STICKY (requires reset to recover)- scheduler_idle asserts in CH_ERROR state- Updated related documentation references
2026-07-02	2.2	**RTL Sync Update:**- Added `cfg_sched_timeout_limit` port

Date	Version	Changes
		(SCHED_TIMEOUT_LIMIT @ 0x208)- Write completion now gated on B-response commit (`r_write_beats_to_commit`); `sched_wr_valid` gated on remaining issue beats- Write-progress timeout is now recoverable: re-arms per window, escalates to CH_ERROR only after `cfg_sched_timeout_limit` consecutive stalled windows- `scheduler_idle` now EXCLUDES CH_ERROR (a faulted channel is no longer reported idle) — corrects the 2.1 note
2026-10-17	2.3	**RTL Sync Update:**- `descriptor_ready` in CH_IDLE now requires `cfg_channel_enable`. Before, a disabled channel accepted the handshake and discarded the descriptor while it stayed in CH_IDLE

Last Updated: 2026-10-17 (matched to current RTL implementation)

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10 Descriptor Engine

Module: `descriptor_engine.sv` **Location:** `projects/components/dma-ip/stream/rtl/fub/` **Category:** FUB (Functional Unit Block) **Parent:** `scheduler_group.sv` (`SHARED_ENGINES=0`); the shared variant `descriptor_engine_shared.sv` sits in `scheduler_group_array.sv` (`SHARED_ENGINES=1`, the default) **Status:** Implemented **Last Updated:** 2026-10-17

10.1 Overview

The `descriptor_engine` module is an autonomous descriptor fetch engine with chaining support. It fetches 256-bit STREAM descriptors from memory via AXI4 master interface and provides two operating modes for kick-off.

10.1.1 Key Features

- **Two Operating Modes:**
  - APB-initiated: Software writes descriptor address, engine fetches
  - Autonomous chaining: Engine automatically fetches next_descriptor_ptr
- **Two-FIFO Architecture:**
  - Descriptor address FIFO: Holds addresses to fetch
  - Descriptor data FIFO: Buffers fetched descriptors for scheduler
- **Address Range Validation:** Two configurable ranges for security
- **MonBus Event Reporting:** Fetch complete and error events

10.2 Architecture

10.2.1 Operating Flow

APB Mode:

```
APB write -> skid buffer -> desc addr FIFO -> AXI fetch -> desc data
FIFO -> scheduler
```

Chaining Mode:

```
Descriptor complete -> extract next_ptr -> validate -> desc addr FIFO
->
AXI fetch -> desc data FIFO -> scheduler (repeat until last=1 or
next_ptr=0)
```

10.2.2 FSM States

```
typedef enum logic [2:0] {
    RD_IDLE       = 3'b000,  // Waiting for descriptor address
    RD_ISSUE_ADDR = 3'b001,  // Issue AXI AR transaction
    RD_WAIT_DATA  = 3'b010,  // Wait for AXI R response
    RD_COMPLETE   = 3'b011,  // Descriptor fetched, push to FIFO
    RD_ERROR      = 3'b100   // Error occurred
} read_engine_state_t;
```

10.2.3 Descriptor Format (256-bit STREAM)

Descriptor Format

Bits	Field	Description
[63:0]	src_addr	Source memory address (64-bit, must be aligned)
[127:64]	dst_addr	Destination memory address (64-bit, must be aligned)
[159:128]	length	Transfer length in BEATS (not bytes!)
[191:160]	next_descriptor_ptr	Address of next descriptor (32-bit, 0 = last)
[192]	valid	Descriptor valid flag
[193]	gen_irq	Generate interrupt on completion
[194]	last	Last descriptor in chain (explicit termination)
[195]	error	Descriptor error flag
[199:196]	channel_id	Channel identifier

Bits	Field	Description
[207:200]	priority	Descriptor priority
[255:208]	reserved	Reserved for future use

10.3 Parameters

Parameters

Parameter	Type	Default	Description
CHANNEL_ID	int	0	Channel identifier
NUM_CHANNELS	int	32	Total number of channels
CHAN_WIDTH	int	$clog2(NUM_CHANNELS)	Channel ID width
ADDR_WIDTH	int	64	Address bus width
AXI_ID_WIDTH	int	8	AXI transaction ID width
FIFO_DEPTH	int	8	Descriptor data FIFO depth
DESC_ADDR_FIFO_DEPTH	int	2	Descriptor address FIFO depth
TIMEOUT_CYCLES	int	1000	AXI timeout threshold

10.3.1 Monitor Bus Parameters

Monitor Bus Parameters

Parameter	Type	Default	Description
MON_AGENT_ID	8-bit	0x10	Descriptor Engine Agent ID
MON_UNIT_ID	4-bit	0x1	Unit identifier
MON_CHANNEL_ID	6-bit	0x0	Base channel ID

10.4 Port List

10.4.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low asynchronous reset

10.4.2 APB Programming Interface

APB Programming Interface

Signal	Direction	Width	Description
apb_valid	input	1	Descriptor address valid
apb_ready	output	1	Ready to accept address
apb_addr	input	ADDR_WIDTH	Descriptor address

10.4.3 Scheduler Interface

Scheduler Interface

Signal	Direction	Width	Description
channel_idle	input	1	Scheduler idle (enables APB)
descriptor_valid	output	1	Descriptor available
descriptor_ready	input	1	Scheduler ready to accept
descriptor_packet	output	256	FIXED 256-bit descriptor
descriptor_error	output	1	Fetch error flag

10.4.4 Enhanced Control Outputs

Enhanced Control Outputs

Signal	Direction	Width	Description
descriptor_eos	output	1	End of Stream (future use)
descriptor_eol	output	1	End of Line (future use)
descriptor_eod	output	1	End of Data (future use)
descriptor_type	output	2	Packet type (future use)

10.4.5 AXI4 AR Channel

AXI4 AR Channel

Signal	Direction	Width	Description
ar_valid	output	1	Address valid
ar_ready	input	1	Address ready
ar_addr	output	ADDR_WIDTH	Address
ar_len	output	8	Burst length - 1
ar_size	output	3	Burst size (log2 bytes)
ar_burst	output	2	Burst type (INCR)
ar_id	output	AXI_ID_WIDTH	Transaction ID
ar_lock	output	1	Lock type
ar_cache	output	4	Cache attributes
ar_prot	output	3	Protection attributes
ar_qos	output	4	QoS value
ar_region	output	4	Region identifier

10.4.6 AXI4 R Channel (FIXED 256-bit)

AXI4 R Channel

Signal	Direction	Width	Description
r_valid	input	1	Read data valid
r_ready	output	1	Read data ready
r_data	input	256	Read data (FIXED 256-bit)
r_resp	input	2	Response
r_last	input	1	Last beat
r_id	input	AXI_ID_WIDTH	Transaction ID

10.4.7 Configuration Interface

Configuration Interface

Signal	Direction	Width	Description
cfg_prefetch_enable	input	1	Enable descriptor prefetch (functional): 0 = on-demand (≈1 buffered ahead), 1 = buffer up to cfg_fifo_threshold ahead
cfg_fifo_threshold	input	4	Max descriptors buffered ahead when prefetch enabled (min 1; 0 is clamped to 1)
cfg_addr0_base	input	ADDR_WIDTH	Address range 0 base
cfg_addr0_limit	input	ADDR_WIDTH	Address range 0 limit
cfg_addr1_base	input	ADDR_WIDTH	Address range 1 base
cfg_addr1_limit	input	ADDR_WIDTH	Address range 1 limit

Signal	Direction	Width	Description
cfg_channel_reset	input	1	Channel reset

10.4.8 Status Interface

Status Interface

Signal	Direction	Width	Description
descriptor_engine_idle	output	1	Engine idle

10.4.9 Monitor Bus Interface

Monitor Bus Interface

Signal	Direction	Width	Description
mon_valid	output	1	Monitor packet valid
mon_ready	input	1	Monitor packet ready
mon_packet	output	64	Monitor packet data

10.5 Operation

10.5.1 APB Acceptance Policy

APB writes are accepted ONLY when ALL conditions met: 1. `!r_channel_reset_active` - Not in channel reset 2. `w_desc_addr_fifo_empty` - No pending descriptor fetches 3. `channel_idle` - Scheduler completed all prior descriptors 4. `!r_apb_ip` - No APB transaction currently in progress 5. `apb_addr != 0` - Address 0 is invalid

10.5.2 Autonomous Chaining Decision Tree

Level 1: Basic Eligibility (w_chain_condition) - `next_descriptor_ptr != 0` (non-null pointer) - `descriptor.last == 0` (not explicitly marked as last) - `descriptor.valid == 1` (valid descriptor)

Level 2: Address Validation (w_next_addr_valid) - next_addr within `cfg_addr0_base..cfg_addr0_limit` OR - next_addr within `cfg_addr1_base..cfg_addr1_limit`

Level 3: Prefetch Throttle Gate (w_should_chain)

`cfg_prefetch_enable` and `cfg_fifo_threshold` are **functional** and bound how many descriptors the engine buffers ahead of the scheduler:

- **Prefetch disabled (`cfg_prefetch_enable = 0`):** on-demand mode. The prefetch limit is 1, so the engine keeps roughly one descriptor buffered ahead — it fetches the next descriptor only after the scheduler drains the current one.
- **Prefetch enabled (`cfg_prefetch_enable = 1`):** the engine buffers up to `cfg_fifo_threshold` descriptors ahead (minimum 1; a threshold of 0 is clamped to 1 to avoid stalling chaining).

An immediate chain push (`w_should_chain`) fires at descriptor commit only when Levels 1-2 hold, the descriptor FIFO has space (`w_desc_fifo_wr_ready`), and the throttle allows another fetch (`w_desc_fifo_count < prefetch_limit`).

Deferred chaining: A chain fetch that is eligible at commit but blocked by the throttle (or a full address FIFO) is **not dropped** — the engine latches the pending `next_descriptor_ptr` (`r_chain_pending` / `r_pending_chain_addr`) and reissues it later, once the output descriptor FIFO drains below the prefetch limit.

10.5.3 AXI Transaction

```
ar_len = 8'h00;           // Single beat transfer
ar_size = 3'b110;         // 64 bytes (512-bit for 256-bit descriptor)
ar_burst = 2'b01;         // INCR burst type
ar_id = {{padding}, CHANNEL_ID};  // Channel ID in lower bits
```

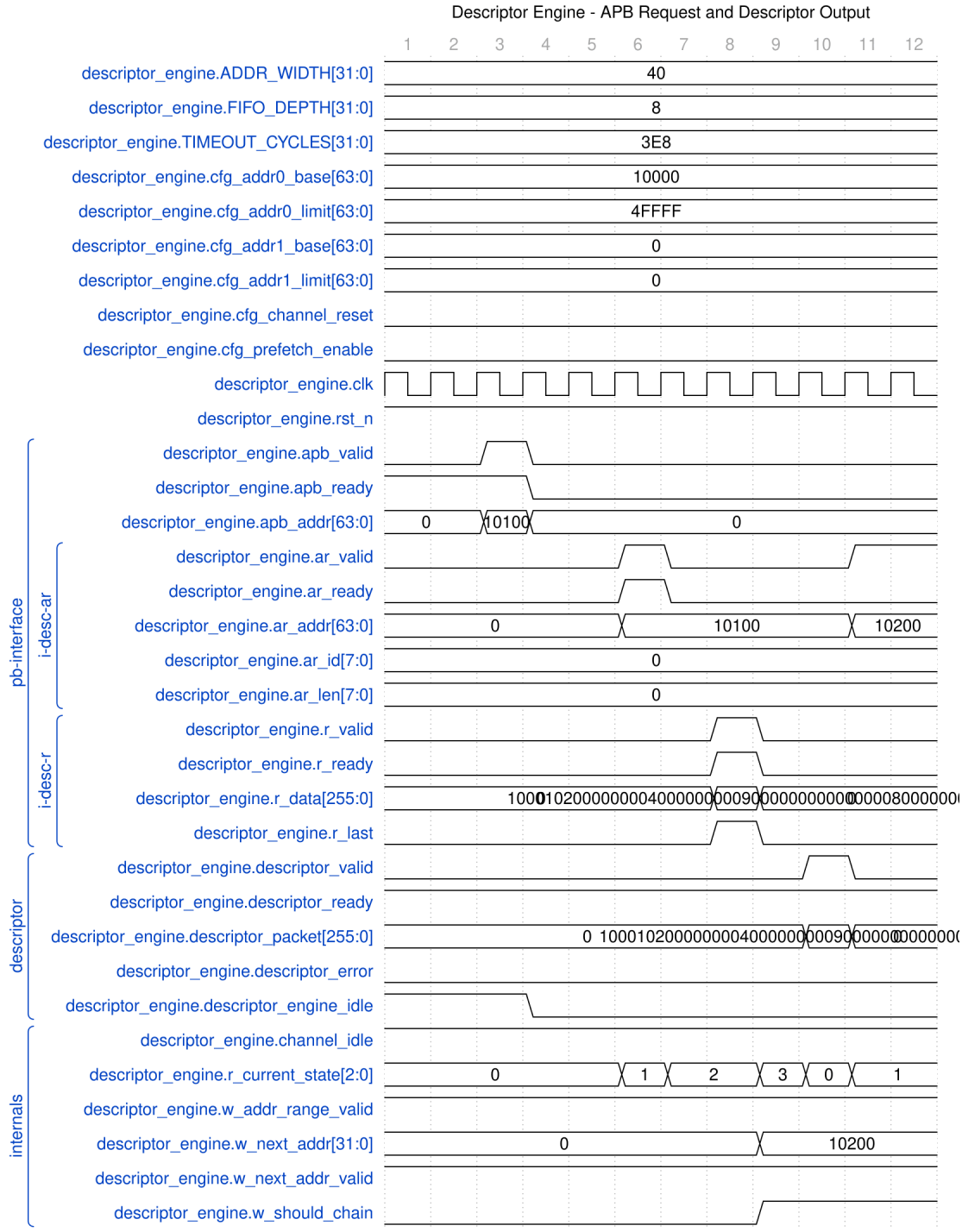
```
ar_cache = 4'b0010;         // Normal non-cacheable bufferable
ar_prot = 3'b000;           // Data, secure, unprivileged
```

10.6 Timing Diagrams

10.6.1 APB Kick-off Sequence

The following diagram shows the APB-initiated descriptor kick-off sequence:

10.6.1.1 Waveform 2.5.1: Descriptor Engine APB Basic Kick-off



Descriptor Engine APB Basic

Source: descriptor_engine_apb_basic.json

10.7 Integration Example

```
descriptor_engine #(
    .CHANNEL_ID         (ch),
    .NUM_CHANNELS       (8),
    .ADDR_WIDTH         (64),
    .AXI_ID_WIDTH       (8),
    .FIFO_DEPTH         (8),
    .MON_AGENT_ID       (8'h10 + ch)
) u_descriptor_engine (
    .clk                (clk),
    .rst_n              (rst_n),

    // APB interface
    .apb_valid          (apb_valid),
    .apb_ready          (apb_ready),
    .apb_addr           (apb_addr),

    // Scheduler interface
    .channel_idle       (scheduler_idle),
    .descriptor_valid   (desc_valid),
    .descriptor_ready   (desc_ready),
    .descriptor_packet  (desc_packet),
    .descriptor_error   (desc_error),

    // AXI interface
    .ar_valid           (ar_valid),
    .ar_ready           (ar_ready),
    .ar_addr            (ar_addr),
    // ... full AR/R interface

    // Configuration
    .cfg_prefetch_enable    (cfg_desceng_prefetch),
    .cfg_addr0_base         (cfg_desceng_addr0_base),
    .cfg_addr0_limit        (cfg_desceng_addr0_limit),
    .cfg_addr1_base         (cfg_desceng_addr1_base),
    .cfg_addr1_limit        (cfg_desceng_addr1_limit),
    .cfg_channel_reset      (cfg_channel_reset),

    // Status
    .descriptor_engine_idle (desc_engine_idle),

    // Monitor bus
    .mon_valid          (desceng_mon_valid),
    .mon_ready          (desceng_mon_ready),
    .mon_packet         (desceng_mon_packet)
);
```

10.8 Common Issues

10.8.1 Issue 1: APB Write Not Accepted

Symptom: `apb_ready` never asserts

Root Causes: 1. Channel not idle (`channel_idle = 0`) 2. Previous APB transaction still in progress (`r_apb_ip = 1`) 3. Descriptor address FIFO not empty 4. Channel reset active

Solution: Wait for channel to complete current work before next kick-off.

10.8.2 Issue 2: Descriptor Chain Stops Early

Symptom: Chaining stops before expected

Root Causes: 1. `next_descriptor_ptr` is 0 or out of valid range 2. `last` flag set in descriptor 3. AXI error during fetch

Debug: Check descriptor memory contents and address range configuration.

10.9 Monitoring (BOTH-End Descriptor Path Observability)

10.9.1 Descriptor Fetch Monitoring (Master Side)

The descriptor engine's AXI4 master interface is monitored by a dedicated AXI monitor wrapper with: - **Agent ID:** 8 (descriptor fetch monitor) - **Unit ID:** 1 (STREAM subsystem) - **Monitored Port:** `m_axi_desc_*` (AXI4 read burst for descriptor fetch)

This monitor tracks: - Descriptor fetch initiation (address phase) - Fetch completion (data phase) - Fetch errors or timeouts (AXI SLVERR, no response)

10.9.2 Descriptor Consume Monitoring (Slave Side)

The scheduler's consumption of descriptors (as a "slave" to the descriptor engine) is monitored separately with: - **Agent ID:** 48 + channel index (descriptor consume monitor, per-channel) - **Unit ID:** 1 (STREAM subsystem) - **Monitored Port:** `descriptor_valid` / `descriptor_ready` handshake

This monitor tracks: - When the scheduler pops descriptors from the engine's output FIFO - Consume latency (time from engine ready to scheduler accept) - Consume stalls (scheduler not ready while descriptors pending)

10.9.3 Why Both Ends?

With dual-end monitoring, software can distinguish: - **Fetch latency** (master-side agent): Time from kick-off to descriptor received - **Consume stall** (slave-side agent): Time descriptor sits in FIFO awaiting scheduler

A slow transfer can be attributed to either: - Network/memory delay (fetch stall visible at master-side agent) - Scheduler queue full (consume stall visible at slave-side agent)

Both monitors emit MonBus packets with identical protocol-independent structure but distinct agent IDs for granular root-cause analysis.

10.10 Shared Variant: descriptor_engine_shared

With SHARED_ENGINES=1 (the default), scheduler_group_array builds one descriptor_engine_shared instead of NUM_CHANNELS copies of descriptor_engine. Each scheduler sees the same behaviour, channel for channel: the same kick policy, range check on the kicked address, EXT second-chunk fetch, prefetch throttle, autonomous chaining, error pulse, channel reset and MonBus completion and error packets. The per-channel interface signals are the ones in the Port List with a [NC-1:0] dimension added.

Shared by all channels:

- One AR issuer. It picks round-robin among the channels that have an address to fetch and owns the descriptor AXI master outright, so no arbiter sits behind it.
- The address-range comparators, used for the kick address at issue and the next pointer at chaining time.
- The R path. Each beat is routed by RID to its channel and written straight into that channel's descriptor FIFO, so no channel needs a 256-bit holding register. The exception is EXT chunk 0, which must wait for chunk 1.
- The MonBus reporter. The agent is MON_AGENT_BASE + channel and the channel_id is the channel.

Per channel: a state code, one pending fetch address, the kick-in-progress flag, the descriptor FIFO, and the EXT chunk-1 FIFO.

Fetches carry ARID = channel. Different channels' fetches overlap on the bus, while each channel keeps one fetch in flight and stays in order.

Difference from descriptor_engine: an R beat that arrives for a channel that is not waiting (one reset mid-fetch) is accepted and dropped. Leaving it on the shared R channel would stall every other channel.

descriptor_engine_shared Parameters (differences from descriptor_engine)

Parameter	Default	Description
NUM_CHANNELS	8	Channels served (8 or 16 in the shipped tops)
FIFO_DEPTH	8	Per-channel descriptor FIFO depth
MON_AGENT_BASE	16'h0010	MonBus agent of channel 0; channel n reports MON_AGENT_BASE + n
GEN_MON	1	0 = no MonBus packets

The engine is ported from RAPIDS (`rapids/rtl/fub/descriptor_engine_shared.sv`) without the RAPIDS tile injection port. The EXT type field is STREAM's `[210:208]`, and the descriptor type sideband is always 0 because STREAM has no opcode. Area and timing for the shared and per-channel builds are in Scheduler Group Array.

10.11 Related Documentation

- **Parent:** `03_scheduler_group.md` - Channel wrapper
- **Consumer:** `04_scheduler.md` - Scheduler that receives descriptors

- **Arbiter: `02_scheduler_group_array.md` - Descriptor AXI arbitration (per-channel build) and the shared engine's placement**

10.12 Revision History

Descriptor Engine Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers
0.93	2026-07-02	seang	`cfg_prefetch_enable` / `cfg_fifo_threshold` are now functional; documented prefetch-throttle gate (on-demand vs. buffer-ahead) and deferred-chain (`r_chain_pending`) behavior
0.94	2026-10-17	seang	Added the shared variant, `descriptor_engine_shared` (`SHARED_ENGINES=1`, the default)

Last Updated: 2026-10-17

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11 AXI Read Engine

Module: `axi_read_engine.sv` **Location:** `projects/components/dma-ip/stream/rtl/fub/` **Category:** FUB (Functional Unit Block) **Parent:** `stream_core.sv` **Status:** Implemented **Last Updated:** 2026-09-29

11.1 Overview

The `axi_read_engine` module is a high-performance multi-channel AXI4 read engine with space-aware arbitration. It serves all 8 STREAM channels through a single AXI master interface with intelligent flow control.

11.1.1 Key Features

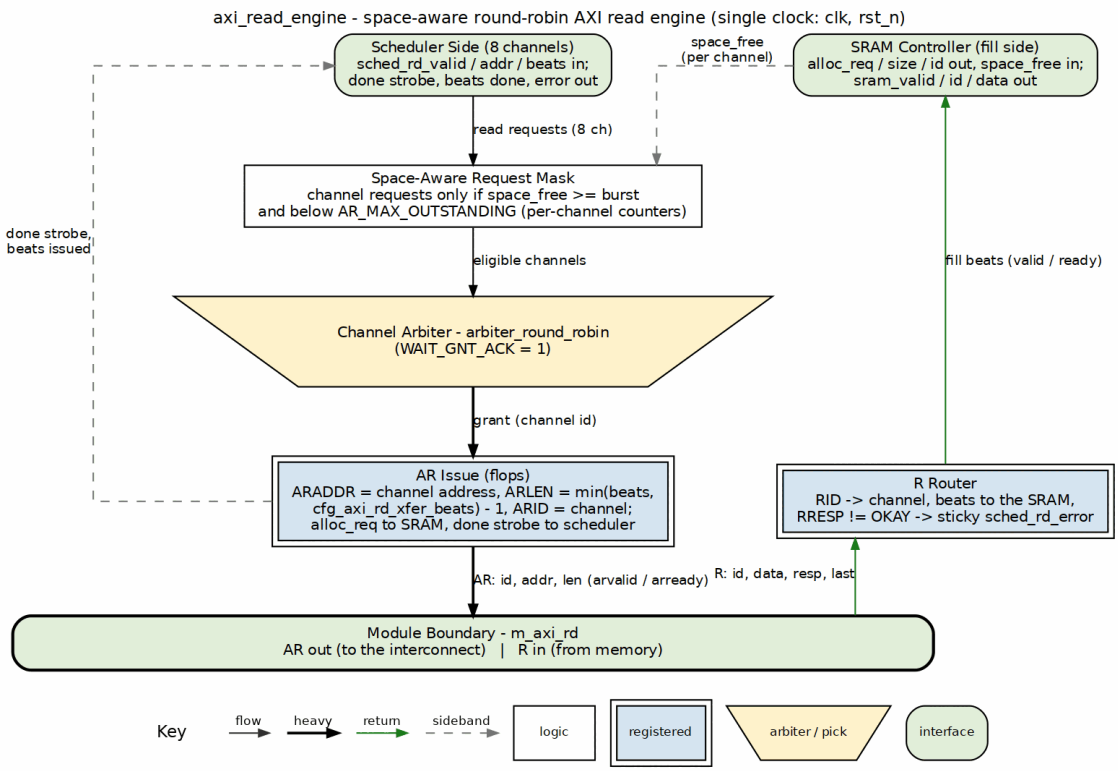
- **Round-Robin Arbitration:** Fair scheduling across channels
- **Space-Aware Masking:** Only arbitrate channels with sufficient SRAM space
- **Pre-Allocation Handshake:** Reserve SRAM space before data arrives
- **Streaming Data Path:** Direct passthrough to SRAM controller
- **Channel ID in AXI ID:** Enables per-channel response routing
- **Pipelined/Non-Pipelined Modes:** Configurable outstanding transaction depth

---

11.2 Architecture

11.2.1 Block Diagram

11.2.2 Figure 2.6.1: AXI Read Engine Block Diagram



AXI Read Engine Block Diagram

Source: 08_axi_read_engine.dot

11.2.3 Operation Flow

```
1. Scheduler Interface: Each channel can request read bursts
2. Space Checking: Mask channels without sufficient SRAM space
3. Arbitration: Round-robin arbiter selects next channel to service
4. AXI AR Issue: Issue read command to AXI, assert rd_alloc to SRAM
controller
5. AXI R Response: Stream read data directly to SRAM controller
```

11.2.4 Key Design Decisions

Combinational AR Outputs: AR outputs are driven combinationally from the arbiter to avoid 1-cycle delay. When `axi_rd_alloc_space_free` goes to 0, `arvalid` drops in the same cycle.

No Internal Buffering: The engine is a streaming pipeline with no internal data storage. Data flows directly from AXI R channel to SRAM controller.

11.2.5 System Idle Behavior

When ALL channels complete their work (`w_arb_request == 0`), the engine becomes idle until new requests arrive. This is NOT a bubble - it's legitimate system idle.

Why This Matters for Testing: - With few active channels and short bursts, there are brief periods when all channels are in their WRITE phase - During these periods, `m_axi_arvalid = 0` (no R channel activity) - This is EXPECTED behavior, not a performance bug - The `dbg_arb_request` signal exposes `w_arb_request` to help testbenches distinguish: - **TRUE BUBBLE:** arvalid=0 while arb_request!=0 (channels waiting, arbiter stalled) - **SYSTEM IDLE:** arvalid=0 while arb_request==0 (no channels need service)

11.3 Parameters

Parameters

Parameter	Type	Default	Description
NUM_CHANNELS	int	8	Number of channels
ADDR_WIDTH	int	64	AXI address width
DATA_WIDTH	int	512	AXI data width
ID_WIDTH	int	8	AXI ID width
SEG_COUNT_WIDTH	int	8	Width of space/count signals
PIPELINE	int	1	0: non-pipelined, 1: pipelined (default 1 since 2026-09-28)
AR_MAX_OUTSTANDING	int	8	Maximum outstanding AR requests (PIPELINE=1)
STROBE_EVERY_BEAT	int	0	0: strobe on last beat, 1: strobe every beat

11.3.1 Derived Parameters

Derived Parameters

Parameter	Derivation	Description
NC	NUM_CHANNELS	Short alias
AW	ADDR_WIDTH	Short alias
DW	DATA_WIDTH	Short alias
IW	ID_WIDTH	Short alias
SCW	SEG_COUNT_WIDTH	Segment count width
CIW	$clog2(NC)	Channel ID width (min 1 bit)

11.4 Port List

11.4.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low asynchronous reset

11.4.2 Configuration Interface

Configuration Interface

Signal	Direction	Width	Description
`cfg_axi_rd_xfer_beats`	input	8	Transfer size in beats (all channels)

11.4.3 Scheduler Interface (Per-Channel)

Scheduler Interface

Signal	Direction	Width	Description
`sched_rd_valid[ch]`	input	NC	Channel requests read
`sched_rd_addr[ch]`	input	NC x AW	Source addresses
`sched_rd_beats[ch]`	input	NC x 32	Beats remaining to read

11.4.4 Completion Interface (Per-Channel)

Completion Interface

Signal	Direction	Width	Description
`sched_rd_done_strobe[ch]`	output	NC	Burst completed (1 cycle pulse)
`sched_rd_beats_done[ch]`	output	NC x 32	Number of beats completed

11.4.5 SRAM Allocation Interface

SRAM Allocation Interface

Signal	Direction	Width	Description
axi_rd_alloc_req	output	1	Request space allocation
axi_rd_alloc_size	output	8	Beats to reserve
axi_rd_alloc_id	output	IW	Transaction ID (channel)
axi_rd_alloc_space_free[ch]	input	NC x SCW	Free space per channel

11.4.6 SRAM Write Interface

SRAM Write Interface

Signal	Direction	Width	Description
axi_rd_sram_valid	output	1	Read data valid
axi_rd_sram_ready	input	1	Ready to accept data
axi_rd_sram_id	output	IW	Transaction ID (channel)
axi_rd_sram_data	output	DW	Read data payload

11.4.7 AXI4 AR Channel

AXI4 AR Channel

Signal	Direction	Width	Description
m_axi_arvalid	output	1	Address valid
m_axi_arready	input	1	Address ready
m_axi_arid	output	IW	Transaction ID
m_axi_araddr	output	AW	Address
m_axi_arlen	output	8	Burst length - 1
m_axi_arsize	output	3	Burst size (log2 bytes)

Signal	Direction	Width	Description
m_axi_arburst	output	2	Burst type (INCR)

11.4.8 AXI4 R Channel

AXI4 R Channel

Signal	Direction	Width	Description
m_axi_rvalid	input	1	Read data valid
m_axi_rready	output	1	Read data ready
m_axi_rid	input	IW	Transaction ID
m_axi_rdata	input	DW	Read data
m_axi_rresp	input	2	Response
m_axi_rlast	input	1	Last beat of burst

11.4.9 Error Interface

Error Interface

Signal	Direction	Width	Description
sched_rd_error[ch]	output	NC	Sticky error flag per channel

11.4.10 Debug Interface

Debug Interface

Signal	Direction	Width	Description
dbg_rd_all_complete[ch]	output	NC	All reads complete
dbg_r_beats_rcvd	output	32	Total R beats received
dbg_sram_writes	output	32	Total writes to SRAM
dbg_arb_request[ch]	output	NC	Arbiter request

Signal	Direction	Width	Description
			vector

11.5 Operation

11.5.1 Burst Length Cap (stream BUG-018)

`cfg_axi_rd_xfer_beats` is an ARLEN (0..255), but a burst can never be larger than the SRAM it lands in: the space mask below pre-allocates the whole burst before the AR, and a burst the buffer cannot hold would never be allocatable. The engine clamps the configured value to `XFER_MAX = min(2^(SCW-1) - 1, 254)` – the buffer depth `SEG_COUNT_WIDTH` encodes, minus one, and 254 beyond that so every beat count fits the 8-bit size port shared with `sram_controller`. On the 512-deep default ARLEN 255 runs as 255-beat bursts.

Before the clamp (2026-09-29, found as rapids BUG-009 on the identical lines), ARLEN 255 was taken literally and the 8-bit `ARLEN + 1` in the gate wrapped to 0: `w_space_ok` passed on any free count, the AR for 256 beats allocated 0, and the engine over-fetched past the buffer.

11.5.2 Space-Aware Request Masking

```
// Only request arbitration if:
// 1. Scheduler is requesting (sched_rd_valid)
// 2. Sufficient SRAM space available (w_space_ok)
// 3. Below outstanding limit (w_below_outstanding_limit)
w_arb_request[i] = sched_rd_valid[i] && w_space_ok[i] &&
w_below_outstanding_limit[i];
```

11.5.3 Transfer Size Calculation

```
// Calculate actual transfer size for this channel
// Min of remaining beats or configured max
w_transfer_size[i] = (sched_rd_beats[i] <= cfg_axi_rd_xfer_beats +
1) ?
                     (sched_rd_beats[i] - 1) : cfg_axi_rd_xfer_beats;
```

11.5.4 Outstanding Transaction Tracking

PIPELINE=0 (Non-Pipelined): - Binary flag per channel (0 or 1 outstanding) - Set when AR issues for channel - Clear when R last arrives for channel

PIPELINE=1 (Pipelined): - Counter per channel (0 to AR_MAX_OUTSTANDING) - Increment on AR handshake - Decrement on R last handshake

11.5.5 Completion Strobe

Strobes fire when AR transaction is accepted (not when R completes):

```
// Pulse when AR transaction is accepted (arvalid && arready)
// This tells scheduler that request was issued to AXI bus
if (m_axi_arvalid && m_axi_arready) begin
    r_done_strobe[w_arb_grant_id] <= 1'b1;
    r_beats_done[w_arb_grant_id] <= w_transfer_size[w_arb_grant_id] +
```

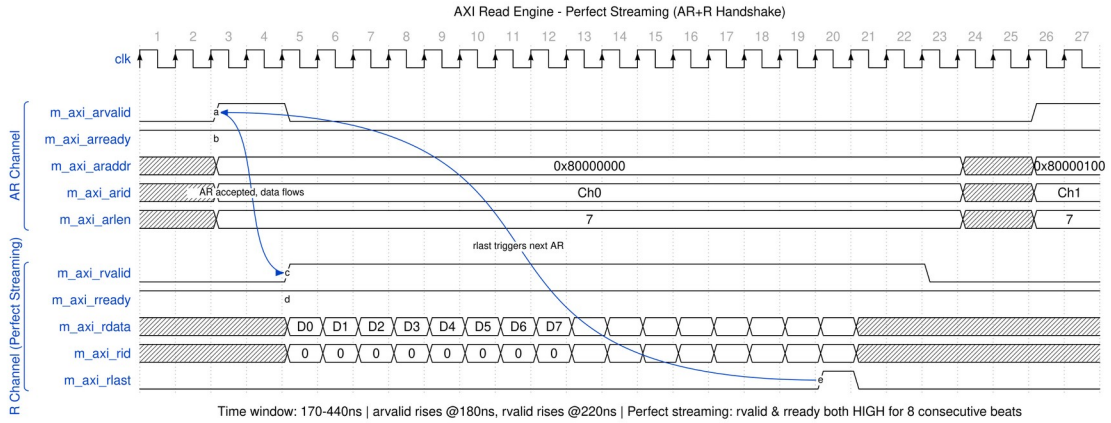
```
1;
end
```

11.6 Timing Diagrams

11.6.1 Perfect Streaming - AXI Read Transaction

The following timing diagram shows the AXI read engine operating at maximum throughput with **perfect streaming** - where both `rvalid` and `rready` remain HIGH for consecutive clock cycles, achieving one data beat per cycle.

11.6.1.1 Waveform 2.6.1: AXI Read Engine - Perfect Streaming



AXI Read Engine - Perfect Streaming

Transaction Flow:

1. **AR Channel Handshake (Address Phase)**
   - `m_axi_arvalid` rises to initiate a read request
   - `m_axi_arready` is already HIGH (slave ready)
   - Address (`araddr`), ID (`arid`), and length (`arlen=7` for 8 beats) are captured
   - Handshake completes in a single cycle when both valid and ready are HIGH
2. **R Channel Streaming (Data Phase)**
   - After AR acceptance, the AXI slave begins returning data
   - `m_axi_rvalid` rises and **stays HIGH** for all 8 beats
   - `m_axi_rready` remains HIGH throughout (no backpressure from SRAM controller)
   - **Perfect streaming**: One data beat transferred every clock cycle
   - `m_axi_rlast` rises on the final beat (beat 7) to mark burst completion
3. **Next Transaction**

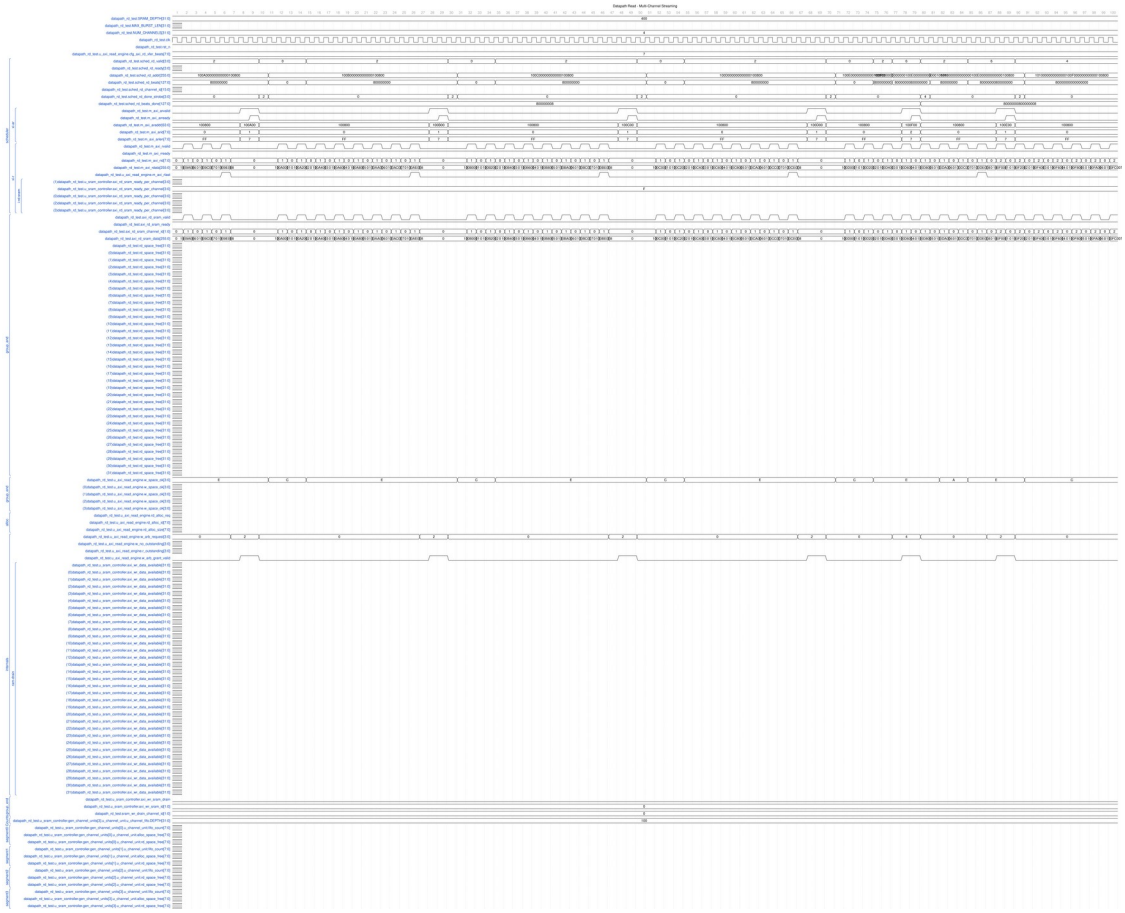
- After `rlast`, the engine can immediately issue the next AR request
- The cycle repeats for the next channel or next burst

Key Performance Indicators: - **No bubbles**: `rvalid` and `rready` both HIGH during data phase - **Full bandwidth**: Data width (256/512 bits) x clock frequency - **Zero-wait states**: SRAM controller accepts data at line rate

11.6.2 Multi-Channel Streaming

For multi-channel operation showing channel switching while maintaining streaming performance, see:

11.6.2.1 Waveform 2.6.2: Datapath Read - Multi-Channel



Datapath Read - Multi-Channel

This diagram shows how the engine arbitrates between channels while maintaining high throughput.

11.7 Integration Example

```
axi_read_engine #(
    .NUM_CHANNELS       (8),
    .ADDR_WIDTH         (64),
    .DATA_WIDTH         (512),
    .ID_WIDTH           (8),
    .SEG_COUNT_WIDTH    (10),
    .PIPELINE           (0),
    .AR_MAX_OUTSTANDING (8)
) u_axi_read_engine (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // Configuration
    .cfg_axi_rd_xfer_beats  (cfg_axi_rd_xfer_beats),

    // Scheduler interface
    .sched_rd_valid         (sched_rd_valid),
    .sched_rd_addr          (sched_rd_addr),
    .sched_rd_beats         (sched_rd_beats),

    // Completion interface
    .sched_rd_done_strobe   (sched_rd_done_strobe),
    .sched_rd_beats_done    (sched_rd_beats_done),

    // SRAM allocation interface
    .axi_rd_alloc_req       (axi_rd_alloc_req),
    .axi_rd_alloc_size      (axi_rd_alloc_size),
    .axi_rd_alloc_id        (axi_rd_alloc_id),
    .axi_rd_alloc_space_free(axi_rd_alloc_space_free),

    // SRAM write interface
    .axi_rd_sram_valid      (axi_rd_sram_valid),
    .axi_rd_sram_ready      (axi_rd_sram_ready),
    .axi_rd_sram_id         (axi_rd_sram_id),
    .axi_rd_sram_data       (axi_rd_sram_data),

    // AXI master
    .m_axi_arvalid          (m_axi_rd_arvalid),
    .m_axi_arready          (m_axi_rd_arready),
    .m_axi_arid             (m_axi_rd_arid),
    .m_axi_araddr           (m_axi_rd_araddr),
    .m_axi_arlen            (m_axi_rd_arlen),
    .m_axi_arsize           (m_axi_rd_arsize),
    .m_axi_arburst          (m_axi_rd_arburst),
```

```
    .m_axi_rvalid           (m_axi_rd_rvalid),
    .m_axi_rready           (m_axi_rd_rready),
    .m_axi_rid              (m_axi_rd_rid),
    .m_axi_rdata            (m_axi_rd_rdata),
    .m_axi_rresp            (m_axi_rd_rresp),
    .m_axi_rlast            (m_axi_rd_rlast),

    // Error and debug
    .sched_rd_error         (sched_rd_error),
    .dbg_rd_all_complete    (dbg_rd_all_complete),
    .dbg_arb_request        (dbg_arb_request)
);
```

11.8 Bug Fixes (v0.92)

11.8.1 w_arb_request Timing Closure (Commit 4e8f9e02)

At 8 channels, the arbiter request combinational path from scheduler to AXI master became critical. The `w_arb_request` signal is now **registered** to add a single pipeline stage, closing timing at the cost of 1 cycle of latency on the scheduler→engine request path.

Impact: ~1% throughput penalty in worst case, 100% timing closure margin gain. Highly beneficial for high-frequency designs.

11.9 Related Documentation

- **Parent:** `01_stream_core.md` - Top-level integration
- **Scheduler Array:** `02_scheduler_group_array.md` - Provides sched_rd_* signals
- **Allocation Controller:** `07_stream_alloc_ctrl.md` - Space tracking for read engine
- **SRAM Controller:** `08_sram_controller.md` - Receives read data

- **Write Engine: `12_axi_write_engine.md` - Complementary write datapath**

11.10 Revision History

AXI Read Engine Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers

Last Updated: 2026-09-29

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12 Stream Allocation Controller

Module: `stream_alloc_ctrl.sv` **Location:** `projects/components/dma-ip/stream/rtl/fub/` **Category:** FUB (Functional Unit Block)
Parent: `sram_controller_unit.sv` **Status:** Implemented **Last Updated:** 2025-11-30

12.1 Overview

The `stream_alloc_ctrl` module is a **virtual FIFO without data storage** that tracks space allocation and availability using FIFO pointer logic. It provides pre-allocation support for AXI read engines to prevent race conditions between burst requests and data arrival.

12.1.1 What Makes This a "Virtual FIFO"

Virtual FIFO: - Has write pointer (allocation pointer) - Has read pointer (actual write pointer) - Calculates full/empty/space_free - **NO data storage** - only pointer arithmetic

Use Case: Reserve FIFO space BEFORE AXI read data arrives

12.2 The Allocation Problem

12.2.1 Without Allocation Controller

Problem: Race condition between space check and data arrival

```
Cycle 0: Read engine checks FIFO space
         space_free = 100 beats → OK to issue 16-beat burst

Cycle 5: AR handshake completes, AXI read starts

Cycle 10: Meanwhile, write engine drains 90 beats from FIFO
          space_free = 10 beats (NOT ENOUGH!)

Cycle 15: AXI read data starts arriving (16 beats)
          → OVERFLOW! Only 10 beats of space available
```

12.2.2 With Allocation Controller

Solution: Reserve space when issuing AR, not when data arrives

```
Cycle 0: Read engine checks space_free = 100 beats
         Allocate 16 beats:
           rd_alloc_req = 1, rd_alloc_size = 16
           → space_free = 84 beats (reserved!)

Cycle 5: AR handshake completes

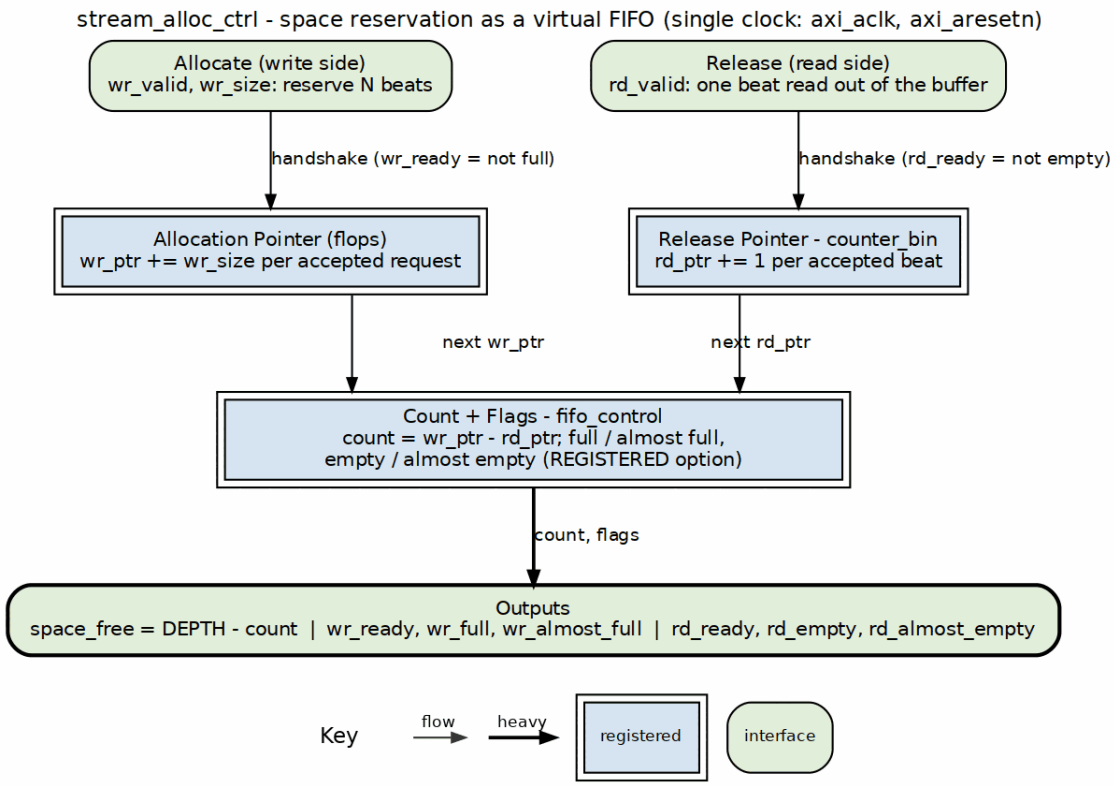
Cycle 10: Write engine checks space_free = 84 beats
          → Sees reduced space, won't overdrain

Cycle 15: AXI read data arrives, enters FIFO
          → Space was reserved, guaranteed to fit
```

12.3 Architecture

12.3.1 Two-Pointer System

12.3.2 Figure 2.7.1: Stream Allocation Controller Block Diagram



Diagram

Source: 07_stream_alloc_ctrl.dot

–>

Write Pointer (Allocation Pointer): - Advances when space is **allocated** (burst request) - Variable-size increment (rd_alloc_size) - Represents “reserved space”

Read Pointer (Actual Write Pointer): - Advances when data **exits the controller** (output handshake) - Single-beat increment - Represents “released space”

Space Calculation:

```
space_free = DEPTH - (wr_ptr - rd_ptr)
```

12.3.3 CRITICAL: Confusing Naming Convention

WARNING: Allocation controller uses OPPOSITE naming from normal FIFO!

Normal FIFO: | Signal | Meaning | |———|———| | `wr_*` | Write data → Consumes space → space_free decreases | | `rd_*` | Read data → Frees space → space_free increases |

: CRITICAL: Confusing Naming Convention

Allocation Controller: | Signal | Meaning | |———|———| | `wr_*` | **ALLOCATE** space → Reserves space → space_free **decreases** | | `rd_*` | **RELEASE** space → Data exits controller → space_free **increases** |

: CRITICAL: Confusing Naming Convention

Why This Matters:

```
// "Write" side = ALLOCATE (reserve space for upcoming burst)
// Read engine says "I need 16 beats"
.wr_valid (rd_alloc_req),
.wr_size  (rd_alloc_size),  // Variable size (16 in this example)

// "Read" side = RELEASE (data exits controller, free space)
// Must monitor OUTPUT handshake (after latency bridge!)
.rd_valid (axi_wr_sram_valid && axi_wr_sram_ready),  // Output side!
```

Key Insight: The “read” side of allocation controller connects to the OUTPUT of the FIFO + latency bridge, NOT the FIFO input!

12.4 Parameters

Parameters

Parameter	Type	Default	Description
DEPTH	int	512	Virtual FIFO depth (must match physical FIFO)
ALMOST_WR_MARGIN	int	1	Almost full threshold
ALMOST_RD_MARGIN	int	1	Almost empty threshold
REGISTERED	int	1	Register outputs for timing

12.5 Port List

12.5.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
axi_aclk	input	1	System clock
axi_aresetn	input	1	Active-low asynchronous reset

12.5.2 Write Interface (Allocation Requests)

Write Interface

Signal	Direction	Width	Description
wr_valid	input	1	Allocate space request
wr_size	input	8	Number of entries to allocate
wr_ready	output	1	Space available (! wr_full)

12.5.3 Read Interface (Actual Data Written)

Read Interface

Signal	Direction	Width	Description
rd_valid	input	1	Data exits controller (release space)
rd_ready	output	1	Not empty (! rd_empty)

12.5.4 Status Outputs

Status Outputs

Signal	Direction	Width	Description
space_free	output	AW+1	Available

Signal	Direction	Width	Description
			space (beats)
wr_full	output	1	Full flag (no space available)
wr_almost_full	output	1	Almost full flag
rd_empty	output	1	Empty flag (no allocations)
rd_almost_empty	output	1	Almost empty flag

12.6 Interfaces

12.6.1 Write Interface (Allocation Requests)

Write Interface

Signal	Direction	Width	Description
wr_valid	Input	1	Allocate space
wr_size	Input	8	Number of entries to allocate
wr_ready	Output	1	Space available (! wr_full)

Usage:

```
// Read engine allocates space before issuing AR
rd_alloc_req = (space_check_ok && !ar_pending);
rd_alloc_size = cfg_axi_rd_xfer_beats;
```

12.6.2 Read Interface (Actual Data Written)

Read Interface

Signal	Direction	Width	Description
rd_valid	Input	1	Data exits controller
rd_ready	Output	1	Not empty (! rd_empty)

Usage:

```
// Connect to OUTPUT handshake (after latency bridge)
assign rd_valid = (axi_wr_sram_valid && axi_wr_sram_ready);
```

12.6.3 Status Outputs

Status Outputs

Signal	Direction	Width	Description
space_free	Output	AW+1	Available unreserved space

Signal	Direction	Width	Description
wr_full	Output	1	No space available
wr_almost_full	Output	1	Almost full
rd_empty	Output	1	No allocations pending
rd_almost_empty	Output	1	Almost empty

Note: space_free is the most important output - used by read engine for space checking.

12.7 Operation

12.7.1 Allocation Flow

Step 1: Check Space

```
// Read engine checks space availability
if (space_free >= (cfg_axi_rd_xfer_beats << 1)) begin  // 2x margin
    // Space OK, proceed to allocation
end
```

Step 2: Allocate Space

```
rd_alloc_req = 1'b1;
rd_alloc_size = 8'd16;  // Reserve 16 beats

// Next cycle: wr_ptr advances
r_wr_ptr_bin <= r_wr_ptr_bin + 16;
space_free decreases by 16
```

Step 3: AXI Read Executes

```
// Some cycles later, AXI AR handshake completes
m_axi_arvalid = 1, m_axi_arready = 1
// AXI read starts
```

Step 4: Data Arrives at FIFO

```
// Many cycles later, AXI read data arrives
axi_rd_sram_valid = 1, axi_rd_sram_ready = 1
// Data enters FIFO (allocation controller doesn't see this directly)
```

Step 5: Data Exits Controller (After Latency Bridge)

```
// Eventually, data traverses FIFO + latency bridge
axi_wr_sram_valid = 1, axi_wr_sram_ready = 1

// THIS triggers allocation controller release!
rd_valid = 1
r_rd_ptr_bin <= r_rd_ptr_bin + 1
space_free increases by 1
```

12.7.2 Pointer Arithmetic

Write Pointer (Allocation):

```
// Variable-size increment
if (w_write && !r_wr_full) begin
    r_wr_ptr_bin <= r_wr_ptr_bin + (AW+1)'(wr_size);
end
```

Read Pointer (Release):

```
// Single-beat increment (uses counter_bin utility)
counter_bin #(
    .WIDTH (AW + 1),
    .MAX   (D)
) read_pointer_inst (
    .clk              (axi_aclk),
    .rst_n            (axi_aresetn),
    .enable           (w_read && !r_rd_empty),
    .counter_bin_curr (r_rd_ptr_bin),
    .counter_bin_next (w_rd_ptr_bin_next)
);
```

Space Calculation:

```
// Occupancy = wr_ptr - rd_ptr
w_count = w_wr_ptr_bin_next - w_rd_ptr_bin_next;

// Space free = total depth - occupancy
space_free = (AW+1)'(D) - w_count;
```

12.8 Timing Behavior

12.8.1 Allocation Latency

Allocation is IMMEDIATE (combinational + 1 cycle):

```
Cycle N:   rd_alloc_req = 1, rd_alloc_size = 16
Cycle N+1: r_wr_ptr_bin = old_value + 16
           space_free = old_value - 16
```

Read engine sees updated space_free on next cycle.

12.8.2 Release Latency

Release is IMMEDIATE (combinational + 1 cycle):

```
Cycle N:   axi_wr_sram_valid = 1, axi_wr_sram_ready = 1
           → rd_valid = 1
Cycle N+1: r_rd_ptr_bin = old_value + 1
           space_free = old_value + 1
```

Write engine sees updated space_free on next cycle.

12.9 Integration Example

12.9.1 In sram_controller_unit.sv

```
stream_alloc_ctrl #(
    .DEPTH(SD),
    .REGISTERED(1)
) u_alloc_ctrl (
    .axi_aclk           (clk),
    .axi_aresetn        (rst_n),

    // ALLOCATE (reserve space for upcoming burst)
    .wr_valid           (rd_alloc_req),      // From read engine
    .wr_size            (rd_alloc_size),     // Burst size
    .wr_ready           (),                  // Unused (space check
uses space_free)

    // RELEASE (data exits controller - OUTPUT handshake!)
    .rd_valid           (axi_wr_sram_valid && axi_wr_sram_ready),  //
After latency bridge!
    .rd_ready           (),                  // Unused

    // Space tracking
    .space_free         (alloc_space_free),  // To read engine (via
register)

    // Unused status
    .wr_full            (),
    .wr_almost_full     (),
    .rd_empty           (),
    .rd_almost_empty    ()
);

// Register output to break long paths
`ALWAYS_FF_RST(clk, rst_n,
    if (`RST_ASSERTED(rst_n)) begin
        rd_space_free <= SCW'(SD);  // Full space on reset
    end else begin
        rd_space_free <= alloc_space_free;
    end
)
```

12.10 Debug Support

12.10.1 Display Statements

Allocation:

```
$display("ALLOC @ %t: allocated %0d beats, wr_ptr: %0d -> %0d,
space_free will be %0d",
        $time, wr_size, r_wr_ptr_bin, r_wr_ptr_bin + wr_size,
        D - (r_wr_ptr_bin + wr_size - r_rd_ptr_bin));
```

Release (Drain):

```
$display("DRAIN @ %t: drained 1 beat, rd_ptr: %0d -> %0d, space_free
will be %0d",
        $time, r_rd_ptr_bin, w_rd_ptr_bin_next,
        D - (r_wr_ptr_bin - w_rd_ptr_bin_next));
```

12.10.2 Waveform Analysis

Key Signals to Monitor: - `r_wr_ptr_bin` - Allocation pointer - `r_rd_ptr_bin` - Release pointer - `space_free` - Available space - `rd_alloc_req`, `rd_alloc_size` - Allocation requests - `axi_wr_sram_valid`, `axi_wr_sram_ready` - Release handshake

12.11 Common Issues

12.11.1 Issue 1: Space Not Released

Symptom: `space_free` decreases but never increases

Root Cause: `rd_valid` not connected to output handshake

Wrong:

```
.rd_valid (axi_rd_sram_valid && axi_rd_sram_ready)  // FIFO input - WRONG!
```

Correct:

```
.rd_valid (axi_wr_sram_valid && axi_wr_sram_ready)  // After latency bridge - CORRECT!
```

12.11.2 Issue 2: Overflow Despite Allocation

Symptom: FIFO overflows even with allocation controller

Root Cause: Read engine uses wrong space value

Wrong:

```
if (space_free >= cfg_axi_rd_xfer_beats) begin  // Exact match - WRONG!
    allocate();
end
```

Correct:

```
if (space_free >= (cfg_axi_rd_xfer_beats << 1)) begin  // 2x margin - CORRECT!
    allocate();
end
```

Why 2x: Accounts for in-flight allocations and pipeline delays.

12.11.3 Issue 3: Allocation Pointer Overflow

Symptom: `r_wr_ptr_bin` wraps at unexpected value

Root Cause: Pointer width insufficient

Check:

```
parameter int AW = $clog2(D);  // Address width
logic [AW:0] r_wr_ptr_bin;     // AW+1 bits (for full detection)
```

Extra bit allows distinguishing full (wr_ptr = rd_ptr + DEPTH) from empty (wr_ptr = rd_ptr).

12.12 Comparison with Drain Controller

Comparison with Drain Controller

Aspect	Allocation Controller	Drain Controller
Purpose	Reserve FIFO space	Reserve FIFO data
User	AXI read engine	AXI write engine
Write side	Allocation request (reserve)	Data enters FIFO (increment)
Read side	Data exits controller (release)	Drain request (reserve)
Naming	OPPOSITE of normal FIFO	SAME as normal FIFO
Output	`space_free`	`data_available`

12.13 Resource Utilization

Per Instance: - 2 × (AW+1)-bit counters (wr_ptr, rd_ptr) - 1 × (AW+1)-bit space_free calculation - FIFO control block (full/empty logic) - ~50-100 flip-flops total

Example (DEPTH=512, AW=9): - 2 × 10-bit counters = 20 FFs - Control logic = ~30 FFs - **Total:** ~50 FFs

Very lightweight - pointer logic only, no data storage.

12.14 Related Modules

- **Counterpart:** `stream_drain_ctrl.sv` - Drain-side flow control
- **Parent:** `sram_controller_unit.sv` - Instantiates allocation controller
- **User:** `axi_read_engine.sv` - Checks `rd_space_free` before issuing AR

12.15 Related Documentation

- **Drain Controller:** `11_stream_drain_ctrl.md`
- **SRAM Controller Unit:** `09_sram_controller_unit.md`
- **SRAM Controller:** `08_sram_controller.md`

- **AXI Read Engine: `06_axi_read_engine.md`**

12.16 Revision History

Stream Allocation Controller Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers

Last Updated: 2026-01-02

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13 SRAM Controller Specification

Module: `sram_controller.sv` **Location:** `projects/components/dma-ip/stream/rtl/fub/`
Status: Implemented **Last Updated:** 2025-11-30

13.1 Overview

The SRAM Controller provides per-channel buffering between AXI read and write engines using independent FIFO structures. Each channel has its own FIFO with dedicated allocation controller (write side) and drain controller (read side).

13.1.1 Key Features

- **Per-channel FIFOs:** Independent gaxi_fifo_sync per channel (no segmentation complexity)
- **ID-based routing:** Transaction ID selects channel for write/read operations
- **Allocation controller:** Reserves space before AXI read data arrives
- **Drain controller:** Manages data availability for AXI write engine
- **Latency bridge:** Aligns FIFO read latency (registered output)
- **Saturating counters:** 8-bit space/count reporting per channel

13.1.2 Design Rationale

Why Per-Channel FIFOs Instead of Segmented SRAM?

The implementation uses **independent FIFOs** rather than a monolithic SRAM divided into segments:

Advantages of Per-Channel FIFOs: - **Simpler logic** - FIFOs are standard, well-tested components - **Better isolation** - Channel failures don't affect others - **Easier timing** - No cross-channel paths or arbitration - **Modular** - Easy to add/remove channels - **No pointer arithmetic** - FIFO handles internally

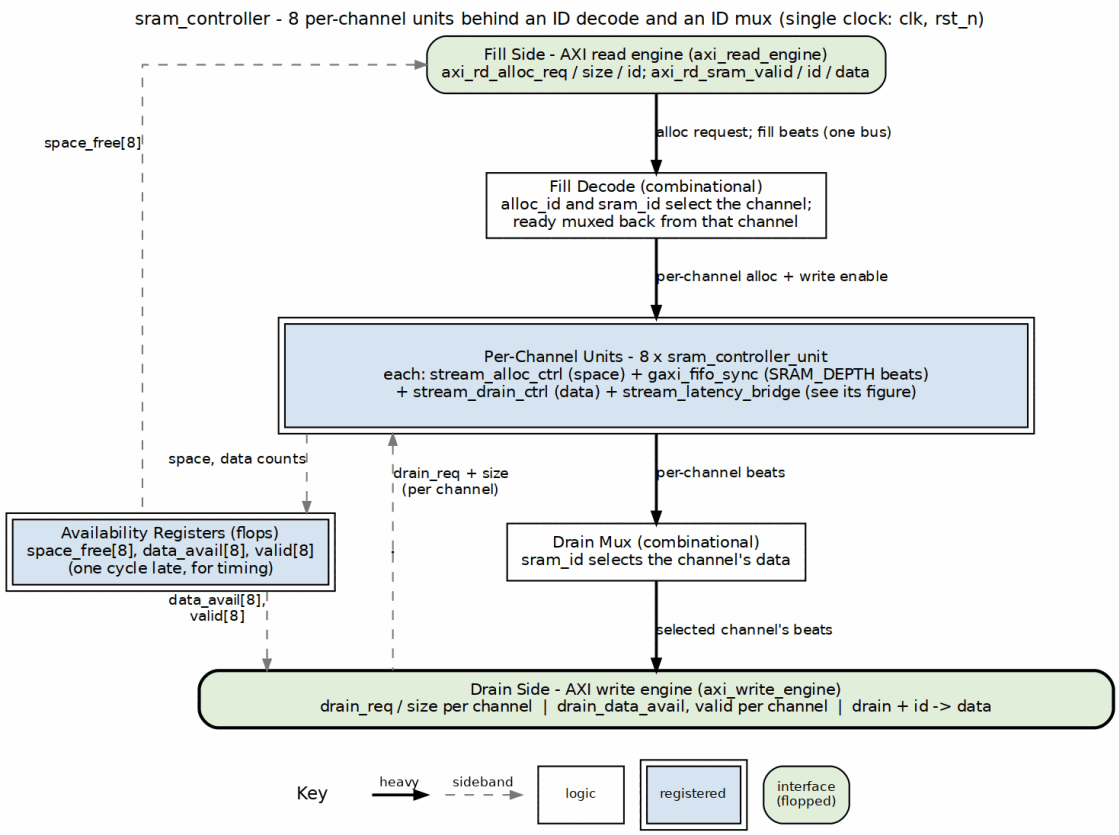
Trade-offs: - More SRAM resources (NC × DEPTH vs. shared pool) - Potential waste if channels idle (unused FIFO space)

Decision: Simplicity and isolation outweigh SRAM efficiency for STREAM's tutorial focus.

13.2 Architecture

13.2.1 Block Diagram

13.2.2 Figure 2.8.1: SRAM Controller Block Diagram



Diagram

Source: 05_sram_controller.dot

13.2.3 Per-Channel Architecture

Each channel contains three components (in `sram_controller_unit`):

1. **Allocation Controller (`stream_alloc_ctrl`):**
   - Receives `rd_alloc_req` from read engine
   - Tracks reserved vs. committed space
   - Provides `rd_space_free` to read engine
2. **FIFO (`gaxi_fifo_sync`):**
   - Stores data between read and write engines

- Depth = SRAM_DEPTH parameter
- Standard valid/ready handshaking

3. **Drain Controller + Latency Bridge (`stream_drain_ctrl`):**
   - Receives `wr_drain_req` from write engine
   - Provides `wr_drain_data_avail` to write engine
   - Latency bridge aligns FIFO read latency

13.3 Parameters

```
parameter int NUM_CHANNELS = 8;                    // Number of
independent channels
parameter int DATA_WIDTH = 512;                    // Data width in bits
parameter int SRAM_DEPTH = 512;                    // Depth per channel
FIFO
parameter int SEG_COUNT_WIDTH = $clog2(SRAM_DEPTH) + 1;  // Width of
count signals

// Short aliases (internal use)
parameter int NC = NUM_CHANNELS;
parameter int DW = DATA_WIDTH;
parameter int SD = SRAM_DEPTH;
parameter int SCW = SEG_COUNT_WIDTH;               // FIFO depth counter
width
parameter int CIW = (NC > 1) ? $clog2(NC) : 1;     // Channel ID width
(min 1 bit)
```

Note: "SEG_COUNT_WIDTH" refers to the FIFO depth counter width (historical name from segmented design consideration).

13.4 Port List

13.4.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low asynchronous reset

13.4.2 Allocation Interface

AXI Read Engine Flow Control (Space Reservation):

Allocation Interface

Signal	Direction	Width	Description
`axi_rd_alloc_req`	input	1	Allocation request (single request with ID)
`axi_rd_alloc_size`	input	8	Number of beats to allocate
`axi_rd_alloc_id`	input	CIW	Channel ID for allocation
`axi_rd_alloc_space_free[ch]`	output	NUM_CHANNELS × SEG_COUNT_WIDTH	Free space per channel FIFO

13.4.3 Write Interface

AXI Read Engine → FIFO (ID-based routing):

Write Interface

Signal	Direction	Width	Description
`axi_rd_sram_valid`	input	1	Write data valid (single valid with ID)
`axi_rd_sram_ready`	output	1	Ready (muxed from selected channel)
`axi_rd_sram_`	input	CIW	Channel ID select for

Signal	Direction	Width	Description
id			write
axi_rd_sram_data	input	DATA_WIDTH	Write data (common bus)

13.4.4 Drain Interface

Write Engine Flow Control (Data Availability):

Drain Interface

Signal	Direction	Width	Description
axi_wr_drain_data_avail[ch]	output	NUM_CHANNELS × SEG_COUNT_WIDTH	Available data after reservations per channel
axi_wr_drain_req[ch]	input	NUM_CHANNELS	Per-channel drain request
axi_wr_drain_size[ch]	input	NUM_CHANNELS × 8	Per-channel drain size (beats to reserve)

13.4.5 Read Interface

FIFO → AXI Write Engine (ID-based routing):

Read Interface

Signal	Direction	Width	Description
axi_wr_sram_valid[ch]	output	NUM_CHANNELS	Per-channel valid (data available)
axi_wr_sram_drain	input	1	Drain request (consumer ready)
axi_wr_sram_id	input	CIW	Channel ID select for read
axi_wr_sram_data	output	DATA_WIDTH	Read data from selected channel (muxed)

13.4.6 Debug Interface

Debug Interface

Signal	Direction	Width	Description
dbg_bridge_pending[ch]	output	NUM_CHANNELS	Latency bridge pending per channel
dbg_bridge_out_valid[ch]	output	NUM_CHANNELS	Latency bridge output valid per channel

13.5 Interface

13.5.1 Clock and Reset

```
input  logic                    clk;
input  logic                    rst_n;      // Active-low asynchronous
reset
```

13.5.2 Allocation Interface (AXI Read Engine Flow Control)

Space Reservation:

```
input  logic                                axi_rd_alloc_req;       //
Single allocation request
input  logic [7:0]                          axi_rd_alloc_size;      //
Beats to allocate
input  logic [CIW-1:0]                      axi_rd_alloc_id;        //
Channel ID for allocation
output logic [NC-1:0][SCW-1:0]              axi_rd_alloc_space_free;
// Free space per channel
```

Allocation Protocol: 1. Read engine issues AR transaction 2. **Before R data arrives:** Engine asserts `axi_rd_alloc_req` with size and ID 3. Allocation controller reserves space in selected channel 4. Engine tracks reserved space, prevents over-issuing AR commands 5. Space commits when actual data arrives (FIFO write)

Why Allocation is Critical:

Without pre-allocation, multiple AR commands could be issued before data arrives, causing FIFO overflow:

```
Problem without allocation:
  Cycle 0: Issue AR (16 beats) - FIFO has 32 free
  Cycle 1: Issue AR (16 beats) - FIFO still shows 32 free (data hasn't
arrived!)
  Cycle 2: Issue AR (16 beats) - FIFO still shows 32 free
  Cycle 10: R data starts arriving (48 beats total)
  → OVERFLOW! Only 32 beats of space

Solution with allocation:
  Cycle 0: Issue AR (16 beats), allocate 16 - FIFO shows 16 free
(reserved)
  Cycle 1: Issue AR (16 beats), allocate 16 - FIFO shows 0 free
(reserved)
  Cycle 2: Cannot issue AR - no space available
  Cycle 10: R data arrives - space is guaranteed
```

13.5.3 Write Interface (AXI Read Engine → FIFO)

ID-Based Write:

```
input  logic                    axi_rd_sram_valid;      // Single
valid for all channels
input  logic [CIW-1:0]          axi_rd_sram_id;         // Channel ID
select
output logic                    axi_rd_sram_ready;      // Ready
(muxed from selected channel)
input  logic [DW-1:0]           axi_rd_sram_data;       // Shared data
bus
```

Write Protocol: 1. Read engine asserts `axi_rd_sram_valid` with data 2. `axi_rd_sram_id` selects which channel FIFO receives data 3. Controller decodes ID to per-channel `valid_decoded[id]` 4. Selected channel's FIFO ready muxed to `axi_rd_sram_ready`

13.5.4 Drain Interface (AXI Write Engine Flow Control)

Data Availability:

```
input  logic [NC-1:0]                   axi_wr_drain_req;       //
Per-channel drain request
input  logic [NC-1:0][7:0]              axi_wr_drain_size;      //
Per-channel drain size
output logic [NC-1:0][SCW-1:0]          axi_wr_drain_data_avail;
// Available data after reservations
```

Drain Protocol: 1. Write engine checks `axi_wr_drain_data_avail[ch]` for available data 2. Engine asserts `axi_wr_drain_req[ch]` to reserve data for W burst 3. Drain controller updates available count (subtracts reserved) 4. Data commits when actually read from FIFO

Why Drain Reservation is Critical:

Similar to allocation, drain prevents under-reporting data availability:

```
Problem without drain reservation:
  Cycle 0: FIFO has 32 beats available
  Cycle 1: Issue AW (16 beats) - FIFO still shows 32 available
  Cycle 2: Issue AW (16 beats) - FIFO still shows 32 available
  Cycle 3: Issue AW (16 beats) - FIFO still shows 32 available
  Cycle 10: W bursts start draining (48 beats expected)
  → UNDERFLOW! Only 32 beats available

Solution with drain reservation:
  Cycle 0: FIFO has 32 beats available
  Cycle 1: Issue AW (16 beats), drain 16 - shows 16 available
(reserved)
  Cycle 2: Issue AW (16 beats), drain 16 - shows 0 available
(reserved)
```

```
  Cycle 3: Cannot issue AW - no data available
  Cycle 10: W bursts drain - data is guaranteed
```

13.5.5 Read Interface (FIFO → AXI Write Engine)

ID-Based Read:

```
output logic [NC-1:0]           axi_wr_sram_valid;      // Per-channel
valid (data available)
input  logic                    axi_wr_sram_drain;      // Single
drain (consumer requests data)
input  logic [CIW-1:0]          axi_wr_sram_id;         // Channel ID
select
output logic [DW-1:0]           axi_wr_sram_data;       // Data from
selected channel
```

Read Protocol: 1. Write engine checks `axi_wr_sram_valid[ch]` (per-channel) 2. Engine asserts `axi_wr_sram_drain` with `axi_wr_sram_id` 3. Controller decodes ID to per-channel `drain_decoded[id]` 4. Selected channel's data muxed to `axi_wr_sram_data`

13.5.6 Debug Interface

```
output logic [NC-1:0]           dbg_bridge_pending;     // Latency
bridge pending per channel
output logic [NC-1:0]           dbg_bridge_out_valid;   // Latency
bridge output valid per channel
```

Purpose: Monitor latency bridge state to catch bugs in read timing.

13.6 ID Decode Logic

13.6.1 Write Valid Decode

Decode `axi_rd_sram_id` to per-channel valid:

```
always_comb begin
    axi_rd_sram_valid_decoded = '0;
    if (axi_rd_sram_valid && axi_rd_sram_id < NC) begin
        axi_rd_sram_valid_decoded[axi_rd_sram_id] = 1'b1;
    end
end
```

Mux ready from selected channel:

```
always_comb begin
    if (axi_rd_sram_id < NC) begin
        axi_rd_sram_ready =
axi_rd_sram_ready_per_channel[axi_rd_sram_id];
    end else begin
        axi_rd_sram_ready = 1'b0;  // Invalid ID → not ready
    end
end
```

13.6.2 Read/Drain Decode

Decode `axi_wr_sram_id` to per-channel drain:

```
always_comb begin
    axi_wr_sram_drain_decoded = '0;
    if (axi_wr_sram_drain && axi_wr_sram_id < NC) begin
        axi_wr_sram_drain_decoded[axi_wr_sram_id] = 1'b1;
    end
end
```

Mux data from selected channel:

```
always_comb begin
    if (axi_wr_sram_id < NC) begin
        axi_wr_sram_data =
axi_wr_sram_data_per_channel[axi_wr_sram_id];
    end else begin
        axi_wr_sram_data = '0;  // Invalid ID → zero data
    end
end
```

13.6.3 Allocation Decode

Decode `axi_rd_alloc_id` to per-channel allocation:

```
always_comb begin
    axi_rd_alloc_req_decoded = '0;
    if (axi_rd_alloc_req && axi_rd_alloc_id < NC) begin
        axi_rd_alloc_req_decoded[axi_rd_alloc_id] = 1'b1;
    end
end
```

13.7 Per-Channel Unit

Each channel instantiates `sram_controller_unit` (separate module):

```
sram_controller_unit #(
    .DATA_WIDTH(DW),
    .SRAM_DEPTH(SRAM_DEPTH),
    .SEG_COUNT_WIDTH(SEG_COUNT_WIDTH)
) u_sram_controller_unit (
    .clk                    (clk),
    .rst_n                  (rst_n),
    .axi_rd_alloc_req       (axi_rd_alloc_req),
    .axi_rd_alloc_size      (axi_rd_alloc_size),
    .axi_rd_alloc_space_free(axi_rd_alloc_space_free),
    .axi_rd_sram_valid      (axi_rd_sram_valid),
    .axi_rd_sram_ready      (axi_rd_sram_ready),
    .axi_rd_sram_data       (axi_rd_sram_data),
    .axi_wr_drain_data_avail(axi_wr_drain_data_avail),
    .axi_wr_drain_req       (axi_wr_drain_req),
    .axi_wr_drain_size      (axi_wr_drain_size),
    .axi_wr_sram_valid      (axi_wr_sram_valid),
    .axi_wr_sram_ready      (axi_wr_sram_ready),
    .axi_wr_sram_data       (axi_wr_sram_data),
    .dbg_bridge_pending     (dbg_bridge_pending),
    .dbg_bridge_out_valid   (dbg_bridge_out_valid)
);
```

Key Points: - `axi_rd_sram_data` is **shared** - all units see same data bus - Only unit with `valid_decoded[i] = 1` writes to its FIFO - `rd_alloc_size` is **shared** - all see same size value - Only unit with `alloc_req_decoded[i] = 1` reserves space

---

13.8 Operation Flows

13.8.1 Write Flow (AXI Read Data → FIFO)

Allocation Phase (before data arrives):

```
1. Read engine issues AR command (araddr, arlen, arid)
2. Engine extracts channel ID from arid[CIW-1:0]
3. Engine asserts:
     axi_rd_alloc_req = 1
     axi_rd_alloc_id = channel_id
     axi_rd_alloc_size = arlen + 1  (burst size in beats)
4. Allocation controller (in sram_controller_unit):
     - Checks rd_space_free[channel_id] >= alloc_size
     - Reserves space (doesn't commit yet)
5. Engine proceeds with AR transaction
```

Data Arrival Phase:

```
1. AXI R data arrives:
     m_axi_rvalid = 1
     m_axi_rid = transaction_id (contains channel_id in lower bits)
     m_axi_rdata = data

2. Read engine forwards to SRAM controller:
     axi_rd_sram_valid = 1
     axi_rd_sram_id = rid[CIW-1:0]  (extract channel ID)
     axi_rd_sram_data = rdata

3. SRAM controller decodes:
     axi_rd_sram_valid_decoded[channel_id] = 1

4. Selected channel FIFO:
     - Writes data if ready
     - Commits 1 beat of reserved space
     - Decrements reserved counter

5. Ready mux:
     axi_rd_sram_ready = fifo_ready[channel_id]
```

13.8.2 Read Flow (FIFO → AXI Write Data)

Drain Reservation Phase (before W data drains):

```
1. Write engine checks drain_data_avail[channel_id]
2. If sufficient data available:
     axi_wr_drain_req[channel_id] = 1
     axi_wr_drain_size[channel_id] = burst_size
```

```
3. Drain controller:
     - Reserves data (decrements available count)
     - Prevents other channels from seeing this data
```

Data Drain Phase:

```
1. Write engine ready to consume data:
     axi_wr_sram_drain = 1
     axi_wr_sram_id = channel_id

2. SRAM controller decodes:
     axi_wr_sram_drain_decoded[channel_id] = 1

3. Selected channel FIFO:
     - Reads data (1-cycle latency through bridge)
     - Commits 1 beat of reserved data
     - Decrements drain reservation

4. Data mux:
     axi_wr_sram_data = fifo_data[channel_id]

5. Valid output:
     axi_wr_sram_valid[channel_id] = bridge_valid
```

13.9 Allocation vs. Drain Controllers

13.9.1 Why Separate Controllers?

Allocation Controller (Write Side): - Problem: AXI read AR issued before R data arrives - Solution: Pre-allocate space when AR issues, commit when R arrives - Prevents: Over-issuing AR commands when FIFO full

Drain Controller (Read Side): - Problem: AXI write AW issued before W data drains from FIFO - Solution: Reserve data when AW issues, commit when W drains - Prevents: Under-reporting available data when burst in progress

13.9.2 Flow Control Comparison

Read Engine (uses allocation):

```
// Check space BEFORE issuing AR
if (rd_space_free[ch] >= burst_size) begin
    issue_ar_command();
    assert_rd_alloc_req();  // Reserve space
end
// Later: R data arrives → commits reservation
```

Write Engine (uses drain):

```
// Check data availability BEFORE issuing AW
if (wr_drain_data_avail[ch] >= burst_size) begin
    issue_aw_command();
    assert_wr_drain_req();  // Reserve data
end
// Later: W beats drain → commits reservation
```

13.10 Timing Diagrams

13.10.1 Combined Write and Read Timing

The following diagram shows the complete SRAM controller write and read flow:

13.10.1.1 Waveform 2.8.1: SRAM Controller Write/Read Operation



SRAM Controller Write/Read

Source: sram_controller_wr_rd.json

13.10.2 Write Path (R Data → FIFO)

13.10.3 Figure 2.8.2: SRAM Controller Write Path (Reserve, Fill, Release)

SRAM controller write path (AXI R data into the channel FIFO): reserve, fill, release

event	1	2	3	4
what happens	**alloc request**	**AR issued**	**R beats arrive**	**beats leave on the W side**
axi_rd_alloc_req	1 cycle, id = 2, size = 4	0		
axi_rd_sram_valid / ready	-		4 beats, id = 2; ready while FIFO has room	-
FIFO[2] occupancy	0		+1 per beat (to 4)	-1 per beat out
alloc_space_free[2]	S - 4 (reserved)	S - 4		+1 per beat out
drain_data_avail[2]	0		+1 per beat in	less any drain reservation
The read engine reserves space BEFORE it issues the AR, so the R beats always find room. Space comes back only when a beat leaves the controller on the write-engine side, not when it lands in the FIFO. Both counters reach the engines through registered outputs. Beat arrival time is set by the memory's read latency, not by the controller.				

SRAM Controller Write Path (Reserve, Fill, Release)

13.10.4 Read Path (FIFO → W Data)

13.10.5 Figure 2.8.3: SRAM Controller Read Path (Reserve, Drain, Release)

SRAM controller read path (channel FIFO out to AXI W data): reserve, drain, release

event	1	2	3
what happens	**drain request**	**AW issued**	**write engine drains**
axi_wr_drain_req[2]	1 cycle, size = 2	0	
drain_data_avail[2]	avail - 2 (reserved)	avail - 2	
axi_wr_sram_drain, id = 2	-		2 beats
axi_wr_sram_data	-		D0, D1 from the latency bridge
alloc_space_free[2]	unchanged		+1 per beat out
The write engine reserves data before it issues the AW, so the W beats are already buffered. The latency bridge hides the FIFO's 1-cycle read latency: it keeps the head beat (and a small prefetch) ready so valid and data are aligned when the engine drains. Each beat out returns one entry to the allocation side.			

SRAM Controller Read Path (Reserve, Drain, Release)

13.11 Error Conditions

13.11.1 Invalid Channel ID

Condition: `axi_rd_sram_id >= NUM_CHANNELS` or `axi_wr_sram_id >= NUM_CHANNELS`

Handling: - Write: `axi_rd_sram_ready = 0` (not ready, data dropped) - Read: `axi_wr_sram_data = 0` (zero data output) - **No error signal** - engines should not generate invalid IDs

13.11.2 FIFO Overflow

Condition: Write when FIFO full

Prevention: - Allocation controller tracks reserved + committed space - Read engine checks `rd_space_free` before issuing AR - Only issues AR when sufficient space available

Should never happen if read engine follows protocol!

13.11.3 FIFO Underflow

Condition: Read when FIFO empty

Prevention: - Drain controller tracks available - reserved data - Write engine checks `wr_drain_data_avail` before issuing AW - Only issues AW when sufficient data available

Should never happen if write engine follows protocol!

13.12 Performance Considerations

13.12.1 Per-Channel FIFO Benefits

Isolation: - Channel failures don't affect others - No cross-channel contention - Simplified debug (each channel independent)

Timing: - No shared arbitration delays - Predictable latency per channel - Easier timing closure

Scalability: - Easy to add/remove channels - Parameterizable depth per channel - No global SRAM redesign needed

13.12.2 SRAM Resource Usage

Total SRAM: NUM_CHANNELS × SRAM_DEPTH × DATA_WIDTH

Example (8 channels, 512 depth, 512-bit data): - Per channel: 512 × 512 bits = 32KB - Total: 8 × 32KB = 256KB

Comparison to Segmented Approach: - Segmented: 1 × (8 × 512) × 512 bits = 256KB (same total) - But segmented requires complex pointer arithmetic and arbitration - Per-channel FIFOs trade complexity for slight area increase (FIFO overhead)

13.13 Testing

Test Location:
`projects/components/dma-ip/stream/dv/tests/fub/test_sram_controller.py` (+ `test_sram_controller_alloc.py`)

Key Test Scenarios:

1. **Single channel fill/drain** - Basic FIFO operation
2. **All channels concurrent** - Independent operation
3. **ID decode** - Correct channel selection
4. **Allocation before data** - Space reserved correctly
5. **Drain before read** - Data reserved correctly
6. **Latency bridge** - 1-cycle delay verified
7. **Invalid ID** - Graceful handling
8. **Back-to-back** - No bubbles between bursts

13.14 Bug Fixes (v0.92)

13.14.1 Registered Avail Outputs Reset (Commit b619eee9)

The registered availability outputs (`w_alloc_avail`, `w_drain_avail`) were missing explicit reset, causing: - X propagation in simulation (metastability) - Unknown post-reset state in synthesis (unmapped flops)

Fix: Added explicit reset to both registered avail outputs. They now initialize to `'0` (no space available, no data available) on reset, preventing garbage values.

13.15 Related Documentation

- **SRAM Controller Unit:** `09_sram_controller_unit.md` - Per-channel implementation
- **Allocation Controller:** `07_stream_alloc_ctrl.md` - Write-side flow control
- **Drain Controller:** `11_stream_drain_ctrl.md` - Read-side flow control
- **AXI Read Engine:** `06_axi_read_engine.md` - Write interface usage
- **AXI Write Engine:** `12_axi_write_engine.md` - Read interface usage
- **Stream Core:** `01_stream_core.md` - Top-level integration

13.16 Revision History

Revision History

Date	Version	Changes
2025-10-18	0.5	Initial draft documenting segmented SRAM approach
2025-11-16	1.0	Updated to document per-channel FIFO implementation (actual RTL)
2025-11-21	1.5	**Merged documentation:**- Consolidated duplicate files- Added design rationale section- Enhanced allocation/drain explanations with examples- Clarified per-channel FIFO architecture- Added performance comparison- Verified all content matches current RTL implementation
2025-11-30	1.6	Updated related documentation references

Last Updated: 2025-11-30 (verified against RTL implementation)

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14 SRAM Controller Unit

Module: sram_controller_unit.sv **Location:** projects/components/dma-ip/stream/rtl/fub/ **Category:** FUB (Functional Unit Block)
Parent: sram_controller.sv **Status:** Implemented **Last Updated:** 2025-11-30

14.1 Overview

The `sram_controller_unit` module is a single-channel SRAM controller unit containing allocation controller, FIFO buffer, and latency bridge. It handles one channel's data flow from AXI read engine through buffering to AXI write engine with proper flow control.

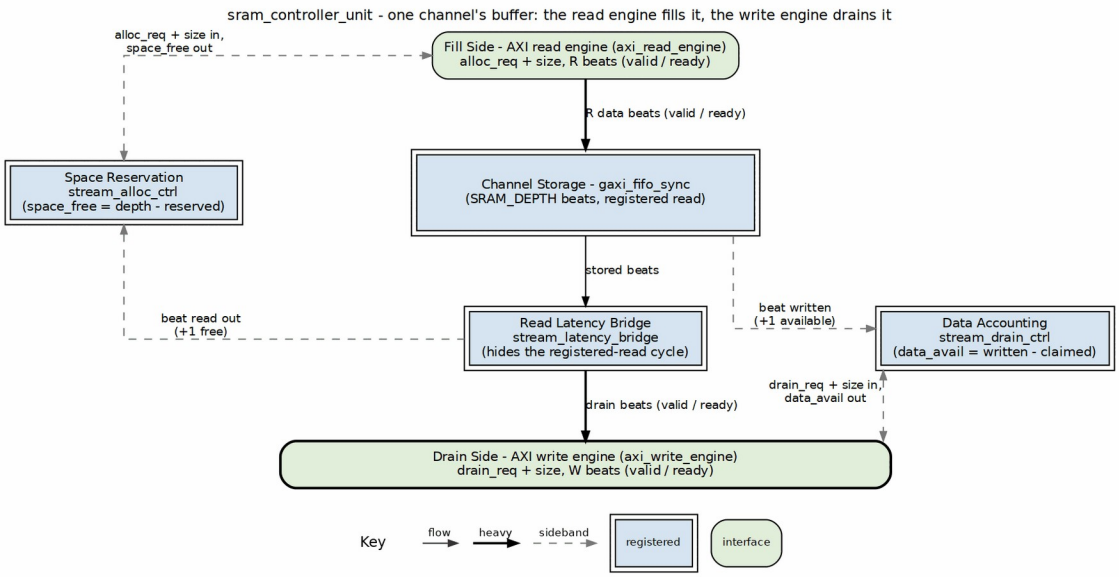
14.1.1 Key Features

- **Three-Component Architecture:**
  - Allocation controller (stream_alloc_ctrl) - Space tracking for reads
  - FIFO buffer (gaxi_fifo_sync) - Physical data storage
  - Latency bridge (stream_latency_bridge) - Timing compensation
- **Drain Controller:** Tracks data availability for write engine
- **Virtual FIFO Pattern:** Pointer arithmetic without data storage for flow control
- **Registered Outputs:** Breaks combinatorial paths for timing closure

14.2 Architecture

14.2.1 Block Diagram

14.2.2 Figure 1: SRAM Controller Unit Block Diagram



SRAM Controller Unit Block Diagram

Source: 06_sram_controller_unit.dot

14.2.3 Component Hierarchy

```
sram_controller_unit
 stream_alloc_ctrl          # Allocation tracking (space availability)
 stream_drain_ctrl          # Drain tracking (data availability)
 gaxi_fifo_sync             # Physical data storage FIFO
 stream_latency_bridge      # 1-cycle latency compensation
```

14.2.4 Data Flow

```
AXI Read Engine → FIFO Write Port → FIFO Storage → FIFO Read Port →
Latency Bridge → AXI Write Engine

         Allocation Controller (space tracking)

                                                Drain Controller (data
tracking)
```

14.2.5 Controller Naming Convention (CRITICAL)

Allocation Controller Perspective: - `wr` side = ALLOCATE (reserve space, advance wr_ptr) - `rd` side = FULFILL (data arrives, advance rd_ptr, FREE space)

Drain Controller Perspective: - `wr` side = DATA WRITTEN (increment occupancy) - `rd` side = DRAIN REQUEST (reserve data for write burst)

This is OPPOSITE of normal FIFO naming conventions!

14.3 Parameters

Parameters

Parameter	Type	Default	Description
DATA_WIDTH	int	512	Data width in bits
SRAM_DEPTH	int	512	FIFO depth in entries
SEG_COUNT_WIDTH	int	$clog2(SRAM_DEPTH)+1	Width for count signals

14.3.1 Derived Parameters

Derived Parameters

Parameter	Derivation	Description
DW	DATA_WIDTH	Short alias
SD	SRAM_DEPTH	Short alias
SCW	SEG_COUNT_WIDTH	Segment count width
ADDR_WIDTH	$clog2(SD)	FIFO address width

14.4 Port List

14.4.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low asynchronous reset

14.4.2 Allocation Interface (Read Engine Flow Control)

Allocation Interface

Signal	Direction	Width	Description
axi_rd_alloc_req	input	1	Request space allocation
axi_rd_alloc_size	input	8	Beats to reserve
axi_rd_alloc_space_free	output	SCW	Free space available

14.4.3 Write Interface (AXI Read Engine to FIFO)

Write Interface

Signal	Direction	Width	Description
axi_rd_sram_valid	input	1	Write data valid
axi_rd_sram_ready	output	1	Ready to accept data
axi_rd_sram_data	input	DW	Write data payload

14.4.4 Drain Interface (Write Engine Flow Control)

Drain Interface

Signal	Direction	Width	Description
axi_wr_drain_data_avail	output	SCW	Data available

Signal	Direction	Width	Description
			for drain
axi_wr_drain_req	input	1	Request to drain data
axi_wr_drain_size	input	8	Beats to drain

14.4.5 Read Interface (FIFO to AXI Write Engine)

Read Interface

Signal	Direction	Width	Description
axi_wr_sram_valid	output	1	Read data valid
axi_wr_sram_ready	input	1	Ready to accept data
axi_wr_sram_data	output	DW	Read data payload

14.4.6 Debug Interface

Debug Interface

Signal	Direction	Width	Description
dbg_bridge_pending	output	1	Data in flight in bridge
dbg_bridge_out_valid	output	1	Bridge output valid

14.5 Operation

14.5.1 Allocation Flow

1. **Space Check:** Read engine checks `axi_rd_alloc_space_free`
2. **Reservation:** Read engine asserts `axi_rd_alloc_req` with burst size
3. **Space Decrement:** Allocation controller decrements space_free
4. **Data Arrival:** Data enters FIFO via `axi_rd_sram_*` interface
5. **Space Release:** When data exits (output handshake), space_free increments

14.5.2 Drain Flow

1. **Data Check:** Write engine checks `axi_wr_drain_data_avail`
2. **Reservation:** Write engine asserts `axi_wr_drain_req` with burst size
3. **Data Decrement:** Drain controller decrements data_available
4. **Data Consumption:** Write engine reads via `axi_wr_sram_*` interface

14.5.3 Data Available Calculation

```
// Total data available = drain controller data + latency bridge
occupancy
assign axi_wr_drain_data_avail = drain_data_available +
SCW'(bridge_occupancy);
```

The latency bridge can hold 1 beat in flight plus up to 4 beats in its skid buffer, which must be accounted for in the data availability count.

14.6 Integration Example

```
sram_controller_unit #(
    .DATA_WIDTH     (512),
    .SRAM_DEPTH     (512),
    .SEG_COUNT_WIDTH(10)
) u_sram_controller_unit (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // Allocation interface
    .axi_rd_alloc_req       (axi_rd_alloc_req[ch]),
    .axi_rd_alloc_size      (axi_rd_alloc_size),
    .axi_rd_alloc_space_free(axi_rd_alloc_space_free[ch]),

    // Write interface (from AXI Read Engine)
    .axi_rd_sram_valid      (axi_rd_sram_valid[ch]),
    .axi_rd_sram_ready      (axi_rd_sram_ready[ch]),
    .axi_rd_sram_data       (axi_rd_sram_data),

    // Drain interface
    .axi_wr_drain_data_avail(axi_wr_drain_data_avail[ch]),
    .axi_wr_drain_req       (axi_wr_drain_req[ch]),
    .axi_wr_drain_size      (axi_wr_drain_size[ch]),

    // Read interface (to AXI Write Engine)
    .axi_wr_sram_valid      (axi_wr_sram_valid[ch]),
    .axi_wr_sram_ready      (axi_wr_sram_ready[ch]),
    .axi_wr_sram_data       (axi_wr_sram_data[ch]),

    // Debug
    .dbg_bridge_pending     (dbg_bridge_pending[ch]),
    .dbg_bridge_out_valid   (dbg_bridge_out_valid[ch])
);
```

14.7 Common Issues

14.7.1 Issue 1: Space Accounting Mismatch

Symptom: Read engine sees space but FIFO overflows

Root Causes: 1. Allocation controller not receiving fulfillment signals 2. Output handshake not connected to allocation controller rd_valid

Solution: Ensure `axi_wr_sram_valid && axi_wr_sram_ready` connects to allocation controller.

14.7.2 Issue 2: Data Available Undercount

Symptom: Write engine stalls despite data in FIFO

Root Causes: 1. Bridge occupancy not included in data_available calculation 2. Drain controller not receiving write handshakes

Solution: Verify `axi_wr_drain_data_avail = drain_data_available + bridge_occupancy`.

14.8 Related Documentation

- **Parent:** `08_sram_controller.md` - Multi-channel SRAM controller
- **Allocation Controller:** `07_stream_alloc_ctrl.md` - Space tracking details
- **Drain Controller:** `11_stream_drain_ctrl.md` - Data tracking details
- **Latency Bridge:** `10_stream_latency_bridge.md` - Timing compensation
- **Read Engine:** `06_axi_read_engine.md` - Data producer

- **Write Engine: `12_axi_write_engine.md` - Data consumer**

14.9 Revision History

SRAM Controller Unit Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers

Last Updated: 2026-01-02

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15 Stream Latency Bridge

Module: `stream_latency_bridge.sv` **Location:** `projects/components/dma-ip/stream/rtl/fub/` **Category:** FUB (Functional Unit Block)
Parent: `sram_controller_unit.sv` **Status:** Implemented **Last Updated:** 2025-11-30

15.1 Overview

The `stream_latency_bridge` module is a simple latency-1 bridge that compensates for registered FIFO output latency. It uses glue logic plus a skid buffer to handle backpressure from downstream consumers.

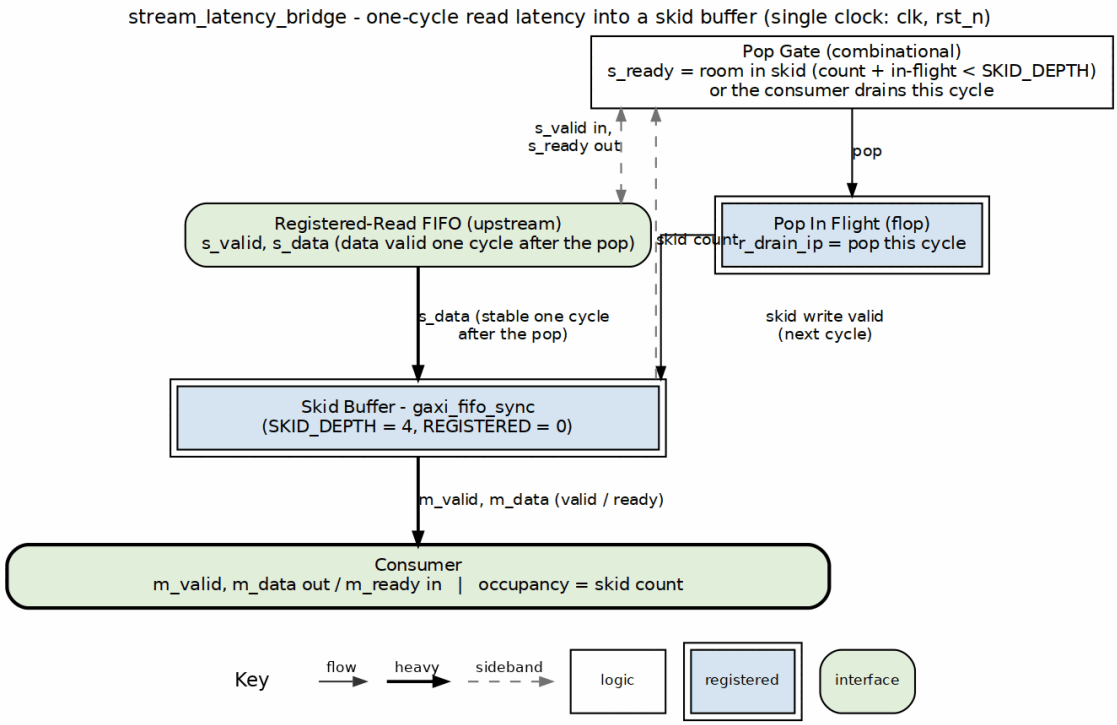
15.1.1 Key Features

- **1-Cycle Latency Compensation:** Bridges registered FIFO output to consumer
- **Skid Buffer Architecture:** 4-deep FIFO absorbs consumer backpressure
- **Full Throughput:** Maintains back-to-back transfers when consumer is ready
- **Simple Design:** No complex ready calculations - skid buffer handles it all
- **Occupancy Tracking:** Reports total beats held for flow control

15.2 Architecture

15.2.1 Block Diagram

15.2.2 Figure 2.10.1: Stream Latency Bridge Block Diagram



Stream Latency Bridge Block Diagram

Source: 06_stream_latency_bridge.dot

15.2.3 Operation Flow

```
Cycle 0: Drain FIFO (s_valid && skid_ready) → r_drain_ip = 1
Cycle 1: Data arrives from FIFO → Push to skid (skid_valid =
r_drain_ip)
Cycle N: Consumer drains skid buffer at its own pace
```

15.2.4 Component Diagram

The glue flop (`r_drain_ip`) and the 4-deep skid buffer are drawn in Figure 2.10.1.

15.2.5 Backpressure Logic

The bridge uses a conservative backpressure approach: - Track stalled writes (wr_valid && ! wr_ready) - Calculate pending count = skid_count + stalled - Allow accept when pending < SKID_DEPTH OR consumer is draining

```
Truth Table (SKID_DEPTH=4):
count | wr_valid | wr_ready | stalled | pending | room | s_ready
------|----------|----------|---------|---------|------|--------
0     | 0        | 1        | 0       | 0       | 1    | 1      
Empty
0     | 1        | 1        | 0       | 0       | 1    | 1      
Writing, completes
3     | 0        | 1        | 0       | 3       | 1    | 1      Room
for 1
3     | 1        | 1        | 0       | 3       | 1    | 1      
Writing, will be 4
3     | 1        | 0        | 1       | 4       | 0    | 0      
Write stalled
4     | 0        | 0        | 0       | 4       | 0    | 0      Full
```

15.3 Parameters

Parameters

Parameter	Type	Default	Description
DATA_WIDTH	int	64	Data width in bits
SKID_DEPTH	int	4	Skid buffer depth (2-4 recommended)

15.3.1 Derived Parameters

Derived Parameters

Parameter	Derivation	Description
DW	DATA_WIDTH	Short alias

15.4 Port List

15.4.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low asynchronous reset

15.4.2 Upstream Interface (from registered FIFO)

Upstream Interface

Signal	Direction	Width	Description
s_valid	input	1	Upstream valid (FIFO not empty)
s_ready	output	1	Ready to accept from FIFO
s_data	input	DW	Data from FIFO (valid 1 cycle after handshake)

15.4.3 Downstream Interface (to consumer)

Downstream Interface

Signal	Direction	Width	Description
m_valid	output	1	Data valid to consumer
m_ready	input	1	Consumer ready
m_data	output	DW	Data to consumer

15.4.4 Status Interface

Status Interface

Signal	Direction	Width	Description
occupancy	output	3	Beats in bridge (0-5: 1 in

Signal	Direction	Width	Description
			flight + 4 in skid)

15.4.5 Debug Interface

Debug Interface

Signal	Direction	Width	Description
dbg_r_pending	output	1	Data in flight (r_drain_ip)
dbg_r_out_valid	output	1	Output valid (m_valid)

15.5 Operation

15.5.1 Glue Logic

The glue logic consists of a single flop (`r_drain_ip`) that tracks when a FIFO drain is in progress:

```
// Drain FIFO when upstream has data AND we can accept
wire w_drain_fifo = s_valid && s_ready;

// Flop the drain signal to track data in flight
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_drain_ip <= 1'b0;
    end else begin
        r_drain_ip <= w_drain_fifo;
    end
end

// When r_drain_ip asserted, data arrived from FIFO  push to skid
assign skid_wr_valid = r_drain_ip;
assign skid_wr_data = s_data;
```

15.5.2 Skid Buffer

A 4-deep `gaxi_fifo_sync` instance with: - Non-registered mode (REGISTERED=0) for minimum latency - Auto memory style selection

15.5.3 Occupancy Calculation

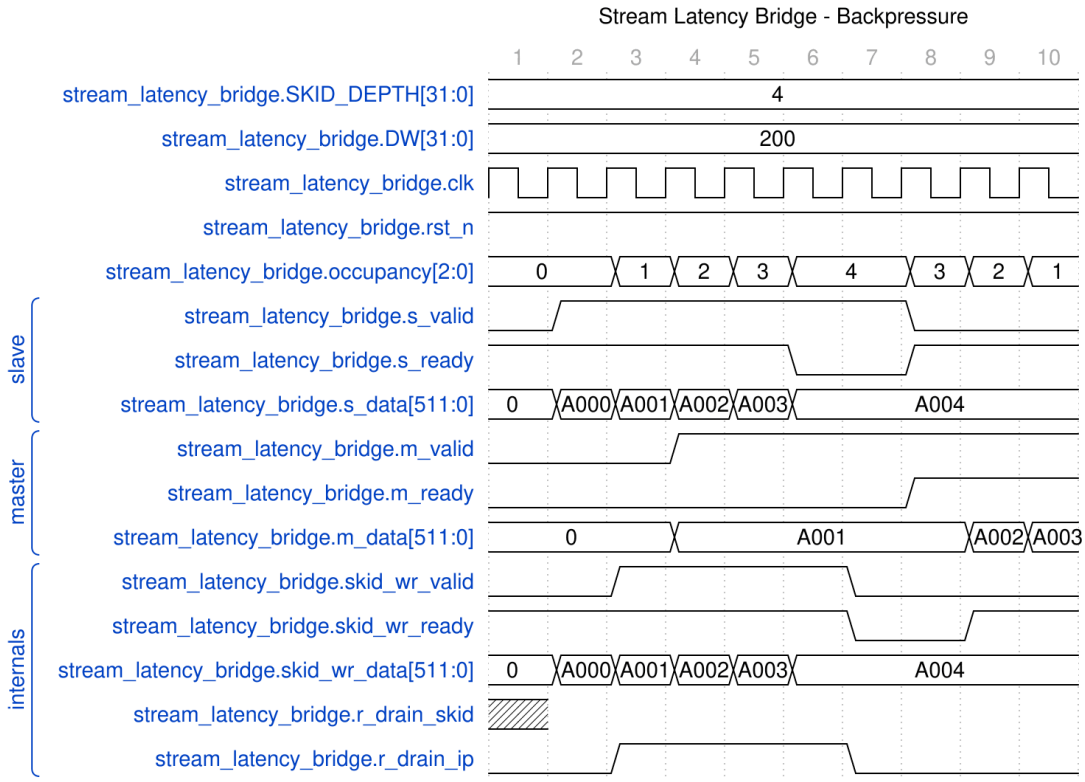
```
// Total occupancy = data in skid buffer
// Note: r_drain_ip not counted separately (conservative)
assign occupancy = skid_count;
```

15.6 Timing Diagrams

15.6.1 Backpressure Handling

The following diagram shows how the latency bridge handles downstream backpressure:

15.6.1.1 Waveform 2.10.1: Stream Latency Bridge Backpressure



Latency Bridge Backpressure

Source: latency_bridge_backpressure.json

15.7 Integration Example

```
stream_latency_bridge #(
    .DATA_WIDTH (512),
    .SKID_DEPTH (4)
) u_latency_bridge (
    .clk            (clk),
    .rst_n          (rst_n),

    // Slave (from FIFO)
    .s_data         (fifo_rd_data),
    .s_valid        (fifo_rd_valid),
    .s_ready        (fifo_rd_ready),

    // Master (to AXI Write Engine)
    .m_data         (axi_wr_sram_data),
    .m_valid        (axi_wr_sram_valid),
    .m_ready        (axi_wr_sram_ready),

    // Status
    .occupancy      (bridge_occupancy),

    // Debug
    .dbg_r_pending  (dbg_bridge_pending),
    .dbg_r_out_valid(dbg_bridge_out_valid)
);
```

15.8 Design Rationale

15.8.1 Why Use a Skid Buffer?

Previous designs attempted complex ready signal calculations to handle backpressure. This led to: - Timing issues from deep combinatorial paths - Edge cases with multi-cycle backpressure - Difficult debugging of flow control bugs

The skid buffer approach: - Simple glue logic (single flop) - FIFO handles all backpressure complexity - Proven, reusable component - Easy to verify and debug

15.8.2 Why 4-Deep?

- **2-deep:** Minimum for back-to-back (1 in flight + 1 buffered)
- **4-deep:** Allows 2-3 cycles of consumer stall without upstream backpressure
- **8-deep:** Overkill for most use cases

4-deep provides good balance between: - Latency tolerance (handles brief stalls) - Resource usage (small FIFO) - Throughput (back-to-back capable)

15.9 Related Documentation

- **Parent:** `09_sram_controller_unit.md` - Single-channel SRAM controller
- **Consumer:** `12_axi_write_engine.md` - Receives bridge output

- **Producer: `06_axi_read_engine.md` - Feeds upstream FIFO**

15.10 Revision History

Stream Latency Bridge Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers

Last Updated: 2026-01-02

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16 Stream Drain Controller

Module: `stream_drain_ctrl.sv` **Location:** `projects/components/dma-ip/stream/rtl/fub/` **Category:** FUB (Functional Unit Block)
Parent: `sram_controller_unit.sv` **Status:** Implemented **Last Updated:** 2025-11-30

16.1 Overview

The `stream_drain_ctrl` module is a virtual FIFO for write engine flow control. It tracks data availability using FIFO pointer logic without storing any data. This allows the write engine to pre-reserve data before issuing AXI write commands.

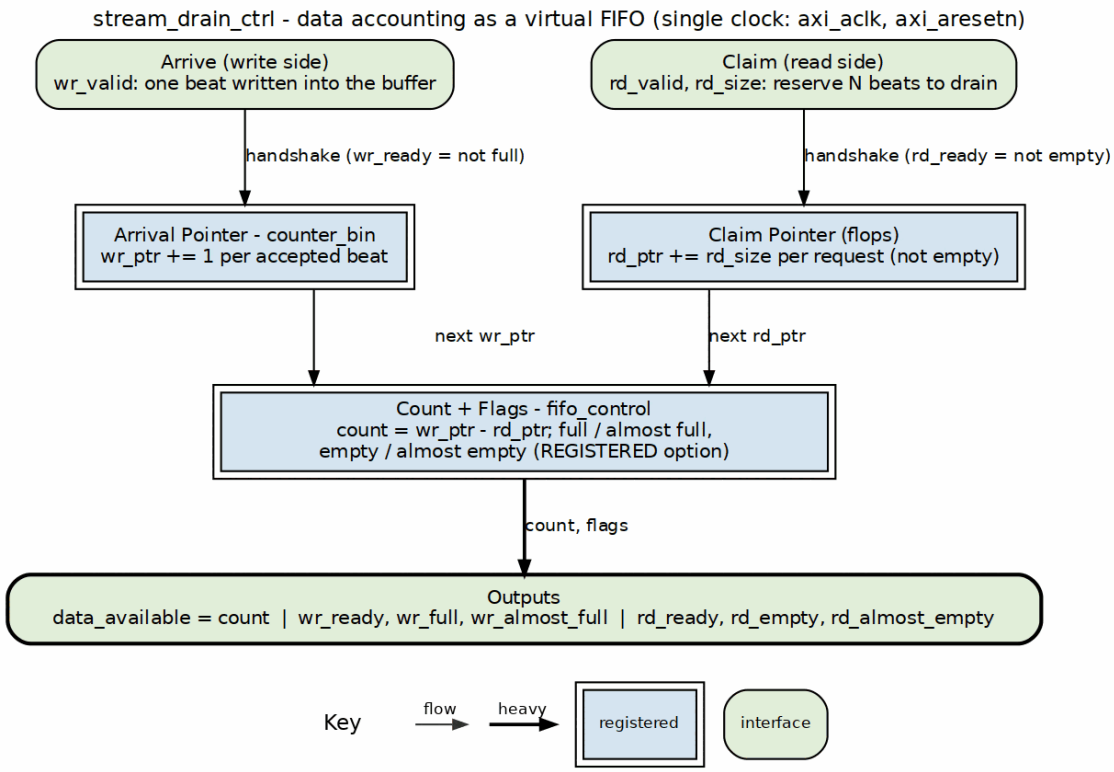
16.1.1 Key Features

- **Virtual FIFO Pattern:** Pointer arithmetic only, no data storage
- **Pre-Reservation:** Write engine reserves data before AXI AW command
- **Variable Size Drains:** Supports burst-sized reservations (not single beat)
- **Registered Outputs:** Optional output registration for timing
- **Underflow Prevention:** Ensures data exists before write engine commits

16.2 Architecture

16.2.1 Block Diagram

16.2.2 Figure 1: Stream Drain Controller Block Diagram



Stream Drain Controller Block Diagram

Source: 09_stream_drain_ctrl.dot

16.2.3 Virtual FIFO Concept

Unlike a real FIFO that stores data, the drain controller only tracks pointers:

```
Real FIFO:              Virtual FIFO (Drain Controller):

 Data[0]              wr_ptr: 5     Tracks entries written
 Data[1]              rd_ptr: 2     Tracks entries drained
 Data[2]              available:3   Difference
 ...


Actual data stored in gaxi_fifo_sync (separate component)
```

16.2.4 Pointer Operations

Write Side (Data Entry):

```
When FIFO receives data:
  wr_ptr += 1
  data_available += 1
```

Read Side (Drain Request):

```
When write engine requests drain:
  rd_ptr += rd_size  (variable burst size!)
  data_available -= rd_size
```

16.3 Parameters

Parameters

Parameter	Type	Default	Description
DEPTH	int	512	Virtual FIFO depth
ALMOST_WR_MARGIN	int	1	Almost full margin
ALMOST_RD_MARGIN	int	1	Almost empty margin
REGISTERED	int	1	Registered outputs

16.3.1 Derived Parameters

Derived Parameters

Parameter	Derivation	Description
D	DEPTH	Short alias
AW	$clog2(D)	Address width

16.4 Port List

16.4.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
`axi_aclk`	input	1	System clock
`axi_aresetn`	input	1	Active-low asynchronous reset

16.4.2 Write Interface (Data Entry)

Write Interface

Signal	Direction	Width	Description
`wr_valid`	input	1	Data written to FIFO
`wr_ready`	output	1	Not full (can accept)

16.4.3 Read Interface (Drain Requests)

Read Interface

Signal	Direction	Width	Description
`rd_valid`	input	1	Request to drain data
`rd_size`	input	8	Number of entries to drain
`rd_ready`	output	1	Data available (not empty)

16.4.4 Status Outputs

Status Outputs

Signal	Direction	Width	Description
`data_available`	output	AW+1	Available data count

Signal	Direction	Width	Description
wr_full	output	1	Full (no space)
wr_almost_full	output	1	Almost full
rd_empty	output	1	Empty (no data)
rd_almost_empty	output	1	Almost empty

16.5 Operation

16.5.1 Data Entry (Write Side)

Data entry is single-beat increments, tracking each beat written to the physical FIFO:

```
// Write pointer uses counter_bin for single-beat increments
counter_bin #(
    .WIDTH (AW + 1),
    .MAX   (D)
) write_pointer_inst (
    .clk              (axi_aclk),
    .rst_n            (axi_aresetn),
    .enable           (w_write && !r_wr_full),
    .counter_bin_curr (r_wr_ptr_bin),
    .counter_bin_next (w_wr_ptr_bin_next)
);
```

16.5.2 Drain Requests (Read Side)

Drain requests support variable-size bursts:

```
// Read pointer advances by rd_size on each drain request
always_ff @(posedge axi_aclk or negedge axi_aresetn) begin
    if (!axi_aresetn) begin
        r_rd_ptr_bin <= '0;
    end else begin
        if (w_read && !r_rd_empty) begin
            r_rd_ptr_bin <= r_rd_ptr_bin + (AW+1)'(rd_size);
        end
    end
end
```

16.5.3 Status Generation

Uses `fifo_control` block for full/empty/almost flags:

```
fifo_control #(
    .DEPTH             (D),
    .ADDR_WIDTH        (AW),
    .ALMOST_RD_MARGIN  (ALMOST_RD_MARGIN),
    .ALMOST_WR_MARGIN  (ALMOST_WR_MARGIN),
    .REGISTERED        (REGISTERED)
) fifo_control_inst (
    // ... pointer inputs
    .count            (w_count),
    .wr_full          (r_wr_full),
    .wr_almost_full   (r_wr_almost_full),
    .rd_empty         (r_rd_empty),
```

```
    .rd_almost_empty  (r_rd_almost_empty)
);
```

16.6 Comparison with stream_alloc_ctrl

Comparison with stream_alloc_ctrl

Aspect	stream_alloc_ctrl	stream_drain_ctrl
Purpose	Track space for reads	Track data for writes
Write Side	Reservation (burst)	Data entry (single beat)
Read Side	Fulfillment (single beat)	Drain request (burst)
Consumer	AXI Read Engine	AXI Write Engine

Both use virtual FIFO pattern but with opposite semantics for read/write operations.

16.7 Integration Example

```
stream_drain_ctrl #(
    .DEPTH            (512),
    .REGISTERED       (1)
) u_drain_ctrl (
    .axi_aclk         (clk),
    .axi_aresetn      (rst_n),

    // Write interface: Connected to FIFO write handshake
    .wr_valid         (fifo_wr_valid && fifo_wr_ready),
    .wr_ready         (),  // Not used (tracks, doesn't control)

    // Read interface: Connected to write engine drain requests
    .rd_valid         (axi_wr_drain_req),
    .rd_size          (axi_wr_drain_size),
    .rd_ready         (),  // Not used (polling interface)

    // Status
    .data_available   (drain_data_available),
    .wr_full          (),
    .wr_almost_full   (),
    .rd_empty         (),
    .rd_almost_empty  ()
);
```

16.8 Common Issues

16.8.1 Issue 1: Data Available Undercount

Symptom: Write engine sees 0 available when FIFO has data

Root Causes: 1. `wr_valid` not connected to FIFO write handshake 2. Bridge occupancy not added to data_available

Solution: Ensure connection: `wr_valid = fifo_wr_valid && fifo_wr_ready`

16.8.2 Issue 2: Drain Request Overflow

Symptom: rd_ptr advances past wr_ptr

Root Causes: 1. Write engine requesting more than available 2. Multiple simultaneous drain requests

Solution: Write engine must check `data_available >= requested_size` before draining.

16.9 Bug Fixes (v0.92)

16.9.1 Stale-View Race Closure (Commit a82627af)

The drain controller was sampling a registered signal one cycle stale, producing `wvalid` before SRAM data had settled. This occurred because:

1. SRAM read data changes on clock edge
2. Drain controller samples that data combinationally in the same cycle
3. Data fed to write engine, wvalid asserted → write engine consumes stale data

Fix: A post-flop register is inserted after the drain mux (SRAM data path), and `wvalid` is only asserted after this register. This guarantees that data has settled before wvalid assertion.

Matching half: Write engine (`12_axi_write_engine.md`) implements the post-flop gating on wvalid in coordination with this fix.

16.10 Related Documentation

- **Parent:** `09_sram_controller_unit.md` - Integration context
- **Counterpart:** `07_stream_alloc_ctrl.md` - Space tracking (read side)
- **Consumer:** `12_axi_write_engine.md` - Uses drain interface

- **Control Block: See `fifo_control` in rtl/amba/gaxi/**

16.11 Revision History

Stream Drain Controller Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers

Last Updated: 2026-01-02

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17 AXI Write Engine

Module: axi_write_engine.sv **Location:** projects/components/dma-ip/stream/rtl/fub/ **Category:** FUB (Functional Unit Block) **Parent:** stream_core.sv **Status:** Implemented **Last Updated:** 2026-10-17

17.1 Overview

The `axi_write_engine` module is a high-performance multi-channel AXI4 write engine with data-aware arbitration. It serves all 8 STREAM channels through a single AXI master interface with intelligent flow control.

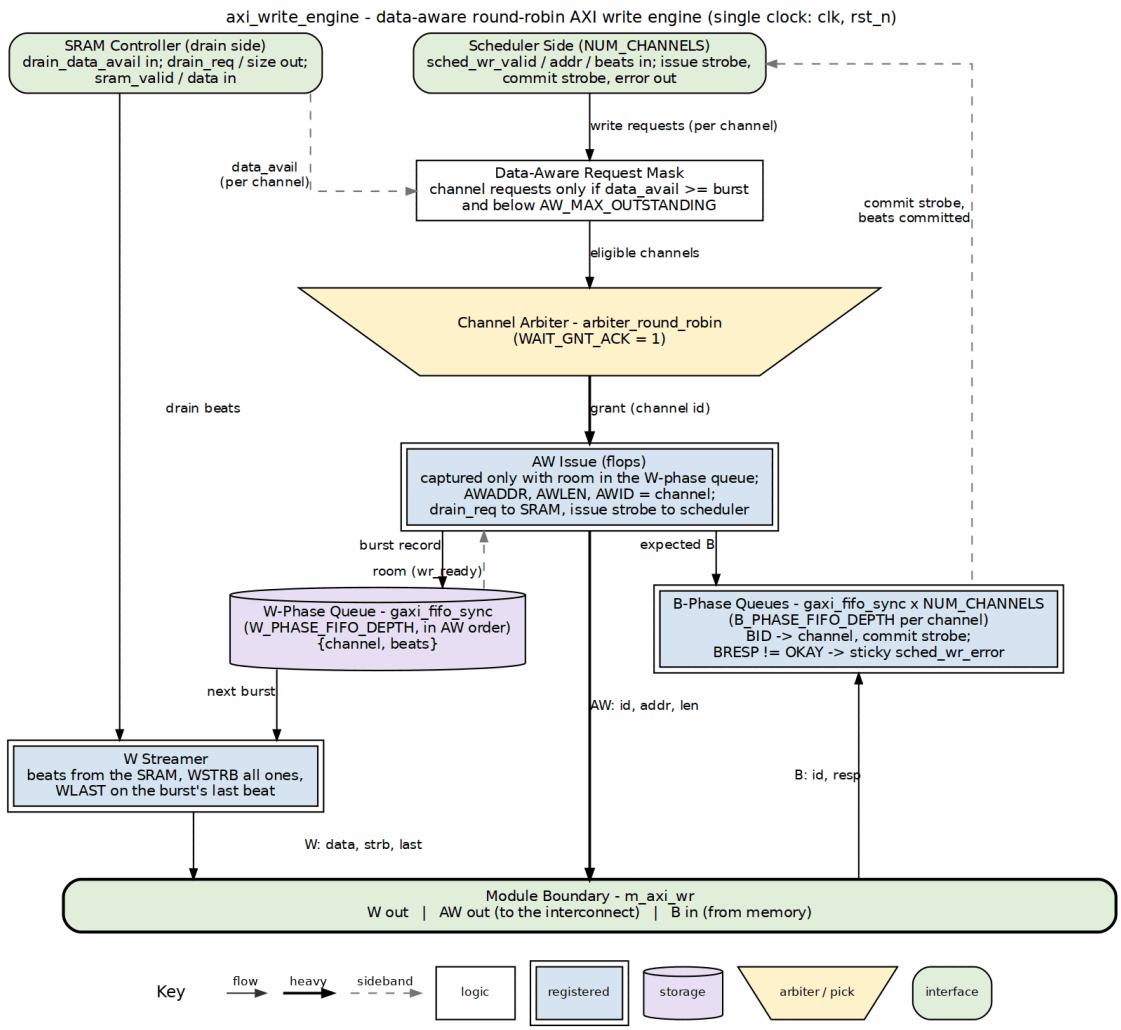
17.1.1 Key Features

- **Round-Robin Arbitration:** Fair scheduling across channels
- **Data-Aware Masking:** Only arbitrate channels with sufficient SRAM data
- **Pre-Drain Handshake:** Reserve SRAM data before AW command issues
- **FIFO-Based W Tracking:** Single shared FIFO for W-phase order preservation
- **Per-Channel B Tracking:** Separate FIFOs for out-of-order B responses
- **Channel ID in AXI ID:** Enables per-channel response routing
- **Pipelined/Non-Pipelined Modes:** Configurable outstanding transaction depth

17.2 Architecture

17.2.1 Block Diagram

17.2.2 Figure 2.12.1: AXI Write Engine Block Diagram



AXI Write Engine Block Diagram

Source: 10_axi_write_engine.dot

17.2.3 Operation Flow

```
1. Scheduler Interface: Each channel can request write bursts
2. Data Checking: Mask channels without sufficient SRAM data
3. Arbitration: Round-robin arbiter selects next channel to service
4. AXI AW Issue: Issue write command to AXI, assert wr_drain to
reserve data
5. W-Phase FIFO: Push transaction to shared FIFO (preserves AW order)
```

```
6. AXI W Stream: Stream write data from SRAM controller (ID-based mux)
7. AXI B Response: Pop per-channel B-phase FIFO, complete transaction
```

17.2.4 Key Design Decisions

FIFO-Based W Tracking (No FSM): Instead of a state machine, uses a single shared FIFO to track pending W-phase transactions. This preserves the AW command order which is critical for W-phase correctness.

Separate W and B Phase FIFOs: - **W-Phase FIFO:** Single shared FIFO (in-order with AW) - **B-Phase FIFOs:** Per-channel FIFOs (out-of-order responses allowed by AXI4)

ID-Based SRAM Interface: The write engine drives `axi_wr_sram_id` to select which channel's data to drain. The SRAM controller muxes the appropriate channel's data to `axi_wr_sram_data`.

17.3 Parameters

Parameters

Parameter	Type	Default	Description
NUM_CHANNELS	int	8	Number of channels
ADDR_WIDTH	int	64	AXI address width
DATA_WIDTH	int	512	AXI data width
ID_WIDTH	int	8	AXI ID width
USER_WIDTH	int	8	AXI USER width
SEG_COUNT_WIDTH	int	8	Width of space/count signals
PIPELINE	int	1	0: non-pipelined, 1: pipelined (default 1 since 2026-09-28: one burst in flight per channel cannot cover the B round trip at 8-beat bursts, rapids BUG-005 / stream BUG-013 aftermath)
AW_MAX_OUTSTANDING	int	8	Maximum outstanding AW requests (PIPELINE=1)
W_PHASE_FIFO_DEPTH	int	64	W-phase transaction FIFO depth
B_PHASE_FIFO_DEPTH	int	16	B-phase transaction FIFO depth per channel

17.3.1 Derived Parameters

Derived Parameters

Parameter	Derivation	Description
NC	NUM_CHANNELS	Short alias
AW	ADDR_WIDTH	Short alias
DW	DATA_WIDTH	Short alias
IW	ID_WIDTH	Short alias

Parameter	Derivation	Description
UW	USER_WIDTH	Short alias
SCW	SEG_COUNT_WIDTH	Segment count width
CIW	$clog2(NC)	Channel ID width (min 1 bit)
AXSIZE	$clog2(DW/8)	AXI burst size

17.4 Port List

17.4.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low asynchronous reset

17.4.2 Configuration Interface

Configuration Interface

Signal	Direction	Width	Description
`cfg_axi_wr_xfer_beats`	input	8	Transfer size in beats (all channels)

17.4.3 Scheduler Interface (Per-Channel)

Scheduler Interface

Signal	Direction	Width	Description
`sched_wr_valid[ch]`	input	NC	Channel requests write
`sched_wr_ready[ch]`	output	NC	Engine ready (descriptor complete)
`sched_wr_addr[ch]`	input	NC x AW	Destination addresses
`sched_wr_beats[ch]`	input	NC x 32	Beats remaining to write
`sched_wr_burst_len[ch]`	input	NC x 8	Requested burst length

17.4.4 Completion Interface (Per-Channel)

Completion Interface

Signal	Direction	Width	Description
`sched_wr_done_strobe[ch]`	output	NC	Burst completed (1 cycle pulse)
`sched_wr_bea`	output	NC x 32	Number of beats

Signal	Direction	Width	Description
ts_done[ch]			completed

17.4.5 SRAM Drain Interface

SRAM Drain Interface

Signal	Direction	Width	Description
axi_wr_drain_req[ch]	output	NC	Request to reserve data
axi_wr_drain_size[ch]	output	NC x 8	Beats to reserve
axi_wr_drain_data_avail[ch]	input	NC x SCW	Data available per channel

17.4.6 SRAM Read Interface

SRAM Read Interface

Signal	Direction	Width	Description
axi_wr_sram_valid[ch]	input	NC	Per-channel valid
axi_wr_sram_drain	output	1	Drain request
axi_wr_sram_id	output	CIW	Channel ID select
axi_wr_sram_data	input	DW	Muxed data from selected channel

17.4.7 AXI4 AW Channel

AXI4 AW Channel

Signal	Direction	Width	Description
m_axi_awvalid	output	1	Address valid
m_axi_awready	input	1	Address ready
m_axi_awid	output	IW	Transaction ID
m_axi_awaddr	output	AW	Address
m_axi_awlen	output	8	Burst length - 1
m_axi_awsize	output	3	Burst size (log2

Signal	Direction	Width	Description
			bytes)
m_axi_awburst	output	2	Burst type (INCR)

17.4.8 AXI4 W Channel

AXI4 W Channel

Signal	Direction	Width	Description
m_axi_wvalid	output	1	Write data valid
m_axi_wready	input	1	Write data ready
m_axi_wdata	output	DW	Write data
m_axi_wstrb	output	DW/8	Write strobes
m_axi_wlast	output	1	Last beat of burst
m_axi_wuser	output	UW	Channel ID for tracking

17.4.9 AXI4 B Channel

AXI4 B Channel

Signal	Direction	Width	Description
m_axi_bvalid	input	1	Response valid
m_axi_bready	output	1	Response ready
m_axi_bid	input	IW	Transaction ID
m_axi_bresp	input	2	Response

17.4.10 Error Interface

Error Interface

Signal	Direction	Width	Description
sched_wr_error[ch]	output	NC	Sticky error flag per channel

17.4.11 Debug Interface

Debug Interface

Signal	Direction	Width	Description
`dbg_wr_all_complete[ch]`	output	NC	All writes complete
`dbg_aw_transactions`	output	32	Total AW transactions issued
`dbg_w_beats`	output	32	Total W beats written to AXI

17.5 Operation

17.5.1 Burst Length Cap (stream BUG-018)

`cfg_axi_wr_xfer_beats` is an AWLEN (0..255), but a burst can never be larger than the SRAM that stages it: the request mask below waits for the whole burst to be in the buffer, and a burst the buffer cannot hold would wait forever. The engine clamps the configured value to `XFER_MAX = min(2^(SCW-1) - 1, 254)` – the buffer depth `SEG_COUNT_WIDTH` encodes, minus one, and 254 beyond that so every beat count fits the 8-bit size ports shared with `sram_controller`. On the 512-deep default AWLEN 255 runs as 255-beat bursts.

Before the clamp (2026-09-29, found as rapids BUG-009 on the identical lines), AWLEN 255 was taken literally and the 8-bit `AWLEN + 1` in the gate wrapped to 0: `w_has_data` passed on an empty buffer, the AW went out, `drain_ctrl` reserved 0 beats and the W phase pulled 256 beats that were never reserved, leaving the channel's drain count wrong until the next full reset.

17.5.2 Data-Aware Request Masking

```
// Only request arbitration if:
// 1. Scheduler is requesting (sched_wr_valid)
// 2. Sufficient SRAM data available (w_data_ok)
// 3. Below outstanding limit (w_no_outstanding)
w_arb_request[i] = sched_wr_valid[i] && w_data_ok[i] &&
w_no_outstanding[i];

// Data check includes final burst handling
w_data_ok[i] = w_has_data[i] || w_final_burst[i];
```

17.5.3 W-Phase Transaction FIFO

Single shared FIFO preserves AW command order:

```
typedef struct packed {
    logic [7:0]    beats;       // Number of beats for this W
transaction
    logic [CIW-1:0] channel_id;  // Channel ID for this transaction
} w_phase_txn_t;

// Push on AW handshake (preserves order)
if (m_axi_awvalid && m_axi_awready) begin
    w_phase_txn_fifo_wr = 1'b1;
    w_phase_txn_fifo_din.beats = m_axi_awlen + 8'd1;
    w_phase_txn_fifo_din.channel_id = r_aw_channel_id;
end
```

Room gate. The AW handshake pushes this FIFO unconditionally, so the engine captures a new AW only while the FIFO has a free entry (`w_phase_txn_fifo_wr_ready`). The engine holds one AW at a time, so an entry that is free at capture is still free at the push. Without the gate, an AW

issued into a full FIFO loses its W burst, and that channel waits for a B response that never comes. In-flight AWs can reach `NUM_CHANNELS x AW_MAX_OUTSTANDING`:

W-Phase FIFO Occupancy Bound

Build	In-flight AWs (8 outstanding)	`W_PHASE_FIFO_DEPTH`	Without the gate
8 channels	64	64	cannot overflow
16 channels	128	64	overflows; the sink hangs

17.5.4 B-Phase Transaction FIFOs

Per-channel FIFOs track beats and last flag for completion:

```
typedef struct packed {
    logic [7:0]  beats;     // Number of beats in this transaction
    logic        last;      // Is this the last transfer for
descriptor?
} b_phase_txn_t;

// last flag determines when sched_wr_ready asserts
if (m_axi_bvalid && m_axi_bready) begin
    if (b_phase_txn_fifo_dout[ch_id].last) begin
        r_sched_ready[ch_id] <= 1'b1;
    end
end
```

17.5.5 Outstanding Transaction Tracking

PIPELINE=0 (Non-Pipelined): - Boolean flag per channel (0 or 1 outstanding) - Set when AW issues, clear when B arrives

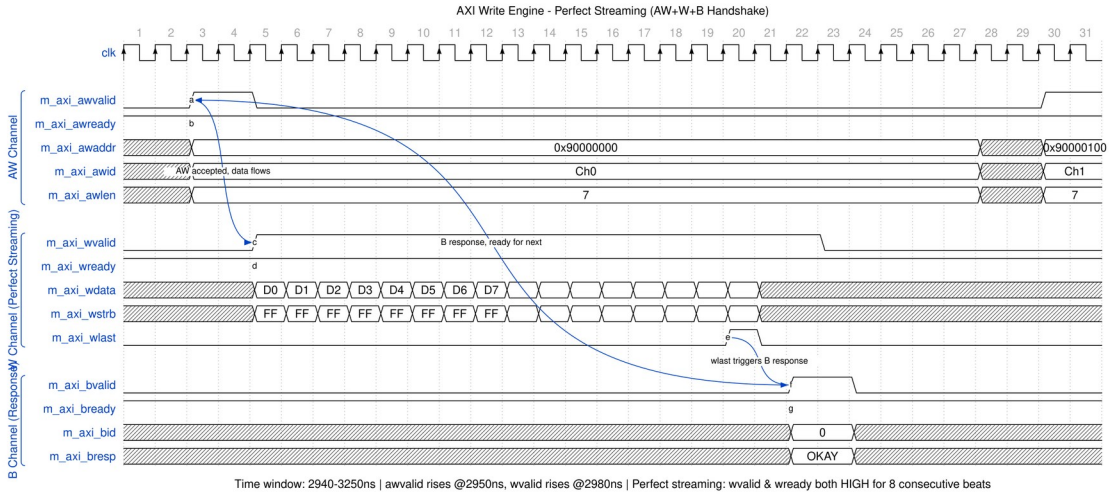
PIPELINE=1 (Pipelined): - Counter per channel (0 to AW_MAX_OUTSTANDING) - Increment on AW, decrement on B

17.6 Timing Diagrams

17.6.1 Perfect Streaming - AXI Write Transaction

The following timing diagram shows the AXI write engine operating at maximum throughput with **perfect streaming** - where both `wvalid` and `wready` remain HIGH for consecutive clock cycles, achieving one data beat per cycle.

17.6.1.1 Waveform 2.12.1: AXI Write Engine - Perfect Streaming



AXI Write Engine - Perfect Streaming

Transaction Flow:

1. **AW Channel Handshake (Address Phase)**
   - `m_axi_awvalid` rises to initiate a write request
   - `m_axi_awready` is already HIGH (slave ready)
   - Address (`awaddr`), ID (`awid`), and length (`awlen=7` for 8 beats) are captured
   - Handshake completes in a single cycle when both valid and ready are HIGH
   - Simultaneously, `axi_wr_drain_req` reserves data from SRAM controller
2. **W Channel Streaming (Data Phase)**
   - After AW acceptance, the engine begins streaming data from SRAM
   - `m_axi_wvalid` rises and **stays HIGH** for all 8 beats
   - `m_axi_wready` remains HIGH throughout (no backpressure from AXI slave)

- **Perfect streaming**: One data beat transferred every clock cycle
- `m_axi_wlast` rises on the final beat (beat 7) to mark burst completion
- `wstrb` is all-ones (0xFF) indicating full data width valid

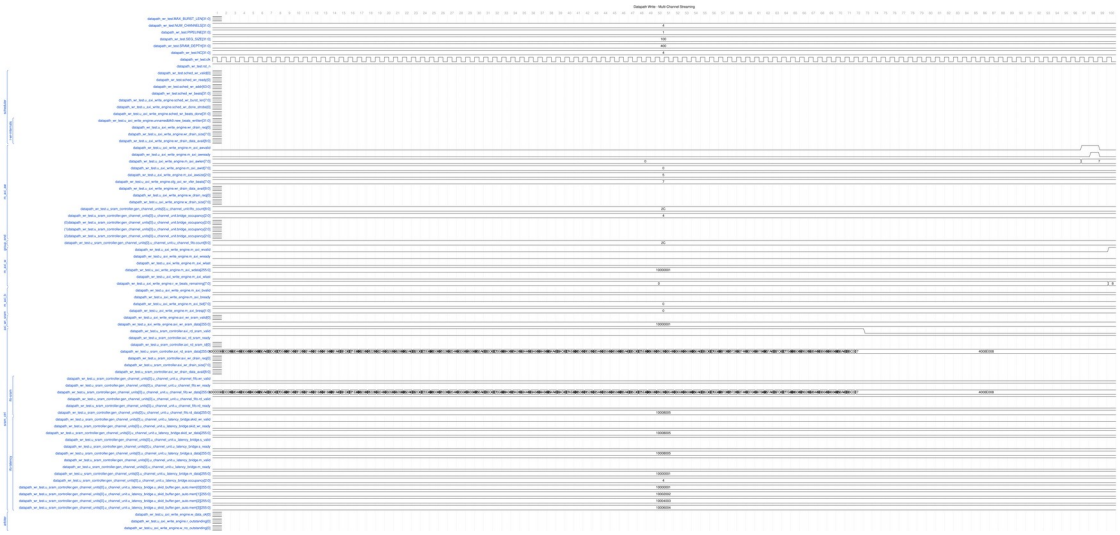
3. **B Channel Response (Completion Phase)**
   - After `wlast`, the AXI slave returns a write response
   - `m_axi_bvalid` rises with `bid` matching the transaction ID
   - `m_axi_bready` is HIGH (engine always ready for responses)
   - `bresp=OKAY` indicates successful write
   - Response triggers `sched_wr_done_strobe` to notify scheduler
4. **Next Transaction**
   - After B response, the engine can immediately issue the next AW request
   - The cycle repeats for the next channel or next burst

Key Performance Indicators: - **No bubbles**: `wvalid` and `wready` both HIGH during data phase - **Full bandwidth**: Data width (256/512 bits) x clock frequency - **Zero-wait states**: SRAM controller provides data at line rate - **Minimal latency**: B response arrives shortly after `wlast`

17.6.2 Multi-Channel Streaming

For multi-channel operation showing channel switching while maintaining streaming performance, see:

17.6.2.1 Waveform 2.12.2: Datapath Write - Multi-Channel



Datapath Write - Multi-Channel

This diagram shows how the engine arbitrates between channels while maintaining high throughput. Note how: - Different channel IDs (`awid`) appear in sequence - W-phase FIFO preserves AW order for correct data association - B responses can arrive out-of-order (per AXI4 spec)

17.7 Integration Example

```
axi_write_engine #(
    .NUM_CHANNELS       (8),
    .ADDR_WIDTH         (64),
    .DATA_WIDTH         (512),
    .ID_WIDTH           (8),
    .SEG_COUNT_WIDTH    (10),
    .PIPELINE           (0),
    .AW_MAX_OUTSTANDING (8),
    .W_PHASE_FIFO_DEPTH (64),
    .B_PHASE_FIFO_DEPTH (16)
) u_axi_write_engine (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // Configuration
    .cfg_axi_wr_xfer_beats  (cfg_axi_wr_xfer_beats),

    // Scheduler interface
    .sched_wr_valid         (sched_wr_valid),
    .sched_wr_ready         (sched_wr_ready),
    .sched_wr_addr          (sched_wr_addr),
    .sched_wr_beats         (sched_wr_beats),
    .sched_wr_burst_len     (sched_wr_burst_len),

    // Completion interface
    .sched_wr_done_strobe   (sched_wr_done_strobe),
    .sched_wr_beats_done    (sched_wr_beats_done),

    // SRAM drain interface
    .axi_wr_drain_req       (axi_wr_drain_req),
    .axi_wr_drain_size      (axi_wr_drain_size),
    .axi_wr_drain_data_avail(axi_wr_drain_data_avail),

    // SRAM read interface
    .axi_wr_sram_valid      (axi_wr_sram_valid),
    .axi_wr_sram_drain      (axi_wr_sram_drain),
    .axi_wr_sram_id         (axi_wr_sram_id),
    .axi_wr_sram_data       (axi_wr_sram_data),

    // AXI master
    .m_axi_awvalid          (m_axi_wr_awvalid),
    .m_axi_awready          (m_axi_wr_awready),
    .m_axi_awid             (m_axi_wr_awid),
    .m_axi_awaddr           (m_axi_wr_awaddr),
    .m_axi_awlen            (m_axi_wr_awlen),
```

```
    .m_axi_awsize           (m_axi_wr_awsize),
    .m_axi_awburst          (m_axi_wr_awburst),

    .m_axi_wvalid           (m_axi_wr_wvalid),
    .m_axi_wready           (m_axi_wr_wready),
    .m_axi_wdata            (m_axi_wr_wdata),
    .m_axi_wstrb            (m_axi_wr_wstrb),
    .m_axi_wlast            (m_axi_wr_wlast),
    .m_axi_wuser            (m_axi_wr_wuser),

    .m_axi_bvalid           (m_axi_wr_bvalid),
    .m_axi_bready           (m_axi_wr_bready),
    .m_axi_bid              (m_axi_wr_bid),
    .m_axi_bresp            (m_axi_wr_bresp),

    // Error and debug
    .sched_wr_error         (sched_wr_error),
    .dbg_wr_all_complete    (dbg_wr_all_complete),
    .dbg_aw_transactions    (dbg_aw_transactions),
    .dbg_w_beats            (dbg_w_beats)
);
```

17.8 Bug Fixes (v0.92)

17.8.1 w_arb_request Timing Closure (Commit 4e8f9e02)

At 8 channels, the arbiter request combinational path became critical. The `w_arb_request` signal is now **registered** to add a single pipeline stage, closing timing. See `06_axi_read_engine.md` for details.

17.8.2 W-Side Post-Flop Gating on wvalid (Commit a82627af)

The write engine now uses **post-flop gating on wvalid** to defeat a stale-view race in the drain controller. Previously, the wvalid signal was generated before SRAM data had settled through the drain path's register, causing the drain controller to sample data one cycle stale.

Fix: A register is placed after the W-path data multiplexer, and `wvalid` is only asserted after this flop. This ensures data stability before wvalid assertion, preventing premature consumption.

Cross-reference: `11_stream_drain_ctrl.md` documents the matching half of this fix (invalidating the stale-view acknowledgment).

17.9 Bug Fixes (v0.93)

17.9.1 W-Phase FIFO Overflow Above 64 In-Flight AWs (2026-10-17)

The W-phase FIFO was pushed on every AW handshake without a room check, so entries were dropped once in-flight AWs passed `W_PHASE_FIFO_DEPTH`. This cannot happen at 8 channels with 8 outstanding (64 = 64). It was found in RAPIDS, whose 16-channel all-channels test hung with 128 AWs on the bus against 776 W beats. AW capture now waits for a free entry; see W-Phase Transaction FIFO above. `test_axi_write_engine` covers it with `wphase` cells, which drive more in-flight AWs than the FIFO holds.

17.10 Related Documentation

- **Parent:** `01_stream_core.md` - Top-level integration
- **Scheduler Array:** `02_scheduler_group_array.md` - Provides sched_wr_* signals
- **SRAM Controller:** `08_sram_controller.md` - Data source
- **Drain Controller:** `11_stream_drain_ctrl.md` - Data availability tracking

- **Read Engine: `06_axi_read_engine.md` - Complementary read datapath**

17.11 Revision History

AXI Write Engine Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers
0.93	2026-10-17	seang	W-phase FIFO room gate on AW capture; occupancy bound for 8 and 16 channels

Last Updated: 2026-10-17

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18 APB Configuration Block

Module: `stream_config_block.sv` **Location:** `projects/components/dma-ip/stream/rtl/top/` **Category:** TOP (Integration) **Parent:** `stream_top.sv` **Status:** Implemented **Last Updated:** 2025-11-30

18.1 Overview

The `stream_config_block` module maps PeakRDL-generated register outputs to STREAM core configuration inputs. It provides a clean interface layer between the register file and the STREAM DMA engine core, handling field extraction, width conversion, and global enable gating.

18.1.1 Key Features

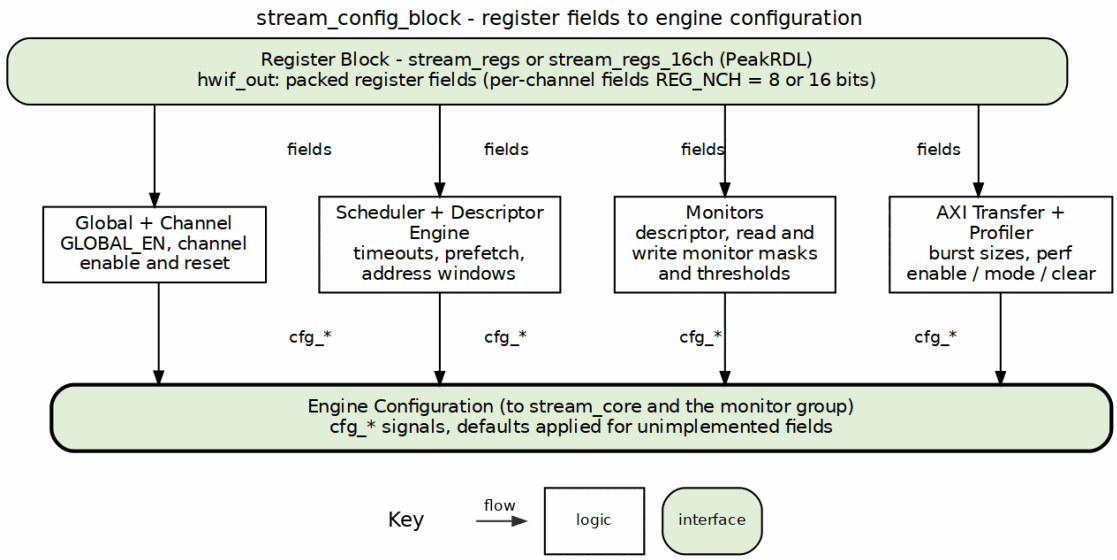
- **Field Extraction:** Extracts fields from packed register values
- **Width Conversion:** Converts register fields to configuration signal widths
- **Global Enable Gating:** Gates enables by global enable register
- **Address Extension:** Zero-extends 32-bit register addresses to 64-bit
- **Clean Interface:** Isolates register naming from core logic

---

18.2 Architecture

18.2.1 Block Diagram

18.2.2 Figure 1: Stream Config Block Diagram



Stream Config Block Diagram

Source: 14_stream_config_block.dot

18.2.3 Configuration Groups

```
stream_config_block
 Global and Channel Control
    cfg_channel_enable (gated by global_en)
    cfg_channel_reset (OR'd with global_rst)
 Scheduler Configuration
    cfg_sched_enable (gated by global_en)
    cfg_sched_timeout_cycles
    cfg_sched_*_enable flags
 Descriptor Engine Configuration
    cfg_desceng_enable (gated by global_en)
    cfg_desceng_prefetch
    cfg_desceng_addr*_base/limit (zero-extended)
 Monitor Configurations (3 sets)
    Descriptor AXI Monitor (cfg_desc_mon_*)
    Read Engine Monitor (cfg_rdeng_mon_*)
    Write Engine Monitor (cfg_wreng_mon_*)
 AXI Transfer Configuration
    cfg_axi_rd_xfer_beats
    cfg_axi_wr_xfer_beats
```

```
Performance Profiler
    cfg_perf_enable (gated by global_en)
    cfg_perf_mode
    cfg_perf_clear
```

18.3 Parameters

Parameters

Parameter	Type	Default	Description
NUM_CHANNELS	int	8	Number of DMA channels
REG_NCH	int	8	Channel bits in the register map: 8 (stream_regs) or 16 (stream_regs_16ch). NUM_CHANNELS <= REG_NCH
ADDR_WIDTH	int	64	Address width for config outputs

18.4 Port List

18.4.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low asynchronous reset

18.4.2 PeakRDL Register Inputs

Global Control:

PeakRDL Register Inputs

Signal	Direction	Width	Description
reg_global_ctrl_global_en	input	1	Global enable
reg_global_ctrl_global_rst	input	1	Global reset
reg_channel_enable_ch_en	input	REG_NCH	Per-channel enable bits
reg_channel_reset_ch_rst	input	REG_NCH	Per-channel reset bits

Scheduler Configuration:

PeakRDL Register Inputs

Signal	Direction	Width	Description
reg_sched_timeout_cycles_*	input	16	Timeout threshold
reg_sched_config_*	input	1	Various enables

Descriptor Engine Configuration:

PeakRDL Register Inputs

Signal	Direction	Width	Description
`reg_desceng_config_*`	input	varies	Enable and threshold
`reg_desceng_addr*_*`	input	32	Address range limits

Monitor Configurations (3 sets):

Each monitor (DAXMON, RDMON, WRMON) has: - `reg_*_enable_*` - Enable flags - `reg_*_timeout_*` - Timeout configuration - `reg_*_latency_thresh_*` - Latency threshold - `reg_*_pkt_mask_*` - Packet type mask - `reg_*_err_cfg_*` - Error configuration - `reg_*_mask*_*` - Event masks

AXI Transfer Configuration:

PeakRDL Register Inputs

Signal	Direction	Width	Description
`reg_axi_xfer_config_rd_xfer_beats`	input	8	Read burst size
`reg_axi_xfer_config_wr_xfer_beats`	input	8	Write burst size

Performance Profiler:

PeakRDL Register Inputs

Signal	Direction	Width	Description
`reg_perf_config_perf_en`	input	1	Profiler enable
`reg_perf_config_perf_mode`	input	1	Profiling mode
`reg_perf_config_perf_clear`	input	1	Clear profiler

18.4.3 Configuration Outputs

Global and Channel:

Configuration Outputs

Signal	Direction	Width	Description
`cfg_channel_enable`	output	NUM_CHANNELS	Gated channel enables
`cfg_channel_reset`	output	NUM_CHANNELS	Combined channel resets

Scheduler:

Configuration Outputs

Signal	Direction	Width	Description
`cfg_sched_enable`	output	1	Gated scheduler enable
`cfg_sched_timeout_cycles`	output	16	Timeout value
`cfg_sched_*_enable`	output	1	Feature enables

Descriptor Engine:

Configuration Outputs

Signal	Direction	Width	Description
`cfg_desceng_enable`	output	1	Gated desceng enable
`cfg_desceng_prefetch`	output	1	Prefetch enable
`cfg_desceng_fifo_thresh`	output	4	FIFO threshold
`cfg_desceng_addr*_base`	output	ADDR_WIDTH	Zero-extended address
`cfg_desceng_addr*_limit`	output	ADDR_WIDTH	Zero-extended address

Monitor Outputs (3 sets):

Each monitor outputs ~16 configuration signals covering: - Enable flags - Timeout values - Thresholds - Masks

AXI Transfer:

Configuration Outputs

Signal	Direction	Width	Description
`cfg_axi_rd_xfer_beats`	output	8	Read burst size
`cfg_axi_wr_xfer_beats`	output	8	Write burst size

Performance Profiler:

Configuration Outputs

Signal	Direction	Width	Description
`cfg_perf_enable`	output	1	Gated profiler enable
`cfg_perf_mode`	output	1	Profiling mode
`cfg_perf_clear`	output	1	Clear signal

18.5 Operation

18.5.1 Global Enable Gating

```
// Gate all channel enables by global enable
assign cfg_channel_enable = reg_channel_enable_ch_en[NUM_CHANNELS-1:0]
                          & {NUM_CHANNELS{reg_global_ctrl_global_en}};

// Gate scheduler enable
assign cfg_sched_enable = reg_sched_config_sched_en &
reg_global_ctrl_global_en;

// Gate monitor enables
assign cfg_desc_mon_enable = reg_daxmon_enable_mon_en &
reg_global_ctrl_global_en;
```

18.5.2 Global Reset OR

```
// Channel resets are OR'd with global reset
assign cfg_channel_reset = reg_channel_reset_ch_rst[NUM_CHANNELS-1:0]
                         | {NUM_CHANNELS{reg_global_ctrl_global_rst}};
```

18.5.3 Address Zero Extension

```
// Zero-extend 32-bit register addresses to ADDR_WIDTH (typically 64-
bit)
assign cfg_desceng_addr0_base = {{(ADDR_WIDTH-32){1'b0}},
reg_desceng_addr0_base_addr0_base};
assign cfg_desceng_addr0_limit = {{(ADDR_WIDTH-32){1'b0}},
reg_desceng_addr0_limit_addr0_limit};
```

18.5.4 Register-Map Width (REG_NCH)

The per-channel CSR fields are `REG_NCH` bits wide, which is 8 or 16 depending on the register map. The block uses the low `NUM_CHANNELS` bits, so an 8-channel core can also sit behind the 16-channel map. The observer channel select (`reg_obs_ctrl_ch_sel` / `cfg_obs_ch_sel`) is 3 bits in the 8-channel map and 4 bits in the 16-channel map (`REG_NCH > 8`). The top passes `REG_NCH` from the register map it was built with, through the `regsel_{8,16}ch` include that the filelist selects.

18.6 Integration Example

```
stream_config_block #(
    .NUM_CHANNELS   (8),
    .ADDR_WIDTH     (64)
) u_config_block (
    .clk            (clk),
    .rst_n          (rst_n),

    // From PeakRDL register file
    .reg_global_ctrl_global_en      (regs_global_ctrl_global_en),
    .reg_global_ctrl_global_rst     (regs_global_ctrl_global_rst),
    .reg_channel_enable_ch_en       (regs_channel_enable_ch_en),
    // ... more register inputs

    // To stream_core
    .cfg_channel_enable             (cfg_channel_enable),
    .cfg_channel_reset              (cfg_channel_reset),
    .cfg_sched_enable               (cfg_sched_enable),
    // ... more config outputs
);
```

18.7 Related Documentation

- **Parent:** `01_stream_core.md` - Receives configuration signals
- **Register File:** PeakRDL-generated `stream_regs.sv`
- **Scheduler:** `04_scheduler.md` - Uses sched_* config
- **Descriptor Engine:** `05_descriptor_engine.md` - Uses desceng_* config

- **Monitors: Monitor documentation for each AXI monitor**

18.8 Revision History

APB Configuration Block Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers
0.92	2026-10-17	seang	`REG_NCH` parameter (8- or 16-channel register map); per-channel input widths; observer channel-select width

Last Updated: 2026-10-17

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19 Performance Profiler

Module: `perf_profiler.sv` **Location:** `projects/components/dma-ip/stream/rtl/fub/` **Category:** FUB (Functional Unit Block) **Parent:** `stream_top.sv` **Status:** Implemented **Last Updated:** 2026-05-11

19.1 Overview

The `perf_profiler` module captures timing information for channel activity to enable performance analysis and bottleneck identification. It monitors channel idle signals and records either timestamps or elapsed times to a FIFO for software retrieval.

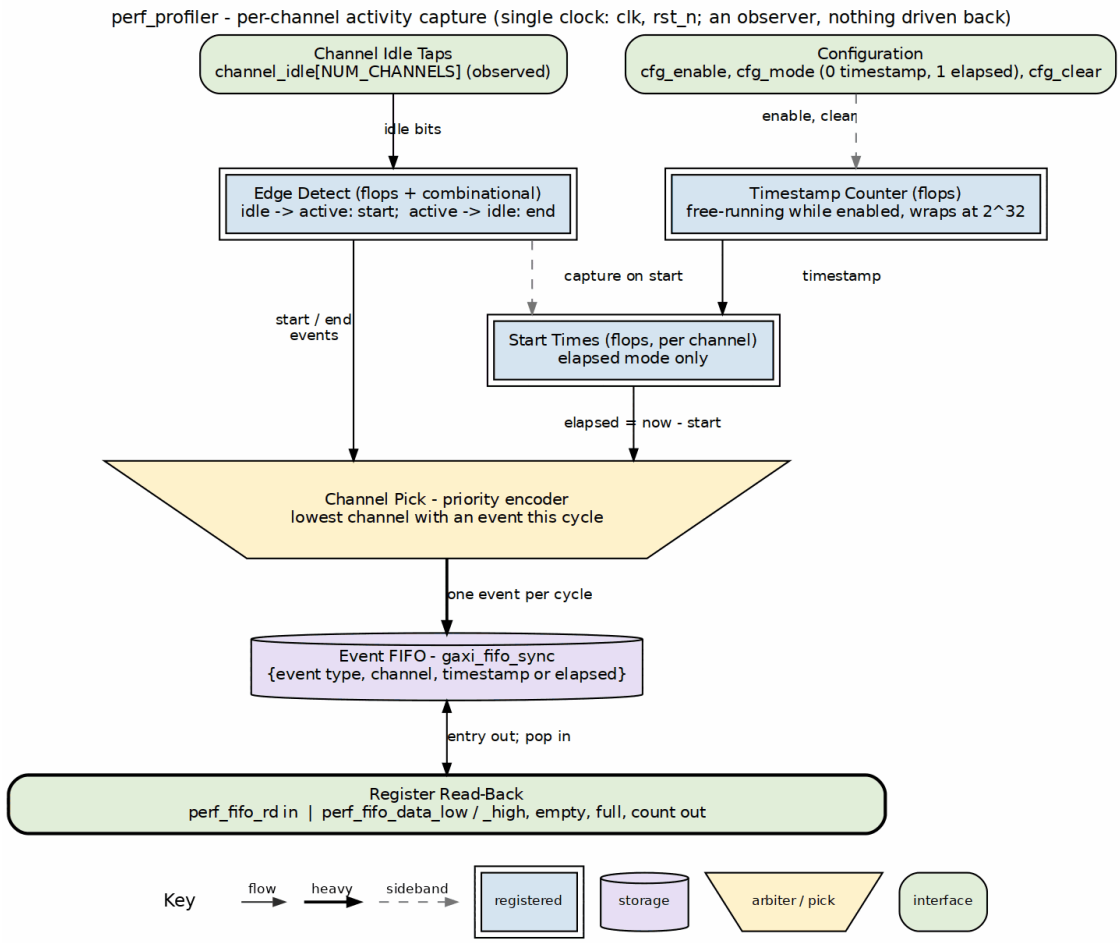
19.1.1 Key Features

- **Two Profiling Modes:**
  - Mode 0 (TIMESTAMP): Captures timestamps on idle signal edges
  - Mode 1 (ELAPSED): Captures elapsed time directly
- **Per-Channel Profiling:** Tracks all NUM_CHANNELS channels (8 or 16) independently
- **256-Entry FIFO:** Buffers performance data for software polling
- **Channel ID Tagging:** Each entry includes source channel
- **Configurable via APB:** Enable, mode, and clear controls

19.2 Architecture

19.2.1 Block Diagram

19.2.2 Figure 2.15.1: Performance Profiler Block Diagram



Performance Profiler Block Diagram

Source: 15_perf_profiler.dot

19.2.3 Profiling Modes

Mode 0 - TIMESTAMP:

```
Records timestamp when:
  - Channel transitions idle → active (start)
  - Channel transitions active → idle (end)

Software calculates: elapsed = timestamp_end - timestamp_start
```

```
Advantages:
  - Simpler hardware
  - More flexible analysis
  - Can detect partial transfers
```

Mode 1 - ELAPSED:

```
Records elapsed time when:
  - Channel returns to idle

elapsed_time = current_timestamp - start_timestamp

Advantages:
  - Hardware calculates duration
  - Simpler software processing
  - Direct performance metric
```

19.2.4 FIFO Entry Format (ENTRY_W bits)

The channel ID field is `CHID_W = max(3, CHANNEL_WIDTH)` bits, and the event type sits directly above it. An 8-channel build keeps the original 36-bit layout. A 16-channel build adds one ID bit, so its entries are 37 bits.

FIFO Entry Format

Bits	Field	8 channels	16 channels
[31:0]	timestamp/ elapsed	[31:0]	[31:0]
channel_id	Source channel	[34:32] (0-7)	[35:32] (0-15)
event_type	0=start, 1=end (timestamp mode)	[35]	[36]

PERF_DATA_HIGH returns the bits above 31, zero-extended: `{event_type[3], channel_id[2:0]}` at 8 channels and `{event_type[4], channel_id[3:0]}` at 16. Software takes the channel count from `VERSION.NUM_CHANNELS`.

19.3 Parameters

Parameters

Parameter	Type	Default	Description
NUM_CHANNELS	int	8	Number of channels to monitor
CHANNEL_WIDTH	int	$clog2(NUM_CHANNELS)	Channel ID width
TIMESTAMP_WIDTH	int	32	Timestamp counter width
FIFO_DEPTH	int	256	Performance FIFO depth
FIFO_ADDR_WIDTH	int	$clog2(FIFO_DEPTH)	FIFO address width

19.4 Port List

19.4.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low asynchronous reset

19.4.2 Channel Monitoring

Channel Monitoring

Signal	Direction	Width	Description
`channel_idle`	input	NUM_CHANNELS	Per-channel idle signals

19.4.3 Configuration Interface

Configuration Interface

Signal	Direction	Width	Description
`cfg_enable`	input	1	Enable profiling
`cfg_mode`	input	1	0=timestamp, 1=elapsed
`cfg_clear`	input	1	Clear FIFO and counters

19.4.4 FIFO Read Interface

FIFO Read Interface

Signal	Direction	Width	Description
`perf_fifo_rd`	input	1	Read strobe (pops FIFO)
`perf_fifo_data_low`	output	32	Timestamp/elapsed [31:0]
`perf_fifo_data_high`	output	32	{0, event_type, channel_id[CHID_W-1:0]}

Signal	Direction	Width	Description
perf_fifo_empty	output	1	FIFO empty flag
perf_fifo_full	output	1	FIFO full flag
perf_fifo_count	output	16	Number of entries

19.5 Operation

19.5.1 Edge Detection

```
// Detect idle signal transitions
assign w_idle_rising  = channel_idle & ~r_idle_prev;  // Active → Idle (end)
assign w_idle_falling = ~channel_idle & r_idle_prev;  // Idle → Active (start)
```

19.5.2 Timestamp Counter

```
// Free-running 32-bit counter
// Increments every cycle when profiling enabled
// Wraps at 2^32 - 1 (software handles rollover)
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_timestamp_counter <= '0;
    end else if (cfg_clear) begin
        r_timestamp_counter <= '0;
    end else if (cfg_enable) begin
        r_timestamp_counter <= r_timestamp_counter + 1'b1;
    end
end
```

19.5.3 Start Time Capture (Elapsed Mode)

```
// Per-channel start time tracking
generate
    for (ch = 0; ch < NUM_CHANNELS; ch++) begin
        always_ff @(posedge clk) begin
            if (cfg_enable && cfg_mode == MODE_ELAPSED) begin
                if (w_idle_falling[ch]) begin
                    r_start_time[ch] <= r_timestamp_counter;
                    r_channel_active[ch] <= 1'b1;
                end
                else if (w_idle_rising[ch]) begin
                    r_channel_active[ch] <= 1'b0;
                end
            end
        end
    end
endgenerate
```

19.5.4 Two-Register Read Sequence (order-independent)

```
Software reads FIFO via APB:
1. Check perf_fifo_empty == 0
2. Read PERF_DATA_LOW (0x2D0) and PERF_DATA_HIGH (0x2D4), in EITHER
order:
```

```
   - Both return the entry at the HEAD of the FIFO
   - LOW  = timestamp/elapsed [31:0]
   - HIGH = {0, event_type, channel_id[CHID_W-1:0]} (CHID_W = 3 at 8
channels, 4 at 16)
3. The FIFO pops once BOTH have been read, so the two halves always
   belong to the same entry and the read order does not matter
4. Parse data and repeat
```

19.6 Integration Example

```
perf_profiler #(
    .NUM_CHANNELS       (8),
    .TIMESTAMP_WIDTH    (32),
    .FIFO_DEPTH         (256)
) u_perf_profiler (
    .clk                (clk),
    .rst_n              (rst_n),

    // From schedulers
    .channel_idle       (scheduler_idle),

    // Configuration
    .cfg_enable         (cfg_perf_enable),
    .cfg_mode           (cfg_perf_mode),
    .cfg_clear          (cfg_perf_clear),

    // FIFO read interface (to APB)
    .perf_fifo_rd       (perf_fifo_rd),
    .perf_fifo_data_low (perf_fifo_data_low),
    .perf_fifo_data_high(perf_fifo_data_high),
    .perf_fifo_empty    (perf_fifo_empty),
    .perf_fifo_full     (perf_fifo_full),
    .perf_fifo_count    (perf_fifo_count)
);
```

19.7 Software Usage

19.7.1 Example: Measure Channel 0 Transfer Time

```
// Enable timestamp mode
write_reg(PERF_CONFIG, PERF_EN | MODE_TIMESTAMP);

// Start transfer on channel 0
write_reg(CH0_CTRL_LOW, desc_addr_low);
write_reg(CH0_CTRL_HIGH, desc_addr_high);

// Wait for completion
while (!(read_reg(AXI_WR_COMPLETE) & (1u << 0)));  // ch0 write
complete

// Read performance data
while (!(read_reg(PERF_STATUS) & FIFO_EMPTY)) {
    uint32_t low  = read_reg(PERF_DATA_LOW);   // head entry, low word
    uint32_t high = read_reg(PERF_DATA_HIGH);  // same entry; second
read pops it

    uint32_t timestamp = low;
    uint8_t  channel   = high & 0x7;
    uint8_t  event     = (high >> 3) & 0x1;

    if (channel == 0) {
        if (event == 0)  // Start
            start_time = timestamp;
        else             // End
            elapsed = timestamp - start_time;
    }
}
```

19.8 Common Issues

19.8.1 Issue 1: FIFO Overflow

Symptom: Performance data lost

Root Cause: Software not polling fast enough

Solution: - Poll more frequently - Use larger FIFO depth - Consider interrupt on half-full

19.8.2 Issue 2: Timestamp Rollover

Symptom: Negative elapsed times calculated

Root Cause: 32-bit counter wrapped during measurement

Solution: Handle rollover in software:

```
if (end_time < start_time)
    elapsed = (0xFFFFFFFF - start_time) + end_time + 1;
```

19.9 Related Documentation

- **Parent:** `01_stream_core.md` - Top-level integration
- **Data Source:** `04_scheduler.md` - Provides idle signals

- **Configuration: `14_apb_config.md` - Config register mapping**

19.10 Revision History

Performance Profiler Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers
0.92	2026-10-17	seang	Channel ID field widens with the channel count (`CHID_W`); 16-channel builds were reporting channels 8-15 as 0-7

Last Updated: 2026-10-17

20 MonBus AXI-Lite Group

Module: `monbus_axil4_axil4_group.sv` (documented here under its historical STREAM name `monbus_axil_group`) **Location:** `projects/components/dma-ip/stream/rtl/macro/` **Category:** MACRO (Integration) **Parent:** `stream_top.sv` **Status:** Implemented **Last Updated:** 2025-11-30

20.1 Overview

The monbus_axil_group module is the canonical shared monitor bus processing block (RTL source: rtl/amba/ocelus/monbus_axil4_axil4_group.sv; formerly rtl/amba/shared/monbus_axil_group.sv) instantiated by STREAM for packet filtering and routing. It receives 128-bit monitor bus packets paired with 64-bit timestamps (192-bit atomic group through AXI-Lite skid), filters them per protocol/packet type, and routes to either error/interrupt FIFO or external memory.

20.1.1 Key Features

- **128-bit Monitor Packets + 64-bit Timestamps:** Carried atomically through 192-bit AXI-Lite record (3 beats: [127:0] pkt, [63:0] ts, [63:0] spare)
- **Per-Protocol Filtering:** Configurable packet type masking per AXI/AXIS/CORE protocol
- **Dual Output Paths:**
  - Error/Interrupt FIFO (64 entries) - generates interrupt when not empty
  - Master Write FIFO (32 entries) - writes packets to configurable memory range
- **Fixed AXI-Lite Record Format:** 3 beats per record (no configurable cfg_ts_append_* settings)
- **Shared Building Block:** Source of truth at rtl/amba/ocelus/monbus_axil4_axil4_group.sv (module renamed monbus_axil4_axil4_group in the monitor-path move); STREAM instantiates this with zero modifications (drift area 1, commit 13706d10)

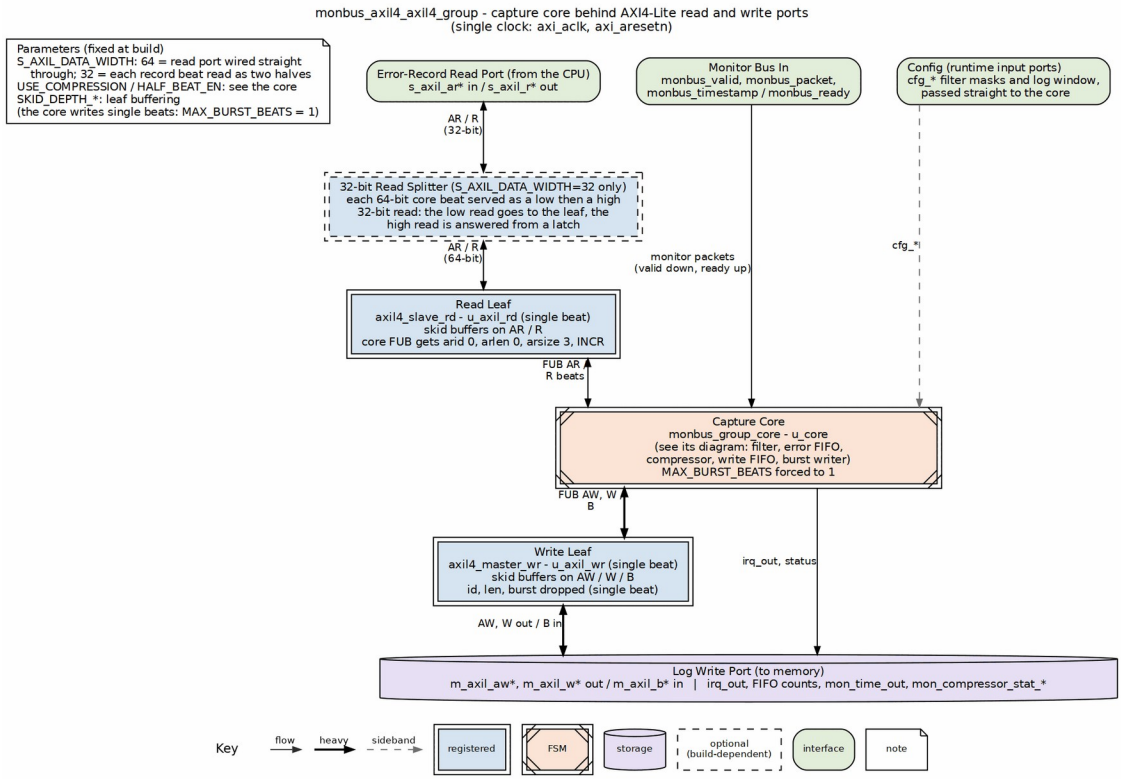
20.1.2 Simplified from RAPIDS

RAPIDS has source + sink data paths requiring TWO monitor buses. STREAM has memory-to-memory only, so this module has ONE monitor bus input (no arbitration needed).

20.2 Architecture

20.2.1 Block Diagram

20.2.2 Figure 2.16.1: MonBus AXI-Lite Group Block Diagram



MonBus AXI-Lite Group Block Diagram

Source: 16_monbus_axil4_axil4_group.dot

20.2.3 Filter Decision Tree

```
For each incoming packet:
1. Extract: pkt_type, pkt_protocol, pkt_event_code
2. Check protocol-specific masks:
   - pkt_mask: Drop if bit[pkt_type] = 1
   - err_select: Route to error FIFO if bit[pkt_type] = 1
   - event_mask: Drop if bit[event_code] = 1
3. Route:
   - Dropped packets: Consumed silently
   - Error-selected: To error/interrupt FIFO
   - Remaining: To master write FIFO
```

20.3 Parameters

Parameters

Parameter	Type	Default	Description
FIFO_DEPTH_ERR	int	64	Error/interrupt FIFO depth
FIFO_DEPTH_WRITE	int	32	Master write FIFO depth
ADDR_WIDTH	int	32	AXI address width
DATA_WIDTH	int	32	AXI data width
NUM_PROTOCOLS	int	3	AXI, AXIS, CORE
USE_COMPRESSION	int	0	Build the bulk-trace compressor into the group. 0 compiles it out; the runtime COMPRESS_EN bit is then inert.
USE_HALFBEAT	int	0	Pack two 30-bit slots per 64-bit beat. Only meaningful with USE_COMPRESSION=1.

20.3.1 Compression

With USE_COMPRESSION=0 the group emits RAW records: three 64-bit beats per 128-bit packet ({tag, timestamp}, packet[127:64], packet[63:0]). With it built and COMPRESS_EN set, it emits variable compressor slots with the tag in bits [63:60] instead.

The two encodings are not interchangeable. A tally reassembles RAW records with a mod-3 beat counter and has no slot decoder, so compressed traffic aimed at a tally silently yields nothing; compressed traffic belongs in a capture memory that the host reads back and decodes. A capture measuring exactly 3.00 slots per packet is uncompressed, whatever the enable bits report.

Measured (cosim, 2026-09-08). With the compressor built and half-beat packing on, a monitors-on DMA workload captured 36 slots for 32 decoded packets: **1.12 slots per packet, 62.5% smaller** than the raw 3-beat encoding. The same workload with the compressor compiled out produced 96 slots for the same 32 packets – exactly 3.00, the raw ratio. Both decoded identically, so the difference is encoding alone.

The encoder, slot format and half-beat packing are specified globally and are not restated here:

- docs/markdown/rtl-amba/monitor/monbus_compressor.md
- docs/markdown/rtl-amba/monitor/monbus_halfbeat_packer.md

- `docs/markdown/rtl-amba/monitor/monbus_group_core.md` (beat layouts for both modes)

20.4 Port List

20.4.1 Clock and Reset

Clock and Reset

Signal	Direction	Width	Description
axi_aclk	input	1	AXI clock
axi_aresetn	input	1	AXI active-low reset

20.4.2 Monitor Bus Input

Monitor Bus Input

Signal	Direction	Width	Description
monbus_valid	input	1	Packet valid
monbus_ready	output	1	Ready to accept
monbus_packet	input	128	128-bit monitor packet (protocol-independent structure)
monbus_timestamp	input	64	64-bit side-band timestamp (cycle count at packet capture)

20.4.3 AXI-Lite Slave Read Interface

AXI-Lite Slave Read Interface

Signal	Direction	Width	Description
s_axil_arvalid	input	1	Read address valid
s_axil_arready	output	1	Read address ready
s_axil_araddr	input	ADDR_WIDTH	Read address
s_axil_arprot	input	3	Protection bits
s_axil_rvalid	output	1	Read data valid
s_axil_rready	input	1	Read data ready

Signal	Direction	Width	Description
s_axil_rdata	output	DATA_WIDTH	Read data
s_axil_rresp	output	2	Read response

20.4.4 AXI-Lite Master Write Interface

AXI-Lite Master Write Interface

Signal	Direction	Width	Description
m_axil_awvalid	output	1	Write address valid
m_axil_awready	input	1	Write address ready
m_axil_awaddr	output	ADDR_WIDTH	Write address
m_axil_awprot	output	3	Protection bits
m_axil_wvalid	output	1	Write data valid
m_axil_wready	input	1	Write data ready
m_axil_wdata	output	DATA_WIDTH	Write data
m_axil_wstrb	output	DATA_WIDTH/8	Write strobes
m_axil_bvalid	input	1	Write response valid
m_axil_bready	output	1	Write response ready
m_axil_bresp	input	2	Write response

20.4.5 Interrupt Output

Interrupt Output

Signal	Direction	Width	Description
irq_out	output	1	Interrupt (error FIFO not empty)

20.4.6 Configuration Interface

Configuration Interface

Signal	Direction	Width	Description
cfg_base_addr	input	ADDR_WIDTH	Base for master writes
cfg_limit_addr	input	ADDR_WIDTH	Limit for master writes

20.4.7 Protocol Configuration (per protocol)

AXI Protocol:

Protocol Configuration

Signal	Direction	Width	Description
cfg_axi_pkt_mask	input	16	Drop mask for packet types
cfg_axi_err_select	input	16	Error FIFO select
cfg_axi_*_mask	input	16	Per-event-type masks

AXIS Protocol:

Protocol Configuration

Signal	Direction	Width	Description
cfg_axis_pkt_mask	input	16	Drop mask
cfg_axis_err_select	input	16	Error FIFO select
cfg_axis_*_mask	input	16	Per-event-type masks

CORE Protocol:

Protocol Configuration

Signal	Direction	Width	Description
cfg_core_pkt_mask	input	16	Drop mask

Signal	Direction	Width	Description
cfg_core_err_select	input	16	Error FIFO select
cfg_core_*_mask	input	16	Per-event-type masks

How STREAM drives these slots (2026-09-28, stream BUG-017). The group filters by the packet's PROTOCOL field, and every in-core monitor (DAXMON, RDMON, WRMON) tags its packets PROTOCOL_AXI, so only the AXI slot ever acts on in-core traffic. stream_top_ch8 drives that slot from all three monitors' register sets at once: the drop masks (PKT_MASK, ERR_MASK, MASK1..3) are ANDed, so a type or event is dropped at the group only if all three monitors drop it, and ERR_SELECT is ORed. Each monitor still applies its own mask inside itself; the group is not a second per-monitor filter. The AXIS and CORE slots are tied to drop-nothing because nothing in-core emits those protocols. Until this change the AXI slot was wired from DAXMON's registers alone, with RDMON's and WRMON's parked on the unused AXIS and CORE slots, so a program that switched DAXMON off (a drop-all DAXMON_PKT_MASK) lost every RDMON and WRMON packet at the group – the order-dependent ADDR_RANGE loss of amba BUG-036.

20.4.8 Debug/Status

Debug/Status

Signal	Direction	Width	Description
err_fifo_full	output	1	Error FIFO full
write_fifo_full	output	1	Write FIFO full
err_fifo_count	output	8	Error FIFO count
write_fifo_count	output	8	Write FIFO count

20.5 Operation

20.5.1 Packet Type Filtering

```
// Protocol-specific filtering (example: AXI)
case (pkt_protocol)
    3'b000: begin // AXI
        // Check if packet type is dropped
        pkt_drop = cfg_axi_pkt_mask[pkt_type];

        // Check if packet goes to error FIFO
        pkt_to_err_fifo = cfg_axi_err_select[pkt_type] && !pkt_drop;

        // Check individual event masking
        case (pkt_type)
            PktTypeError:     pkt_event_masked =
cfg_axi_error_mask[pkt_event_code];
            PktTypeTimeout:   pkt_event_masked =
cfg_axi_timeout_mask[pkt_event_code];
            // ... more packet types
        endcase
    end
endcase
```

20.5.2 Master Write Address Generation

```
// Address wraps within configured range
always_comb begin
    next_write_addr = current_write_addr + (DATA_WIDTH == 64 ? 8 : 4);
    if (next_write_addr > cfg_limit_addr) begin
        next_write_addr = cfg_base_addr;
    end
end
```

20.5.3 Master Write FSM

```
typedef enum logic [2:0] {
    WRITE_IDLE       = 3'b000,
    WRITE_ADDR       = 3'b001,
    WRITE_DATA_LOW   = 3'b010,
    WRITE_DATA_HIGH  = 3'b011,  // For 32-bit data width only
    WRITE_RESP       = 3'b100
} write_state_t;
```

20.6 Integration Example

```
monbus_axil_group #(
    .FIFO_DEPTH_ERR     (64),
    .FIFO_DEPTH_WRITE   (32),
    .ADDR_WIDTH         (32),
    .DATA_WIDTH         (32),
    .NUM_PROTOCOLS      (3)
) u_monbus_axil_group (
    .axi_aclk           (clk),
    .axi_aresetn        (rst_n),

    // Monitor bus input
    .monbus_valid       (stream_mon_valid),
    .monbus_ready       (stream_mon_ready),
    .monbus_packet      (stream_mon_packet),

    // AXI-Lite slave read (for error FIFO access)
    .s_axil_arvalid     (axil_slave_arvalid),
    .s_axil_arready     (axil_slave_arready),
    // ... more slave signals

    // AXI-Lite master write (for logging)
    .m_axil_awvalid     (axil_master_awvalid),
    .m_axil_awready     (axil_master_awready),
    // ... more master signals

    // Interrupt
    .irq_out            (monbus_interrupt),

    // Configuration
    .cfg_base_addr      (cfg_monbus_base_addr),
    .cfg_limit_addr     (cfg_monbus_limit_addr),
    .cfg_axi_pkt_mask   (cfg_axi_pkt_mask),
    // ... more protocol configuration
);
```

20.7 Common Issues

20.7.1 Issue 1: Missing Monitor Packets

Symptom: Expected events not appearing in FIFO

Root Causes: 1. Packet type masked in cfg_*pkt_mask 2. Event masked in cfg_*_mask 3. FIFO full (packets dropped)

Solution: Check mask configuration and FIFO status.

20.7.2 Issue 2: Master Writes Stall

Symptom: Write FIFO fills up, backpressure to monitor bus

Root Causes: 1. AXI-Lite master target not responding 2. Address range exhausted (wrapping too fast)

Solution: Increase address range or reduce logging rate via masks.

20.8 Related Documentation

- **Parent:** `01_stream_core.md` - Provides monitor bus packets
- **Packet Format:** Monitor Bus Protocol documentation

- **Configuration: `14_apb_config.md` - Register mapping**

20.9 Revision History

MonBus AXI-Lite Group Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers

Last Updated: 2026-01-02

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21 Chapter 3: External Interfaces

This chapter documents the external interfaces of the STREAM DMA engine (stream_top_ch8.sv).

21.1 Interface Specifications

STREAM has seven primary external interfaces:

21.1.1 01_axi4_interface_spec.md

- **AXI4 Master Interfaces (3 total)**
  - Descriptor fetch (256-bit, read-only)
  - Data read (parameterizable, default 512-bit)
  - Data write (parameterizable, default 512-bit)
- Protocol specifications and assumptions
- Transfer modes and alignment requirements
- Timing and performance characteristics

21.1.2 02_axil4_interface_spec.md

- **AXI4-Lite Interfaces (2 total, in monbus_axil_group)**
  - Error FIFO slave read (s_axil_err_*): 32-bit, read-only
  - Monitor data master write (m_axil_mon_*): 32-bit, write-only
- Used for monitor bus packet access and logging
- Part of monbus_axil_group integration (USE_AXI_MONITORS=1)

21.1.3 03_apb_interface_spec.md

- **APB4 Slave Interface**
- Full APB4 protocol with PSEL/PENABLE
- Optional CDC for asynchronous clock domains (CDC_ENABLE parameter)
- Address map: 0x000-0x03F kick-off, 0x100-0x3FF registers

21.1.4 05_monbus_interface_spec.md

- **MonBus Internal Protocol**
- Unified 64-bit monitoring bus (stream_core → monbus_axil_group)
- Event packet format and encoding
- Agent ID assignments for STREAM components
- Converted to AXI-Lite at top level

21.2 STREAM Interface Summary (stream_top_ch8.sv)

Interface	Type	Direction	Width	Purpose
APB4	Slave	Input	32-bit data, 12-bit addr	Configuration and kick-off
AXI4 Descriptor	Master Read	Output	256-bit	Descriptor fetch
AXI4 Data Read	Master Read	Output	512-bit (param)	Source data read
AXI4 Data Write	Master Write	Output	512-bit (param)	Destination data write
AXIL Error FIFO	Slave Read	Input	32-bit	Monitor error/interrupt FIFO
AXIL Monitor Write	Master Write	Output	32-bit	Monitor data to memory
IRQ	Output	Output	1-bit	Interrupt (error FIFO not empty)

Note: AXIL interfaces are active when USE_AXI_MONITORS=1. MonBus is internal (stream_core → monbus_axil_group).

Last Updated: 2025-12-01

Dependencies: - AMBA AXI4 Protocol Specification v4.0 - AMBA Monitor Bus Protocol (internal spec) - STREAM Architecture Overview

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22 AXI4 Interface Specification for STREAM

22.1 Overview

This document defines the AXI4 interface specification for the STREAM DMA engine. STREAM uses three AXI4 master interfaces for memory access:

1. **Descriptor Fetch Master** - 256-bit read-only interface for fetching descriptors
2. **Data Read Master** - Parameterizable width (default 512-bit) for reading source data
3. **Data Write Master** - Parameterizable width (default 512-bit) for writing destination data

Note: This document focuses on STREAM-specific implementation details. For generic AXI4 protocol information, refer to the AMBA AXI4 Protocol Specification v4.0.

22.2 STREAM AXI4 Interface Summary

22.2.1 Number of Interfaces

STREAM implements **3 AXI4 Master Interfaces**:

Number of Interfaces

Interface	Type	Width	Channels	Purpose
Descriptor Fetch	Master Read	256-bit (fixed)	AR, R	Fetch descriptors from memory
Data Read	Master Read	512-bit (param)	AR, R	Read source data to SRAM
Data Write	Master Write	512-bit (param)	AW, W, B	Write SRAM data to destination

22.2.2 Interface Parameters

Interface Parameters

Parameter	Description	Valid Values	Default
`DATA_WIDTH`	AXI data bus width in bits	32, 64, 128, 256, 512, 1024	32
`ADDR_WIDTH`	AXI address bus width in bits	32, 64	37
`ID_WIDTH`	AXI ID tag width in bits	1-16	8
`USER_WIDTH`	AXI user signal width in bits (optional)	0-16	1

22.2.3 Interface Types and Transfer Modes

Interface Types and Transfer Modes

Interface Group	Channels	Transfer Mode	Address Alignment	Monitor	DCG	Notes
AXI4 Master	AR, R	**Flexible**	4-byte	Yes	Yes	Data

Interface Group	Channels	Transfer Mode	Address Alignment	Monitor	DCG	Notes
Read-Split						interfaces with chunk enables
AXI4 Master Write-Split	AW, W, B	**Flexible**	4-byte	Yes	Yes	Data interfaces with chunk enables
AXI4 Master Read	AR, R	**Simplified**	Bus-width	No	Yes	Control interfaces, fully aligned
AXI4 Master Write	AW, W, B	**Simplified**	Bus-width	Yes	Yes	Control interfaces, full

Interface Group	Channels	Transfer Mode	Address Alignment	Monitor	DCG	Notes
						y aligned

22.2.4 Interface Group Parameter Settings

Interface Group Parameter Settings

Interface Group	Data Width	Address Width	ID Width	User Width	Transfer Mode
AXI4 Master Read-Split	512 bits	37 bits	8 bits	1 bit	**Flexible**
AXI4 Master Write-Split	512 bits	37 bits	8 bits	1 bit	**Flexible**
AXI4 Master Read	32 bits	37 bits	8 bits	1 bit	**Simplified**
AXI4 Master Write	32 bits	37 bits	8 bits	1 bit	**Simplified**

22.2.5 Interface Configuration Summary

Interface Configuration Summary

Interface Type	Interface Group	Transfer Mode	Alignment	Notes
Descriptor Sink	**AXI4 Master Read**	**Simplified**	32-bit aligned	Control interface
Descriptor Source	**AXI4 Master Read**	**Simplified**	32-bit aligned	Control interface
Data Source	**AXI4 Master Read-Split**	**Flexible**	4-byte aligned	High-bandwidth data
Data Sink	**AXI4 Master**	**Flexible**	4-byte	High-

Interface Type	Interface Group	Transfer Mode	Alignment	Notes
	Write-Split		aligned	bandwidth data
Program Sink	**AXI4 Master Write**	**Simplified**	32-bit aligned	Control interface
Program Source	**AXI4 Master Write**	**Simplified**	32-bit aligned	Control interface
Flag Sink	**AXI4 Master Read**	**Simplified**	32-bit aligned	Control interface
Flag Source	**AXI4 Master Read**	**Simplified**	32-bit aligned	Control interface

22.3 Transfer Mode Specifications

This specification defines two distinct transfer modes to optimize different interface types:

22.3.1 Mode 1: Simplified Transfer Mode (Control Interfaces)

Used for control interfaces (descriptors, programs, flags) that prioritize simplicity and predictable timing.

22.3.1.1 Simplified Mode Assumptions

Simplified Mode Assumptions

Aspect	Requirement
Address Alignment	All addresses aligned to full data bus width
Transfer Size	All transfers use maximum size equal to bus width
Burst Type	Incrementing bursts only (AxBURST = 2'b01)
Transfer Complexity	Maximum simplicity for predictable operation

22.3.2 Mode 2: Flexible Transfer Mode (Data Interfaces)

Used for high-bandwidth data interfaces that need to handle arbitrary address alignment while maintaining efficiency.

22.3.2.1 Flexible Mode Assumptions

Flexible Mode Assumptions

Aspect	Requirement
Address Alignment	4-byte aligned addresses (minimum alignment)
Transfer Sizes	Multiple sizes supported: 4, 8, 16, 32, 64 bytes
Burst Type	Incrementing bursts only (AxBURST = 2'b01)
Alignment Strategy	Progressive alignment to optimize bus utilization

22.4 Mode 1: Simplified Transfer Mode Specification

22.4.1 Assumption 1: Address Alignment to Data Bus Width

Assumption 1: Address Alignment to Data Bus Width

Aspect	Requirement
Alignment Rule	All AXI transactions aligned to data bus width
32-bit bus (4 bytes)	Address[1:0] must be 2'b00
64-bit bus (8 bytes)	Address[2:0] must be 3'b000
128-bit bus (16 bytes)	Address[3:0] must be 4'b0000
256-bit bus (32 bytes)	Address[4:0] must be 5'b00000
512-bit bus (64 bytes)	Address[5:0] must be 6'b000000
1024-bit bus (128 bytes)	Address[6:0] must be 7'b0000000
Rationale	Maximizes bus efficiency and eliminates unaligned access complexity

22.4.2 Assumption 2: Fixed Transfer Size

Assumption 2: Fixed Transfer Size

Aspect	Requirement
Transfer Size Rule	All transfers use maximum size equal to bus width
32-bit bus	AxSIZE = 3'b010 (4 bytes)
64-bit bus	AxSIZE = 3'b011 (8 bytes)
128-bit bus	AxSIZE = 3'b100 (16 bytes)
256-bit bus	AxSIZE = 3'b101 (32 bytes)
512-bit bus	AxSIZE = 3'b110 (64 bytes)
1024-bit bus	AxSIZE = 3'b111 (128 bytes)
Rationale	Maximizes bus utilization and simplifies address alignment

22.5 Mode 2: Flexible Transfer Mode Specification

22.5.1 Assumption 1: 4-Byte Address Alignment

Assumption 1: 4-Byte Address Alignment

Aspect	Requirement
Alignment Rule	All AXI transactions aligned to 4-byte boundaries
Address Constraint	Address[1:0] must be 2'b00
Rationale	Balances flexibility with AXI protocol requirements
Benefit	Supports arbitrary data placement while maintaining AXI compliance

22.5.2 Assumption 2: Multiple Transfer Sizes

Assumption 2: Multiple Transfer Sizes

Transfer Size	AxSIZE Value	Use Case
4 bytes	3'b010	Initial alignment, small transfers
8 bytes	3'b011	Progressive alignment
16 bytes	3'b100	Progressive alignment
32 bytes	3'b101	Progressive alignment
64 bytes	3'b110	Optimal full-width transfers
128 bytes	3'b111	Maximum efficiency (1024-bit bus)

22.5.3 Assumption 3: Progressive Alignment Strategy

Assumption 3: Progressive Alignment Strategy

Aspect	Requirement
Alignment Goal	Align to 64-byte boundaries for optimal bus utilization

Aspect	Requirement
Alignment Sequence	Use progressive sizes: 4 → 8 → 16 → 32 → 64 bytes
Optimization	Choose largest possible transfer size at each step
Example	Address 0x1004: 4-byte transfer → aligned to 0x1008, then larger transfers

22.5.4 Assumption 4: Chunk Enable Support

Assumption 4: Chunk Enable Support

Aspect	Requirement
Chunk Granularity	16 chunks of 32-bits each (512-bit bus)
Write Strobes	Generated from chunk enables for precise byte control
Alignment Transfers	Chunk patterns optimized for alignment sequences
Benefits	Precise data validity, optimal memory utilization

22.6 Common Protocol Assumptions (Both Modes)

22.6.1 Assumption 1: Incrementing Bursts Only

Assumption 1: Incrementing Bursts Only

Aspect	Requirement
Burst Type	All AXI bursts use incrementing address mode (AxBURST = 2'b01)
Excluded Types	No FIXED (2'b00) or WRAP (2'b10) bursts supported
Rationale	Simplifies address generation logic and covers most use cases
Benefit	Eliminates wrap boundary calculations and fixed address handling

22.6.2 Assumption 2: No Address Wraparound

Assumption 2: No Address Wraparound

Aspect	Requirement
Wraparound Rule	Transactions never wrap around top of address space
Example	No 0xFFFFFFFF → 0x00000000 transitions
Rationale	Real systems never allow this due to memory layout
Benefit	Dramatically simplified boundary crossing detection logic

22.7 Flexible Mode: Address Calculation Examples

22.7.1 Progressive Alignment Examples

Example 1: Address 0x1004 → 0x1040 (64-byte boundary)

Progressive Alignment Examples

Step	Address	Size	AxSIZE	Length	Bytes Transferred	Notes
1	0x1004	4 bytes	3’b010	1 beat	4	Initial alignment
2	0x1008	8 bytes	3’b011	1 beat	8	Progressive alignment
3	0x1010	16 bytes	3’b100	1 beat	16	Progressive alignment
4	0x1020	32 bytes	3’b101	1 beat	32	Progressive alignment
5	0x1040	**64 bytes**	3’b110	N beats	64×N	**Optimal transfers**

Example 2: Address 0x1010 → 0x1040 (64-byte boundary)

Progressive Alignment Examples

Step	Address	Size	AxSIZE	Length	Bytes Transferred	Notes
1	0x1010	16 bytes	3’b100	1 beat	16	Optimal initial size
2	0x1020	32 bytes	3’b101	1 beat	32	Progressive

Step	Address	Size	AxSIZE	Length	Bytes Transferred	Notes
						alignment
3	0x1040	**64 bytes**	3’b110	N beats	64×N	**Optimal transfers**

22.7.2 Chunk Enable Pattern Examples

512-bit Bus with 16×32-bit chunks

Chunk Enable Pattern Examples

Transfer Size	Address Offset	Chunk Pattern	Description
4 bytes	0x04	16’h0002	Chunk 1 only
8 bytes	0x08	16’h000C	Chunks 2-3
16 bytes	0x10	16’h00F0	Chunks 4-7
32 bytes	0x20	16’hFF00	Chunks 8-15
64 bytes	0x00	16’hFFFF	All chunks

22.8 Master Read Interface Specification

22.8.1 Read Address Channel (AR)

Read Address Channel

Signal	Width	Direction	Simplified Mode	Flexible Mode	Description
ar_addr	ADDR_WIDTH	Master→Slave	**Bus-width aligned**	**4-byte aligned**	Read address
ar_len	8	Master→Slave	0-255	0-255	Burst length - 1
ar_size	3	Master→Slave	**Fixed per bus**	**Variable: 4-64 bytes**	Transfer size
ar_burst	2	Master→Slave	**2'b01 (INCR only)**	**2'b01 (INCR only)**	Burst type
ar_id	ID_WIDTH	Master→Slave	Any	Any	Transaction ID
ar_lock	1	Master→Slave	1'b0	1'b0	Lock type (normal)
ar_cache	4	Master→Slave	Implementation specific	4'b0011	Cache attributes
ar_prot	3	Master→Slave	Implementation specific	3'b000	Protection attributes
ar_qos	4	Master→Slave	4'b0000	4'b0000	Quality of Service
ar_region	4	Master→Slave	4'b0000	4'b0000	Region identifier
ar_user	USER_WIDTH	Master→Slave	Optional	Optional	User-defined
ar_valid	1	Master→Slave	0 or 1	0 or 1	Address valid
ar_ready	1	Slave→Master	0 or 1	0 or 1	Address ready

22.8.2 Read Data Channel (R)

Read Data Channel

Signal	Width	Direction	Description
r_data	DATA_WIDTH	Slave→Master	Read data
r_id	ID_WIDTH	Slave→Master	Transaction ID
r_resp	2	Slave→Master	Read response
r_last	1	Slave→Master	Last transfer in burst
r_user	USER_WIDTH	Slave→Master	User-defined (optional)
r_valid	1	Slave→Master	Read data valid
r_ready	1	Master→Slave	Read data ready

22.9 Master Write Interface Specification

22.9.1 Write Address Channel (AW)

Write Address Channel

Signal	Width	Direction	Simplified Mode	Flexible Mode	Description
aw_addr	ADDR_WIDTH	Master→Slave	**Bus-width aligned**	**4-byte aligned**	Write address
aw_len	8	Master→Slave	0-255	0-255	Burst length - 1
aw_size	3	Master→Slave	**Fixed per bus**	**Variable: 4-64 bytes**	Transfer size
aw_burst	2	Master→Slave	**2'b01 (INCR only)**	**2'b01 (INCR only)**	Burst type
aw_id	ID_WIDTH	Master→Slave	Any	Any	Transaction ID
aw_lock	1	Master→Slave	1'b0	1'b0	Lock type (normal)
aw_cache	4	Master→Slave	Implementation specific	4'b0011	Cache attributes
aw_prot	3	Master→Slave	Implementation specific	3'b000	Protection attributes
aw_qos	4	Master→Slave	4'b0000	4'b0000	Quality of Service
aw_region	4	Master→Slave	4'b0000	4'b0000	Region identifier
aw_user	USER_WIDTH	Master→Slave	Optional	Optional	User-defined
aw_valid	1	Master→Slave	0 or 1	0 or 1	Address valid
aw_ready	1	Slave→Master	0 or 1	0 or 1	Address ready

22.9.2 Write Data Channel (W)

Write Data Channel

Signal	Width	Direction	Simplified Mode	Flexible Mode	Description
w_data	DATA_WIDTH	Master→Slave	Write data	Write data	Write data
w_strb	DATA_WIDTH/8	Master→Slave	**All 1's**	**From chunk enables**	Write strobes
w_last	1	Master→Slave	Last transfer	Last transfer	Last transfer in burst
w_user	USER_WIDTH	Master→Slave	Optional	Optional	User-defined
w_valid	1	Master→Slave	0 or 1	0 or 1	Write data valid
w_ready	1	Slave→Master	0 or 1	0 or 1	Write data ready

22.9.3 Write Response Channel (B)

Write Response Channel

Signal	Width	Direction	Description
b_id	ID_WIDTH	Slave→Master	Transaction ID
b_resp	2	Slave→Master	Write response
b_user	USER_WIDTH	Slave→Master	User-defined (optional)
b_valid	1	Slave→Master	Response valid
b_ready	1	Master→Slave	Response ready

22.10 Address Calculation Rules

22.10.1 Simplified Mode Address Generation

Simplified Mode Address Generation

Parameter	Formula	Description
First Address	Must be bus-width aligned	Starting address
Address N	First_Address + (N × Bus_Width_Bytes)	Address for beat N
Alignment Check	(Address % Bus_Width_Bytes) == 0	Must always be true

22.10.2 Flexible Mode Address Generation

Flexible Mode Address Generation

Parameter	Formula	Description
First Address	Must be 4-byte aligned	Starting address
Address N	First_Address + (N × Transfer_Size)	Address for beat N
Alignment Check	(Address % 4) == 0	Must always be true
Progressive Alignment	Choose largest size ≤ bytes_to_boundary	Optimization strategy

22.10.3 4KB Boundary Considerations (Both Modes)

4KB Boundary Considerations

Validation Rule	Formula	Description
4KB Boundary	Bursts cannot cross 4KB (0x1000) boundaries	AXI specification
Max Burst Calculation	Max_Beats = (4KB - (Start_Address %	Burst limit

Validation Rule	Formula	Description
	4KB)) / Transfer_Size	
Boundary Check	Verify no 4KB crossings in burst	Mandatory validation

22.11 Write Strobe Generation

22.11.1 Simplified Mode Strobe Generation

Simplified Mode Strobe Generation

Bus Width	Strobe Pattern	Description
32-bit	4'b1111	All bytes valid
512-bit	64'hFFFFFFFFFFFFFFFF	All bytes valid

22.11.2 Flexible Mode Strobe Generation

From Chunk Enables (512-bit bus example):

```
// Convert 16×32-bit chunk enables to 64×8-bit write strobes
for (int chunk = 0; chunk < 16; chunk++) begin
    if (chunk_enable[chunk]) begin
        w_strb[chunk*4 +: 4] = 4'hF; // 4 bytes per chunk
    end
end
```

Alignment Transfer Examples:

Flexible Mode Strobe Generation

Transfer Size	Chunk Pattern	Strobe Pattern	Description
4 bytes	16'h0001	64'h000000000000000F	First 4 bytes
16 bytes	16'h000F	64'h00000000000000FF	First 16 bytes
32 bytes	16'h00FF	64'h0000000000FFFFFF	First 32 bytes
64 bytes	16'hFFFF	64'hFFFFFFFFFFFFFFFF	All 64 bytes

22.12 Response Codes

22.12.1 Response Code Specification

Response Code Specification

Value	Name	Description	Simplified Mode Usage	Flexible Mode Usage
2'b00	OKAY	Normal access success	Bus-width aligned access	4-byte aligned access
2'b01	EXOKAY	Exclusive access success	Bus-width aligned exclusive	4-byte aligned exclusive
2'b10	SLVERR	Slave error	Slave-specific error	Slave-specific error
2'b11	DECERR	Decode error	**Bus-width misalignment**	**4-byte misalignment**

22.13 Implementation Benefits

22.13.1 Simplified Mode Benefits

Simplified Mode Benefits

Benefit Area	Simplification	Impact
Address Generation	Simple increment by bus width	Minimal logic complexity
Size Checking	No dynamic size validation	No validation logic needed
Strobe Generation	All strobes always high	Trivial implementation
Timing	Predictable single-size transfers	Optimal timing closure

22.13.2 Flexible Mode Benefits

Flexible Mode Benefits

Benefit Area	Capability	Impact
Data Placement	Arbitrary 4-byte aligned placement	Maximum flexibility
Bus Utilization	Progressive alignment optimization	High efficiency achieved
Chunk Control	Precise byte-level validity	Optimal memory utilization
Alignment Strategy	Automatic alignment to boundaries	Performance optimization

22.13.3 Mode Selection Guidelines

Mode Selection Guidelines

Interface Type	Recommended Mode	Rationale
High-bandwidth data	**Flexible**	Maximize throughput, handle arbitrary alignment
Control/status	**Simplified**	Predictable

Interface Type	Recommended Mode	Rationale
		timing, minimal complexity
Descriptors	**Simplified**	Fixed-size structures, simple implementation
Programs	**Simplified**	Single-word writes, minimal overhead
Flags	**Simplified**	Fixed-size status, predictable behavior

22.14 Validation Requirements

22.14.1 Simplified Mode Validation

Simplified Mode Validation

Validation Area	Requirements
Address Alignment	Verify all addresses aligned to full bus width
Fixed Size	Verify AxSIZE always matches DATA_WIDTH
Full Strobes	Verify w_strb is always all 1’s
Burst Type	Verify AxBURST is always 2’b01

22.14.2 Flexible Mode Validation

Flexible Mode Validation

Validation Area	Requirements
Address Alignment	Verify all addresses are 4-byte aligned
Size Validation	Verify AxSIZE matches actual transfer size
Chunk Consistency	Verify chunk enables match transfer size
Strobe Generation	Verify strobes generated correctly from chunks
Progressive Alignment	Verify alignment strategy optimization
Boundary Checking	Verify no 4KB boundary crossings

22.14.3 Common Validation

Common Validation

Validation Area	Requirements
No Wraparound	Verify addresses never wrap around
Incrementing Only	Verify AxBURST is always 2’b01
Response Handling	Verify proper response

Validation Area	Requirements
	generation
Error Conditions	Verify alignment violation responses

22.15 Performance Characteristics

22.15.1 Simplified Mode Performance

Simplified Mode Performance

Metric	Typical Value	Description
Latency	3 cycles	Address + Data + Response
Throughput	1 transfer per clock	Sustained rate
Efficiency	100%	Perfect bus utilization
Complexity	Minimal	Simple implementation

22.15.2 Flexible Mode Performance

Flexible Mode Performance

Metric	Alignment Phase	Optimized Phase	Description
Latency	3-15 cycles	3 cycles	Variable based on alignment
Throughput	Variable	1 transfer per clock	Depends on alignment pattern
Efficiency	25-100%	100%	Improves with alignment
Complexity	Moderate	Minimal	Progressive optimization

22.15.3 Performance Optimization Strategy

Flexible Mode Alignment Strategy: 1. **Initial Phase**: Use largest possible transfer size for current alignment 2. **Progressive Phase**: Incrementally align to larger boundaries
3. **Optimized Phase**: Use full bus-width transfers once aligned 4. **Result**: Achieve maximum efficiency while handling arbitrary starting addresses

22.16 This dual-mode approach provides the best of both worlds: simplified, predictable operation for control interfaces and flexible, high-performance operation for data interfaces.

22.17 Revision History

AXI4 Interface Specification for STREAM Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers

Last Updated: 2026-01-02

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23 AXI4-Lite Interface Specification and Assumptions

23.1 Overview

This document defines the formal specification and assumptions for an AXI4-Lite interface implementation. AXI4-Lite is a subset of AXI4 optimized for simple, lightweight control register interfaces with inherent protocol simplifications.

23.2 Interface Summary

23.2.1 Number of Interfaces

- **2 Master Read Interface**: Single read channel for Monitor Packets (one for each Source and Sink)
- **2 Master Write Interface**: Single write channel for Monitor Packets plus a timestamp (one for each Source and Sink)

23.2.2 Interface Parameters

Interface Parameters

Parameter	Description	Valid Values	Default
`DATA_WIDTH`	AXI data bus width in bits	32, 64	32, 64
`ADDR_WIDTH`	AXI address bus width in bits	32, 64	37
`STRB_WIDTH`	Write strobe width	`DATA_WIDTH/8`	8

23.3 Core Protocol Assumptions

23.3.1 Inherent AXI4-Lite Simplifications

AXI4-Lite protocol inherently provides the following constraints:

Inherent AXI4-Lite Simplifications

Constraint	Description
Single Transfers Only	No burst transactions supported
No Transaction IDs	All transactions are in-order
Fixed Transfer Size	Always uses full data bus width
No User Signals	Simplified interface without user-defined extensions

23.3.2 Implementation Assumptions

23.3.2.1 Assumption 1: Address Alignment to Data Bus Width

Assumption 1: Address Alignment to Data Bus Width

Aspect	Requirement
Alignment Rule	All AXI4-Lite transactions aligned to data bus width
32-bit bus alignment	Address[1:0] must be 2'b00 (4-byte aligned)
64-bit bus alignment	Address[2:0] must be 3'b000 (8-byte aligned)
Rationale	Maximizes bus efficiency and eliminates unaligned access complexity
Benefit	Simplifies address decode and data steering logic

23.3.2.2 Assumption 2: Fixed Transfer Size

Assumption 2: Fixed Transfer Size

Aspect	Requirement
Transfer Size Rule	All transfers use maximum size equal to bus width
32-bit bus	AxSIZE = 3'b010 (4 bytes)
64-bit bus	AxSIZE = 3'b011 (8 bytes)

Aspect	Requirement
Rationale	Maximizes bus utilization and simplifies control logic
Benefit	No size decode logic required

23.3.2.3 Assumption 3: No Address Wraparound

Assumption 3: No Address Wraparound

Aspect	Requirement
Wraparound Rule	Transactions never wrap around top of address space
Rationale	Control register accesses never require wraparound behavior
Benefit	Simplified address boundary checking

23.3.2.4 Assumption 4: Standard Protection Attributes

Assumption 4: Standard Protection Attributes

Access Type	AxPROT Value	Description
Normal Access	3'b000	Data, secure, unprivileged
Privileged Access	3'b001	Data, secure, privileged
Rationale		Covers the majority of control register access patterns

23.4 Master Read Interface Specification

23.4.1 Read Address Channel (AR)

Read Address Channel

Signal	Width	Direction	Required Values	Description
ar_addr	ADDR_WIDTH	Master→Slave	**8-byte aligned**	Read address
ar_prot	3	Master→Slave	Implementation specific	Protection attributes
ar_valid	1	Master→Slave	0 or 1	Address valid
ar_ready	1	Slave→Master	0 or 1	Address ready

23.4.2 Read Data Channel (R)

Read Data Channel

Signal	Width	Direction	Description
r_data	64	Slave→Master	Read data
r_resp	2	Slave→Master	Read response
r_valid	1	Slave→Master	Read data valid
r_ready	1	Master→Slave	Read data ready

23.4.3 AXI4-Lite Simplifications (Read)

AXI4-Lite Simplifications

Removed Signal	AXI4 Usage	AXI4-Lite Reason
ar_id	Transaction ID	Single transfers, no transaction IDs
ar_len	Burst length	Single transfers only
ar_size	Transfer size	Fixed to bus width
ar_burst	Burst type	Single transfers only
ar_lock	Lock type	Simplified access model
ar_cache	Cache attributes	Simplified memory

Removed Signal	AXI4 Usage	AXI4-Lite Reason
		model
ar_qos	Quality of Service	Simplified priority model
ar_region	Region identifier	Simplified address space
ar_user	User-defined	Simplified interface
r_id	Transaction ID	No transaction IDs needed
r_last	Last transfer	Single transfers only
r_user	User-defined	Simplified interface

23.5 Master Write Interface Specification

23.5.1 Write Address Channel (AW)

Write Address Channel

Signal	Width	Direction	Required Values	Description
aw_addr	ADDR_WIDTH	Master→Slave	**8-byte aligned**	Write address
aw_prot	3	Master→Slave	Implementation specific	Protection attributes
aw_valid	1	Master→Slave	0 or 1	Address valid
aw_ready	1	Slave→Master	0 or 1	Address ready

23.5.2 Write Data Channel (W)

Write Data Channel

Signal	Width	Direction	Description
w_data	32	Master→Slave	Write data
w_strb	4	Master→Slave	Write strobes (byte enables)
w_valid	1	Master→Slave	Write data valid
w_ready	1	Slave→Master	Write data ready

23.5.3 Write Response Channel (B)

Write Response Channel

Signal	Width	Direction	Description
b_resp	2	Slave→Master	Write response
b_valid	1	Slave→Master	Response valid
b_ready	1	Master→Slave	Response ready

23.5.4 AXI4-Lite Simplifications (Write)

AXI4-Lite Simplifications

Removed Signal	AXI4 Usage	AXI4-Lite Reason
aw_id	Transaction ID	Single transfers, no transaction IDs
aw_len	Burst length	Single transfers only
aw_size	Transfer size	Fixed to bus width
aw_burst	Burst type	Single transfers only
aw_lock	Lock type	Simplified access model
aw_cache	Cache attributes	Simplified memory model
aw_qos	Quality of Service	Simplified priority model
aw_region	Region identifier	Simplified address space
aw_user	User-defined	Simplified interface
w_last	Last transfer	Single transfers only
w_user	User-defined	Simplified interface
b_id	Transaction ID	No transaction IDs needed
b_user	User-defined	Simplified interface

23.6 Address Requirements

23.6.1 Address Alignment Rules

Address Alignment Rules

Alignment Type	Formula	Description
Valid Address	(Address % 4) == 0	Must be 8-byte aligned
Mandatory Alignment	Address[2:0] must be 3'b000	Per Assumption 1

23.6.2 Address Validation Examples

Address Validation Examples

Address Category	Examples	Status
Valid (84byte aligned)	0x1000, 0x1004, 0x1008, 0x100C	Accepted
Invalid (unaligned)	0x1001, 0x1002, 0x1003	DECERR response

23.7 Response Codes

23.7.1 Response Code Specification

Response Code Specification

Value	Name	Description	Usage in Control Registers
2'b00	OKAY	Normal access success	Successful register access
2'b01	EXOKAY	Exclusive access success	**Not used in AXI4-Lite**
2'b10	SLVERR	Slave error	Invalid register access
2'b11	DECERR	Decode error	**Address decode failure or misalignment**

23.7.2 Response Usage Guidelines

Response Usage Guidelines

Response Type	Usage	Description
OKAY	Normal completion	Successful register access
EXOKAY	Not applicable	AXI4-Lite doesn't support exclusive accesses
SLVERR	Register error	Invalid register operation
DECERR	Address error	Misalignment or decode failure per Assumption 1

23.8 Protection Signal Usage

23.8.1 Protection Signal Encoding

Protection Signal Encoding

Bit	Name	Description	Recommended Usage
[0]	Privileged	0=Normal, 1=Privileged	Set based on processor mode
[1]	Non-secure	0=Secure, 1=Non-secure	Set based on security domain
[2]	Instruction	0=Data, 1=Instruction	Always 0 for control registers

23.8.2 Common Protection Patterns

Common Protection Patterns

Pattern	AxPROT Value	Description
Normal Data Access	3'b000	Standard register access
Privileged Data Access	3'b001	Privileged register access
Debug Access	3'b010	Debug register access
Privileged Debug	3'b011	Privileged debug access

23.9 Implementation Benefits

23.9.1 Simplified Control Register Interface

Simplified Control Register Interface

Benefit Area	Simplification	Impact
Address Decode	Simple 8-byte aligned address comparison	Reduced decode logic
Transaction Handling	No burst or ID tracking required	Simplified state machines
Flow Control	Straightforward valid-ready handshakes	Reduced complexity
Response Generation	Simple OKAY/SLVERR/DECERR responses	Minimal response logic
Size Handling	Fixed 64-bit transfers only	No size decode needed

23.9.2 Address Decode Implementation

Address Decode Implementation

Implementation Aspect	Method	Benefit
4-byte Alignment Check	addr[1:0] == 2'b00	Simple bit masking
Address Range Check	addr >= base && addr <= limit	Simple comparisons
Combined Check	alignment_ok && range_ok	Single decode decision

23.9.3 Error Generation Logic

Error Generation Logic

Error Condition	Check	Response
Address Misalignment	addr[1:0] != 2'b00	Generate DECERR

Error Condition	Check	Response
Address Out of Range	! addr_in_range(addr)	Generate DECERR
Register Error	register_error_condition	Generate SLVERR
Normal Access	All checks pass	Generate OKAY

23.10 Timing Requirements

23.10.1 Handshake Protocol

Handshake Protocol

Protocol Rule	Requirement	Description
Valid-Ready Transfer	Transfer occurs when both VALID and READY are high	Standard AXI handshake
Valid Independence	VALID can be asserted independently of READY	Master controls valid
Ready Dependency	READY can depend on VALID state	Slave controls ready
Signal Stability	Once VALID asserted, all signals stable until READY	Data integrity

23.10.2 Channel Dependencies

Channel Dependencies

Dependency	Requirement	Description
Write Channels	AW and W channels are independent	Can be presented in any order
Write Response	B channel waits for both AW and W completion	Response dependency
Read Channels	R channel waits for AR channel completion	Response dependency
Transaction Ordering	Multiple outstanding transactions not supported	Inherent AXI4-Lite limitation

23.10.3 Reset Behavior

Reset Behavior

Reset Phase	Requirement	Description
Active Reset	aresetn is active-low	Standard AXI reset

Reset Phase	Requirement	Description
	reset signal	
Reset Requirements	All VALID signals deasserted during reset	Clean reset state
Reset Recovery	All VALID signals low after reset deassertion	Proper startup

23.11 Validation Requirements

23.11.1 Functional Validation

Functional Validation

Validation Area	Requirements
Address Alignment	Verify all accesses are 8-byte aligned per Assumption 1
Fixed Size	Verify all transfers are full 64-bit width per Assumption 2
Response Correctness	Verify appropriate response codes (DECERR for misaligned access)
Handshake Compliance	Verify all valid-ready handshakes
Register Behavior	Verify read/write register functionality
No Wraparound	Verify no address wraparound scenarios per Assumption 3

23.11.2 Timing Validation

Timing Validation

Validation Area	Requirements
Setup/Hold	Verify signal timing requirements
Reset Behavior	Verify proper reset sequence
Back-pressure	Verify ready signal behavior under load

23.11.3 Error Injection Testing

Error Injection Testing

Test Type	Injection Method	Expected Response
Misaligned Address	Inject addresses with addr[2:0] != 0	DECERR response
Out of Range	Inject addresses outside valid range	DECERR response

Test Type	Injection Method	Expected Response
Register Errors	Inject register-specific errors	SLVERR response

23.12 Example Transactions

23.12.1 64-bit Register Write

64-bit Register Write

Parameter	Value	Description
Bus Width	64 bits (8 bytes)	Data bus configuration
Target Address	0x1000 (8-byte aligned)	Valid aligned address
Write Data	0xDEADBEEFCAFEBABE	64-bit data value
Required Settings	aw_addr=0x1000, aw_prot=3'b000, w_data=0xDEADBEEFCAFEBABE, w_strb=8'b11111111	Transaction configuration

23.12.2 AW Transaction Flow

AW Transaction Flow

Step	Action	Signal States
1	Assert aw_valid with address	aw_valid=1, aw_addr=0x1000
2	Assert w_valid with data	w_valid=1, w_data=0xDEADBEEFCAFEBABE
3	Wait for handshakes	aw_ready=1, w_ready=1
4	Wait for response	b_valid=1, b_resp=OKAY
5	Complete transaction	b_ready=1

23.12.3 64-bit Register Read

64-bit Register Read

Parameter	Value	Description
Bus Width	64 bits (8 bytes)	Data bus configuration
Target Address	0x1008 (8-byte aligned)	Valid aligned address
Required Settings	ar_addr=0x1008, ar_prot=3'b000	Transaction configuration

23.12.4 AR Transaction Flow

AR Transaction Flow

Step	Action	Signal States
1	Assert ar_valid with address	ar_valid=1, ar_addr=0x1008
2	Wait for address handshake	ar_ready=1
3	Wait for data response	r_valid=1, r_resp=OKAY
4	Complete transaction	r_ready=1
5	Capture data	r_data (64 bits)

23.12.5 Misaligned Address Example

Misaligned Address Example

Parameter	Value	Description
Bus Width	64 bits (8 bytes)	Data bus configuration
Target Address	0x1004 (misaligned)	Invalid address
Expected Behavior	Address decode detects misalignment → DECERR response → No register access	Error handling

23.13 Common Use Cases

23.13.1 Typical Applications

Typical Applications

Application	Description
Control/Status Registers	64-bit device configuration and monitoring
Memory-Mapped Peripherals	Simple register-based devices
Debug Interfaces	Debug and trace control registers
Configuration Space	PCIe configuration space access
Performance Counters	64-bit performance monitoring registers

23.13.2 Performance Considerations

Performance Considerations —

Consideration	Impact	Description
Latency	Single-cycle responses preferred	Simple registers
Throughput	Limited by single outstanding transaction	AXI4-Lite constraint
Efficiency	64-bit transfers maximize data efficiency	Modern system optimization

23.14 Revision History

AXI4-Lite Interface Specification and Assumptions Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers

Last Updated: 2026-01-02

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24 APB Programming Interface Specification for STREAM

24.1 Overview

This document defines the APB programming interface for the STREAM DMA engine.

Important: STREAM uses a **simplified programming interface** for descriptor kick-off, NOT a full APB slave. The interface uses standard valid/ready handshaking per channel to initiate descriptor-based transfers.

This interface is separate from the optional APB configuration interface that may be added in future implementations for runtime configuration of STREAM parameters.

24.2 Interface Summary

24.2.1 STREAM Programming Interface

STREAM provides a **simplified per-channel descriptor programming interface**:

STREAM Programming Interface

Signal Group	Per-Channel Signals	Protocol	Purpose
Programming	`apb_valid[ch]`, `apb_ready[ch]`, `apb_addr[ch]`	Valid/Ready handshake	Descriptor address input

Note: Despite the "apb_" prefix, this is NOT a full AMBA APB interface. It's a simplified programming interface using APB naming convention for historical reasons.

24.2.2 Interface Parameters

Interface Parameters

Parameter	Description	Valid Values	Default
`DATA_WIDTH`	APB data bus width in bits	8, 16, 32	32
`ADDR_WIDTH`	APB address bus width in bits	16, 24, 32	32
`STRB_WIDTH`	Write strobe width (APB4 only)	`DATA_WIDTH/8`	4
`NUM_SLAVES`	Number of peripheral slaves	1-32	1
`DEPTH`	Internal buffer depth	2+	2

24.3 Core Protocol Assumptions

24.3.1 Inherent APB Simplifications

APB protocol inherently provides the following constraints:

1. **Single Master Only**: Only one bus master supported
2. **Non-Pipelined**: One transaction completes before next begins
3. **Simple 2-Phase Protocol**: Setup phase followed by access phase
4. **No Burst Transfers**: Only single transfers supported
5. **In-Order Completion**: No out-of-order transaction capability

24.3.2 Implementation Assumptions

24.3.2.1 Assumption 1: Word-Aligned Access Only

Assumption 1: Word-Aligned Access Only

Aspect	Requirement
Alignment Rule	All APB transfers are aligned to natural word boundaries
8-bit transfers	Byte aligned (no restriction)
16-bit transfers	2-byte aligned (PADDR[0] = 0)
32-bit transfers	4-byte aligned (PADDR[1:0] = 2'b00)
Rationale	Simplifies peripheral decode logic and ensures efficient access

24.3.2.2 Assumption 2: Standard Transfer Sizes

Assumption 2: Standard Transfer Sizes

Bus Width	Supported Transfer Sizes	Rationale
32-bit bus	8-bit, 16-bit, 32-bit	Covers typical register access patterns
16-bit bus	8-bit, 16-bit	Keeps decode logic simple
8-bit bus	8-bit only	Minimal complexity

24.3.2.3 Assumption 3: Single-Cycle Default Operation

Assumption 3: Single-Cycle Default Operation

Aspect	Requirement
Default Behavior	Most peripherals respond in a single cycle (PREADY tied high)
Optimization	PREADY optimization for simple registers
Exception	Complex peripherals may use PREADY for wait states
Rationale	Minimizes latency for control register access

24.4 Interface Signal Specification

24.4.1 Clock and Reset Signals

Clock and Reset Signals ### Address and Control Signals

Signal	IO	Description
src_clk	I	Source APB clock signal
snk_rdata[31:0]	I	Read data
src_wdata[31:0]	O	Write data

Address and Control Signals

Signal	Width	Direction	Required Values	Description
src_addr	13 bits	Master→Slave	**Aligned per transfer size**	Peripheral address (8KB space)
src_sel	1	Master→Slave	0 or 1	Peripheral select
src_enable	1	Master→Slave	0 or 1	Enable signal
src_write	1	Master→Slave	0 or 1	Write enable (1=write, 0=read)
snk_addr	13 bits	Master→Slave	**Aligned per transfer size**	Peripheral address (8KB space)
snk_sel	1	Master→Slave	0 or 1	Peripheral select
snk_enable	1	Master→Slave	0 or 1	Enable signal
snk_write	1	Master→Slave	0 or 1	Write enable (1=write, 0=read)

24.4.2 Data Signals

Data Signals

Signal	Width	Direction	Description
src_wdata	32	Master→Slave	Write data
src_rdata	32	Slave→Master	Read data
snk_wdata	32	Master→Slave	Write data
snk_rdata	32	Slave→Master	Read data

24.4.3 Response Signals

Response Signals

Signal	Width	Direction	Description
src_ready	1	Slave→Master	Transfer ready
src_slverr	1	Slave→Master	Slave error
snk_ready	1	Slave→Master	Transfer ready
snk_slverr	1	Slave→Master	Slave error

24.5 Address Space Configuration

24.5.1 Address Range Specification

Address Range Specification

Parameter	Value	Description
Address Width	13 bits (src_addr[12:0])	Full address space
Address Space	8KB (8192 bytes)	Total addressable space
Register Alignment	32-bit aligned (4-byte boundaries)	Address constraints
Usable Addresses	0x000 to 0x1FFF	Valid address range

Note: The address decode is **13 bits (8 KB)** so the STREAM monitor register block — instantiated in a separate `stream_mon_regs` regfile at **0x1000** on the same APB slave — is addressable. An 11-bit / 2 KB decode would not reach the monitor registers. See `stream_mas/ch04_registers/register_map.md`.

24.5.2 Address Decode Implementation

Address Decode Implementation

Register	Address Bits	Address Value	Description
reg0	src_addr[12:2] == 11'h000	0x000	First register
reg1	src_addr[12:2] == 11'h001	0x004	Second register
reg2	src_addr[12:2] == 11'h002	0x008	Third register
...	...	...	Up to 2048 32-bit registers

24.6 Transaction Protocol

24.6.1 2-Phase Transaction States

2-Phase Transaction States

Phase	State	Duration	Signal Requirements
Setup (T1)	SETUP	1 clock cycle	src_sel=1, src_enable=0, address/data stable
Access (T2+)	ACCESS	1+ clock cycles	src_sel=1, src_enable=1, wait for src_ready
Idle	IDLE	Variable	src_sel=0, src_enable=0

24.6.2 State Transitions

State Transitions

Current State	Next State	Condition	Description
IDLE	SETUP	Transaction start	Master drives address and control
SETUP	ACCESS	Clock edge	Master asserts src_enable
ACCESS	IDLE	src_ready=1	Transaction completes
ACCESS	ACCESS	src_ready=0	Wait state continues

24.6.3 Transaction Timing Requirements

Transaction Timing Requirements

Timing Parameter	Requirement	Description
Setup Time	All master signals stable before rising src_clk	Signal stability
Hold Time	All master signals held after rising src_clk	Signal stability
Output Delay	Slave outputs valid	Response timing

Timing Parameter	Requirement	Description
	after rising src_clk	

24.7 Advanced Features

24.7.1 Dynamic Clock Gating

Dynamic Clock Gating

Feature	Description	Benefit
Automatic Power Reduction	Clocks gated when no activity detected	Significant power savings
Configurable Idle Threshold	Programmable idle count before gating	Tunable power/performance
Multi-Domain Support	Independent gating for APB (PCLK) and backend (ACLK)	Flexible power management
Graceful Handoff	Ready signals forced to zero during gating	Protocol compliance
Activity Detection	Monitors valid signals across interfaces	Intelligent gating control

24.7.2 Clock Domain Crossing (CDC) Handshaking

Clock Domain Crossing (CDC) Handshaking

Feature	Description	Benefit
Arbitrary Frequency Support	Works across any clock frequency ratios	Design flexibility
Robust Reliability	Proven handshaking with proper synchronization	Zero data loss
Command/Response Separation	Independent CDC paths	Maximum throughput
Skid Buffer Integration	Internal buffering prevents backpressure	Smooth operation
Deterministic Latency	Predictable timing characteristics	System predictability

24.7.3 Buffering and Flow Control

Buffering and Flow Control

Feature	Description	Benefit
Internal Skid Buffers	Configurable depth buffering	Improved performance
Backpressure Handling	Proper ready/valid handshaking	Flow control
Command Pipelining	Backend processes while APB handles responses	Efficiency
Response Queuing	Responses buffered for varying latencies	Latency tolerance

24.8 Error Handling

24.8.1 Error Response Specification

Error Response Specification

Condition	src_slverr	Description
Successful access	0	Normal completion
Address decode error	1	Invalid address
Protection violation	1	Insufficient privilege
Backend error	1	Downstream error condition

24.8.2 Error Response Timing

Error Response Timing

Timing Requirement	Description
src_slverr Validity	Must be valid when src_ready is asserted
Error Completion	Error response completes transaction normally
Master Responsibility	Master must check src_slverr status

24.9 Reset Behavior

24.9.1 Reset Requirements

Reset Requirements

Reset Phase	Signal Requirements	Description
Reset Assertion	All outputs to known states	Deterministic reset
Reset Deassertion	src_sel=0, src_enable=0 in first cycle	Clean startup
Reset Values	src_ready=0, src_slverr=0, src_rdata=0	Default states

24.10 Implementation Variants

24.10.1 Available Implementations

Available Implementations

Variant	Features	Use Case
Basic APB Slave	Single clock domain, internal buffering	Simple peripherals
CDC-Enabled APB Slave	Dual clock domain (PCLK/ACLK)	Mixed-frequency systems
Power-Optimized APB Slave	Dynamic clock gating	Power-sensitive designs

24.11 Performance Characteristics

24.11.1 Latency Specifications

Latency Specifications

Metric	Value	Description
Minimum Read Latency	2 clock cycles	Setup + access phases
Minimum Write Latency	2 clock cycles	Setup + access phases
CDC Additional Latency	2-6 clock cycles	Depends on clock relationships
Buffer Latency	Minimal	Skid buffer design

24.11.2 Throughput Specifications

Throughput Specifications

Metric	Value	Description
Single Transaction	2 clocks minimum	Basic transaction time
Back-to-back Transactions	Limited by src_ready	Depends on peripheral
CDC Throughput	Maintained across domains	Proper buffering

24.11.3 Power Consumption

Power Consumption

Power Category	Characteristics	Description
Active Power	Standard CMOS logic	Normal operation
Idle Power	Significantly reduced with clock gating	Power optimization
Gating Overhead	Minimal additional logic	Efficient implementation

24.12 Validation Requirements

24.12.1 Functional Validation

Functional Validation

Validation Area	Requirements
Protocol Compliance	Verify 2-phase setup/access sequence
Address Decode	Verify 13-bit address handling and alignment
Error Response	Verify src_slverr generation and handling
Wait States	Verify src_ready functionality
CDC Operation	Verify cross-clock domain transfers
Clock Gating	Verify power management behavior
Buffer Operation	Verify skid buffer and flow control

24.12.2 Timing Validation

Timing Validation

Validation Area	Requirements
Setup/Hold	Verify signal timing requirements
Reset Sequence	Verify proper reset behavior
Multi-Clock	Verify CDC timing across frequency ranges
Gating Transitions	Verify clock enable/disable timing

24.13 Example Transactions

24.13.1 32-bit Register Write

32-bit Register Write

Clock Cycle	src_sel	src_enable	src_write	src_addr	src_wdata	src_ready	Phase
1	1	0	1	0x100	0xABCD	0	Setup
2	1	1	1	0x100	0xABCD	1	Access
3	0	0	-	-	-	0	Idle

24.13.2 32-bit Register Read with Wait State

32-bit Register Read with Wait State —

Clock Cycle	src_sel	src_enable	src_write	src_addr	src_rdata	src_ready	Phase
1	1	0	0	0x104	-	0	Setup
2	1	1	0	0x104	-	0	Access
3	1	1	0	0x104	0x1234	1	Complete
4	0	0	-	-	-	0	Idle

24.14 Revision History

APB Programming Interface Specification for STREAM Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers
0.93	2026-07-02	seang	Address decode widened to 13 bits / 8 KB so the monitor register block at 0x1000 (separate `stream_mon_regs` regfile, same APB slave) is addressable

Last Updated: 2026-07-02

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25 Monitor Bus Architecture and Event Code Organization

25.1 STREAM-Specific Context

For STREAM DMA Engine: This document describes the generic Monitor Bus (MonBus) protocol used across all AMBA-based components. STREAM uses MonBus for unified event reporting from: - Descriptor engines (8 sources, agent IDs 16-23) - Schedulers (8 sources, agent IDs 48-55) - Descriptor AXI monitor (agent ID 8) - Read/Write AXI monitors (configurable agent IDs)

All STREAM events use Unit ID = 1 and follow the standard 64-bit packet format defined below.

25.2 Overview

The Monitor Bus architecture provides a unified, scalable framework for monitoring and error reporting across multiple bus protocols in complex SoC designs. This system supports AXI, APB, MNOC (Mesh Network on Chip), ARB (Arbiter), CORE, and custom protocols through a standardized **128-bit packet format** paired with **64-bit side-band timestamps**, both carried atomically through a single handshake interface.

STREAM Integration: STREAM uses this unified MonBus for all event reporting (descriptor fetch, scheduler state, AXI transaction monitoring). See `docs/markdown/rtl-amba/includes/ocelus_package_spec.md` for the canonical 128-bit packet layout.

25.3 Interface Summary

25.3.1 Number of Interfaces

- **1 Monitor Bus Output Interface**: Unified 64-bit packet stream
- **Multiple Protocol Input Interfaces**: AXI, APB, MNOC, ARB, CORE, Custom protocol monitors
- **Local Memory Interface**: Error/interrupt packet storage
- **External Memory Interface**: Bulk packet storage

25.3.2 Interface Parameters

Interface Parameters

Parameter	Description	Valid Values	Default
PACKET_WIDTH	Monitor bus packet width (FIXED)	128	128
TIMESTAMP_WIDTH	Side-band timestamp width (FIXED)	64	64
PROTOCOL_WIDTH	Protocol identifier width	3	3
EVENT_CODE_WIDTH	Event code width	4	4
PACKET_TYPE_WIDTH	Packet type width	4	4
CHANNEL_ID_WIDTH	Channel identifier width	9	9
UNIT_ID_WIDTH	Unit identifier width	8	8
AGENT_ID_WIDTH	Agent identifier width	16	16
EVENT_DATA_WIDTH	Event data width	64	64

Note: PACKET_WIDTH, TIMESTAMP_WIDTH, and other core dimensions are now **locked** across the codebase (see rtl/amba/includes/ocelus_common_pkg.sv). Do not override these with per-module parameters.

25.4 Core Design Assumptions

25.4.1 Assumption 1: Hierarchical Event Organization

Assumption 1: Hierarchical Event Organization

Aspect	Requirement
Organization Rule	Protocol → Packet Type → Event Code hierarchy
Event Space	Each protocol × packet type combination has exactly 16 event codes
Mapping	1:1 mapping between packet types and event codes
Rationale	Provides clear, scalable event organization

25.4.2 Assumption 2: Protocol Isolation

Assumption 2: Protocol Isolation

Aspect	Requirement
Isolation Rule	Each protocol owns its event space
Conflict Prevention	No cross-protocol event conflicts
Independent Evolution	Protocols can evolve independently
Rationale	Prevents interference and enables protocol-specific optimization

25.4.3 Assumption 3: Two-Tier Memory Architecture

Assumption 3: Two-Tier Memory Architecture

Aspect	Requirement
Local Storage	Critical events (errors/interrupts) stored locally
External Storage	Non-critical events routed to external memory
Routing Decision	Based on packet type configuration
Rationale	Balances immediate access with bulk storage needs

25.4.4 Assumption 4: Configurable Packet Routing

Assumption 4: Configurable Packet Routing

Aspect	Requirement
Routing Rule	Different packet types can route to different destinations
Configuration	Base/limit registers define routing per packet type
Priority Support	Configurable priority levels per packet type
Rationale	Enables flexible memory allocation and access patterns

25.5 Interface Signal Specification

25.5.1 Monitor Bus Output Interface

Monitor Bus Output Interface

Signal	Width	Direction	Description
mon_packet	64	Monitor→System	Monitor packet data
mon_valid	1	Monitor→System	Packet valid signal
mon_ready	1	System→Monitor	Ready to accept packet
mon_error	1	Monitor→System	Monitor error condition

25.5.2 Protocol Input Interfaces

Protocol Input Interfaces

Signal	Width	Direction	Description
axi_event	64	AXI Monitor→Bus	AXI event packet
axi_event_valid	1	AXI Monitor→Bus	AXI event valid
axi_event_ready	1	Bus→AXI Monitor	Ready for AXI event
apb_event	64	APB Monitor→Bus	APB event packet
apb_event_valid	1	APB Monitor→Bus	APB event valid
apb_event_ready	1	Bus→APB Monitor	Ready for APB event
mnoc_event	64	MNOC Monitor→Bus	MNOC event packet
mnoc_event_valid	1	MNOC Monitor→Bus	MNOC event valid

Signal	Width	Direction	Description
mnoc_event_ready	1	Bus→MNOC Monitor	Ready for MNOC event
arb_event	64	ARB Monitor→Bus	ARB event packet
arb_event_valid	1	ARB Monitor→Bus	ARB event valid
arb_event_ready	1	Bus→ARB Monitor	Ready for ARB event
core_event	64	CORE Monitor→Bus	CORE event packet
core_event_valid	1	CORE Monitor→Bus	CORE event valid
core_event_ready	1	Bus→CORE Monitor	Ready for CORE event

25.5.3 Control and Status Signals

Control and Status Signals

Signal	Width	Direction	Description
clk	1	Input	System clock
resetn	1	Input	Active-low reset
monitor_enable	1	Input	Global monitor enable
packet_type_enables	16	Input	Per-type enable bits
local_memory_full	1	Output	Local memory full flag
external_memory_error	1	Output	External memory error

25.6 Packet Format and Field Allocation

25.6.1 128-bit Monitor Bus Packet Structure

The 128-bit packet is the standard across all AMBA/STREAM monitoring. Full field definitions are in `docs/markdown/rtl-amba/includes/ocelus_package_spec.md` and the SystemVerilog typedef `monitor_packet_t` in `rtl/amba/includes/ocelus_common_pkg.sv`.

Quick Reference:

128-bit Monitor Bus Packet Structure

Field	Bits	Width	Description
Packet Type	[127:124]	4	Event category (Error=0x0, Completion=0x1, etc.)
Reserved	[123:109]	15	Forward-compatibility slack
Protocol	[108:105]	4	Bus protocol (AXI=0, AXIS=1, APB=2, ARB=3, CORE=4)
Event Code	[104:97]	8	Specific events within category
Channel ID	[96:88]	9	Transaction/channel identifier
Agent ID	[87:72]	16	Module identifier (descriptor fetch=8, scheduler CH0=48, etc.)
Unit ID	[71:64]	8	Subsystem identifier (STREAM=0x01)
Event Data	[63:0]	64	Event-specific payload (addresses, counts, etc.)

25.6.2 64-bit Side-Band Timestamp

The timestamp is carried in parallel with the packet (not within it) through the same valid/ready handshake:

64-bit Side-Band Timestamp

Field	Bits	Width	Description
Timestamp	[63:0]	64	Cycle count at packet capture (hardware clock counter)

25.6.3 Packet Type Definitions

Packet Type Definitions

Value	Name	Purpose	Applicable Protocols
0x0	Error	Protocol violations, response errors	All
0x1	Completion	Successful transaction completion	All
0x2	Threshold	Threshold crossed events	All
0x3	Timeout	Timeout conditions	All
0x4	Performance	Performance metrics	All
0x5	Credit	Credit management	MNOC only
0x6	Channel	Channel status	MNOC only
0x7	Stream	Stream events	MNOC only
0x8	Address Match	Address matching	AXI only
0x9	APB Specific	APB protocol events	APB only
0xA-0xE	Reserved	Future expansion	-
0xF	Debug	Debug and trace events	All

25.7 Protocol-Specific Event Codes

25.7.1 AXI Protocol Events

25.7.1.1 Error Events (PktTypeError + PROTOCOL_AXI)

Error Events

Code	Event Name	Description
0x0	AXI_ERR_RESP_SLVERR	Slave error response
0x1	AXI_ERR_RESP_DECERR	Decode error response
0x2	AXI_ERR_DATA_ORPHAN	Data without command
0x3	AXI_ERR_RESP_ORPHAN	Response without transaction
0x4	AXI_ERR_PROTOCOL	Protocol violation
0x5	AXI_ERR_BURST_LENGTH	Invalid burst length
0x6	AXI_ERR_BURST_SIZE	Invalid burst size
0x7	AXI_ERR_BURST_TYPE	Invalid burst type
0x8	AXI_ERR_ID_COLLISION	ID collision detected
0x9	AXI_ERR_WRITE_BEFORE_ADDR	Write data before address
0xA	AXI_ERR_RESP_BEFORE_DATA	Response before data complete
0xB	AXI_ERR_LAST_MISSING	Missing LAST signal
0xC	AXI_ERR_STROBE_ERROR	Write strobe error
0xD	AXI_ERR_RESERVED_D	Reserved
0xE	AXI_ERR_EVENT_DROPPED	monitor-lite: events lost to monbus backpressure (data = count)
0xF	AXI_ERR_USER_DEFINED	User-defined error

25.7.1.2 Timeout Events (PktTypeTimeout + PROTOCOL_AXI)

Timeout Events

Code	Event Name	Description
0x0	AXI_TIMEOUT_CMD	Command/Address timeout

Code	Event Name	Description
0x1	AXI_TIMEOUT_DATA	Data timeout
0x2	AXI_TIMEOUT_RESP	Response timeout
0x3	AXI_TIMEOUT_HANDSHAKE	Handshake timeout
0x4	AXI_TIMEOUT_BURST	Burst completion timeout
0x5	AXI_TIMEOUT_EXCLUSIVE	Exclusive access timeout
0x6-0xE	Reserved	Future expansion
0xF	AXI_TIMEOUT_USER_DEFINED	User-defined timeout

25.7.1.3 Performance Events (PktTypePerf + PROTOCOL_AXI)

Performance Events

Code	Event Name	Description
0x0	AXI_PERF_ADDR_LATENCY	Address phase latency
0x1	AXI_PERF_DATA_LATENCY	Data phase latency
0x2	AXI_PERF_RESP_LATENCY	Response phase latency
0x3	AXI_PERF_TOTAL_LATENCY	Total transaction latency
0x4	AXI_PERF_THROUGHPUT	Transaction throughput
0x5	AXI_PERF_ERROR_RATE	Error rate
0x6	AXI_PERF_ACTIVE_COUNT	Active transaction count
0x7	AXI_PERF_BANDWIDTH_UTIL	Bandwidth utilization
0x8	AXI_PERF_QUEUE_DEPTH	Average queue depth

Code	Event Name	Description
0x9	AXI_PERF_BURST_EFFICIENCY	Burst efficiency metric
0xA-0xE	Reserved	Future expansion
0xF	AXI_PERF_USER_DEFINED	User-defined performance

25.7.2 APB Protocol Events

25.7.2.1 Error Events (PktTypeError + PROTOCOL_APB)

Error Events

Code	Event Name	Description
0x0	APB_ERR_PSLVERR	Peripheral slave error
0x1	APB_ERR_SETUP_VIOLATION	Setup phase protocol violation
0x2	APB_ERR_ACCESS_VIOLATION	Access phase protocol violation
0x3	APB_ERR_STROBE_ERROR	Write strobe error
0x4	APB_ERR_ADDR_DECODE	Address decode error
0x5	APB_ERR_PROT_VIOLATION	Protection violation (PPROT)
0x6	APB_ERR_ENABLE_ERROR	Enable phase error
0x7	APB_ERR_READY_ERROR	PREADY protocol error
0x8-0xE	Reserved	Future expansion
0xF	APB_ERR_USER_DEFINED	User-defined error

25.7.2.2 Timeout Events (PktTypeTimeout + PROTOCOL_APB)

Timeout Events

Code	Event Name	Description
0x0	APB_TIMEOUT_SETUP	Setup phase timeout
0x1	APB_TIMEOUT_ACCESS	Access phase timeout
0x2	APB_TIMEOUT_ENABLE	Enable phase timeout (PREADY stuck)
0x3	APB_TIMEOUT_PREADY_STUCK	PREADY stuck low
0x4	APB_TIMEOUT_TRANSFER	Overall transfer timeout
0x5-0xE	Reserved	Future expansion
0xF	APB_TIMEOUT_USER_DEFINED	User-defined timeout

25.7.2.3 Completion Events (PktTypeCompletion + PROTOCOL_APB)

Completion Events

Code	Event Name	Description
0x0	APB_COMPL_TRANS_COMPLETE	Transaction completed
0x1	APB_COMPL_READ_COMPLETE	Read transaction complete
0x2	APB_COMPL_WRITE_COMPLETE	Write transaction complete
0x3-0xE	Reserved	Future expansion
0xF	APB_COMPL_USER_DEFINED	User-defined completion

25.7.2.4 Threshold Events (PktTypeThreshold + PROTOCOL_APB)

Threshold Events

Code	Event Name	Description
0x0	APB_THRESH_LATENCY	APB latency threshold
0x1	APB_THRESH_ERROR_RATE	APB error rate threshold
0x2	APB_THRESH_ACCESS_COUNT	Access count threshold
0x3	APB_THRESH_BANDWIDTH	Bandwidth threshold
0x4-0xE	Reserved	Future expansion
0xF	APB_THRESH_USER_DEFINED	User-defined threshold

25.7.2.5 Performance Events (PktTypePerf + PROTOCOL_APB)

Performance Events

Code	Event Name	Description
0x0	APB_PERF_READ_LATENCY	Read transaction latency
0x1	APB_PERF_WRITE_LATENCY	Write transaction latency
0x2	APB_PERF_THROUGHPUT	Transaction throughput
0x3	APB_PERF_ERROR_RATE	Error rate
0x4	APB_PERF_ACTIVE_COUNT	Active transaction count
0x5	APB_PERF_COMPLETED_COUNT	Completed transaction count
0x6-0xE	Reserved	Future expansion
0xF	APB_PERF_USER_DEFINED	User-defined performance

25.7.2.6 Debug Events (PktTypeDebug + PROTOCOL_APB)

Debug Events

Code	Event Name	Description
0x0	APB_DEBUG_STATE_CHANGE	APB state changed
0x1	APB_DEBUG_SETUP_PHASE	Setup phase event
0x2	APB_DEBUG_ACCESS_PHASE	Access phase event
0x3	APB_DEBUG_ENABLE_PHASE	Enable phase event
0x4	APB_DEBUG_PSEL_TRACE	PSEL trace
0x5	APB_DEBUG_PENABLE_TRACE	PENABLE trace
0x6	APB_DEBUG_PREADY_TRACE	PREADY trace
0x7	APB_DEBUG_PPROT_TRACE	PPROT trace
0x8	APB_DEBUG_PSTRB_TRACE	PSTRB trace
0x9-0xE	Reserved	Future expansion
0xF	APB_DEBUG_USER_DEFINED	User-defined debug

25.7.3 MNOC Protocol Events

25.7.3.1 Error Events (PktTypeError + PROTOCOL_MNOC)

Error Events

Code	Event Name	Description
0x0	MNOC_ERR_PARITY	Parity error
0x1	MNOC_ERR_PROTOCOL	Protocol violation
0x2	MNOC_ERR_OVERFL	Buffer/Credit

Code	Event Name	Description
	OW	overflow
0x3	MNOC_ERR_UNDERFLOW	Buffer/Credit underflow
0x4	MNOC_ERR_ORPHAN	Orphaned packet/ACK
0x5	MNOC_ERR_INVALID	Invalid type/channel/payload
0x6	MNOC_ERR_HEADER_CRC	Header CRC error
0x7	MNOC_ERR_PAYLOAD_CRC	Payload CRC error
0x8	MNOC_ERR_SEQUENCE	Sequence number error
0x9	MNOC_ERR_ROUTE	Routing error
0xA	MNOC_ERR_DEADLOCK	Deadlock detected
0xB-0xE	Reserved	Future expansion
0xF	MNOC_ERR_USER_DEFINED	User-defined error

25.7.3.2 Credit Events (PktTypeCredit + PROTOCOL_MNOC)

Credit Events

Code	Event Name	Description
0x0	MNOC_CREDIT_ALLOCATED	Credits allocated
0x1	MNOC_CREDIT_CONSUMED	Credits consumed
0x2	MNOC_CREDIT_RETURNED	Credits returned
0x3	MNOC_CREDIT_OVERFLOW	Credit overflow detected
0x4	MNOC_CREDIT_UND	Credit underflow

Code	Event Name	Description
	ERFLOW	detected
0x5	MNOC_CREDIT_EXHAUSTED	All credits exhausted
0x6	MNOC_CREDIT_RESTORED	Credits restored
0x7	MNOC_CREDIT_EFFICIENCY	Credit efficiency metric
0x8	MNOC_CREDIT_LEAK	Credit leak detected
0x9-0xE	Reserved	Future expansion
0xF	MNOC_CREDIT_USER_DEFINED	User-defined credit event

25.7.3.3 Channel Events (PktTypeChannel + PROTOCOL_MNOC)

Channel Events

Code	Event Name	Description
0x0	MNOC_CHANNEL_OPEN	Channel opened
0x1	MNOC_CHANNEL_CLOSE	Channel closed
0x2	MNOC_CHANNEL_STALL	Channel stalled
0x3	MNOC_CHANNEL_RESUME	Channel resumed
0x4	MNOC_CHANNEL_CONGESTION	Channel congestion detected
0x5	MNOC_CHANNEL_PRIORITY	Channel priority change
0x6-0xE	Reserved	Future expansion
0xF	MNOC_CHANNEL_USER_DEFINED	User-defined channel event

25.7.3.4 Stream Events (PktTypeStream + PROTOCOL_MNOC)

Stream Events

Code	Event Name	Description
0x0	MNOC_STREAM_START	Stream started
0x1	MNOC_STREAM_END	Stream ended (EOS)
0x2	MNOC_STREAM_PAUSE	Stream paused
0x3	MNOC_STREAM_RESUME	Stream resumed
0x4	MNOC_STREAM_OVERFLOW	Stream buffer overflow
0x5	MNOC_STREAM_UNDERFLOW	Stream buffer underflow
0x6-0xE	Reserved	Future expansion
0xF	MNOC_STREAM_USER_DEFINED	User-defined stream event

25.7.4 ARB Protocol Events

25.7.4.1 Error Events (PktTypeError + PROTOCOL_ARB)

Error Events

Code	Event Name	Description
0x0	ARB_ERR_STARVATION	Client request starvation
0x1	ARB_ERR_ACK_TIMEOUT	Grant ACK timeout
0x2	ARB_ERR_PROTOCOL_VIOLATION	ACK protocol violation
0x3	ARB_ERR_CREDIT_VIOLATION	Credit system violation
0x4	ARB_ERR_FAIRNESS_VIOLATION	Weighted fairness violation

Code	Event Name	Description
0x5	ARB_ERR_WEIGHT_UNDERFLOW	Weight credit underflow
0x6	ARB_ERR_CONCURRENT_GRANTS	Multiple simultaneous grants
0x7	ARB_ERR_INVALID_GRANT_ID	Invalid grant ID detected
0x8	ARB_ERR_ORPHAN_ACK	ACK without pending grant
0x9	ARB_ERR_GRANT_OVERLAP	Overlapping grant periods
0xA	ARB_ERR_MASK_ERROR	Round-robin mask error
0xB	ARB_ERR_STATE_MACHINE	FSM state error
0xC	ARB_ERR_CONFIGURATION	Invalid configuration
0xD-0xE	Reserved	Future expansion
0xF	ARB_ERR_USER_DEFINED	User-defined error

25.7.4.2 Timeout Events (PktTypeTimeout + PROTOCOL_ARB)

Timeout Events

Code	Event Name	Description
0x0	ARB_TIMEOUT_GRANT_ACK	Grant ACK timeout
0x1	ARB_TIMEOUT_REQUEST_HOLD	Request held too long
0x2	ARB_TIMEOUT_WEIGHT_UPDATE	Weight update timeout
0x3	ARB_TIMEOUT_BLOCK_RELEASE	Block release timeout
0x4	ARB_TIMEOUT_CREDIT_UPDATE	Credit update timeout

Code	Event Name	Description
0x5	ARB_TIMEOUT_STATE_CHANGE	State machine timeout
0x6-0xE	Reserved	Future expansion
0xF	ARB_TIMEOUT_USER_DEFINED	User-defined timeout

25.7.4.3 Completion Events (PktTypeCompletion + PROTOCOL_ARB)

Completion Events

Code	Event Name	Description
0x0	ARB_COMPL_GRANT_ISSUED	Grant successfully issued
0x1	ARB_COMPL_ACK_RECEIVED	ACK successfully received
0x2	ARB_COMPL_TRANSACTION	Complete transaction (grant+ack)
0x3	ARB_COMPL_WEIGHT_UPDATE	Weight update completed
0x4	ARB_COMPL_CREDIT_CYCLE	Credit cycle completed
0x5	ARB_COMPL_FAIRNESS_PERIOD	Fairness analysis period
0x6	ARB_COMPL_BLOCK_PERIOD	Block period completed
0x7-0xE	Reserved	Future expansion
0xF	ARB_COMPL_USER_DEFINED	User-defined completion

25.7.4.4 Threshold Events (PktTypeThreshold + PROTOCOL_ARB)

Threshold Events

Code	Event Name	Description
0x0	ARB_THRESH_REQUEST_LATENCY	Request-to-grant latency threshold
0x1	ARB_THRESH_ACK_LATENCY	Grant-to-ACK latency threshold

Code	Event Name	Description
0x2	ARB_THRESH_FAIRNESS_DEV	Fairness deviation threshold
0x3	ARB_THRESH_ACTIVE_REQUESTS	Active request count threshold
0x4	ARB_THRESH_GRANT_RATE	Grant rate threshold
0x5	ARB_THRESH_EFFICIENCY	Grant efficiency threshold
0x6	ARB_THRESH_CREDIT_LOW	Low credit threshold
0x7	ARB_THRESH_WEIGHT_IMBALANCE	Weight imbalance threshold
0x8	ARB_THRESH_STARVATION_TIME	Starvation time threshold
0x9-0xE	Reserved	Future expansion
0xF	ARB_THRESH_USER_DEFINED	User-defined threshold

25.7.4.5 Performance Events (PktTypePerf + PROTOCOL_ARB)

Performance Events

Code	Event Name	Description
0x0	ARB_PERF_GRANT_ISSUED	Grant issued event
0x1	ARB_PERF_ACK_RECEIVED	ACK received event
0x2	ARB_PERF_GRANT_EFFICIENCY	Grant completion efficiency
0x3	ARB_PERF_FAIRNESS_METRIC	Fairness compliance metric
0x4	ARB_PERF_THROUGHPUT	Arbitration throughput
0x5	ARB_PERF_LATENCY_AVG	Average latency measurement
0x6	ARB_PERF_WEIGHT_COMPLIANCE	Weight compliance metric
0x7	ARB_PERF_CREDIT_UTILIZATION	Credit utilization efficiency
0x8	ARB_PERF_CLIENT_ACTIVITY	Per-client activity metric

Code	Event Name	Description
0x9	ARB_PERF_STARVATION_COUNT	Starvation event count
0xA	ARB_PERF_BLOCK_EFFICIENCY	Block/unblock efficiency
0xB-0xE	Reserved	Future expansion
0xF	ARB_PERF_USER_DEFINED	User-defined performance

25.7.4.6 Debug Events (PktTypeDebug + PROTOCOL_ARB)

Debug Events

Code	Event Name	Description
0x0	ARB_DEBUG_STATE_CHANGE	Arbiter state machine change
0x1	ARB_DEBUG_MASK_UPDATE	Round-robin mask update
0x2	ARB_DEBUG_WEIGHT_CHANGE	Weight configuration change
0x3	ARB_DEBUG_CREDIT_UPDATE	Credit level update
0x4	ARB_DEBUG_CLIENT_MASK	Client enable/disable mask
0x5	ARB_DEBUG_PRIORITY_CHANGE	Priority level change
0x6	ARB_DEBUG_BLOCK_EVENT	Block/unblock event
0x7	ARB_DEBUG_QUEUE_STATUS	Request queue status
0x8	ARB_DEBUG_COUNTER_SNAPSHOT	Counter values snapshot
0x9	ARB_DEBUG_FIFO_STATUS	FIFO status change
0xA	ARB_DEBUG_FAIRNESS_STATE	Fairness tracking state

Code	Event Name	Description
0xB	ARB_DEBUG_ACK_STATE	ACK protocol state
0xC-0xE	Reserved	Future expansion
0xF	ARB_DEBUG_USER_DEFINED	User-defined debug

25.7.5 CORE Protocol Events

25.7.5.1 Error Events (PktTypeError + PROTOCOL_CORE)

Error Events

Code	Event Name	Description
0x0	CORE_ERR_DESCRIPTOR_MALFORMED	Missing magic number (0x900dc0de)
0x1	CORE_ERR_DESCRIPTOR_BAD_ADDR	Invalid descriptor address
0x2	CORE_ERR_DATA_BAD_ADDR	Invalid data address (fetch or runtime)
0x3	CORE_ERR_FLAG_COMPARISON	Flag mask/compare mismatch
0x4	CORE_ERR_CREDIT_UNDERFLOW	Credit system violation
0x5	CORE_ERR_STATE_MACHINE	Invalid FSM state transition
0x6	CORE_ERR_DESCRIPTOR_ENGINE	Descriptor engine FSM error
0x7	CORE_ERR_FLAG_ENGINE	Flag engine FSM error
0x8	CORE_ERR_PROGRAM_ENGINE	Program engine FSM error
0x9	CORE_ERR_DATA_ENGINE	Data engine error
0xA	CORE_ERR_CHANNEL_INVALID	Invalid channel ID
0xB	CORE_ERR_CONTROL_VIOLATION	Control register violation
0xC-0xE	Reserved	Future expansion

Code	Event Name	Description
0xF	CORE_ERR_USER_DEFINED	User-defined error

25.7.5.2 Timeout Events (PktTypeTimeout + PROTOCOL_CORE)

Timeout Events

Code	Event Name	Description
0x0	CORE_TIMEOUT_DESCRIPTOR_FETCH	Descriptor fetch timeout
0x1	CORE_TIMEOUT_FLAG_RETRY	Flag comparison retry timeout
0x2	CORE_TIMEOUT_PROGRAM_WRITE	Program write timeout
0x3	CORE_TIMEOUT_DATA_TRANSFER	Data transfer timeout
0x4	CORE_TIMEOUT_CREDIT_WAIT	Credit wait timeout
0x5	CORE_TIMEOUT_CONTROL_WAIT	Control enable wait timeout
0x6	CORE_TIMEOUT_ENGINE_RESPONSE	Sub-engine response timeout
0x7	CORE_TIMEOUT_STATE_TRANSITION	FSM state transition timeout
0x8-0xE	Reserved	Future expansion
0xF	CORE_TIMEOUT_USER_DEFINED	User-defined timeout

25.7.5.3 Completion Events (PktTypeCompletion + PROTOCOL_CORE)

Completion Events

Code	Event Name	Description
0x0	CORE_COMPL_DESCRIPTOR_LOADED	Descriptor successfully loaded
0x1	CORE_COMPL_DESCRIPTOR_CHAIN	Descriptor chain completed
0x2	CORE_COMPL_FLAG_MATCH	Flag comparison successful

Code	Event Name	Description
	ED	
0x3	CORE_COMPL_PROGRAM_COMPLETED	Post-programming completed
0x4	CORE_COMPL_DATA_TRANSFER	Data transfer completed
0x5	CORE_COMPL_CREDIT_CYCLE	Credit cycle completed
0x6	CORE_COMPL_CHANNEL_COMPLETE	Channel processing complete
0x7	CORE_COMPL_ENGINE_READY	Sub-engine ready
0x8-0xE	Reserved	Future expansion
0xF	CORE_COMPL_USER_DEFINED	User-defined completion

25.7.5.4 Threshold Events (PktTypeThreshold + PROTOCOL_CORE)

Threshold Events

Code	Event Name	Description
0x0	CORE_THRESH_DESCRIPTOR_QUEUE	Descriptor queue depth threshold
0x1	CORE_THRESH_CREDIT_LOW	Credit low threshold
0x2	CORE_THRESH_FLAG_RETRY_COUNT	Flag retry count threshold
0x3	CORE_THRESH_LATENCY	Processing latency threshold
0x4	CORE_THRESH_ERROR_RATE	Error rate threshold
0x5	CORE_THRESH_THROUGHPUT	Throughput threshold
0x6	CORE_THRESH_ACTIVE_CHANNELS	Active channel count threshold
0x7	CORE_THRESH_PROGRAM_LATENCY	Program write latency threshold
0x8	CORE_THRESH_DATA_RATE	Data transfer rate threshold
0x9-0xE	Reserved	Future expansion

Code	Event Name	Description
0xF	CORE_THRESH_USER_DEFINED	User-defined threshold

25.7.5.5 Performance Events (PktTypePerf + PROTOCOL_CORE)

Performance Events

Code	Event Name	Description
0x0	CORE_PERF_END_OF_DATA	Stream continuation signal
0x1	CORE_PERF_END_OF_STREAM	Stream termination signal
0x2	CORE_PERF_ENTERING_IDLE	FSM returning to idle
0x3	CORE_PERF_CREDIT_INCREMENTED	Credit added by software
0x4	CORE_PERF_CREDIT_EXHAUSTED	Credit blocking execution
0x5	CORE_PERF_STATE_TRANSITION	FSM state change
0x6	CORE_PERF_DESCRIPTOR_ACTIVE	Data processing started
0x7	CORE_PERF_FLAG_RETRY	Flag comparison retry
0x8	CORE_PERF_CHANNEL_ENABLE	Channel enabled by software
0x9	CORE_PERF_CHANNEL_DISABLE	Channel disabled by software
0xA	CORE_PERF_CREDIT_UTILIZATION	Credit utilization metric
0xB	CORE_PERF_PROCESSING_LATENCY	Total processing latency
0xC	CORE_PERF_QUEUE_DEPTH	Current queue depth
0xD-0xE	Reserved	Future expansion

Code	Event Name	Description
0xF	CORE_PERF_USER_DEFINED	User-defined performance

25.7.5.6 Debug Events (PktTypeDebug + PROTOCOL_CORE)

Debug Events

Code	Event Name	Description
0x0	CORE_DEBUG_FSM_STATE_CHANGE	Descriptor FSM state change
0x1	CORE_DEBUG_DESCRIPTOR_CONTENT	Descriptor content trace
0x2	CORE_DEBUG_FLAG_ENGINE_STATE	Flag engine state trace
0x3	CORE_DEBUG_PROGRAM_ENGINE_STATE	Program engine state trace
0x4	CORE_DEBUG_CREDIT_OPERATION	Credit system operation
0x5	CORE_DEBUG_CONTROL_REGISTER	Control register access
0x6	CORE_DEBUG_ENGINE_HANDSHAKE	Engine handshake trace
0x7	CORE_DEBUG_QUEUE_STATUS	Queue status change
0x8	CORE_DEBUG_COUNTER_SNAPSHOT	Counter values snapshot
0x9	CORE_DEBUG_ADDRESS_TRACE	Address progression trace
0xA	CORE_DEBUG_PAYLOAD_TRACE	Payload content trace
0xB-0xE	Reserved	Future expansion
0xF	CORE_DEBUG_USER_DEFINED	User-defined debug

25.8 Memory Architecture and Packet Routing

25.8.1 Two-Tier Memory Architecture

25.8.1.1 Local Error/Interrupt Memory

Local Error/Interrupt Memory

Characteristic	Description
Storage Types	Error Packets (Type 0x0) and Timeout Packets (Type 0x3)
Access Method	Immediate CPU access without memory subsystem delays
Capacity	Large enough to prevent overflow during error bursts
Priority	Critical events requiring immediate attention
Indexing	Fast search and retrieval mechanisms

25.8.1.2 Configurable External Memory

Configurable External Memory

Characteristic	Description
Storage Types	Performance, Completion, Threshold, Debug packets
Access Method	Base and limit registers define memory regions
Capacity	Bulk storage for non-critical events
DMA Support	Can be accessed via DMA for efficient transfer
Time Stamping	32-bit timestamp appended when routing externally

25.8.2 Routing Configuration

25.8.2.1 Base and Limit Registers

Base and Limit Registers

Register Set	Purpose	Configuration
Completion Config	Type 0x1 routing	base_addr, limit_addr, enable, priority
Threshold Config	Type 0x2 routing	base_addr, limit_addr, enable, priority
Performance Config	Type 0x4 routing	base_addr, limit_addr, enable, priority
Debug Config	Type 0xF routing	base_addr, limit_addr, enable, priority

25.8.2.2 Routing Decision Logic

Routing Decision Logic

Packet Type	Destination	Address Calculation
Error (0x0)	Local Memory	local_error_write_pointer
Timeout (0x3)	Local Memory	local_error_write_pointer
Completion (0x1)	External Memory	completion_config.base_addr + offset
Performance (0x4)	External Memory	performance_config.base_addr + offset
Debug (0xF)	External Memory	debug_config.base_addr + offset

25.8.3 Address Space Management

25.8.3.1 Memory Layout Example

Memory Layout Example

Address Range	Usage	Description
0x1000_0000 - 0x1000_FFFF	Local Error Memory	Immediate access storage
0x2000_0000 - 0x2001_FFFF	Performance Packets	External bulk storage
0x2010_0000 - 0x2011_FFFF	Completion	External bulk storage

Address Range	Usage	Description
	Packets	
0x2020_0000 - 0x202F_FFFF	Debug Packets	External bulk storage

25.8.4 Transaction State and Bus Transaction Structure

25.8.4.1 Transaction State Enumeration

Transaction State Enumeration

State	Value	Description	Usage
TRANS_EMPTY	3'b000	Unused entry	Available slot
TRANS_ADDR_PHASE	3'b001	Address phase active (AXI) / Packet sent (MNOC) / Setup phase (APB)	Initial phase
TRANS_DATA_PHASE	3'b010	Data phase active (AXI) / Waiting for ACK (MNOC) / Access phase (APB)	Data transfer
TRANS_RESP_PHASE	3'b011	Response phase active (AXI) / ACK received (MNOC) / Enable phase (APB)	Response handling
TRANS_COMPLETE	3'b100	Transaction complete	Successful completion
TRANS_ERROR	3'b101	Transaction has error	Error condition
TRANS_ORPHANED	3'b110	Orphaned transaction	Missing components
TRANS_CREDIT_STALL	3'b111	Credit stall (MNOC only)	MNOC-specific stall

25.8.4.2 Enhanced Transaction Structure

Enhanced Transaction Structure

Field	Width	Description	Protocol Usage
valid	1	Entry is valid	All protocol's
protocol	3	Protocol type (AXI/MNOC/APB/ARB/C	All protocols

Field	Width	Description	Protocol Usage
		ORE)	
state	3	Transaction state	All protocols
id	32	Transaction ID (AXI) / Sequence (MNOC) / PSEL encoding (APB)	All protocols
addr	64	Transaction address / Channel addr / PADDR	All protocols
len	8	Burst length (AXI) / Packet count (MNOC) / Always 0 (APB)	AXI, MNOC
size	3	Access size (AXI) / Reserved (MNOC) / Transfer size (APB)	AXI, APB
burst	2	Burst type (AXI) / Payload type (MNOC) / PPROT[1:0] (APB)	All protocols

25.8.4.3 Phase Completion Flags

Phase Completion Flags

Flag	Description	Protocol Usage
cmd_received	Address phase received / Packet sent / Setup phase	All protocols
data_started	Data phase started / ACK expected / Access phase	All protocols
data_completed	Data phase completed / ACK received / Enable phase	All protocols
resp_received	Response received / Final ACK / PREADY asserted	All protocols

25.8.4.4 Protocol-Specific Tracking Fields

Protocol-Specific Tracking Fields

Field	Width	Description	Protocol
channel	6	Channel ID (AXI ID / MNOC channel / PSEL bit	All protocols

Field	Width	Description	Protocol
		position)	
eos_seen	1	EOS marker seen	MNOC only
parity_error	1	Parity error detected	MNOC only
credit_at_start	8	Credits available at start	MNOC only
retry_count	3	Number of retries	MNOC only
desc_addr_match	1	Descriptor address match detected	AXI only
data_addr_match	1	Data address match detected	AXI only
apb_phase	2	Current APB phase	APB only
pslverr_seen	1	PSLVERR detected	APB only
pprot_value	3	PPROT value	APB only
pstrb_value	4	PSTRB value for writes	APB only
arb_grant_id	8	Current grant ID	ARB only
arb_weight	8	Current weight value	ARB only
core_fsm_state	3	Current CORE FSM state	CORE only
core_channel_id	6	CORE channel identifier	CORE only

25.8.5 APB Transaction Phases

APB Transaction Phases

Phase	Value	Description
APB_PHASE_IDLE	2'b00	Bus idle
APB_PHASE_SETUP	2'b01	Setup phase (PSEL asserted)
APB_PHASE_ACCESS	2'b10	Access phase (PENABLE asserted)
APB_PHASE_ENABLE	2'b11	Enable phase (waiting for PREADY)

25.8.6 APB Protection Types

APB Protection Types

Protection	Value	Description
APB_PROT_NORMAL	3'b000	Normal access
APB_PROT_PRIVILEGED	3'b001	Privileged access
APB_PROT_SECURE	3'b010	Secure access
APB_PROT_INSTRUCTION	3'b100	Instruction access

25.8.7 MNOC Payload Types

MNOC Payload Types

Payload	Value	Description
MNOC_PAYLOAD_CONFIG	2'b00	CONFIG_PKT
MNOC_PAYLOAD_TS	2'b01	TS_PKT
MNOC_PAYLOAD_RAW	2'b11	RAW_PKT

25.8.8 MNOC ACK Types

MNOC ACK Types

ACK Type	Value	Description
MNOC_ACK_STOP	2'b00	MSAP_STOP
MNOC_ACK_START	2'b01	MSAP_START
MNOC_ACK_CREDIT_ON	2'b10	MSAP_CREDIT_ON
MNOC_ACK_STOP_AT_EOS	2'b11	MSAP_STOP_AT_EOS

25.8.9 ARB State Types

ARB State Types

State	Value	Description
ARB_STATE_IDLE	3'b000	Idle state
ARB_STATE_ARBITRATE	3'b001	Performing arbitration
ARB_STATE_GRANT	3'b010	Grant issued, waiting for ACK
ARB_STATE_BLOCKED	3'b011	Arbitration blocked
ARB_STATE_WEIGHT_UPD	3'b100	Weight update in progress
ARB_STATE_ERROR	3'b101	Error state

25.8.10 CORE State Types

CORE State Types

State	Value	Description
CORE_STATE_IDLE	3'b000	Idle state
CORE_STATE_DESC_FETCH	3'b001	Fetching descriptor
CORE_STATE_FLAG_CHECK	3'b010	Checking flag condition
CORE_STATE_PROGRAM_WRITE	3'b011	Writing program
CORE_STATE_DATA_TRANSFER	3'b100	Transferring data
CORE_STATE_CREDIT_WAIT	3'b101	Waiting for credits
CORE_STATE_ERROR	3'b110	Error state

25.9 Configuration and Control

25.9.1 Monitor Configuration Registers

25.9.1.1 Global Configuration

Global Configuration

Field	Width	Description
monitor_enable	1	Global monitor enable
error_local_enable	1	Enable local error storage
external_route_enable	1	Enable external routing
unit_id	4	Unit identifier
agent_id	8	Agent identifier
packet_type_enables	16	Per-type enable bits

25.9.1.2 Packet Type Enable Mapping

Packet Type Enable Mapping

Bit	Enable	Description
0	PKT_ENABLE_ERROR	Enable error packets
1	PKT_ENABLE_COMPLETION	Enable completion packets
2	PKT_ENABLE_THRESHOLD	Enable threshold packets
3	PKT_ENABLE_TIMEOUT	Enable timeout packets
4	PKT_ENABLE_PERF	Enable performance packets
5	PKT_ENABLE_CREDIT	Enable credit packets (MNOC)
6	PKT_ENABLE_CHAN	Enable channel

Bit	Enable	Description
	NEL	packets (MNOC)
7	PKT_ENABLE_STREAM	Enable stream packets (MNOC)
8	PKT_ENABLE_ADDR_MATCH	Enable address match (AXI)
9	PKT_ENABLE_APB	Enable APB packets
15	PKT_ENABLE_DEBUG	Enable debug packets

25.9.2 Protocol-Specific Configuration

25.9.2.1 AXI Monitor Configuration

AXI Monitor Configuration

Field	Width	Description
active_trans_threshold	16	Active transaction threshold
latency_threshold	32	Latency threshold (cycles)
addr_timeout_cnt	4	Address timeout count
data_timeout_cnt	4	Data timeout count
resp_timeout_cnt	4	Response timeout count
burst_boundary_check	1	Enable burst boundary checking
address_match_enable	1	Enable address matching
desc_addr_match_base	64	Descriptor address match base
desc_addr_match_mask	64	Descriptor address match mask
data_addr_match_base	64	Data address match base
data_addr_match_	64	Data address match

Field	Width	Description
mask		mask

25.9.2.2 MNOC Monitor Configuration

MNOC Monitor Configuration

Field	Width	Description
credit_low_threshold	8	Credit low threshold
packet_rate_threshold	16	Packet rate threshold
max_route_hops	8	Maximum routing hops
enable_credit_tracking	1	Enable credit tracking
enable_deadlock_detect	1	Enable deadlock detection
deadlock_timeout	4	Deadlock detection timeout

25.9.2.3 ARB Monitor Configuration

ARB Monitor Configuration

Field	Width	Description
grant_timeout_cnt	16	Grant ACK timeout count
fairness_window	32	Fairness analysis window
weight_update_enable	1	Enable weight tracking
starvation_threshold	16	Starvation detection threshold
efficiency_threshold	8	Grant efficiency threshold

25.9.2.4 CORE Monitor Configuration

CORE Monitor Configuration

Field	Width	Description
descriptor_timeout_cnt	16	Descriptor fetch timeout count
flag_retry_limit	8	Maximum flag retry count
credit_low_threshold	8	Credit low threshold
processing_timeout_cnt	32	Processing timeout count
enable_descriptor_trace	1	Enable descriptor content tracing
enable_fsm_trace	1	Enable FSM state tracing

25.10 Validation Requirements

25.10.1 Functional Validation

Functional Validation

Validation Area	Requirements
Packet Format	Verify 64-bit packet structure and field encoding
Event Organization	Verify hierarchical event code organization
Protocol Isolation	Verify independent protocol event spaces
Routing Logic	Verify packet routing based on type and configuration
Memory Management	Verify local and external memory operations
Configuration	Verify register configuration and enable controls

25.10.2 Performance Validation

Performance Validation

Validation Area	Requirements
Throughput	Verify monitor bus can handle peak event rates
Latency	Verify low-latency path for critical events
Memory Efficiency	Verify efficient memory usage patterns
Power Consumption	Verify power-efficient operation

25.10.3 Error Handling Validation

Error Handling Validation

Validation Area	Requirements
Error Injection	Verify error detection and

Validation Area	Requirements
	reporting
Overflow Handling	Verify behavior when memories fill
Configuration Errors	Verify invalid configuration detection
Recovery Mechanisms	Verify error recovery procedures

25.11 Usage Examples

25.11.1 Creating Monitor Packets

Creating Monitor Packets

Packet Type	Example Usage
AXI Error	Protocol=AXI, Type=Error, Code=AXI_ERR_RESP_SLVERR
MNOC Credit	Protocol=MNOC, Type=Credit, Code=MNOC_CREDIT_EXHAUSTED
APB Performance	Protocol=APB, Type=Performance, Code=APB_PERF_TOTAL_LATENCY
ARB Threshold	Protocol=ARB, Type=Threshold, Code=ARB_THRESH_FAIRNESS_DEV
CORE Completion	Protocol=CORE, Type=Completion, Code=CORE_COMPL_DESCRIPTOR_LOADED

25.11.2 Packet Decoding

Packet Decoding

Decoding Step	Method
Extract Type	packet[63:60]
Extract Protocol	packet[59:57]
Extract Event Code	packet[56:53]
Extract Channel ID	packet[52:47]
Extract Event Data	packet[34:0]

25.11.3 Monitor Bus Packet Helper Functions

25.11.3.1 Packet Field Extraction

Packet Field Extraction

Function	Return Type	Description
get_packet_type(pkt)	logic [3:0]	Extract packet type [63:60]
get_protocol_type(	protocol_type_t	Extract protocol [59:57]

Function	Return Type	Description
pkt)		
get_event_code(pkt)	logic [3:0]	Extract event code [56:53]
get_channel_id(pkt)	logic [5:0]	Extract channel ID [52:47]
get_unit_id(pkt)	logic [3:0]	Extract unit ID [46:43]
get_agent_id(pkt)	logic [7:0]	Extract agent ID [42:35]
get_event_data(pkt)	logic [34:0]	Extract event data [34:0]

25.11.3.2 Packet Creation Function

Packet Creation Function

Function	Parameters	Description
create_monitor_packet()	packet_type, protocol, event_code, channel_id, unit_id, agent_id, event_data	Create complete 64-bit packet

25.11.3.3 Event Code Creation Functions

Event Code Creation Functions

Function	Parameter	Description
create_axi_error_event()	axi_error_code_t	Create AXI error event code
create_axi_timeout_event()	axi_timeout_code_t	Create AXI timeout event code
create_axi_completion_event()	axi_completion_code_t	Create AXI completion event code
create_axi_threshold_event()	axi_threshold_code_t	Create AXI threshold event code
create_axi_performance_event()	axi_performance_code_t	Create AXI performance event code
create_axi_addr_match_event()	axi_addr_match_code_t	Create AXI address match event code
create_axi_debug_ev	axi_debug_code_t	Create AXI debug event code

Function	Parameter	Description
ent()		
create_apb4_error_event()	apb_error_code_t	Create APB error event code
create_apb4_timeout_event()	apb_timeout_code_t	Create APB timeout event code
create_apb4_completion_event()	apb_completion_code_t	Create APB completion event code
create_mnoc_error_event()	mnoc_error_code_t	Create MNOC error event code
create_mnoc_timeout_event()	mnoc_timeout_code_t	Create MNOC timeout event code
create_mnoc_completion_event()	mnoc_completion_code_t	Create MNOC completion event code
create_mnoc_credit_event()	mnoc_credit_code_t	Create MNOC credit event code
create_mnoc_channel_event()	mnoc_channel_code_t	Create MNOC channel event code
create_mnoc_stream_event()	mnoc_stream_code_t	Create MNOC stream event code
create_arb_error_event()	arb_error_code_t	Create ARB error event code
create_arb_timeout_event()	arb_timeout_code_t	Create ARB timeout event code
create_arb_completion_event()	arb_completion_code_t	Create ARB completion event code
create_arb_threshold_event()	arb_threshold_code_t	Create ARB threshold event code
create_arb_performance_event()	arb_performance_code_t	Create ARB performance event code
create_arb_debug_event()	arb_debug_code_t	Create ARB debug event code
create_core_error_event()	core_error_code_t	Create CORE error event code

Function	Parameter	Description
create_core_timeout_event()	core_timeout_code_t	Create CORE timeout event code
create_core_completion_event()	core_completion_code_t	Create CORE completion event code
create_core_threshold_event()	core_threshold_code_t	Create CORE threshold event code
create_core_performance_event()	core_performance_code_t	Create CORE performance event code
create_core_debug_event()	core_debug_code_t	Create CORE debug event code

25.11.3.4 Validation Functions

Validation Functions

Function	Parameters	Description
is_valid_event_for_packet_type()	packet_type, protocol, event_code	Validate event code for packet type and protocol

25.11.3.5 String Functions for Debugging

String Functions for Debugging

Function	Parameter	Description
get_axi_error_name()	axi_error_code_t	Get human-readable AXI error name
get_arb_error_name()	arb_error_code_t	Get human-readable ARB error name
get_core_error_name()	core_error_code_t	Get human-readable CORE error name
get_packet_type_name()	logic [3:0]	Get packet type name string
get_protocol_name()	protocol_type_t	Get protocol name string
get_event_name()	packet_type, protocol, event_code	Get comprehensive event name

25.12 Debug and Monitoring Signals

25.12.1 Essential Debug Signals

Essential Debug Signals

Signal	Width	Purpose
debug_packet_counts	32 × 16	Packet count per type
debug_protocol_counts	32 × 5	Packet count per protocol
debug_error_counts	32	Total error packet count
debug_local_memory_level	16	Local memory usage level
debug_external_memory_level	16	External memory usage level

25.12.2 Performance Counters

Performance Counters

Counter	Width	Purpose
total_packets_processed	32	Total packets processed
packets_dropped	32	Packets dropped due to overflow
routing_errors	32	Routing configuration errors
memory_full_events	32	Memory full occurrences

25.13 Protocol Coverage Summary

25.13.1 Complete Protocol Event Matrix

Complete Protocol Event Matrix

Protocol	Error	Timeout	Completion	Threshold	Performance	Debug	Protocol-Specific
AXI	YES 16	YES 16	YES 16	YES 16	YES 16	YES 16	AddrMatch YES 16
MNOC	YES 16	YES 16	YES 16	NO 0	NO 0	NO 0	Credit/ Channel/ Stream YES 48
APB	YES 16	YES 16	YES 16	YES 16	YES 16	YES 16	None
ARB	YES 16	YES 16	YES 16	YES 16	YES 16	YES 16	None
CORE	YES 16	YES 16	YES 16	YES 16	YES 16	YES 16	None

25.14 Total Event Codes: 544 defined across all protocols and packet types.

25.15 Revision History

Monitor Bus Architecture and Event Code Organization Revision History

Version	Date	Author	Description
0.90	2025-11-22	seang	Initial block specification
0.91	2026-01-02	seang	Added table captions and figure numbers

Last Updated: 2026-01-02

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26 STREAM Register Map

Scope of this chapter. What follows is the narrative view: the register groups, what they are for, and how a host drives them. It is NOT the complete map and must not be read as one – it hand-lists 47 of the 140 registers the RDL defines, and the 93 it omits are not obscure: they are the whole monitor programming surface the board campaigns depend on (the DAXMON/RDMON/WRMON perf-window banks, the address-range checker, MON_GROUP routing, OBS_, *HIST_*, KICK_ENABLE, CH_STATEn).

The authoritative map is generated from `rtl/macro/stream_regs.rdl` into `regs/generated/docs/stream_regs.md` and is stitched into this document immediately after this chapter. Every address in the narrative below was verified against that generated map (47/47 correct); when they disagree in future, the generated one is right.

26.1 Overview

This chapter documents the 8-channel map (`stream_regs`, used by `stream_top_ch8`). The 16-channel map (`stream_regs_16ch`, used by `stream_top_ch16`) differs only where a channel count shows; see 16-Channel Register Map at the end of the chapter.

The STREAM DMA engine register interface consists of three distinct regions:

1. **Channel Kick-off Registers** (0x000 - 0x03F) - Direct routing to descriptor engines
2. **Configuration and Status Registers** (0x100 - 0x3FF) - PeakRDL-generated base register file
3. **Monitor / Performance Registers** (0x1000+) - Separate `stream_mon_regs` regfile instantiated at 0x1000 under the same addrmap (one APB slave)

Note: The monitor block lives in a relocatable `stream_mon_regs` regfile instantiated at offset **0x1000** within the same PeakRDL addrmap and served by the same APB slave. The APB address decode must therefore be **at least 13 bits (8 KB address space)** so the monitor registers at 0x1000+ are reachable (the generated register file uses `s_cpuif_addr[12:0]`).

26.2 Address Space Layout

```
Base Address (configurable parameter)
│
├─ 0x000 - 0x03F: Channel descriptor addresses (staged; see
KICK_ENABLE)
│   ├─ 0x000: CH0_CTRL_LOW  - Channel 0 descriptor address [31:0]
│   ├─ 0x004: CH0_CTRL_HIGH - Channel 0 descriptor address [63:32]
│   ├─ 0x008: CH1_CTRL_LOW  - Channel 1 descriptor address [31:0]
│   ├─ 0x00C: CH1_CTRL_HIGH - Channel 1 descriptor address [63:32]
│   ├─ 0x010: CH2_CTRL_LOW  - Channel 2 descriptor address [31:0]
│   ├─ 0x014: CH2_CTRL_HIGH - Channel 2 descriptor address [63:32]
│   ├─ 0x018: CH3_CTRL_LOW  - Channel 3 descriptor address [31:0]
│   ├─ 0x01C: CH3_CTRL_HIGH - Channel 3 descriptor address [63:32]
│   ├─ 0x020: CH4_CTRL_LOW  - Channel 4 descriptor address [31:0]
│   ├─ 0x024: CH4_CTRL_HIGH - Channel 4 descriptor address [63:32]
│   ├─ 0x028: CH5_CTRL_LOW  - Channel 5 descriptor address [31:0]
│   ├─ 0x02C: CH5_CTRL_HIGH - Channel 5 descriptor address [63:32]
│   ├─ 0x030: CH6_CTRL_LOW  - Channel 6 descriptor address [31:0]
│   ├─ 0x034: CH6_CTRL_HIGH - Channel 6 descriptor address [63:32]
│   ├─ 0x038: CH7_CTRL_LOW  - Channel 7 descriptor address [31:0]
│   └─ 0x03C: CH7_CTRL_HIGH - Channel 7 descriptor address [63:32]
│
├─ 0x040 - 0x07F: CH8..CH15_CTRL_{LOW,HIGH} (16-channel map only;
reserved in the 8-channel map)
│
├─ 0x128: KICK_ENABLE - write bit[ch]=1 to launch that channel (self-
clearing)
├─ 0x080 - 0x0FF: Reserved
│
├─ 0x100 - 0x3FF: Configuration and Status Registers (base regfile)
│   ├─ 0x100 - 0x11F: Global Control and Status
│   ├─ 0x120 - 0x13F: Per-Channel Control
│   ├─ 0x140 - 0x16F: Per-Channel Status
│   ├─ 0x170 - 0x17F: Engine Completion and Error Status
│   ├─ 0x200 - 0x21F: Scheduler Configuration (incl.
SCHED_TIMEOUT_LIMIT @ 0x208)
│   ├─ 0x220 - 0x23F: Descriptor Engine Configuration
│   ├─ 0x2A0 - 0x2AF: AXI Transfer Configuration
│   ├─ 0x2B0 - 0x2BF: Performance Profiler Configuration
│   ├─ 0x2C0 - 0x2CF: Channel Observation
(OBS_CTRL/OBS_FLAGS/OBS_DATA0/1)
│   └─ 0x2D0 - 0x2DF: Performance Profiler Data and Status
│
└─ 0x1000+: Monitor / Performance Registers (stream_mon_regs regfile,
same APB slave)
    ├─ 0x1000 - 0x1007: Monitor FIFO Status / Count
    ├─ 0x10C0 - 0x10DF: Descriptor AXI Monitor Configuration
```

```
├─ 0x10E0 - 0x10FF: Read Engine AXI Monitor Configuration
├─ 0x1100 - 0x111F: Write Engine AXI Monitor Configuration
└─ 0x1150 - 0x11F4: Per-Monitor Performance Counters
```

26.3 Register Details

26.3.1 Channel Descriptor Address Registers (0x000 - 0x03F)

Ordinary read/write storage. A write STAGES an address; it does **not** start a transfer. Launch is a separate write to KICK_ENABLE (0x128).

This changed: these were previously write-only address-map placeholders with no storage, and the write itself kicked – decoded off the raw APB command stream by a separate kick block, which sat beside the register block rather than in it. Two consequences drove the change. The descriptor address existed nowhere as readable state, so it could not be verified before launch. And because the write *was* the launch, starting N channels cost N APB transactions: over UART that is milliseconds between the first and last channel, long enough to bias any cross-channel measurement. Staging the addresses first and then writing KICK_ENABLE once starts every channel on the same cycle.

Note: Descriptor addresses are 64-bit (ADDR_WIDTH parameter, default 64). On 32-bit APB bus, each channel requires TWO registers (LOW/HIGH).

Channel Kick-off Registers

Offset	Register	Type	Reset	Description
0x000	CH0_CTRL_LOW	RW	0x0	Channel 0 descriptor address [31:0] (staged)
0x004	CH0_CTRL_HIGH	WO	N/A	Channel 0 descriptor address [63:32]
0x008	CH1_CTRL_LOW	WO	N/A	Channel 1 descriptor address [31:0]
0x00C	CH1_CTRL_HIGH	WO	N/A	Channel 1 descriptor address [63:32]
0x010	CH2_CTRL_LOW	WO	N/A	Channel 2 descriptor address [31:0]
0x014	CH2_CTRL_HIGH	WO	N/A	Channel 2 descriptor address [63:32]
0x018	CH3_CTRL_LOW	WO	N/A	Channel 3 descriptor address [31:0]
0x01C	CH3_CTRL_HIGH	WO	N/A	Channel 3 descriptor address [63:32]

Offset	Register	Type	Reset	Description
0x020	CH4_CTRL_LOW	WO	N/A	Channel 4 descriptor address [31:0]
0x024	CH4_CTRL_HIGH	WO	N/A	Channel 4 descriptor address [63:32]
0x028	CH5_CTRL_LOW	WO	N/A	Channel 5 descriptor address [31:0]
0x02C	CH5_CTRL_HIGH	WO	N/A	Channel 5 descriptor address [63:32]
0x030	CH6_CTRL_LOW	WO	N/A	Channel 6 descriptor address [31:0]
0x034	CH6_CTRL_HIGH	WO	N/A	Channel 6 descriptor address [63:32]
0x038	CH7_CTRL_LOW	WO	N/A	Channel 7 descriptor address [31:0]
0x03C	CH7_CTRL_HIGH	WO	N/A	Channel 7 descriptor address [63:32]

Write Behavior: - Descriptor address is 64-bit, split across LOW and HIGH registers - Write to HIGH register triggers descriptor engine kick-off - LOW register write is buffered, HIGH register write initiates transfer - Both registers must be written in order (LOW then HIGH) - Write blocks until descriptor engine accepts (back-pressure) - Read not supported (returns error)

Example:

```
// Start DMA transfer on channel 0
// Descriptor at physical address 0x0000_0001_8000_0000 (64-bit)

// Write lower 32 bits first (buffered)
write32(BASE + CH0_CTRL_LOW, 0x8000_0000);

// Write upper 32 bits second (triggers kick-off)
write32(BASE + CH0_CTRL_HIGH, 0x0000_0001);  // Blocks until accepted

// For descriptors in lower 4GB (typical case):
write32(BASE + CH0_CTRL_LOW, 0x8000_0000);
write32(BASE + CH0_CTRL_HIGH, 0x0000_0000);  // Upper 32 bits = 0
```

26.3.2 Global Control and Status (0x100 - 0x11F)

26.3.2.1 GLOBAL_CTRL (0x100)

Master control register for entire STREAM engine.

GLOBAL_CTRL

Bits	Field	Type	Reset	Description
31:2	Reserved	RO	0x0	Reserved
1	GLOBAL_RST	RW	0	Global reset (self-clearing)
0	GLOBAL_EN	RW	0	Global enable (1=enabled, 0=disabled)

Usage:

```
// Enable STREAM engine
write32(BASE + GLOBAL_CTRL, 0x1);

// Reset all channels
write32(BASE + GLOBAL_CTRL, 0x3);  // Set both EN and RST
// RST self-clears after one cycle
```

26.3.2.2 GLOBAL_STATUS (0x104)

Overall system status.

GLOBAL_STATUS

Bits	Field	Type	Description
31:1	Reserved	RO	Reserved
0	SYSTEM_IDLE	RO	System idle (all channels idle)

26.3.2.3 VERSION (0x108)

Version and configuration information (read-only).

VERSION

Bits	Field	Type	Value	Description
31:24	Reserved	RO	0x00	Reserved
23:16	NUM_CHANNELS	RO	0x08	Number of channels (8)
15:8	MAJOR	RO	0x00	Major version (0)

Bits	Field	Type	Value	Description
7:0	MINOR	RO	0x5A	Minor version (90 decimal = 0.90)

26.3.3 Per-Channel Control and Status (0x120 - 0x17F)

26.3.3.1 CHANNEL_ENABLE (0x120)

Per-channel enable control (bit vector).

CHANNEL_ENABLE

Bits	Field	Type	Reset	Description
31:8	Reserved	RO	0x0	Reserved
7:0	CH_EN	RW	0x00	Channel enable [7:0] (1=enabled)

Usage:

```
// Enable channels 0, 1, 2
write32(BASE + CHANNEL_ENABLE, 0x07);

// Disable channel 1, keep others
uint32_t val = read32(BASE + CHANNEL_ENABLE);
val &= ~(1 << 1);  // Clear bit 1
write32(BASE + CHANNEL_ENABLE, val);
```

26.3.3.2 CHANNEL_RESET (0x124)

Per-channel reset control (bit vector, self-clearing).

CHANNEL_RESET

Bits	Field	Type	Reset	Description
31:8	Reserved	RO	0x0	Reserved
7:0	CH_RST	RW	0x00	Channel reset [7:0] (write 1 to reset)

Usage:

```
// Reset channels 0 and 3
write32(BASE + CHANNEL_RESET, 0x09);  // Bits 0 and 3
// Self-clears after reset completes
```

26.3.3.3 CHANNEL_IDLE (0x140)

Per-channel idle status (bit vector, read-only).

CHANNEL_IDLE

Bits	Field	Type	Description
31:8	Reserved	RO	Reserved
7:0	CH_IDLE	RO	Channel idle [7:0] (1=idle, 0=active)

26.3.3.4 DESC_ENGINE_IDLE (0x144)

Per-channel descriptor engine idle status.

DESC_ENGINE_IDLE

Bits	Field	Type	Description
31:8	Reserved	RO	Reserved
7:0	DESC_IDLE	RO	Descriptor engine idle [7:0] (1=idle, 0=active)

26.3.3.5 SCHEDULER_IDLE (0x148)

Per-channel scheduler idle status.

SCHEDULER_IDLE

Bits	Field	Type	Description
31:8	Reserved	RO	Reserved
7:0	SCHED_IDLE	RO	Scheduler idle [7:0] (1=idle, 0=active)

26.3.3.6 CH_STATE[0..7] (0x150 - 0x16C)

Per-channel scheduler FSM state (8 registers, stride 0x4).

CH_STATE[0..7]

Offset	Register	Bits	Field	Type	Description
0x150	CH_STATE0_STATE	31:7	Reserved	RO	Reserved
		6:0	STATE	RO	Channel 0 scheduler state (one-hot)
0x154	CH_STATE1_STATE	6:0	STATE	RO	Channel 1 scheduler state (one-hot)
0x158	CH_STATE2_STATE	6:0	STATE	RO	Channel 2 scheduler state (one-hot)
0x15C	CH_STATE	6:0	STATE	RO	Channel 3 scheduler state (one-

Offset	Register	Bits	Field	Type	Description
	3_STATE				hot)
0x160	CH_STATE 4_STATE	6:0	STATE	RO	Channel 4 scheduler state (one-hot)
0x164	CH_STATE 5_STATE	6:0	STATE	RO	Channel 5 scheduler state (one-hot)
0x168	CH_STATE 6_STATE	6:0	STATE	RO	Channel 6 scheduler state (one-hot)
0x16C	CH_STATE 7_STATE	6:0	STATE	RO	Channel 7 scheduler state (one-hot)

State Encoding (One-Hot):

```
Bit 0 (0x01) = CH_IDLE         - Channel idle, waiting for descriptor
Bit 1 (0x02) = CH_FETCH_DESC   - Fetching descriptor from memory
Bit 2 (0x04) = CH_XFER_DATA    - Concurrent read AND write transfer
Bit 3 (0x08) = CH_COMPLETE     - Transfer complete
Bit 4 (0x10) = CH_NEXT_DESC    - Fetching next chained descriptor
Bit 5 (0x20) = CH_ERROR        - Error state
Bit 6 (0x40) = CH_RESERVED     - Reserved for future use
```

Note: Only ONE bit should be set at a time (one-hot encoding). Multiple bits set indicates a logic error.

26.3.4 Engine Completion and Error Status (0x170 - 0x17F)

26.3.4.1 SCHED_ERROR (0x170)

Per-channel scheduler error flags.

SCHED_ERROR

Bits	Field	Type	Description
31:8	Reserved	RO	Reserved
7:0	SCHED_ER R	RO	Scheduler error bits [7:0] (1=error, 0=no error)

26.3.4.2 AXI_RD_COMPLETE (0x174)

Per-channel AXI read engine completion status.

AXI_RD_COMPLETE

Bits	Field	Type	Description
31:8	Reserved	RO	Reserved
7:0	RD_COMPLETE	RO	Read completion bits [7:0] (1=complete, 0=pending)

26.3.4.3 AXI_WR_COMPLETE (0x178)

Per-channel AXI write engine completion status.

AXI_WR_COMPLETE

Bits	Field	Type	Description
31:8	Reserved	RO	Reserved
7:0	WR_COMPLETE	RO	Write completion bits [7:0] (1=complete, 0=pending)

26.3.5 Monitor FIFO Status (0x1000 - 0x1007)

`USE_AXI_MONITORS` gates the monitor-lite logic and the latency histograms, but not the bus meters. Some registers in this block are therefore active only when `USE_AXI_MONITORS=1`: the DAXMON / RDMON / WRMON enable / mask / timeout configuration and the monitor-group window (`MON_GROUP_*`). The latency histograms (`HIST_DATA` / `HIST_TOTAL`, outside this window) are also built only with monitors and read 0 otherwise. The FIFO status registers report the monitor error FIFO, which a monitors-off build does not have; there they read `MON_FIFO_STATUS = 0x2` (EMPTY) and `MON_FIFO_COUNT = 0`. The **datapath perf-window registers** are populated by the in-core `axi_bus_meter` instances, which are always built. These are the RDMON / WRMON `PERF_CTRL` registers and their `PROD/BP/STARV/IDLE_CYCLES`, `BEAT/BYTE/BURST_COUNT` and `WINDOW_*` readbacks. The DAXMON `PERF_*` readbacks read 0 in every build, because the descriptor monitor-lite has no perf window.

`USE_MON_REGS` defaults to 1, independent of `USE_AXI_MONITORS`, so a default monitors-off build answers the whole MON window and the meters can be armed and read. On such a build the monitor configuration registers accept writes and read them back but drive nothing. Setting `USE_MON_REGS = 0` straps the monitor configuration outputs off and makes the whole MON window answer every access with `PSLVERR`, so “not built” is distinguishable from “set to zero”; the meters are then unreachable.

Monitor register block relocation (v0.93): All monitor and per-monitor performance registers now live in a separate `stream_mon_regs` regfile instantiated at **0x1000** within the same PeakRDL addrmap (served by the same APB slave). Addresses below reflect the new base; the previous layout placed these registers at 0x180-0x2FF. The mapping is `new = old - 0x180 + 0x1000`.

26.3.5.1 MON_FIFO_STATUS (0x1000)

Monitor bus FIFO status indicators. These report the monitor error/interrupt FIFO (the one drained over `s_axil_err`). The monitor group exposes no overflow or underflow flag, so `MON_FIFO_OVFL` and `MON_FIFO_UNFL` read 0. With `USE_AXI_MONITORS = 0` the register reads 0x2 (EMPTY).

MON_FIFO_STATUS

Bits	Field	Type	Description
31:4	Reserved	RO	Reserved
3	MON_FIFO_UNFL	RO	FIFO underflow detected (1=error)
2	MON_FIFO_OVFL	RO	FIFO overflow detected (1=error)
1	MON_FIFO_EMPTY	RO	FIFO empty (1=empty, 0=data available)
0	MON_FIFO_FULL	RO	FIFO full (1=full, 0=space available)

26.3.5.2 MON_FIFO_COUNT (0x1004)

Monitor bus FIFO entry count.

MON_FIFO_COUNT

Bits	Field	Type	Description
31:16	Reserved	RO	Reserved
15:0	FIFO_COUNT	RO	Number of entries in FIFO [15:0]

26.3.6 Scheduler Configuration (0x200 - 0x21F)

26.3.6.1 SCHED_TIMEOUT_CYCLES (0x200)

Timeout threshold for scheduler (global for all channels).

SCHED_TIMEOUT_CYCLES

Bits	Field	Type	Reset	Description
31:0	TIMEOUT_CYCLES	RW	1000	Timeout in clock cycles (32-bit)

26.3.6.2 SCHED_CONFIG (0x204)

Scheduler feature enables (global for all channels).

SCHED_CONFIG

Bits	Field	Type	Reset	Description
31:6	Reserved	RO	0x0	Reserved
5	RD_PREFETCH_EN	RW	1	Read-ahead descriptor prefetch (cross-descriptor streaming)
4	PERF_EN	RW	0	Performance monitoring enable
3	COMPL_EN	RW	1	Completion reporting enable
2	ERR_EN	RW	1	Error reporting enable
1	TIMEOUT_EN	RW	1	Timeout detection enable
0	SCHED_EN	RW	1	Scheduler enable

`RD_PREFETCH_EN` (default **enabled**) turns on read-ahead descriptor prefetch: on a chained legacy descriptor the scheduler read side loads the next descriptor from the descriptor-FIFO head and keeps filling the SRAM while the write side drains the current descriptor, collapsing the per-descriptor boundary bubble to zero. Clear it for lockstep behaviour (useful for an on-silicon A/B on one bitstream). Ignored for EXT (row/col-major) descriptors, which always run lockstep. See the Scheduler block (Section 2.4) and HAS Section 5.1.

26.3.6.3 SCHED_TIMEOUT_LIMIT (0x208)

Consecutive write-progress-timeout windows before the scheduler escalates a channel to a sticky `CH_ERROR` (global for all channels).

SCHED_TIMEOUT_LIMIT

Bits	Field	Type	Reset	Description
31:8	Reserved	RO	0x0	Reserved
7:0	LIMIT	RW	0x4	Consecutive timeout windows before fatal escalation (0 = never escalate)

Behavior: The write-progress timeout is a recoverable liveness fault. It re-arms each window and only escalates to a fatal, sticky `CH_ERROR` after `LIMIT` consecutive windows with no write progress. `LIMIT = 0` disables escalation (pure soft timeout). Any write progress clears the strike counter. See `stream_mas/ch02_blocks/04_scheduler.md`.

26.3.7 Descriptor Engine Configuration (0x220 - 0x23F)

26.3.7.1 DESCENG_CONFIG (0x220)

Descriptor engine feature enables (global for all channels).

DESCENG_CONFIG

Bits	Field	Type	Reset	Description
31:6	Reserved	RO	0x0	Reserved
5:2	FIFO_THRESH	RW	0x8	Prefetch FIFO threshold (4 bits)
1	PREFETCH_EN	RW	0	Prefetch enable
0	DESCENG_EN	RW	1	Descriptor engine enable

26.3.7.2 DESCENG_ADDR0_BASE (0x224)

Base address for descriptor address range 0 (lower 32 bits).

DESCENG_ADDR0_BASE

Bits	Field	Type	Reset	Description
31:0	ADDR0_BASE	RW	0x00000000	Address range 0 base

26.3.7.3 DESCENG_ADDR0_LIMIT (0x228)

Limit address for descriptor address range 0 (lower 32 bits).

DESCENG_ADDR0_LIMIT

Bits	Field	Type	Reset	Description
31:0	ADDR0_LIMIT	RW	0xFFFFFFFF	Address range 0 limit

26.3.7.4 DESCENG_ADDR1_BASE (0x22C)

Base address for descriptor address range 1 (lower 32 bits).

DESCENG_ADDR1_BASE

Bits	Field	Type	Reset	Description
31:0	ADDR1_BASE	RW	0x00000000	Address range 1 base

26.3.7.5 DESCENG_ADDR1_LIMIT (0x230)

Limit address for descriptor address range 1 (lower 32 bits).

DESCENG_ADDR1_LIMIT

Bits	Field	Type	Reset	Description
31:0	ADDR1_LIMIT	RW	0xFFFFFFFF	Address range 1 limit

26.3.8 Descriptor AXI Monitor Configuration (0x10C0 - 0x10DF)

26.3.8.1 DAXMON_ENABLE (0x10C0)

Descriptor AXI master monitor enable controls.

DAXMON_ENABLE

Bits	Field	Type	Reset	Description
31:5	Reserved	RO	0x0	Reserved
4	PERF_EN	RW	0	Performance packet enable
3	TIMEOUT_EN	RW	1	Timeout detection enable
2	COMPL_EN	RW	0	Completion packet enable
1	ERR_EN	RW	1	Error detection enable
0	MON_EN	RW	1	Master enable for descriptor monitor

26.3.8.2 DAXMON_TIMEOUT (0x10C4)

Descriptor AXI monitor timeout threshold.

DAXMON_TIMEOUT

Bits	Field	Type	Reset	Description
31:0	TIMEOUT_CYCLES	RW	10000	Timeout in clock cycles

26.3.8.3 DAXMON_LATENCY_THRESH (0x10C8)

Descriptor AXI monitor latency threshold.

DAXMON_LATENCY_THRESH

Bits	Field	Type	Reset	Description
31:0	LATENCY_THRESH	RW	5000	Latency threshold in clock cycles

26.3.8.4 DAXMON_PKT_MASK (0x10CC)

Descriptor AXI monitor packet type filtering.

DAXMON_PKT_MASK

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:0	PKT_MASK	RW	0xFFFF	Packet type drop mask: bit[type]=1 masks (drops) that type; reset all-masked

Scope note (stream BUG-017, 2026-09-28): this mask acts inside DAXMON. At the monbus group, which filters by protocol and sees every in-core packet as AXI, the effective drop mask is `DAXMON_PKT_MASK & RDMON_PKT_MASK & WRMON_PKT_MASK` (dropped only if all three drop), so switching one monitor off with a drop-all mask no longer silences the other two. Before this note the group used `DAXMON_PKT_MASK` alone for every in-core packet.

26.3.8.5 DAXMON_ERR_CFG (0x10D0)

Descriptor AXI monitor error selection and filtering.

DAXMON_ERR_CFG

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:8	ERR_MASK	RW	0xFF	Error type filtering mask
7:4	Reserved	RO	0x0	Reserved
3:0	ERR_SELECT	RW	0x0	Error type selection

26.3.8.6 DAXMON_MASK1 (0x10D4)

Descriptor AXI monitor timeout and completion masks.

DAXMON_MASK1

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:8	COMPL_MASK	RW	0x00	Completion mask
7:0	TIMEOUT_MASK	RW	0xFF	Timeout mask

26.3.8.7 DAXMON_MASK2 (0x10D8)

Descriptor AXI monitor threshold and performance masks.

DAXMON_MASK2

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:8	PERF_MASK	RW	0x00	Performance mask
7:0	THRESH_MASK	RW	0xFF	Threshold mask

26.3.8.8 DAXMON_MASK3 (0x10DC)

Descriptor AXI monitor address and debug masks.

DAXMON_MASK3

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:8	DEBUG_MASK	RW	0x00	Debug mask
7:0	ADDR_MASK	RW	0xFF	Address mask

26.3.9 Read Engine AXI Monitor Configuration (0x10E0 - 0x10FF)

26.3.9.1 RDMON_ENABLE (0x10E0)

Read engine AXI master monitor enable controls.

RDMON_ENABLE

Bits	Field	Type	Reset	Description
31:5	Reserved	RO	0x0	Reserved
4	PERF_EN	RW	0	Performance packet enable
3	TIMEOUT_EN	RW	1	Timeout detection enable
2	COMPL_EN	RW	0	Completion packet enable
1	ERR_EN	RW	1	Error detection enable
0	MON_EN	RW	1	Master enable for read monitor

26.3.9.2 RDMON_TIMEOUT (0x10E4)

Read engine AXI monitor timeout threshold.

RDMON_TIMEOUT

Bits	Field	Type	Reset	Description
31:0	TIMEOUT_CYCLES	RW	10000	Timeout in clock cycles

26.3.9.3 RDMON_LATENCY_THRESH (0x10E8)

Read engine AXI monitor latency threshold.

RDMON_LATENCY_THRESH

Bits	Field	Type	Reset	Description
31:0	LATENCY_THRESH	RW	5000	Latency threshold in clock cycles

26.3.9.4 RDMON_PKT_MASK (0x10EC)

Read engine AXI monitor packet type filtering.

RDMON_PKT_MASK

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:0	PKT_MASK	RW	0xFFFF	Packet type drop mask: bit[type]=1 masks (drops) that type; reset all-masked

26.3.9.5 RDMON_ERR_CFG (0x10F0)

Read engine AXI monitor error selection and filtering.

RDMON_ERR_CFG

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:8	ERR_MASK	RW	0xFF	Error type filtering mask
7:4	Reserved	RO	0x0	Reserved
3:0	ERR_SELECT	RW	0x0	Error type selection

26.3.9.6 RDMON_MASK1 (0x10F4)

Read engine AXI monitor timeout and completion masks.

RDMON_MASK1

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:8	COMPL_MASK	RW	0x00	Completion mask
7:0	TIMEOUT_MASK	RW	0xFF	Timeout mask

26.3.9.7 RDMON_MASK2 (0x10F8)

Read engine AXI monitor threshold and performance masks.

RDMON_MASK2

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:8	PERF_MASK	RW	0x00	Performance mask
7:0	THRESH_MASK	RW	0xFF	Threshold mask

26.3.9.8 RDMON_MASK3 (0x10FC)

Read engine AXI monitor address and debug masks.

RDMON_MASK3

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:8	DEBUG_MASK	RW	0x00	Debug mask
7:0	ADDR_MASK	RW	0xFF	Address mask

26.3.10 Write Engine AXI Monitor Configuration (0x1100 - 0x111F)

26.3.10.1 WRMON_ENABLE (0x1100)

Write engine AXI master monitor enable controls.

WRMON_ENABLE

Bits	Field	Type	Reset	Description
31:5	Reserved	RO	0x0	Reserved
4	PERF_EN	RW	0	Performance packet enable
3	TIMEOUT_EN	RW	1	Timeout detection enable
2	COMPL_EN	RW	0	Completion packet enable
1	ERR_EN	RW	1	Error detection enable
0	MON_EN	RW	1	Master enable for write monitor

26.3.10.2 WRMON_TIMEOUT (0x1104)

Write engine AXI monitor timeout threshold.

WRMON_TIMEOUT

Bits	Field	Type	Reset	Description
31:0	TIMEOUT_CYCLES	RW	10000	Timeout in clock cycles

26.3.10.3 WRMON_LATENCY_THRESH (0x1108)

Write engine AXI monitor latency threshold.

WRMON_LATENCY_THRESH

Bits	Field	Type	Reset	Description
31:0	LATENCY_THRESH	RW	5000	Latency threshold in clock cycles

26.3.10.4 WRMON_PKT_MASK (0x110C)

Write engine AXI monitor packet type filtering.

WRMON_PKT_MASK

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:0	PKT_MASK	RW	0xFFFF	Packet type drop mask: bit[type]=1 masks (drops) that type; reset all-masked

26.3.10.5 WRMON_ERR_CFG (0x1110)

Write engine AXI monitor error selection and filtering.

WRMON_ERR_CFG

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:8	ERR_MASK	RW	0xFF	Error type filtering mask
7:4	Reserved	RO	0x0	Reserved
3:0	ERR_SELECT	RW	0x0	Error type selection

26.3.10.6 WRMON_MASK1 (0x1114)

Write engine AXI monitor timeout and completion masks.

WRMON_MASK1

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:8	COMPL_MASK	RW	0x00	Completion mask
7:0	TIMEOUT_MASK	RW	0xFF	Timeout mask

26.3.10.7 WRMON_MASK2 (0x1118)

Write engine AXI monitor threshold and performance masks.

WRMON_MASK2

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:8	PERF_MASK	RW	0x00	Performance mask
7:0	THRESH_MASK	RW	0xFF	Threshold mask

26.3.10.8 WRMON_MASK3 (0x111C)

Write engine AXI monitor address and debug masks.

WRMON_MASK3

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:8	DEBUG_MASK	RW	0x00	Debug mask
7:0	ADDR_MASK	RW	0xFF	Address mask

26.3.11 AXI Transfer Configuration (0x2A0 - 0x2AF)

26.3.11.1 AXI_XFER_CONFIG (0x2A0)

AXI read and write transfer burst sizes.

AXI_XFER_CONFIG

Bits	Field	Type	Reset	Description
31:16	Reserved	RO	0x0	Reserved
15:8	WR_XFER_BEATS	RW	15	AXI write transfer beats (AWLEN: 0-255 = 1-256)
7:0	RD_XFER_BEATS	RW	15	AXI read transfer beats (ARLEN: 0-255 = 1-256)

Usage:

```
// Configure for 16-beat bursts (default)
write32(BASE + AXI_XFER_CONFIG, 0x0F0F);

// Configure for 64-beat bursts
write32(BASE + AXI_XFER_CONFIG, 0x3F3F);

// Configure for maximum 256-beat bursts
write32(BASE + AXI_XFER_CONFIG, 0xFFFF);
```

26.3.12 Performance Profiler Configuration (0x2B0 - 0x2BF)

26.3.12.1 PERF_CONFIG (0x2B0)

Performance profiler enable and mode controls.

PERF_CONFIG

Bits	Field	Type	Reset	Description
31:3	Reserved	RO	0x0	Reserved
2	PERF_CLEAR	RW	0	Clear counters (write 1, self-clearing)
1	PERF_MODE	RW	0	Mode: 0=count, 1=histogram
0	PERF_EN	RW	0	Performance profiler enable

Usage:

```
// Enable performance profiler in count mode
write32(BASE + PERF_CONFIG, 0x01);

// Enable in histogram mode
write32(BASE + PERF_CONFIG, 0x03);

// Clear counters
write32(BASE + PERF_CONFIG, 0x05);  // EN + CLEAR
// PERF_CLEAR self-clears after one cycle
```

26.4 Typical Usage Flow

26.4.1 Initialization

```
// 1. Global enable
write32(BASE + GLOBAL_CTRL, 0x1);

// 2. Configure scheduler
write32(BASE + SCHED_TIMEOUT_CYCLES, 10000);
write32(BASE + SCHED_CONFIG, 0x1F);  // All features enabled

// 3. Configure descriptor engine
write32(BASE + DESCENG_CONFIG, 0x01);  // Enable, no prefetch
write32(BASE + DESCENG_ADDR0_BASE, 0x8000_0000);
write32(BASE + DESCENG_ADDR0_LIMIT, 0x8FFF_FFFF);

// 4. Configure monitors (minimal reporting)
write32(BASE + DAXMON_ENABLE, 0x05);  // Error + timeout only
write32(BASE + RDMON_ENABLE, 0x05);
write32(BASE + WRMON_ENABLE, 0x05);

// 5. Enable desired channels
write32(BASE + CHANNEL_ENABLE, 0xFF);  // All 8 channels
```

26.4.2 Start Transfer

```
// Write 64-bit descriptor address to channel kick-off registers
// Descriptor at address 0x0000_0000_8000_0100 (64-bit)
write32(BASE + CH0_CTRL_LOW, 0x8000_0100);   // Lower 32 bits
(buffered)
write32(BASE + CH0_CTRL_HIGH, 0x0000_0000);  // Upper 32 bits
(triggers kick-off)
// Transfer starts immediately (blocks until descriptor engine ready)
```

26.4.3 Poll for Completion

```
// Check channel 0 idle status
while (!(read32(BASE + CHANNEL_IDLE) & 0x01)) {
    // Wait for channel 0 to become idle
}

// Or check scheduler state (one-hot encoding)
while ((read32(BASE + CH_STATE0_STATE) & 0x7F) != 0x01) {
    // Wait for CH_IDLE state (bit 0 = 0x01)
}
```

26.4.4 Error Handling

```
// Check all channel states for errors (one-hot encoding)
for (int ch = 0; ch < 8; ch++) {
    uint32_t state = read32(BASE + CH_STATE0_STATE + (ch * 4)) & 0x7F;
    if (state & 0x20) {  // CH_ERROR (bit 5)
        // Reset channel
        write32(BASE + CHANNEL_RESET, 1 << ch);
    }
}
```

26.5 Register Summary Table

Register Summary Table

Offset Range	Description	Count	Type
0x000-0x03F	Channel kick-off registers (LOW/HIGH)	16	Write-routing
0x100-0x11F	Global control and status	3	RW/RO
0x120-0x13F	Per-channel control	2	RW
0x140-0x16F	Per-channel status	11	RO
0x170-0x17F	Engine completion and error status	3	RO
0x200-0x21F	Scheduler configuration	3	RW
0x220-0x23F	Descriptor engine configuration	5	RW
0x2A0-0x2AF	AXI transfer configuration	1	RW
0x2B0-0x2BF	Performance profiler configuration	1	RW
0x2C0-0x2CF	Channel observation	4	RW/RO
0x2D0-0x2DF	Performance profiler data and status	3	RO
0x1000-0x1007	Monitor FIFO status (mon regfile)	2	RO
0x10C0-0x10DF	Descriptor AXI monitor configuration	8	RW
0x10E0-0x10FF	Read engine AXI monitor configuration	8	RW
0x1100-0x111F	Write engine AXI monitor configuration	8	RW

Note: The monitor block (0x1000+) resides in the separate `stream_mon_regs` regfile on the same APB slave; per-monitor performance counters (0x1150-0x11F4) are additionally present but omitted from this summary. Base config/status (0x100-0x3FF) also includes PERF_CH_SEL and the HIST_* histogram registers.

26.6 PeakRDL Generation

To generate SystemVerilog from the register definition:

```
cd projects/components/dma-ip/stream/rtl/macro/
../../../../../../bin/peakrdl_generate.py stream_regs.rdl
```

(Always regenerate via `bin/peakrdl_generate.py` - it emits RTL, docs, and the Python regmap in lockstep; raw `peakrdl regblock` desyncs the regmap.)

This generates: - `stream_regs_pkg.sv` - Register definitions package - `stream_regs.sv` - APB slave register interface

---

26.6.1 Channel Observation Registers (0x2C0 - 0x2CF)

New in v0.92 — Enables software to observe scheduler state without halting transfers (commit 4467554e).

Channel Observation Registers

Offset	Register	Type	Reset	Description
0x2C0	OBS_CTRL	RW	0x0	CH_SEL [2:0] = channel index, CAT_SEL [4:3] = data category (16-channel map: CH_SEL [3:0], CAT_SEL [5:4])
0x2C4	OBS_FLAGS	RO	0x0	Observation flags for selected channel (error sticky, timeout)
0x2C8	OBS_DATA0	RO	0x0	Per-channel diagnostic data 0 (low 32 bits)
0x2CC	OBS_DATA1	RO	0x0	Per-channel diagnostic data 1 (high 32 bits)

Usage: 1. Write channel index to OBS_CTRL.CH_SEL ([2:0], channels 0-7; [3:0], channels 0-15 in the 16-channel map) 2. Read OBS_FLAGS, OBS_DATA0, OBS_DATA1 (muxed views of selected channel) 3. OBS_FLAGS[0] = `SCH_ERROR_STICKY` (persists until cfg_channel_reset) 4. OBS_FLAGS[1] = `SCH_TIMEOUT` (current timeout state)

Implementation: `stream_config_block.sv` implements the NUM_CHANNELS:1 mux for per-channel state observation (commit 0c979c35 exposes scheduler error/timeout signals into OBS_FLAGS).

26.7 16-Channel Register Map

`stream_top_ch16` uses `stream_regs_16ch` and `stream_mon_regs_16ch`, which `bin/gen_stream_regs_16ch.py` derives from the 8-channel RDL (`rtl/macro/generated_16ch/`). Never edit them by hand: edit the 8-channel RDL, rerun the script, then regenerate both regblocks with `bin/peakrdl_generate.py`. `bin/check_rdl_regen.py` checks both maps. Software uses `rtl/stream_regmap_16ch.py` (registers by name), so the moved offsets below do not reach test code.

Everything not listed here is identical to the 8-channel map.

16-Channel Register Map Differences

Register	8-channel map	16-channel map
CH8..CH15_CTRL_{LOW,HIGH}	reserved (0x040-0x07F)	0x040-0x07C, same layout as CH0..CH7
KICK_ENABLE (0x128)	KICK0..KICK7	KICK0..KICK15 (offset unchanged)
CHANNEL_ENABLE, CHANNEL_RESET, CHANNEL_IDLE, DESC_ENGINE_IDLE, SCHEDULER_IDLE, SCHED_ERROR, AXI_RD_COMPLETE, AXI_WR_COMPLETE	fields [7:0]	fields [15:0]
CH_STATE	8 entries at 0x150-0x16C	16 entries at 0x180-0x1BC (16 entries at 0x150 would overrun SCHED_ERROR at 0x170)
VERSION.NUM_CHANNELS	8	16
OBS_CTRL (0x2C0)	CH_SEL [2:0], CAT_SEL [4:3]	CH_SEL [3:0], CAT_SEL [5:4]
PERF_CH_SEL.CH	[2:0]	[3:0]

Register	8-channel map	16-channel map
_SEL (0x35C)		
RDMON_PERF_CH_OVERFLOW_HI / WRMON_PERF_CH_OVERFLOW_HI	not present	0x11F8 / 0x11FC: channels 8..15, 4 bits per channel, channel 8 in the low nibble

Software that reads OBS_DATA must use the CAT_SEL position for its map: the field moved up one bit to make room for the fourth CH_SEL bit.

Revision History:

PeakRDL Generation

Version	Date	Author	Description
1.0	2025-10-20	sean galloway	Initial creation
1.1	2025-12-01	sean galloway	Added complete monitor registers from RDL Added engine completion/error status (0x170) Added monitor FIFO status (0x180) Added AXI transfer config (0x2A0) Added performance profiler config (0x2B0)
1.2	2026-07-02	sean galloway	Monitor block relocated into separate stream_mon_regs regfile at 0x1000 (same APB slave, 13-bit/8 KB decode); added SCHED_TIMEOUT_LIMIT @ 0x208
1.3	2026-10-17	sean galloway	16-channel register map (stream_regs_16ch); OBS_CTRL CH_SEL/CAT_SEL layout per map

26.8 stream_regs address map

- Absolute Address: 0x0
- Base Offset: 0x0
- Size: 0x126C

Configuration and status registers for 8-channel STREAM DMA engine with full monitor control

Offset	Identifier	Name
0x0000	CH0_CTRL_LOW	Ch0 desc addr LOW
0x0004	CH0_CTRL_HIGH	Ch0 desc addr HIGH
0x0008	CH1_CTRL_LOW	Ch1 desc addr LOW
0x000C	CH1_CTRL_HIGH	Ch1 desc addr HIGH
0x0010	CH2_CTRL_LOW	Ch2 desc addr LOW
0x0014	CH2_CTRL_HIGH	Ch2 desc addr HIGH
0x0018	CH3_CTRL_LOW	Ch3 desc addr LOW
0x001C	CH3_CTRL_HIGH	Ch3 desc addr HIGH
0x0020	CH4_CTRL_LOW	Ch4 desc addr LOW
0x0024	CH4_CTRL_HIGH	Ch4 desc addr HIGH
0x0028	CH5_CTRL_LOW	Ch5 desc addr LOW
0x002C	CH5_CTRL_HIGH	Ch5 desc addr HIGH
0x0030	CH6_CTRL_LOW	Ch6 desc addr LOW
0x0034	CH6_CTRL_HIGH	Ch6 desc addr HIGH
0x0038	CH7_CTRL_LOW	Ch7 desc addr LOW
0x003C	CH7_CTRL_HIGH	Ch7 desc addr HIGH
0x0100	GLOBAL_CTRL	Global Control Register
0x0104	GLOBAL_STATUS	Global Status Register
0x0108	VERSION	Version Register
0x0120	CHANNEL_ENABLE	Channel Enable Register
0x0124	CHANNEL_RESET	Channel Reset Register

Offset	Identifier	Name
0x0128	KICK_ENABLE	Kick enable
0x0140	CHANNEL_IDLE	Channel Idle Status
0x0144	DESC_ENGINE_IDLE	Descriptor Engine Idle Status
0x0148	SCHEDULER_IDLE	Scheduler Idle Status
0x0150	CH_STATE[0]	Per-Channel State Registers
0x0154	CH_STATE[1]	Per-Channel State Registers
0x0158	CH_STATE[2]	Per-Channel State Registers
0x015C	CH_STATE[3]	Per-Channel State Registers
0x0160	CH_STATE[4]	Per-Channel State Registers
0x0164	CH_STATE[5]	Per-Channel State Registers
0x0168	CH_STATE[6]	Per-Channel State Registers
0x016C	CH_STATE[7]	Per-Channel State Registers
0x0170	SCHED_ERROR	Scheduler Error Status
0x0174	AXI_RD_COMPLETE	AXI Read Engine Completion Status
0x0178	AXI_WR_COMPLETE	AXI Write Engine Completion Status
0x0200	SCHED_TIMEOUT_CYCLES	Scheduler Timeout Cycles
0x0204	SCHED_CONFIG	Scheduler Configuration
0x0208	SCHED_TIMEOUT_LIMIT	Scheduler Timeout Escalation Limit

Offset	Identifier	Name
0x0220	DESCENG_CONFIG	Descriptor Engine Configuration
0x0224	DESCENG_ADDR0_BASE	Descriptor Address Range 0 Base
0x0228	DESCENG_ADDR0_LIMIT	Descriptor Address Range 0 Limit
0x022C	DESCENG_ADDR1_BASE	Descriptor Address Range 1 Base
0x0230	DESCENG_ADDR1_LIMIT	Descriptor Address Range 1 Limit
0x02A0	AXI_XFER_CONFIG	AXI Transfer Configuration
0x02B0	PERF_CONFIG	Performance Profiler Configuration
0x02C0	OBS_CTRL	Observation Mux Control
0x02C4	OBS_FLAGS	Observation Flags
0x02C8	OBS_DATA0	Observation Data 0
0x02CC	OBS_DATA1	Observation Data 1
0x02D0	PERF_DATA_LOW	Perf FIFO Data Low
0x02D4	PERF_DATA_HIGH	Perf FIFO Data High
0x02D8	PERF_STATUS	Perf FIFO Status
0x035C	PERF_CH_SEL	Per-Channel Perf Bucket Readout Select
0x0378	HIST_SEL	Latency Histogram Readout Select
0x037C	HIST_DATA	Latency Histogram Bin Count
0x0380	HIST_TOTAL	Latency Histogram Metric Total
0x1000	MON	STREAM Monitor Registers

26.8.1 CH0_CTRL_LOW register

- Absolute Address: 0x0
- Base Offset: 0x0
- Size: 0x4

Ch0 descriptor addr [31:0] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_LOW	rw	0x0	—

26.8.2 CH0_CTRL_HIGH register

- Absolute Address: 0x4
- Base Offset: 0x4
- Size: 0x4

Ch0 descriptor addr [63:32] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_HIGH	rw	0x0	—

26.8.3 CH1_CTRL_LOW register

- Absolute Address: 0x8
- Base Offset: 0x8
- Size: 0x4

Ch1 descriptor addr [31:0] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_LOW	rw	0x0	—

26.8.4 CH1_CTRL_HIGH register

- Absolute Address: 0xC
- Base Offset: 0xC
- Size: 0x4

Ch1 descriptor addr [63:32] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_HIGH	rw	0x0	—

26.8.5 CH2_CTRL_LOW register

- Absolute Address: 0x10
- Base Offset: 0x10
- Size: 0x4

Ch2 descriptor addr [31:0] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_LOW	rw	0x0	—

26.8.6 CH2_CTRL_HIGH register

- Absolute Address: 0x14
- Base Offset: 0x14
- Size: 0x4

Ch2 descriptor addr [63:32] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_HIGH	rw	0x0	—

26.8.7 CH3_CTRL_LOW register

- Absolute Address: 0x18
- Base Offset: 0x18
- Size: 0x4

Ch3 descriptor addr [31:0] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_LOW	rw	0x0	—

26.8.8 CH3_CTRL_HIGH register

- Absolute Address: 0x1C

- Base Offset: 0x1C
- Size: 0x4

Ch3 descriptor addr [63:32] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_HIGH	rw	0x0	—

26.8.9 CH4_CTRL_LOW register

- Absolute Address: 0x20
- Base Offset: 0x20
- Size: 0x4

Ch4 descriptor addr [31:0] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_LOW	rw	0x0	—

26.8.10 CH4_CTRL_HIGH register

- Absolute Address: 0x24
- Base Offset: 0x24
- Size: 0x4

Ch4 descriptor addr [63:32] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_HIGH	rw	0x0	—

26.8.11 CH5_CTRL_LOW register

- Absolute Address: 0x28
- Base Offset: 0x28
- Size: 0x4

Ch5 descriptor addr [31:0] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADD	rw	0x0	—

Bits	Identifier	Access	Reset	Name
	R_LOW			

26.8.12 CH5_CTRL_HIGH register

- Absolute Address: 0x2C
- Base Offset: 0x2C
- Size: 0x4

Ch5 descriptor addr [63:32] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_HIGH	rw	0x0	—

26.8.13 CH6_CTRL_LOW register

- Absolute Address: 0x30
- Base Offset: 0x30
- Size: 0x4

Ch6 descriptor addr [31:0] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_LOW	rw	0x0	—

26.8.14 CH6_CTRL_HIGH register

- Absolute Address: 0x34
- Base Offset: 0x34
- Size: 0x4

Ch6 descriptor addr [63:32] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_HIGH	rw	0x0	—

26.8.15 CH7_CTRL_LOW register

- Absolute Address: 0x38
- Base Offset: 0x38

- Size: 0x4

Ch7 descriptor addr [31:0] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_LOW	rw	0x0	—

26.8.16 CH7_CTRL_HIGH register

- Absolute Address: 0x3C
- Base Offset: 0x3C
- Size: 0x4

Ch7 descriptor addr [63:32] (staged; launch via KICK_ENABLE)

Bits	Identifier	Access	Reset	Name
31:0	DESC_ADDR_HIGH	rw	0x0	—

26.8.17 GLOBAL_CTRL register

- Absolute Address: 0x100
- Base Offset: 0x100
- Size: 0x4

Master enable and global configuration

Bits	Identifier	Access	Reset	Name
0	GLOBAL_EN	rw	0x0	—
1	GLOBAL_RST	rw	0x0	—
31:2	RSVD	r	0x0	—

26.8.17.1 GLOBAL_EN field

Global enable - master switch for entire STREAM engine

26.8.17.2 GLOBAL_RST field

Global reset - resets all channels and state machines

26.8.17.3 RSVD field

Reserved

26.8.18 GLOBAL_STATUS register

- Absolute Address: 0x104
- Base Offset: 0x104
- Size: 0x4

Overall system status and error flags

Bits	Identifier	Access	Reset	Name
0	SYSTEM_IDLE	r	—	—
31:1	RSVD	r	0x0	—

26.8.18.1 SYSTEM_IDLE field

System idle - all channels idle

26.8.18.2 RSVD field

Reserved

26.8.19 VERSION register

- Absolute Address: 0x108
- Base Offset: 0x108
- Size: 0x4

STREAM version and configuration information

Bits	Identifier	Access	Reset	Name
7:0	MINOR	r	0x5A	—
15:8	MAJOR	r	0x0	—
23:16	NUM_CHANNELS	r	0x8	—
31:24	RSVD	r	0x0	—

26.8.19.1 MINOR field

Minor version

26.8.19.2 MAJOR field

Major version

26.8.19.3 NUM_CHANNELS field

Number of channels

26.8.19.4 RSVD field

Reserved

26.8.20 CHANNEL_ENABLE register

- Absolute Address: 0x120
- Base Offset: 0x120
- Size: 0x4

Per-channel enable bits (one bit per channel)

Bits	Identifier	Access	Reset	Name
7:0	CH_EN	rw	0x0	—
31:8	RSVD	r	0x0	—

26.8.20.1 CH_EN field

Channel enable bits [7:0] - 1=enabled, 0=disabled

26.8.20.2 RSVD field

Reserved

26.8.21 CHANNEL_RESET register

- Absolute Address: 0x124
- Base Offset: 0x124
- Size: 0x4

Per-channel reset bits (one bit per channel, self-clearing)

Bits	Identifier	Access	Reset	Name
7:0	CH_RST	rw	0x0	—
31:8	RSVD	r	0x0	—

26.8.21.1 CH_RST field

Channel reset bits [7:0] - write 1 to reset channel

26.8.21.2 RSVD field

Reserved

26.8.22 KICK_ENABLE register

- Absolute Address: 0x128
- Base Offset: 0x128
- Size: 0x4

Write 1 to bit[ch] to launch that channel from its staged CHx_CTRL address

Bits	Identifier	Access	Reset	Name
0	KICK0	w	0x0	—
1	KICK1	w	0x0	—
2	KICK2	w	0x0	—
3	KICK3	w	0x0	—
4	KICK4	w	0x0	—
5	KICK5	w	0x0	—
6	KICK6	w	0x0	—
7	KICK7	w	0x0	—
31:8	RSVD	r	0x0	—

26.8.23 CHANNEL_IDLE register

- Absolute Address: 0x140
- Base Offset: 0x140
- Size: 0x4

Per-channel idle status (one bit per channel)

Bits	Identifier	Access	Reset	Name
7:0	CH_IDLE	r	—	—
31:8	RSVD	r	0x0	—

26.8.23.1 CH_IDLE field

Channel idle bits [7:0] - 1=idle, 0=active

26.8.23.2 RSVD field

Reserved

26.8.24 DESC_ENGINE_IDLE register

- Absolute Address: 0x144
- Base Offset: 0x144
- Size: 0x4

Per-channel descriptor engine idle status

Bits	Identifier	Access	Reset	Name
7:0	DESC_IDLE	r	—	—
31:8	RSVD	r	0x0	—

26.8.24.1 DESC_IDLE field

Descriptor engine idle bits [7:0] - 1=idle, 0=active

26.8.24.2 RSVD field

Reserved

26.8.25 SCHEDULER_IDLE register

- Absolute Address: 0x148
- Base Offset: 0x148
- Size: 0x4

Per-channel scheduler idle status

Bits	Identifier	Access	Reset	Name
7:0	SCHED_IDLE	r	—	—
31:8	RSVD	r	0x0	—

26.8.25.1 SCHED_IDLE field

Scheduler idle bits [7:0] - 1=idle, 0=active

26.8.25.2 RSVD field

Reserved

26.9 CH_STATE register file

- Absolute Address: 0x150
- Base Offset: 0x150
- Size: 0x4
- Array Dimensions: [8]
- Array Stride: 0x4
- Total Size: 0x20

Detailed state for individual channel

Offset	Identifier	Name
0x0	STATE	Channel State

26.9.1 STATE register

- Absolute Address: 0x150
- Base Offset: 0x0
- Size: 0x4

Current FSM state of scheduler (one-hot 7-bit encoding)

Bits	Identifier	Access	Reset	Name
6:0	STATE	r	—	—
31:7	RSVD	r	0x0	—

26.9.1.1 STATE field

Scheduler state [6:0] - one-hot encoding

26.9.1.2 RSVD field

Reserved

26.10 CH_STATE register file

- Absolute Address: 0x154
- Base Offset: 0x150
- Size: 0x4
- Array Dimensions: [8]
- Array Stride: 0x4
- Total Size: 0x20

Detailed state for individual channel

Offset	Identifier	Name
0x0	STATE	Channel State

26.10.1 STATE register

- Absolute Address: 0x154
- Base Offset: 0x0
- Size: 0x4

Current FSM state of scheduler (one-hot 7-bit encoding)

Bits	Identifier	Access	Reset	Name
6:0	STATE	r	—	—
31:7	RSVD	r	0x0	—

26.10.1.1 STATE field

Scheduler state [6:0] - one-hot encoding

26.10.1.2 RSVD field

Reserved

26.11 CH_STATE register file

- Absolute Address: 0x158
- Base Offset: 0x150
- Size: 0x4
- Array Dimensions: [8]
- Array Stride: 0x4
- Total Size: 0x20

Detailed state for individual channel

Offset	Identifier	Name
0x0	STATE	Channel State

26.11.1 STATE register

- Absolute Address: 0x158
- Base Offset: 0x0
- Size: 0x4

Current FSM state of scheduler (one-hot 7-bit encoding)

Bits	Identifier	Access	Reset	Name
6:0	STATE	r	—	—
31:7	RSVD	r	0x0	—

26.11.1.1 STATE field

Scheduler state [6:0] - one-hot encoding

26.11.1.2 RSVD field

Reserved

26.12 CH_STATE register file

- Absolute Address: 0x15C
- Base Offset: 0x150
- Size: 0x4
- Array Dimensions: [8]
- Array Stride: 0x4
- Total Size: 0x20

Detailed state for individual channel

Offset	Identifier	Name
0x0	STATE	Channel State

26.12.1 STATE register

- Absolute Address: 0x15C
- Base Offset: 0x0
- Size: 0x4

Current FSM state of scheduler (one-hot 7-bit encoding)

Bits	Identifier	Access	Reset	Name
6:0	STATE	r	—	—
31:7	RSVD	r	0x0	—

26.12.1.1 STATE field

Scheduler state [6:0] - one-hot encoding

26.12.1.2 RSVD field

Reserved

26.13 CH_STATE register file

- Absolute Address: 0x160
- Base Offset: 0x150
- Size: 0x4
- Array Dimensions: [8]
- Array Stride: 0x4
- Total Size: 0x20

Detailed state for individual channel

Offset	Identifier	Name
0x0	STATE	Channel State

26.13.1 STATE register

- Absolute Address: 0x160
- Base Offset: 0x0
- Size: 0x4

Current FSM state of scheduler (one-hot 7-bit encoding)

Bits	Identifier	Access	Reset	Name
6:0	STATE	r	—	—
31:7	RSVD	r	0x0	—

26.13.1.1 STATE field

Scheduler state [6:0] - one-hot encoding

26.13.1.2 RSVD field

Reserved

26.14 CH_STATE register file

- Absolute Address: 0x164
- Base Offset: 0x150
- Size: 0x4
- Array Dimensions: [8]
- Array Stride: 0x4
- Total Size: 0x20

Detailed state for individual channel

Offset	Identifier	Name
0x0	STATE	Channel State

26.14.1 STATE register

- Absolute Address: 0x164
- Base Offset: 0x0
- Size: 0x4

Current FSM state of scheduler (one-hot 7-bit encoding)

Bits	Identifier	Access	Reset	Name
6:0	STATE	r	—	—
31:7	RSVD	r	0x0	—

26.14.1.1 STATE field

Scheduler state [6:0] - one-hot encoding

26.14.1.2 RSVD field

Reserved

26.15 CH_STATE register file

- Absolute Address: 0x168
- Base Offset: 0x150
- Size: 0x4
- Array Dimensions: [8]
- Array Stride: 0x4
- Total Size: 0x20

Detailed state for individual channel

Offset	Identifier	Name
0x0	STATE	Channel State

26.15.1 STATE register

- Absolute Address: 0x168
- Base Offset: 0x0
- Size: 0x4

Current FSM state of scheduler (one-hot 7-bit encoding)

Bits	Identifier	Access	Reset	Name
6:0	STATE	r	—	—
31:7	RSVD	r	0x0	—

26.15.1.1 STATE field

Scheduler state [6:0] - one-hot encoding

26.15.1.2 RSVD field

Reserved

26.16 CH_STATE register file

- Absolute Address: 0x16C
- Base Offset: 0x150
- Size: 0x4
- Array Dimensions: [8]
- Array Stride: 0x4
- Total Size: 0x20

Detailed state for individual channel

Offset	Identifier	Name
0x0	STATE	Channel State

26.16.1 STATE register

- Absolute Address: 0x16C
- Base Offset: 0x0
- Size: 0x4

Current FSM state of scheduler (one-hot 7-bit encoding)

Bits	Identifier	Access	Reset	Name
6:0	STATE	r	—	—
31:7	RSVD	r	0x0	—

26.16.1.1 STATE field

Scheduler state [6:0] - one-hot encoding

26.16.1.2 RSVD field

Reserved

26.16.2 SCHED_ERROR register

- Absolute Address: 0x170
- Base Offset: 0x170
- Size: 0x4

Per-channel scheduler error flags

Bits	Identifier	Access	Reset	Name
7:0	SCHED_ERR	r	—	—

Bits	Identifier	Access	Reset	Name
31:8	RSVD	r	0x0	—

26.16.2.1 SCHED_ERR field

Scheduler error bits [7:0] - 1=error detected, 0=no error

26.16.2.2 RSVD field

Reserved

26.16.3 AXI_RD_COMPLETE register

- Absolute Address: 0x174
- Base Offset: 0x174
- Size: 0x4

Per-channel read engine all_complete flags

Bits	Identifier	Access	Reset	Name
7:0	RD_COMPLETE	r	—	—
31:8	RSVD	r	0x0	—

26.16.3.1 RD_COMPLETE field

Read completion bits [7:0] - 1=all reads complete, 0=reads pending

26.16.3.2 RSVD field

Reserved

26.16.4 AXI_WR_COMPLETE register

- Absolute Address: 0x178
- Base Offset: 0x178
- Size: 0x4

Per-channel write engine all_complete flags

Bits	Identifier	Access	Reset	Name
7:0	WR_COMPLETE	r	—	—
31:8	RSVD	r	0x0	—

26.16.4.1 WR_COMPLETE field

Write completion bits [7:0] - 1=all writes complete, 0=writes pending

26.16.4.2 RSVD field

Reserved

26.16.5 SCHED_TIMEOUT_CYCLES register

- Absolute Address: 0x200
- Base Offset: 0x200
- Size: 0x4

Timeout threshold in clock cycles (global for all channels)

Bits	Identifier	Access	Reset	Name
31:0	TIMEOUT_CYCLES	rw	0x3E8	—

26.16.5.1 TIMEOUT_CYCLES field

Timeout cycles [31:0] - number of cycles before timeout

26.16.6 SCHED_CONFIG register

- Absolute Address: 0x204
- Base Offset: 0x204
- Size: 0x4

Scheduler feature enables (global for all channels)

Bits	Identifier	Access	Reset	Name
0	SCHED_EN	rw	0x1	—
1	TIMEOUT_EN	rw	0x1	—
2	ERR_EN	rw	0x1	—
3	COMPL_EN	rw	0x1	—
4	PERF_EN	rw	0x0	—
5	RD_PREFETCH_EN	rw	0x1	—
31:6	RSVD	r	0x0	—

26.16.6.1 SCHED_EN field

Scheduler enable - master scheduler enable

26.16.6.2 TIMEOUT_EN field

Timeout enable - enable timeout detection

26.16.6.3 ERR_EN field

Error enable - enable error reporting

26.16.6.4 COMPL_EN field

Completion enable - enable completion reporting

26.16.6.5 PERF_EN field

Performance enable - enable performance monitoring

26.16.6.6 RD_PREFETCH_EN field

Read-ahead descriptor prefetch enable. When set, on a chained legacy descriptor the scheduler read side loads the next descriptor from the descriptor-FIFO head and keeps filling SRAM while the write side drains the current one – collapsing the per-descriptor boundary bubble to zero (perfect cross-descriptor streaming). Default enabled; clear for lockstep A/B on the same bitstream. Ignored for EXT (row/col) descriptors.

26.16.6.7 RSVD field

Reserved

26.16.7 SCHED_TIMEOUT_LIMIT register

- Absolute Address: 0x208
- Base Offset: 0x208
- Size: 0x4

Number of consecutive write-progress timeout windows a channel may tolerate before the (recoverable) timeout escalates to a fatal, sticky CH_ERROR. 0 = never escalate (pure soft timeout: report and keep waiting). Total time to escalate = LIMIT x TIMEOUT_CYCLES.

Bits	Identifier	Access	Reset	Name
7:0	LIMIT	rw	0x4	—

26.16.7.1 LIMIT field

Consecutive-timeout strike limit before fatal escalation (0 = never)

26.16.8 DESCENG_CONFIG register

- Absolute Address: 0x220
- Base Offset: 0x220
- Size: 0x4

Descriptor engine feature enables (global for all channels)

Bits	Identifier	Access	Reset	Name
0	DESCENG_EN	rw	0x1	—
1	PREFETCH_EN	rw	0x1	—
5:2	FIFO_THRESH	rw	0x8	—
31:6	RSVD	r	0x0	—

26.16.8.1 DESCENG_EN field

Descriptor engine enable - master enable

26.16.8.2 PREFETCH_EN field

Prefetch enable - enable descriptor prefetch. Default ENABLED: with prefetch off the descriptor engine is on-demand (fetch the next descriptor only after the current one drains), which inserts a per-descriptor pipeline drain/refill bubble (~40 cycles) on chains. Prefetch buffers FIFO_THRESH descriptors ahead so the datapath streams continuously across descriptor boundaries.

26.16.8.3 FIFO_THRESH field

FIFO threshold [5:2] - prefetch threshold (4 bits)

26.16.8.4 RSVD field

Reserved

26.16.9 DESCENG_ADDR0_BASE register

- Absolute Address: 0x224
- Base Offset: 0x224
- Size: 0x4

Base address for descriptor address range 0 (lower 32 bits)

Bits	Identifier	Access	Reset	Name
31:0	ADDR0_BASE	rw	0x0	—

26.16.9.1 ADDR0_BASE field

Address range 0 base [31:0]

26.16.10 DESCENG_ADDR0_LIMIT register

- Absolute Address: 0x228
- Base Offset: 0x228
- Size: 0x4

Limit address for descriptor address range 0 (lower 32 bits)

Bits	Identifier	Access	Reset	Name
31:0	ADDR0_LIMIT	rw	0xFFFFFFFF	—

26.16.10.1 ADDR0_LIMIT field

Address range 0 limit [31:0]

26.16.11 DESCENG_ADDR1_BASE register

- Absolute Address: 0x22C
- Base Offset: 0x22C
- Size: 0x4

Base address for descriptor address range 1 (lower 32 bits)

Bits	Identifier	Access	Reset	Name
31:0	ADDR1_BASE	rw	0x0	—

26.16.11.1 ADDR1_BASE field

Address range 1 base [31:0]

26.16.12 DESCENG_ADDR1_LIMIT register

- Absolute Address: 0x230
- Base Offset: 0x230
- Size: 0x4

Limit address for descriptor address range 1 (lower 32 bits)

Bits	Identifier	Access	Reset	Name
31:0	ADDR1_LIMIT	rw	0xFFFFFFFF	—

26.16.12.1 ADDR1_LIMIT field

Address range 1 limit [31:0]

26.16.13 AXI_XFER_CONFIG register

- Absolute Address: 0x2A0
- Base Offset: 0x2A0
- Size: 0x4

AXI read and write transfer burst sizes

Bits	Identifier	Access	Reset	Name
7:0	RD_XFER_BEATS	rw	0xF	—
15:8	WR_XFER_BEATS	rw	0xF	—
31:16	RSVD	r	0x0	—

26.16.13.1 RD_XFER_BEATS field

AXI read transfer beats [7:0] - ARLEN value (0-255 represents 1-256 beats). The engine clamps it to the per-channel SRAM depth minus one, and to 254 beyond that (stream BUG-018): a burst is never larger than the buffer that stages it

26.16.13.2 WR_XFER_BEATS field

AXI write transfer beats [15:8] - AWLEN value (0-255 represents 1-256 beats). The engine clamps it to the per-channel SRAM depth minus one, and to 254 beyond that (stream BUG-018): a burst is never larger than the buffer that stages it

26.16.13.3 RSVD field

Reserved

26.16.14 PERF_CONFIG register

- Absolute Address: 0x2B0
- Base Offset: 0x2B0
- Size: 0x4

Performance profiler enable and mode controls

Bits	Identifier	Access	Reset	Name
0	PERF_EN	rw	0x0	—
1	PERF_MODE	rw	0x0	—
2	PERF_CLEAR	rw	0x0	—
31:3	RSVD	r	0x0	—

26.16.14.1 PERF_EN field

Performance profiler enable

26.16.14.2 PERF_MODE field

Performance profiler mode - 0=count, 1=histogram

26.16.14.3 PERF_CLEAR field

Performance profiler clear - write 1 to clear counters

26.16.14.4 RSVD field

Reserved

26.16.15 OBS_CTRL register

- Absolute Address: 0x2C0
- Base Offset: 0x2C0
- Size: 0x4

Selects which channel and which category drive OBS_FLAGS / OBS_DATA0 / OBS_DATA1

Bits	Identifier	Access	Reset	Name
2:0	CH_SEL	rw	0x0	—
4:3	CAT_SEL	rw	0x0	—
31:5	RSVD	r	0x0	—

26.16.15.1 CH_SEL field

Channel select (0..NUM_CHANNELS-1)

26.16.15.2 CAT_SEL field

Category select: 0=status (data0=sched_rd_beats, data1=sched_wr_beats), 1=rd_addr (data0=lo, data1=hi), 2=wr_addr (data0=lo, data1=hi), 3=sram (data0=rd_space_free, data1=wr_data_avail)

26.16.15.3 RSVD field

Reserved

26.16.16 OBS_FLAGS register

- Absolute Address: 0x2C4
- Base Offset: 0x2C4
- Size: 0x4

Combinational status vector for the channel selected by OBS_CTRL.CH_SEL. See stream_core.sv for the bit layout (scheduler_state, sched_rd/wr_valid, sched_wr_ready, error stickies, idle bits, cfg_channel_enable, axi_*_all_complete).

Bits	Identifier	Access	Reset	Name
31:0	FLAGS	r	—	—

26.16.16.1 FLAGS field

Observation flags

26.16.17 OBS_DATA0 register

- Absolute Address: 0x2C8
- Base Offset: 0x2C8
- Size: 0x4

Category-muxed data word 0 for the selected channel. Semantics depend on OBS_CTRL.CAT_SEL.

Bits	Identifier	Access	Reset	Name
31:0	DATA	r	—	—

26.16.17.1 DATA field

Observation data 0

26.16.18 OBS_DATA1 register

- Absolute Address: 0x2CC
- Base Offset: 0x2CC
- Size: 0x4

Category-muxed data word 1 for the selected channel. Semantics depend on OBS_CTRL.CAT_SEL.

Bits	Identifier	Access	Reset	Name
31:0	DATA	r	—	—

26.16.18.1 DATA field

Observation data 1

26.16.19 PERF_DATA_LOW register

- Absolute Address: 0x2D0
- Base Offset: 0x2D0
- Size: 0x4

Timestamp or elapsed time [31:0] of the entry at the head of the capture FIFO. The FIFO is popped once BOTH this register and DATA_HIGH have been read, so the two halves always belong to the same entry and the read order does not matter.

Bits	Identifier	Access	Reset	Name
31:0	DATA	r	0x0	—

26.16.19.1 DATA field

Timestamp or elapsed cycles

26.16.20 PERF_DATA_HIGH register

- Absolute Address: 0x2D4
- Base Offset: 0x2D4
- Size: 0x4

Upper bits of the entry at the head of the capture FIFO: {28'b0, event_type, channel_id[2:0]}. The FIFO is popped once BOTH this register and DATA_LOW have been read.

Bits	Identifier	Access	Reset	Name
31:0	DATA	r	0x0	—

26.16.20.1 DATA field

Channel id [2:0], event_type [3]

26.16.21 PERF_STATUS register

- Absolute Address: 0x2D8
- Base Offset: 0x2D8

- Size: 0x4

FIFO occupancy and flags. Layout matches the retired hand-rolled register at 0x04C exactly, so host code porting over needs no field changes.

Bits	Identifier	Access	Reset	Name
0	EMPTY	r	0x0	—
1	FULL	r	0x0	—
15:2	RSVD	r	0x0	—
31:16	COUNT	r	0x0	—

26.16.21.1 EMPTY field

FIFO empty

26.16.21.2 FULL field

FIFO full

26.16.21.3 RSVD field

Reserved

26.16.21.4 COUNT field

Entries in FIFO

26.16.22 PERF_CH_SEL register

- Absolute Address: 0x35C
- Base Offset: 0x35C
- Size: 0x4

Selects which channel's buckets appear in RD/WRMON_PERF_CH_*

Bits	Identifier	Access	Reset	Name
2:0	CH_SEL	rw	0x0	—
31:3	RSVD	r	0x0	—

26.16.22.1 CH_SEL field

Channel select (0..NUM_CHANNELS-1)

26.16.22.2 RSVD field

Reserved

26.16.23 HIST_SEL register

- Absolute Address: 0x378
- Base Offset: 0x378
- Size: 0x4

Selects bus/metric/bin presented in HIST_DATA / HIST_TOTAL

Bits	Identifier	Access	Reset	Name
0	BUS	rw	0x0	—
1	METRIC	rw	0x0	—
5:2	BIN	rw	0x0	—
31:6	RSVD	r	0x0	—

26.16.23.1 BUS field

Bus: 0=data-read R bus, 1=data-write W bus

26.16.23.2 METRIC field

Metric: 0=AR->firstR (rd) / AW->B (wr), 1=AR->RLAST (rd only)

26.16.23.3 BIN field

Latency bin index (0..15, log2)

26.16.23.4 RSVD field

Reserved

26.16.24 HIST_DATA register

- Absolute Address: 0x37C
- Base Offset: 0x37C
- Size: 0x4

Transaction count in the HIST_SEL-selected bus/metric/bin

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.16.24.1 VAL field

Selected histogram bin count [31:0]

26.16.25 HIST_TOTAL register

- Absolute Address: 0x380
- Base Offset: 0x380
- Size: 0x4

Total transactions for the HIST_SEL-selected bus/metric (= sum over all bins). Equals the burst/transaction count, an acceptance cross-check against the perf-window burst counters.

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.16.25.1 VAL field

Selected metric total transaction count [31:0]

26.17 MON register file

- Absolute Address: 0x1000
- Base Offset: 0x1000
- Size: 0x26C

AXI monitor + perf registers (relocatable block, instantiated at 0x1000)

Offset	Identifier	Name
0x000	MON_FIFO_STATUS	Monitor FIFO Status
0x004	MON_FIFO_COUNT	Monitor FIFO Count
0x0C0	DAXMON_ENABLE	Descriptor AXI Monitor Enable
0x0C4	DAXMON_TIMEOUT	Descriptor AXI Monitor Timeout
0x0C8	DAXMON_LATENCY_THRESH	Descriptor AXI Monitor Latency Threshold
0x0CC	DAXMON_PKT_MASK	Descriptor AXI Monitor Packet Mask
0x0D0	DAXMON_ERR_CFG	Descriptor AXI Monitor Error Select and Mask
0x0D4	DAXMON_MASK1	Descriptor AXI Monitor Masks 1
0x0D8	DAXMON_MASK2	Descriptor AXI Monitor Masks 2
0x0DC	DAXMON_MASK3	Descriptor AXI Monitor Masks 3
0x0E0	RDMON_ENABLE	Read Engine AXI Monitor Enable
0x0E4	RDMON_TIMEOUT	Read Engine AXI Monitor Timeout
0x0E8	RDMON_LATENCY_THRESH	Read Engine AXI Monitor Latency Threshold

Offset	Identifier	Name
0x0EC	RDMON_PKT_MASK	Read Engine AXI Monitor Packet Mask
0x0F0	RDMON_ERR_CFG	Read Engine AXI Monitor Error Select and Mask
0x0F4	RDMON_MASK1	Read Engine AXI Monitor Masks 1
0x0F8	RDMON_MASK2	Read Engine AXI Monitor Masks 2
0x0FC	RDMON_MASK3	Read Engine AXI Monitor Masks 3
0x100	WRMON_ENABLE	Write Engine AXI Monitor Enable
0x104	WRMON_TIMEOUT	Write Engine AXI Monitor Timeout
0x108	WRMON_LATENCY_THRESH	Write Engine AXI Monitor Latency Threshold
0x10C	WRMON_PKT_MASK	Write Engine AXI Monitor Packet Mask
0x110	WRMON_ERR_CFG	Write Engine AXI Monitor Error Select and Mask
0x114	WRMON_MASK1	Write Engine AXI Monitor Masks 1
0x118	WRMON_MASK2	Write Engine AXI Monitor Masks 2
0x11C	WRMON_MASK3	Write Engine AXI Monitor Masks 3
0x150	DAXMON_PERF_CTRL	Descriptor AXI Monitor Perf Window Control
0x154	DAXMON_PERF_STATUS	Descriptor AXI Monitor Perf Window Status
0x158	DAXMON_PERF_WINDOW_CYCLES	Descriptor AXI Monitor Perf Window Cycles

Offset	Identifier	Name
0x15C	DAXMON_PERF_PROD_CYCLES	Descriptor AXI Monitor Perf Productive Cycles
0x160	DAXMON_PERF_BP_CYCLES	Descriptor AXI Monitor Perf Backpressure Cycles
0x164	DAXMON_PERF_STARV_CYCLES	Descriptor AXI Monitor Perf Starvation Cycles
0x168	DAXMON_PERF_IDLE_CYCLES	Descriptor AXI Monitor Perf Idle Cycles
0x16C	DAXMON_PERF_BEAT_COUNT	Descriptor AXI Monitor Perf Beat Count
0x170	DAXMON_PERF_BYTE_COUNT_LO	Descriptor AXI Monitor Perf Byte Count Low
0x174	DAXMON_PERF_BYTE_COUNT_HI	Descriptor AXI Monitor Perf Byte Count High
0x178	DAXMON_PERF_BURST_COUNT	Descriptor AXI Monitor Perf Burst Count
0x180	RDMON_PERF_CTRL	Read Datapath Monitor Perf Window Control
0x184	RDMON_PERF_STATUS	Read Datapath Monitor Perf Window Status
0x188	RDMON_PERF_WINDOW_CYCLES	Read Datapath Monitor Perf Window Cycles
0x18C	RDMON_PERF_PROD_CYCLES	Read Datapath Monitor Perf Productive Cycles
0x190	RDMON_PERF_BP_CYCLES	Read Datapath Monitor Perf Backpressure Cycles
0x194	RDMON_PERF_STARV_CYCLES	Read Datapath Monitor Perf Starvation Cycles
0x198	RDMON_PERF_IDLE_CYCLES	Read Datapath Monitor Perf Idle Cycles
0x19C	RDMON_PERF_BEAT_COUNT	Read Datapath Monitor Perf Beat Count

Offset	Identifier	Name
0x1A0	RDMON_PERF_BYTE_COUNT_LO	Read Datapath Monitor Perf Byte Count Low
0x1A4	RDMON_PERF_BYTE_COUNT_HI	Read Datapath Monitor Perf Byte Count High
0x1A8	RDMON_PERF_BURST_COUNT	Read Datapath Monitor Perf Burst Count
0x1B0	WRMON_PERF_CTRL	Write Datapath Monitor Perf Window Control
0x1B4	WRMON_PERF_STATUS	Write Datapath Monitor Perf Window Status
0x1B8	WRMON_PERF_WINDOW_CYCLES	Write Datapath Monitor Perf Window Cycles
0x1BC	WRMON_PERF_PROD_CYCLES	Write Datapath Monitor Perf Productive Cycles
0x1C0	WRMON_PERF_BP_CYCLES	Write Datapath Monitor Perf Backpressure Cycles
0x1C4	WRMON_PERF_STARV_CYCLES	Write Datapath Monitor Perf Starvation Cycles
0x1C8	WRMON_PERF_IDLE_CYCLES	Write Datapath Monitor Perf Idle Cycles
0x1CC	WRMON_PERF_BEAT_COUNT	Write Datapath Monitor Perf Beat Count
0x1D0	WRMON_PERF_BYTE_COUNT_LO	Write Datapath Monitor Perf Byte Count Low
0x1D4	WRMON_PERF_BYTE_COUNT_HI	Write Datapath Monitor Perf Byte Count High
0x1D8	WRMON_PERF_BURST_COUNT	Write Datapath Monitor Perf Burst Count
0x1E0	RDMON_PERF_CH_PROD_BP	Read Datapath Per-Channel Productive/Backpressure
0x1E4	RDMON_PERF_CH_STARV_IDLE	Read Datapath Per-Channel Starvation/Idle

Offset	Identifier	Name
0x1E8	WRMON_PERF_CH_PROD_BP	Write Datapath Per-Channel Productive/Backpressure
0x1EC	WRMON_PERF_CH_STARV_IDLE	Write Datapath Per-Channel Starvation/Idle
0x1F0	RDMON_PERF_CH_OVERFLOW	Read Datapath Per-Channel Overflow Stickies
0x1F4	WRMON_PERF_CH_OVERFLOW	Write Datapath Per-Channel Overflow Stickies
0x200	RDMON_ADDR_RANGE0_LOW	RD addr range0 low
0x204	RDMON_ADDR_RANGE0_HIGH	RD addr range0 high
0x208	RDMON_ADDR_RANGE1_LOW	RD addr range1 low
0x20C	RDMON_ADDR_RANGE1_HIGH	RD addr range1 high
0x210	RDMON_ADDR_RANGE2_LOW	RD addr range2 low
0x214	RDMON_ADDR_RANGE2_HIGH	RD addr range2 high
0x218	RDMON_ADDR_RANGE3_LOW	RD addr range3 low
0x21C	RDMON_ADDR_RANGE3_HIGH	RD addr range3 high
0x220	RDMON_ADDR_RANGE_CTRL	RD addr range control
0x230	WRMON_ADDR_RANGE0_LOW	WR addr range0 low
0x234	WRMON_ADDR_RANGE0_HIGH	WR addr range0 high
0x238	WRMON_ADDR_RANGE1_LOW	WR addr range1 low

Offset	Identifier	Name
0x23C	WRMON_ADDR_RANGE1_HIGH	WR addr range1 high
0x240	WRMON_ADDR_RANGE2_LOW	WR addr range2 low
0x244	WRMON_ADDR_RANGE2_HIGH	WR addr range2 high
0x248	WRMON_ADDR_RANGE3_LOW	WR addr range3 low
0x24C	WRMON_ADDR_RANGE3_HIGH	WR addr range3 high
0x250	WRMON_ADDR_RANGE_CTRL	WR addr range control
0x260	MON_GROUP_BASE_ADDR	Monbus group base addr
0x264	MON_GROUP_LIMIT_ADDR	Monbus group limit addr
0x268	MON_GROUP_FLUSH_WATERMARK	Monbus group flush wmark

26.17.1 MON_FIFO_STATUS register

- Absolute Address: 0x1000
- Base Offset: 0x0
- Size: 0x4

Monitor bus FIFO status indicators

Bits	Identifier	Access	Reset	Name
0	MON_FIFO_FULL	r	—	—
1	MON_FIFO_EMPTY	r	—	—
2	MON_FIFO_OVFL	r	—	—
3	MON_FIFO_	r	—	—

Bits	Identifier	Access	Reset	Name
	UNFL			
31:4	RSVD	r	0x0	—

26.17.1.1 MON_FIFO_FULL field

Monitor FIFO full - 1=FIFO full, 0=space available

26.17.1.2 MON_FIFO_EMPTY field

Monitor FIFO empty - 1=FIFO empty, 0=data available

26.17.1.3 MON_FIFO_OVFL field

Monitor FIFO overflow - 1=overflow detected, 0=normal

26.17.1.4 MON_FIFO_UNFL field

Monitor FIFO underflow - 1=underflow detected, 0=normal

26.17.1.5 RSVD field

Reserved

26.17.2 MON_FIFO_COUNT register

- Absolute Address: 0x1004
- Base Offset: 0x4
- Size: 0x4

Monitor bus FIFO entry count

Bits	Identifier	Access	Reset	Name
15:0	FIFO_COUNT	r	—	—
31:16	RSVD	r	0x0	—

26.17.2.1 FIFO_COUNT field

FIFO count [15:0] - number of entries in FIFO

26.17.2.2 RSVD field

Reserved

26.17.3 DAXMON_ENABLE register

- Absolute Address: 0x10C0

- Base Offset: 0xC0
- Size: 0x4

Descriptor AXI master monitor enable controls

Bits	Identifier	Access	Reset	Name
0	MON_EN	rw	0x0	—
1	ERR_EN	rw	0x0	—
2	COMPL_EN	rw	0x0	—
3	TIMEOUT_EN	rw	0x0	—
4	PERF_EN	rw	0x0	—
5	RSVD5	r	0x0	—
6	THRESH_EN	rw	0x0	—
31:7	RSVD	r	0x0	—

26.17.3.1 MON_EN field

Monitor enable - master enable for descriptor monitor

26.17.3.2 ERR_EN field

Error enable - enable error detection

26.17.3.3 COMPL_EN field

Completion enable - enable completion packets

26.17.3.4 TIMEOUT_EN field

Timeout enable - enable timeout detection

26.17.3.5 PERF_EN field

Performance enable - enable performance packets

26.17.3.6 RSVD5 field

Reserved. Bit 5 is COMPRESS_EN on WRMON_ENABLE only; held reserved here so all three monitor ENABLE registers share one bit layout.

26.17.3.7 THRESH_EN field

Threshold enable - enable latency-threshold packets. Before this field existed the threshold cone was gated by PERF_EN, so a host that left PERF_EN clear saw no threshold packets however low it set LATENCY_THRESH.

26.17.3.8 RSVD field

Reserved

26.17.4 DAXMON_TIMEOUT register

- Absolute Address: 0x10C4
- Base Offset: 0xC4
- Size: 0x4

Descriptor AXI monitor timeout threshold (cycles)

Bits	Identifier	Access	Reset	Name
31:0	TIMEOUT_CYCLES	rw	0x2710	—

26.17.4.1 TIMEOUT_CYCLES field

Timeout cycles [31:0]

26.17.5 DAXMON_LATENCY_THRESH register

- Absolute Address: 0x10C8
- Base Offset: 0xC8
- Size: 0x4

Descriptor AXI monitor latency threshold (cycles)

Bits	Identifier	Access	Reset	Name
31:0	LATENCY_THRESH	rw	0x1388	—

26.17.5.1 LATENCY_THRESH field

Latency threshold cycles [31:0]

26.17.6 DAXMON_PKT_MASK register

- Absolute Address: 0x10CC
- Base Offset: 0xCC
- Size: 0x4

Descriptor AXI monitor packet type filtering (16-bit mask)

Bits	Identifier	Access	Reset	Name
15:0	PKT_MASK	rw	0xFFFF	—
31:16	RSVD	r	0x0	—

26.17.6.1 PKT_MASK field

Packet type mask [15:0] - bit[type] = 1 MASKS (drops) that packet type, at the monitor and again at the monbus group. Reset = all masked: the host clears the bits of the classes it wants to see (stream BUG-016 corrected this description; the polarity and the reset are as intended)

26.17.6.2 RSVD field

Reserved

26.17.7 DAXMON_ERR_CFG register

- Absolute Address: 0x10D0
- Base Offset: 0xD0
- Size: 0x4

Descriptor AXI monitor error selection and filtering

Bits	Identifier	Access	Reset	Name
15:0	ERR_SELECT	rw	0x0	—
31:16	ERR_MASK	rw	0xFFFF	—

26.17.7.1 ERR_SELECT field

Error select - per-packet-type route: 1=err FIFO/IRQ, 0=bulk trace

26.17.7.2 ERR_MASK field

Error mask - per-event-code drop mask, indexed by event_code[3:0]

26.17.8 DAXMON_MASK1 register

- Absolute Address: 0x10D4
- Base Offset: 0xD4
- Size: 0x4

Descriptor AXI monitor timeout and completion masks

Bits	Identifier	Access	Reset	Name
15:0	TIMEOUT_MASK	rw	0xFFFF	—
31:16	COMPL_MASK	rw	0x0	—

26.17.8.1 TIMEOUT_MASK field

Timeout mask

26.17.8.2 COMPL_MASK field

Completion mask

26.17.9 DAXMON_MASK2 register

- Absolute Address: 0x10D8
- Base Offset: 0xD8
- Size: 0x4

Descriptor AXI monitor threshold and performance masks

Bits	Identifier	Access	Reset	Name
15:0	THRESH_MASK	rw	0xFFFF	—
31:16	PERF_MASK	rw	0x0	—

26.17.9.1 THRESH_MASK field

Threshold mask

26.17.9.2 PERF_MASK field

Performance mask

26.17.10 DAXMON_MASK3 register

- Absolute Address: 0x10DC
- Base Offset: 0xDC
- Size: 0x4

Descriptor AXI monitor address and debug masks

Bits	Identifier	Access	Reset	Name
15:0	ADDR_MASK	rw	0xFFFF	—
31:16	DEBUG_MASK	rw	0x0	—

26.17.10.1 ADDR_MASK field

Address mask

26.17.10.2 DEBUG_MASK field

Debug mask

26.17.11 RDMON_ENABLE register

- Absolute Address: 0x10E0
- Base Offset: 0xE0
- Size: 0x4

Read engine AXI master monitor enable controls

Bits	Identifier	Access	Reset	Name
0	MON_EN	rw	0x0	—
1	ERR_EN	rw	0x0	—
2	COMPL_EN	rw	0x0	—
3	TIMEOUT_EN	rw	0x0	—
4	PERF_EN	rw	0x0	—
5	RSVD5	r	0x0	—
6	THRESH_EN	rw	0x0	—
31:7	RSVD	r	0x0	—

26.17.11.1 MON_EN field

Monitor enable - master enable for read monitor

26.17.11.2 ERR_EN field

Error enable - enable error detection

26.17.11.3 COMPL_EN field

Completion enable - enable completion packets

26.17.11.4 TIMEOUT_EN field

Timeout enable - enable timeout detection

26.17.11.5 PERF_EN field

Performance enable - enable performance packets

26.17.11.6 RSVD5 field

Reserved. Bit 5 is COMPRESS_EN on WRMON_ENABLE only; held reserved here so all three monitor ENABLE registers share one bit layout.

26.17.11.7 THRESH_EN field

Threshold enable - enable latency-threshold packets. Before this field existed the threshold cone was gated by PERF_EN, so a host that left PERF_EN clear saw no threshold packets however low it set LATENCY_THRESH.

26.17.11.8 RSVD field

Reserved

26.17.12 RDMON_TIMEOUT register

- Absolute Address: 0x10E4
- Base Offset: 0xE4
- Size: 0x4

Read engine AXI monitor timeout threshold (cycles)

Bits	Identifier	Access	Reset	Name
31:0	TIMEOUT_CYCLES	rw	0x2710	—

26.17.12.1 TIMEOUT_CYCLES field

Timeout cycles [31:0]

26.17.13 RDMON_LATENCY_THRESH register

- Absolute Address: 0x10E8
- Base Offset: 0xE8
- Size: 0x4

Read engine AXI monitor latency threshold (cycles)

Bits	Identifier	Access	Reset	Name
31:0	LATENCY_THRESH	rw	0x1388	—

26.17.13.1 LATENCY_THRESH field

Latency threshold cycles [31:0]

26.17.14 RDMON_PKT_MASK register

- Absolute Address: 0x10EC
- Base Offset: 0xEC
- Size: 0x4

Read engine AXI monitor packet type filtering (16-bit mask)

Bits	Identifier	Access	Reset	Name
15:0	PKT_MASK	rw	0xFFFF	—
31:16	RSVD	r	0x0	—

26.17.14.1 PKT_MASK field

Packet type mask [15:0] - bit[type] = 1 MASKS (drops) that packet type, at the monitor and again at the monbus group. Reset = all masked: the host clears the bits of the classes it wants to see (stream BUG-016 corrected this description; the polarity and the reset are as intended)

26.17.14.2 RSVD field

Reserved

26.17.15 RDMON_ERR_CFG register

- Absolute Address: 0x10F0
- Base Offset: 0xF0
- Size: 0x4

Read engine AXI monitor error selection and filtering

Bits	Identifier	Access	Reset	Name
15:0	ERR_SELECT	rw	0x0	—
31:16	ERR_MASK	rw	0xFFFF	—

26.17.15.1 ERR_SELECT field

Error select - per-packet-type route: 1=err FIFO/IRQ, 0=bulk trace

26.17.15.2 ERR_MASK field

Error mask - per-event-code drop mask, indexed by event_code[3:0]

26.17.16 RDMON_MASK1 register

- Absolute Address: 0x10F4
- Base Offset: 0xF4
- Size: 0x4

Read engine AXI monitor timeout and completion masks

Bits	Identifier	Access	Reset	Name
15:0	TIMEOUT_MASK	rw	0xFFFF	—
31:16	COMPL_MASK	rw	0x0	—

26.17.16.1 TIMEOUT_MASK field

Timeout mask

26.17.16.2 COMPL_MASK field

Completion mask

26.17.17 RDMON_MASK2 register

- Absolute Address: 0x10F8
- Base Offset: 0xF8
- Size: 0x4

Read engine AXI monitor threshold and performance masks

Bits	Identifier	Access	Reset	Name
15:0	THRESH_MASK	rw	0xFFFF	—
31:16	PERF_MASK	rw	0x0	—

26.17.17.1 THRESH_MASK field

Threshold mask

26.17.17.2 PERF_MASK field

Performance mask

26.17.18 RDMON_MASK3 register

- Absolute Address: 0x10FC
- Base Offset: 0xFC
- Size: 0x4

Read engine AXI monitor address and debug masks

Bits	Identifier	Access	Reset	Name
15:0	ADDR_MASK	rw	0xFFFF	—
31:16	DEBUG_MASK	rw	0x0	—

26.17.18.1 ADDR_MASK field

Address mask

26.17.18.2 DEBUG_MASK field

Debug mask

26.17.19 WRMON_ENABLE register

- Absolute Address: 0x1100
- Base Offset: 0x100
- Size: 0x4

Write engine AXI master monitor enable controls

Bits	Identifier	Access	Reset	Name
0	MON_EN	rw	0x0	—
1	ERR_EN	rw	0x0	—
2	COMPL_EN	rw	0x0	—
3	TIMEOUT_EN	rw	0x0	—

Bits	Identifier	Access	Reset	Name
4	PERF_EN	rw	0x0	—
5	COMPRESS_EN	rw	0x1	—
6	THRESH_EN	rw	0x0	—
31:7	RSVD	r	0x0	—

26.17.19.1 MON_EN field

Monitor enable - master enable for write monitor

26.17.19.2 ERR_EN field

Error enable - enable error detection

26.17.19.3 COMPL_EN field

Completion enable - enable completion packets

26.17.19.4 TIMEOUT_EN field

Timeout enable - enable timeout detection

26.17.19.5 PERF_EN field

Performance enable - enable performance packets

26.17.19.6 COMPRESS_EN field

Compression enable - 1=compress the monbus write stream, 0=raw 3-beat records. Only effective when the monbus group is built with USE_COMPRESSION=1. Program once before monitoring starts (must be stable while the write path is active).

26.17.19.7 THRESH_EN field

Threshold enable - enable latency-threshold packets. Before this field existed the threshold cone was gated by PERF_EN, so a host that left PERF_EN clear saw no threshold packets however low it set LATENCY_THRESH.

26.17.19.8 RSVD field

Reserved

26.17.20 WRMON_TIMEOUT register

- Absolute Address: 0x1104
- Base Offset: 0x104

- Size: 0x4

Write engine AXI monitor timeout threshold (cycles)

Bits	Identifier	Access	Reset	Name
31:0	TIMEOUT_CYCLES	rw	0x2710	—

26.17.20.1 TIMEOUT_CYCLES field

Timeout cycles [31:0]

26.17.21 WRMON_LATENCY_THRESH register

- Absolute Address: 0x1108
- Base Offset: 0x108
- Size: 0x4

Write engine AXI monitor latency threshold (cycles)

Bits	Identifier	Access	Reset	Name
31:0	LATENCY_THRESH	rw	0x1388	—

26.17.21.1 LATENCY_THRESH field

Latency threshold cycles [31:0]

26.17.22 WRMON_PKT_MASK register

- Absolute Address: 0x110C
- Base Offset: 0x10C
- Size: 0x4

Write engine AXI monitor packet type filtering (16-bit mask)

Bits	Identifier	Access	Reset	Name
15:0	PKT_MASK	rw	0xFFFF	—
31:16	RSVD	r	0x0	—

26.17.22.1 PKT_MASK field

Packet type mask [15:0] - bit[type] = 1 MASKS (drops) that packet type, at the monitor and again at the monbus group. Reset = all masked: the host clears the bits of the classes it wants to see (stream BUG-016 corrected this description; the polarity and the reset are as intended)

26.17.22.2 RSVD field

Reserved

26.17.23 WRMON_ERR_CFG register

- Absolute Address: 0x1110
- Base Offset: 0x110
- Size: 0x4

Write engine AXI monitor error selection and filtering

Bits	Identifier	Access	Reset	Name
15:0	ERR_SELECT	rw	0x0	—
31:16	ERR_MASK	rw	0xFFFF	—

26.17.23.1 ERR_SELECT field

Error select - per-packet-type route: 1=err FIFO/IRQ, 0=bulk trace

26.17.23.2 ERR_MASK field

Error mask - per-event-code drop mask, indexed by event_code[3:0]

26.17.24 WRMON_MASK1 register

- Absolute Address: 0x1114
- Base Offset: 0x114
- Size: 0x4

Write engine AXI monitor timeout and completion masks

Bits	Identifier	Access	Reset	Name
15:0	TIMEOUT_MASK	rw	0xFFFF	—
31:16	COMPL_MASK	rw	0x0	—

26.17.24.1 TIMEOUT_MASK field

Timeout mask

26.17.24.2 COMPL_MASK field

Completion mask

26.17.25 WRMON_MASK2 register

- Absolute Address: 0x1118
- Base Offset: 0x118
- Size: 0x4

Write engine AXI monitor threshold and performance masks

Bits	Identifier	Access	Reset	Name
15:0	THRESH_MASK	rw	0xFFFF	—
31:16	PERF_MASK	rw	0x0	—

26.17.25.1 THRESH_MASK field

Threshold mask

26.17.25.2 PERF_MASK field

Performance mask

26.17.26 WRMON_MASK3 register

- Absolute Address: 0x111C
- Base Offset: 0x11C
- Size: 0x4

Write engine AXI monitor address and debug masks

Bits	Identifier	Access	Reset	Name
15:0	ADDR_MASK	rw	0xFFFF	—
31:16	DEBUG_MASK	rw	0x0	—

26.17.26.1 ADDR_MASK field

Address mask

26.17.26.2 DEBUG_MASK field

Debug mask

26.17.27 DAXMON_PERF_CTRL register

- Absolute Address: 0x1150
- Base Offset: 0x150
- Size: 0x4

Perf-window run control for the descriptor AXI monitor

Bits	Identifier	Access	Reset	Name
0	RUN	rw	0x0	—
31:1	RSVD	r	0x0	—

26.17.27.1 RUN field

Perf window run - 1=open window and accumulate cycle buckets, 0=close window and freeze counters. Rising edge clears all counters.

26.17.27.2 RSVD field

Reserved

26.17.28 DAXMON_PERF_STATUS register

- Absolute Address: 0x1154
- Base Offset: 0x154
- Size: 0x4

Perf-window live status

Bits	Identifier	Access	Reset	Name
0	WIN_ACTIVE	r	—	—
31:1	RSVD	r	0x0	—

26.17.28.1 WIN_ACTIVE field

Window active - 1=window open and accumulating

26.17.28.2 RSVD field

Reserved

26.17.29 DAXMON_PERF_WINDOW_CYCLES register

- Absolute Address: 0x1158
- Base Offset: 0x158

- Size: 0x4

LIVE free-running window-cycle counter. Valid only while DAXMON_PERF_STATUS.WIN_ACTIVE=1; the monitor ZEROES this when the window closes (RUN->0). For a closed window, use the sum of the four bucket registers (PROD+BP+STARV+IDLE), which HOLD their values after close.

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.29.1 VAL field

Window cycles [31:0] (live-only; reads 0 after close)

26.17.30 DAXMON_PERF_PROD_CYCLES register

- Absolute Address: 0x115C
- Base Offset: 0x15C
- Size: 0x4

Cycles with R data valid && ready (data delivered)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.30.1 VAL field

Productive cycles [31:0]

26.17.31 DAXMON_PERF_BP_CYCLES register

- Absolute Address: 0x1160
- Base Offset: 0x160
- Size: 0x4

Cycles with R data valid && !ready (back-pressure)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.31.1 VAL field

Backpressure cycles [31:0]

26.17.32 DAXMON_PERF_STARV_CYCLES register

- Absolute Address: 0x1164

- Base Offset: 0x164
- Size: 0x4

Cycles with !R data valid && ready (starvation)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.32.1 VAL field

Starvation cycles [31:0]

26.17.33 DAXMON_PERF_IDLE_CYCLES register

- Absolute Address: 0x1168
- Base Offset: 0x168
- Size: 0x4

Cycles with !R data valid && !ready (idle)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.33.1 VAL field

Idle cycles [31:0]

26.17.34 DAXMON_PERF_BEAT_COUNT register

- Absolute Address: 0x116C
- Base Offset: 0x16C
- Size: 0x4

R beats transferred in window (= productive cycles)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.34.1 VAL field

Beat count [31:0]

26.17.35 DAXMON_PERF_BYTE_COUNT_LO register

- Absolute Address: 0x1170
- Base Offset: 0x170

- Size: 0x4

Bytes transferred in window (lower 32 bits)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.35.1 VAL field

Byte count [31:0]

26.17.36 DAXMON_PERF_BYTE_COUNT_HI register

- Absolute Address: 0x1174
- Base Offset: 0x174
- Size: 0x4

Bytes transferred in window (upper 32 bits)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.36.1 VAL field

Byte count [63:32]

26.17.37 DAXMON_PERF_BURST_COUNT register

- Absolute Address: 0x1178
- Base Offset: 0x178
- Size: 0x4

AR handshakes (bursts issued) in window

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.37.1 VAL field

Burst count [31:0]

26.17.38 RDMON_PERF_CTRL register

- Absolute Address: 0x1180
- Base Offset: 0x180
- Size: 0x4

Perf-window run control for the data-read datapath monitor

Bits	Identifier	Access	Reset	Name
0	RUN	rw	0x0	—
31:1	RSVD	r	0x0	—

26.17.38.1 RUN field

Perf window run - 1=open window and accumulate cycle buckets, 0=close window and freeze counters. Rising edge clears all counters.

26.17.38.2 RSVD field

Reserved

26.17.39 RDMON_PERF_STATUS register

- Absolute Address: 0x1184
- Base Offset: 0x184
- Size: 0x4

Perf-window live status

Bits	Identifier	Access	Reset	Name
0	WIN_ACTIVE	r	—	—
31:1	RSVD	r	0x0	—

26.17.39.1 WIN_ACTIVE field

Window active - 1=window open and accumulating

26.17.39.2 RSVD field

Reserved

26.17.40 RDMON_PERF_WINDOW_CYCLES register

- Absolute Address: 0x1188
- Base Offset: 0x188
- Size: 0x4

LIVE free-running window-cycle counter. Valid only while RDMON_PERF_STATUS.WIN_ACTIVE=1; the monitor ZEROES this when the window closes (RUN->0). For a closed window, use the sum of the four bucket registers (PROD+BP+STARV+IDLE), which HOLD their values after close.

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.40.1 VAL field

Window cycles [31:0] (live-only; reads 0 after close)

26.17.41 RDMON_PERF_PROD_CYCLES register

- Absolute Address: 0x118C
- Base Offset: 0x18C
- Size: 0x4

Cycles with R data valid && ready (data delivered)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.41.1 VAL field

Productive cycles [31:0]

26.17.42 RDMON_PERF_BP_CYCLES register

- Absolute Address: 0x1190
- Base Offset: 0x190
- Size: 0x4

Cycles with R data valid && !ready (back-pressure)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.42.1 VAL field

Backpressure cycles [31:0]

26.17.43 RDMON_PERF_STARV_CYCLES register

- Absolute Address: 0x1194
- Base Offset: 0x194
- Size: 0x4

Cycles with !R data valid && ready (starvation)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.43.1 VAL field

Starvation cycles [31:0]

26.17.44 RDMON_PERF_IDLE_CYCLES register

- Absolute Address: 0x1198
- Base Offset: 0x198
- Size: 0x4

Cycles with !R data valid && !ready (idle)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.44.1 VAL field

Idle cycles [31:0]

26.17.45 RDMON_PERF_BEAT_COUNT register

- Absolute Address: 0x119C
- Base Offset: 0x19C
- Size: 0x4

R beats transferred in window (= productive cycles)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.45.1 VAL field

Beat count [31:0]

26.17.46 RDMON_PERF_BYTE_COUNT_LO register

- Absolute Address: 0x11A0
- Base Offset: 0x1A0
- Size: 0x4

Bytes transferred in window (lower 32 bits)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.46.1 VAL field

Byte count [31:0]

26.17.47 RDMON_PERF_BYTE_COUNT_HI register

- Absolute Address: 0x11A4
- Base Offset: 0x1A4
- Size: 0x4

Bytes transferred in window (upper 32 bits)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.47.1 VAL field

Byte count [63:32]

26.17.48 RDMON_PERF_BURST_COUNT register

- Absolute Address: 0x11A8
- Base Offset: 0x1A8
- Size: 0x4

AR handshakes (bursts issued) in window

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.48.1 VAL field

Burst count [31:0]

26.17.49 WRMON_PERF_CTRL register

- Absolute Address: 0x11B0
- Base Offset: 0x1B0
- Size: 0x4

Perf-window run control for the data-write datapath monitor

Bits	Identifier	Access	Reset	Name
0	RUN	rw	0x0	—
31:1	RSVD	r	0x0	—

26.17.49.1 RUN field

Perf window run - 1=open window and accumulate cycle buckets, 0=close window and freeze counters. Rising edge clears all counters.

26.17.49.2 RSVD field

Reserved

26.17.50 WRMON_PERF_STATUS register

- Absolute Address: 0x11B4
- Base Offset: 0x1B4
- Size: 0x4

Perf-window live status

Bits	Identifier	Access	Reset	Name
0	WIN_ACTIVE	r	—	—
31:1	RSVD	r	0x0	—

26.17.50.1 WIN_ACTIVE field

Window active - 1=window open and accumulating

26.17.50.2 RSVD field

Reserved

26.17.51 WRMON_PERF_WINDOW_CYCLES register

- Absolute Address: 0x11B8
- Base Offset: 0x1B8
- Size: 0x4

LIVE free-running window-cycle counter. Valid only while WRMON_PERF_STATUS.WIN_ACTIVE=1; the monitor ZEROES this when the window closes (RUN->0). For a closed window, use the sum of the four bucket registers (PROD+BP+STARV+IDLE), which HOLD their values after close.

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.51.1 VAL field

Window cycles [31:0] (live-only; reads 0 after close)

26.17.52 WRMON_PERF_PROD_CYCLES register

- Absolute Address: 0x11BC
- Base Offset: 0x1BC
- Size: 0x4

Cycles with W data valid && ready (beat delivered)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.52.1 VAL field

Productive cycles [31:0]

26.17.53 WRMON_PERF_BP_CYCLES register

- Absolute Address: 0x11C0
- Base Offset: 0x1C0
- Size: 0x4

Cycles with W data valid && !ready (back-pressure)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.53.1 VAL field

Backpressure cycles [31:0]

26.17.54 WRMON_PERF_STARV_CYCLES register

- Absolute Address: 0x11C4
- Base Offset: 0x1C4
- Size: 0x4

Cycles with !W data valid && ready (starvation)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.54.1 VAL field

Starvation cycles [31:0]

26.17.55 WRMON_PERF_IDLE_CYCLES register

- Absolute Address: 0x11C8
- Base Offset: 0x1C8
- Size: 0x4

Cycles with !W data valid && !ready (idle)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.55.1 VAL field

Idle cycles [31:0]

26.17.56 WRMON_PERF_BEAT_COUNT register

- Absolute Address: 0x11CC
- Base Offset: 0x1CC
- Size: 0x4

W beats transferred in window (= productive cycles)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.56.1 VAL field

Beat count [31:0]

26.17.57 WRMON_PERF_BYTE_COUNT_LO register

- Absolute Address: 0x11D0
- Base Offset: 0x1D0
- Size: 0x4

Bytes transferred in window (lower 32 bits)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.57.1 VAL field

Byte count [31:0]

26.17.58 WRMON_PERF_BYTE_COUNT_HI register

- Absolute Address: 0x11D4
- Base Offset: 0x1D4
- Size: 0x4

Bytes transferred in window (upper 32 bits)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.58.1 VAL field

Byte count [63:32]

26.17.59 WRMON_PERF_BURST_COUNT register

- Absolute Address: 0x11D8
- Base Offset: 0x1D8
- Size: 0x4

AW handshakes (bursts issued) in window

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.59.1 VAL field

Burst count [31:0]

26.17.60 RDMON_PERF_CH_PROD_BP register

- Absolute Address: 0x11E0
- Base Offset: 0x1E0
- Size: 0x4

Selected channel: {backpressure[31:16], productive[15:0]} (16-bit each)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.60.1 VAL field

{bp[15:0], prod[15:0]} for PERF_CH_SEL channel

26.17.61 RDMON_PERF_CH_STARV_IDLE register

- Absolute Address: 0x11E4
- Base Offset: 0x1E4
- Size: 0x4

Selected channel: {idle[31:16], starvation[15:0]} (16-bit each)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.61.1 VAL field

{idle[15:0], starv[15:0]} for PERF_CH_SEL channel

26.17.62 WRMON_PERF_CH_PROD_BP register

- Absolute Address: 0x11E8
- Base Offset: 0x1E8
- Size: 0x4

Selected channel: {backpressure[31:16], productive[15:0]} (16-bit each)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.62.1 VAL field

{bp[15:0], prod[15:0]} for PERF_CH_SEL channel

26.17.63 WRMON_PERF_CH_STARV_IDLE register

- Absolute Address: 0x11EC
- Base Offset: 0x1EC
- Size: 0x4

Selected channel: {idle[31:16], starvation[15:0]} (16-bit each)

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.63.1 VAL field

{idle[15:0], starv[15:0]} for PERF_CH_SEL channel

26.17.64 RDMON_PERF_CH_OVERFLOW register

- Absolute Address: 0x11F0
- Base Offset: 0x1F0
- Size: 0x4

All channels, {prod,bp,starv,idle} sticky overflow per channel (4 bits/channel, channel 0 in the low nibble). A set bit means that 16-bit per-channel bucket wrapped during the window.

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.64.1 VAL field

Per-channel overflow mask (NUM_CHANNELS*4 bits)

26.17.65 WRMON_PERF_CH_OVERFLOW register

- Absolute Address: 0x11F4
- Base Offset: 0x1F4
- Size: 0x4

All channels, {prod,bp,starv,idle} sticky overflow per channel.

Bits	Identifier	Access	Reset	Name
31:0	VAL	r	—	—

26.17.65.1 VAL field

Per-channel overflow mask (NUM_CHANNELS*4 bits)

26.17.66 RDMON_ADDR_RANGE0_LOW register

- Absolute Address: 0x1200
- Base Offset: 0x200
- Size: 0x4

Read monitor range0 inclusive low bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0x0	—

26.17.67 RDMON_ADDR_RANGE0_HIGH register

- Absolute Address: 0x1204
- Base Offset: 0x204
- Size: 0x4

Read monitor range0 inclusive high bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0xFFFFFFFF	—

26.17.68 RDMON_ADDR_RANGE1_LOW register

- Absolute Address: 0x1208
- Base Offset: 0x208
- Size: 0x4

Read monitor range1 inclusive low bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0x0	—

26.17.69 RDMON_ADDR_RANGE1_HIGH register

- Absolute Address: 0x120C
- Base Offset: 0x20C
- Size: 0x4

Read monitor range1 inclusive high bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0xFFFFFFFF	—

26.17.70 RDMON_ADDR_RANGE2_LOW register

- Absolute Address: 0x1210
- Base Offset: 0x210
- Size: 0x4

Read monitor range2 inclusive low bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0x0	—

26.17.71 RDMON_ADDR_RANGE2_HIGH register

- Absolute Address: 0x1214
- Base Offset: 0x214
- Size: 0x4

Read monitor range2 inclusive high bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0xFFFFFFFF	—

26.17.72 RDMON_ADDR_RANGE3_LOW register

- Absolute Address: 0x1218
- Base Offset: 0x218
- Size: 0x4

Read monitor range3 inclusive low bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0x0	—

26.17.73 RDMON_ADDR_RANGE3_HIGH register

- Absolute Address: 0x121C
- Base Offset: 0x21C
- Size: 0x4

Read monitor range3 inclusive high bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0xFFFFFFFF	—

26.17.74 RDMON_ADDR_RANGE_CTRL register

- Absolute Address: 0x1220
- Base Offset: 0x220

- Size: 0x4

Read monitor address-range checker enables

Bits	Identifier	Access	Reset	Name
3:0	RANGE_EN	rw	0x0	—
4	CHECK_EN	rw	0x0	—
5	MATCH_EN	rw	0x0	—
6	MISS_EN	rw	0x0	—

26.17.74.1 RANGE_EN field

Per-range enable mask (bit i enables range i)

26.17.74.2 CHECK_EN field

Master addr-check enable

26.17.74.3 MATCH_EN field

DEBUG/match path enable (drives cfg_debug_enable)

26.17.74.4 MISS_EN field

ERROR/miss path enable (drives cfg_error_enable)

26.17.75 WRMON_ADDR_RANGE0_LOW register

- Absolute Address: 0x1230
- Base Offset: 0x230
- Size: 0x4

Write monitor range0 inclusive low bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0x0	—

26.17.76 WRMON_ADDR_RANGE0_HIGH register

- Absolute Address: 0x1234
- Base Offset: 0x234
- Size: 0x4

Write monitor range0 inclusive high bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0xFFFFFFFF	—

26.17.77 WRMON_ADDR_RANGE1_LOW register

- Absolute Address: 0x1238
- Base Offset: 0x238
- Size: 0x4

Write monitor range1 inclusive low bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0x0	—

26.17.78 WRMON_ADDR_RANGE1_HIGH register

- Absolute Address: 0x123C
- Base Offset: 0x23C
- Size: 0x4

Write monitor range1 inclusive high bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0xFFFFFFFF	—

26.17.79 WRMON_ADDR_RANGE2_LOW register

- Absolute Address: 0x1240
- Base Offset: 0x240
- Size: 0x4

Write monitor range2 inclusive low bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0x0	—

26.17.80 WRMON_ADDR_RANGE2_HIGH register

- Absolute Address: 0x1244
- Base Offset: 0x244
- Size: 0x4

Write monitor range2 inclusive high bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0xFFFFFFFF	—

26.17.81 WRMON_ADDR_RANGE3_LOW register

- Absolute Address: 0x1248
- Base Offset: 0x248
- Size: 0x4

Write monitor range3 inclusive low bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0x0	—

26.17.82 WRMON_ADDR_RANGE3_HIGH register

- Absolute Address: 0x124C
- Base Offset: 0x24C
- Size: 0x4

Write monitor range3 inclusive high bound

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0xFFFFFFFF	—

26.17.83 WRMON_ADDR_RANGE_CTRL register

- Absolute Address: 0x1250
- Base Offset: 0x250
- Size: 0x4

Write monitor address-range checker enables

Bits	Identifier	Access	Reset	Name
3:0	RANGE_EN	rw	0x0	—
4	CHECK_EN	rw	0x0	—
5	MATCH_EN	rw	0x0	—
6	MISS_EN	rw	0x0	—

26.17.83.1 RANGE_EN field

Per-range enable mask (bit i enables range i)

26.17.83.2 CHECK_EN field

Master addr-check enable

26.17.83.3 MATCH_EN field

DEBUG/match path enable (drives cfg_debug_enable)

26.17.83.4 MISS_EN field

ERROR/miss path enable (drives cfg_error_enable)

26.17.84 MON_GROUP_BASE_ADDR register

- Absolute Address: 0x1260
- Base Offset: 0x260
- Size: 0x4

Master-write window inclusive low bound (monbus bulk-trace)

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0x40000	—

26.17.85 MON_GROUP_LIMIT_ADDR register

- Absolute Address: 0x1264
- Base Offset: 0x264
- Size: 0x4

Master-write window inclusive high bound (monbus bulk-trace)

Bits	Identifier	Access	Reset	Name
31:0	VALUE	rw	0x7FFFF	—

26.17.86 MON_GROUP_FLUSH_WATERMARK register

- Absolute Address: 0x1268
- Base Offset: 0x268
- Size: 0x4

Flush burst once this many beats buffered; 0 = flush every complete record

Bits	Identifier	Access	Reset	Name
15:0	VALUE	rw	0x0	—

27 Chapter 5: Programming Models

This chapter provides software developer guidance for using the STREAM DMA engine.

27.1 Contents

27.1.1 01_initialization.md

- Power-on initialization sequence
- Register configuration
- Channel setup
- Configuration presets (minimal, high-performance)

27.1.2 02_single_transfer.md

- Descriptor format (256-bit)
- Kick-off address map and write sequence
- Simple single-descriptor transfer
- C code examples with complete workflow

27.1.3 03_chained_transfers.md

- Multi-descriptor chains
- Descriptor linking via next_descriptor_ptr
- Chain termination methods
- Scatter-gather operations
- Prefetch mode for performance

27.1.4 04_multi_channel.md

- 8-channel concurrent operations
- Priority-based scheduling
- Resource sharing strategies
- Channel pooling and load balancing
- Performance considerations

27.1.5 05_error_handling.md

- Error types (AXI, timeout, internal)
- Error detection registers
- Recovery procedures
- Interrupt-based error handling
- Debug monitoring

27.2 Planned (Future)

27.2.1 06_performance_tuning.md

- Burst size selection
- Priority tuning
- SRAM depth considerations
- Maximizing throughput

27.2.2 07_software_examples.md

- Complete working examples
- Linux driver skeleton
- Bare-metal usage
- Common use cases

Status: Core programming guides complete (01-05)

Target Audience: - Software engineers integrating STREAM - Driver developers - System architects - Application developers

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28 Chapter 6: Configuration Reference

Version: 0.90 **Last Updated:** 2025-11-22 **Purpose:** Complete reference for all STREAM configuration signals

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28.1 Overview

STREAM provides comprehensive runtime configuration through APB registers and compile-time parameters. This chapter documents all configuration signals, their valid ranges, default values, and recommended settings for different use cases.

28.1.1 Configuration Categories

Configuration Categories

Category	Signals	Purpose
Channel Control	2	Enable/reset individual channels
Scheduler	6	Transfer scheduling and timeouts
Descriptor Engine	7	Descriptor fetch behavior
AXI Monitors	45 (15 per monitor × 3)	Debug/trace filtering
AXI Transfer	2	Burst configuration
Performance	3	Profiling and metrics
Total	**65 configuration signals**	Full system control

28.2 1. Channel Control Configuration

28.2.1 cfg_channel_enable[NUM_CHANNELS-1:0]

Type: Per-channel enable **Width:** 1 bit × NUM_CHANNELS (default 8) **Default:** 8'hFF (all enabled) **Register:** CHANNEL_ENABLE @ 0x120

Description: Enables or disables individual DMA channels. When disabled, the channel: - Ignores descriptor kick-off requests - Completes current transfer if in progress - Enters idle state - Remains in idle until re-enabled

Valid Values: - `1'b1`: Channel enabled (can accept transfers) - `1'b0`: Channel disabled (ignores new transfers)

Use Cases:

```
// Enable only channels 0, 2, 4 (even channels)
cfg_channel_enable = 8'b01010101;

// Disable all channels (emergency stop)
cfg_channel_enable = 8'b00000000;

// Enable all channels (normal operation)
cfg_channel_enable = 8'b11111111;
```

Interaction with Global Enable: The final channel enable is: `cfg_channel_enable[i] & reg_global_ctrl_global_en`

28.2.2 cfg_channel_reset[NUM_CHANNELS-1:0]

Type: Per-channel reset **Width:** 1 bit × NUM_CHANNELS (default 8) **Default:** 8'h00 (no resets active) **Register:** CHANNEL_RESET @ 0x124

Description: Asserts reset for individual channels without affecting other channels or global state. When asserted: - Channel FSM returns to IDLE - Pending transfers aborted - SRAM buffers flushed - Descriptor engine reset

Valid Values: - `1'b1`: Channel in reset (clears state) - `1'b0`: Channel operating normally

Use Cases:

```
// Reset channel 3 after error
cfg_channel_reset = 8'b00001000;
wait_cycles(10);
cfg_channel_reset = 8'b00000000;  // Release reset

// Reset all channels (soft reset)
```

```
cfg_channel_reset = 8'hFF;
wait_cycles(10);
cfg_channel_reset = 8'h00;
```

IMPORTANT: Assert reset for minimum 10 clock cycles to ensure complete FSM reset.

28.3 2. Scheduler Configuration

28.3.1 cfg_sched_timeout_cycles[15:0]

Type: Timeout threshold **Width:** 16 bits **Default:** 16'd1000 (1,000 cycles) **Register:** SCHED_TIMEOUT_CYCLES @ 0x200[15:0]

Description: Number of clock cycles before a channel operation times out. Applies to: - Descriptor fetch latency - AXI read/write response latency - Scheduler state transitions

Valid Range: 100 to 65535 cycles **Typical Values:** - Fast SRAM: 100-500 cycles - DDR3/DDR4: 1000-10000 cycles - High-latency: 10000-65535 cycles

Calculation:

```
Timeout cycles = (Expected latency × 10) + margin

Example for DDR4 @ 200 MHz:
- Expected read latency: 100 cycles (500 ns)
- Safety margin: 10x
- Timeout = 100 × 10 = 1000 cycles
```

28.3.2 cfg_sched_enable

Type: Global scheduler enable **Width:** 1 bit **Default:** 1'b1 (enabled) **Register:** SCHED_CONFIG @ 0x204[0]

Description: Master enable for all schedulers. When disabled, all channels stop scheduling new operations.

28.3.3 cfg_sched_timeout_enable

Type: Timeout detection enable **Width:** 1 bit **Default:** 1'b1 (enabled) **Register:** SCHED_CONFIG @ 0x204[1]

Description: Enables timeout detection for scheduler operations. Disable for simulation or known slow memory.

28.3.4 cfg_sched_err_enable

Type: Error detection enable **Width:** 1 bit **Default:** 1'b1 (enabled) **Register:** SCHED_CONFIG @ 0x204[2]

Description: Enables error packet generation for scheduler errors (AXI SLVERR, DECERR, timeouts).

28.3.5 cfg_sched_compl_enable

Type: Completion event enable **Width:** 1 bit **Default:** 1'b1 (enabled by default) **Register:** SCHED_CONFIG @ 0x204[3]

Description: Enables MonBus packets for transfer completion events. Generate high traffic, use sparingly.

28.3.6 cfg_sched_perf_enable

Type: Performance monitoring enable **Width:** 1 bit **Default:** 1'b0 (disabled by default) **Register:** SCHED_CONFIG @ 0x204[4]

Description: Enables performance profiling packets (latency, throughput metrics).

28.3.7 cfg_rd_prefetch_enable

Type: Read-ahead descriptor prefetch enable **Width:** 1 bit **Default:** 1'b1 (enabled) **Register:** SCHED_CONFIG @ 0x204[5]

Description: Enables read-ahead descriptor prefetch. On a chained legacy descriptor the scheduler read side loads the next descriptor from the descriptor-FIFO head and keeps filling the SRAM while the write side drains the current descriptor, collapsing the per-descriptor boundary bubble to zero (cross-descriptor streaming). Clear it for lockstep behaviour — useful for an on-silicon A/B on a single bitstream. Ignored for EXT (row/col-major) descriptors, which always run lockstep. See MAS Section 2.4 (Scheduler) and HAS Section 5.1.

28.4 3. Descriptor Engine Configuration

28.4.1 cfg_desceng_enable

Type: Descriptor engine global enable **Width:** 1 bit **Default:** 1'b1 (enabled) **Register:** DESCENG_CONFIG.DESCENG_EN @ 0x220[0]

Description: Global enable for all descriptor engines. When disabled, descriptor fetch stops.

28.4.2 cfg_desceng_prefetch

Type: Prefetch enable **Width:** 1 bit **Default:** 1'b0 (disabled) **Register:** DESCENG_CONFIG.PREFETCH_EN @ 0x220[1]

Description: Enables descriptor prefetching to hide fetch latency.

Prefetch Behavior: - Enabled: Fetch next descriptor while current transfer executes - Disabled: Wait for current transfer completion before fetching next

Performance Impact: - Prefetch ON: +15-30% throughput for chained descriptors - Prefetch OFF: Simpler logic, lower area

---

28.4.3 cfg_desceng_fifo_thresh[3:0]

Type: FIFO threshold **Width:** 4 bits **Default:** 4'h8 (8 entries) **Register:** DESCENG_CONFIG.FIFO_THRESH @ 0x220[5:2]

Description: Number of entries in descriptor FIFO before asserting backpressure.

Valid Range: 1-15 entries **Typical Values:** - Low latency: 2-4 entries - Balanced: 8 entries (default) - High throughput: 12-15 entries

---

28.4.4 cfg_desceng_addr0_base[31:0]

Type: Base address for descriptor region 0 **Width:** 32 bits **Default:** 32'h0000_0000 **Register:** DESCENG_ADDR0_BASE @ 0x224

Description: Base address of first descriptor memory region. Descriptors fetched from this region if within range.

Alignment: Must be aligned to descriptor size (256 bits = 32 bytes)

28.4.5 cfg_desceng_addr0_limit[31:0]

Type: Limit address for descriptor region 0 **Width:** 32 bits **Default:** 32'hFFFF_FFFF (no limit)
Register: DESCENG_ADDR0_LIMIT @ 0x228

Description: Upper limit of first descriptor region. Descriptors beyond this address use region 1.

28.4.6 cfg_desceng_addr1_base[31:0]

Type: Base address for descriptor region 1 **Width:** 32 bits **Default:** 32'h0000_0000 **Register:** DESCENG_ADDR1_BASE @ 0x22C

Description: Base address of second descriptor memory region (optional).

28.4.7 cfg_desceng_addr1_limit[31:0]

Type: Limit address for descriptor region 1 **Width:** 32 bits **Default:** 32'hFFFF_FFFF **Register:** DESCENG_ADDR1_LIMIT @ 0x230

Description: Upper limit of second descriptor region.

28.5 4. AXI Monitor Configuration

STREAM includes three independent AXI monitors with identical configuration sets:

1. **Descriptor AXI Monitor** (`cfg_desc_mon_*`) - Monitors descriptor fetch AXI master
2. **Read Engine Monitor** (`cfg_rdeng_mon_*`) - Monitors data read AXI master
3. **Write Engine Monitor** (`cfg_wreng_mon_*`) - Monitors data write AXI master

Each monitor has 15 configuration signals with the same structure.

28.5.1 4.1 Descriptor AXI Monitor (cfg_desc_mon_*)

Register Base: 0x10C0-0x10DF (stream_mon_regs regfile)

28.5.1.1 cfg_desc_mon_enable

Type: Monitor master enable **Width:** 1 bit **Default:** 1'b1 (enabled) **Register:** DAXMON_ENABLE.MON_EN @ 0x10C0[0]

Description: Master enable for descriptor AXI monitor. All monitor packets disabled when this is 0.

28.5.1.2 cfg_desc_mon_err_enable

Type: Error packet enable **Width:** 1 bit **Default:** 1'b1 (enabled) **Register:** DAXMON_ENABLE.ERR_EN @ 0x10C0[1]

Description: Enables error packet generation (SLVERR, DECERR, protocol violations).

28.5.1.3 cfg_desc_mon_perf_enable

Type: Performance packet enable **Width:** 1 bit **Default:** 1'b0 (disabled) **Register:** DAXMON_ENABLE.COMPL_EN @ 0x10C0[2]

Description: Enables performance monitoring packets (latency, bandwidth).

WARNING: High packet rate - use only during debug.

28.5.1.4 cfg_desc_mon_timeout_enable

Type: Timeout detection enable **Width:** 1 bit **Default:** 1'b1 (enabled) **Register:** DAXMON_ENABLE.TIMEOUT_EN @ 0x10C0[3]

Description: Enables timeout packet generation when transactions exceed threshold.

28.5.1.5 cfg_desc_mon_timeout_cycles[31:0]

Type: Timeout threshold **Width:** 32 bits **Default:** 32'd10000 **Register:** DAXMON_TIMEOUT.TIMEOUT_CYCLES @ 0x10C4[31:0]

Description: Number of cycles before transaction times out.

28.5.1.6 cfg_desc_mon_latency_thresh[31:0]

Type: Latency threshold **Width:** 32 bits **Default:** 32'd1000 **Register:** DAXMON_LATENCY_THRESH.LATENCY_THRESH @ 0x10C8[31:0]

Description: Latency threshold for performance warnings.

28.5.1.7 cfg_desc_mon_pkt_mask[15:0]

Type: Packet type filter **Width:** 16 bits (1 bit per packet type) **Default:** 16'hFFFF (all types masked; the host opens the classes it wants) **Register:** DAXMON_PKT_MASK.PKT_MASK @ 0x10CC[15:0]

At the monbus group the effective AXI-protocol drop mask is the AND of `DAXMON_PKT_MASK`, `RDMON_PKT_MASK` and `WRMON_PKT_MASK` (stream BUG-017): a drop-all mask on one monitor switches that monitor off without silencing the other two.

Description: Drop mask over packet types: a SET bit masks (drops) that type, a clear bit lets it through. The reset value masks everything.

Packet Type Mapping:

```
Bit  | Packet Type
-----|-------------
0    | AR command
1    | AW command
2    | R completion
3    | W data
4    | B response
5    | Error
6    | Timeout
```

```
7    | Completion
8-15 | Reserved
```

Examples:

```
// Only errors
cfg_desc_mon_pkt_mask = 16'h0020;  // Bit 5

// Errors + timeouts
cfg_desc_mon_pkt_mask = 16'h0060;  // Bits 5-6

// All packets
cfg_desc_mon_pkt_mask = 16'hFFFF;
```

28.5.1.8 cfg_desc_mon_err_select[3:0]

Type: Error type selector **Width:** 4 bits **Default:** 4'hF (all errors) **Register:** DAXMON_ERR_CFG.ERR_SELECT @ 0x10D0[15:0]

Description: Selects which error types to monitor.

Error Type Bits:

```
Bit | Error Type
----|------------
0   | SLVERR
1   | DECERR
2   | Protocol violation
3   | Reserved
```

28.5.1.9 cfg_desc_mon_err_mask[7:0]

Type: Error event filter **Width:** 8 bits **Default:** 8'hFF (all errors) **Register:** DAXMON_ERR_CFG.ERR_MASK @ 0x10D0[31:16]

Description: Bit mask for error packet generation (channel-specific filtering).

28.5.1.10 cfg_desc_mon_timeout_mask[7:0]

Type: Timeout channel filter **Width:** 8 bits **Default:** 8'hFF (all channels) **Register:** DAXMON_MASK1.TIMEOUT_MASK @ 0x10D4[15:0]

Description: Channel mask for timeout packets. Set bit enables timeout detection for corresponding channel.

28.5.1.11 cfg_desc_mon_compl_mask[7:0]

Type: Completion channel filter **Width:** 8 bits **Default:** 8'h00 (no channels) **Register:** DAXMON_MASK1.COMPL_MASK @ 0x10D4[31:16]

Description: Channel mask for completion packets.

WARNING: Completion packets are high volume - enable only for specific channels.

28.5.1.12 cfg_desc_mon_thresh_mask[7:0]

Type: Threshold event filter **Width:** 8 bits **Default:** 8'hFF (all channels) **Register:** DAXMON_MASK2.THRESH_MASK @ 0x10D8[15:0]

Description: Channel mask for latency threshold exceedance packets.

28.5.1.13 cfg_desc_mon_perf_mask[7:0]

Type: Performance packet filter **Width:** 8 bits **Default:** 8'h00 (no channels) **Register:** DAXMON_MASK2.PERF_MASK @ 0x10D8[31:16]

Description: Channel mask for performance monitoring packets.

28.5.1.14 cfg_desc_mon_addr_mask[7:0]

Type: Address-based filter **Width:** 8 bits **Default:** 8'hFF (all addresses) **Register:** DAXMON_MASK3.ADDR_MASK @ 0x10DC[15:0]

Description: Channel mask for address-range-based packet filtering.

28.5.1.15 cfg_desc_mon_debug_mask[7:0]

Type: Debug event filter **Width:** 8 bits **Default:** 8'h00 (no debug packets) **Register:** DAXMON_MASK3.DEBUG_MASK @ 0x10DC[31:16]

Description: Channel mask for debug-level packets (verbose trace).

28.5.2 4.2 Read Engine Monitor (cfg_rdeng_mon_*)

Register Base: 0x10E0-0x10FF (stream_mon_regs regfile)

All signals identical to Descriptor Monitor (Section 4.1), but applied to data read AXI master.

Key Differences: - Monitors data transfers (not descriptors) - Higher throughput → more packets - Recommended: Keep `cfg_rdeng_mon_compl_enable = 0` unless debugging

---

28.5.3 4.3 Write Engine Monitor (cfg_wreng_mon_*)

Register Base: 0x1100-0x111F (stream_mon_regs regfile)

All signals identical to Descriptor Monitor (Section 4.1), but applied to data write AXI master.

Key Differences: - Monitors write transactions (AW/W/B channels) - B response timing critical for performance - Recommended: Enable only error packets by default

---

28.6 5. AXI Transfer Configuration

28.6.1 cfg_axi_rd_xfer_beats[7:0]

Type: Read transfer size **Width:** 8 bits **Default:** 8'd16 (16 beats) **Register:** AXI_XFER_CONFIG.RD_XFER_BEATS @ 0x2A0[7:0]

Description: Default number of beats per AXI read burst. Actual burst size may be less to respect 4KB boundaries.

Valid Range: 1-256 beats (AXI4 standard) **Typical Values:** - Small transfers: 4-8 beats - Balanced: 16 beats (default) - Large transfers: 32-64 beats - Maximum throughput: 128-256 beats

Calculation:

```
Transfer size (bytes) = beats × (DATA_WIDTH / 8)

Example for DATA_WIDTH = 512 bits:
- 16 beats = 16 × 64 = 1024 bytes (1 KB)
- 64 beats = 64 × 64 = 4096 bytes (4 KB)
```

28.6.2 cfg_axi_wr_xfer_beats[7:0]

Type: Write transfer size **Width:** 8 bits **Default:** 8'd16 (16 beats) **Register:** AXI_XFER_CONFIG.WR_XFER_BEATS @ 0x2A0[15:8]

Description: Default number of beats per AXI write burst.

Same constraints as `cfg_axi_rd_xfer_beats`

28.7 6. Performance Profiler Configuration

28.7.1 cfg_perf_enable

Type: Profiler enable **Width:** 1 bit **Default:** 1'b0 (disabled) **Register:** PERF_CONFIG.PERF_EN @ 0x2B0[0]

Description: Enables performance profiling for all channels. When enabled, profiler captures: - Transfer start/end timestamps - Latency per channel - Throughput measurements - Channel utilization

---

28.7.2 cfg_perf_mode

Type: Profiling mode **Width:** 1 bit **Default:** 1'b0 (timestamp mode) **Register:** PERF_CONFIG.PERF_MODE @ 0x2B0[1]

Description: Selects profiling mode: - `1'b0`: Timestamp mode - Record absolute timestamps - `1'b1`: Elapsed time mode - Record delta times

Use Cases: - Timestamp: Correlate events across multiple blocks - Elapsed: Measure operation latencies

---

28.7.3 cfg_perf_clear

Type: Clear profiler state **Width:** 1 bit (write-only) **Default:** 1'b0 **Register:** PERF_CONFIG.PERF_CLEAR @ 0x2B0[2]

Description: Write `1'b1` to clear profiler FIFOs and counters. Self-clearing (automatically returns to 0).

28.8 7. Configuration Presets

28.8.17.1 Minimal Configuration (Tutorial/Embedded)

Use Case: Educational, minimal logic, single-channel operation

```
// Channel control
cfg_channel_enable = 8'b00000001;  // Only channel 0

// Scheduler
cfg_sched_enable = 1'b1;
cfg_sched_timeout_cycles = 16'd500;      // Short timeout (SRAM)
cfg_sched_timeout_enable = 1'b1;
cfg_sched_err_enable = 1'b1;
cfg_sched_compl_enable = 1'b0;           // Disable completion packets
cfg_sched_perf_enable = 1'b0;            // Disable performance
packets

// Descriptor engine
cfg_desceng_enable = 1'b1;
cfg_desceng_prefetch = 1'b0;             // No prefetch (simpler)
cfg_desceng_fifo_thresh = 4'h4;          // Small FIFO

// All monitors DISABLED (reduce logic)
cfg_desc_mon_enable = 1'b0;
cfg_rdeng_mon_enable = 1'b0;
cfg_wreng_mon_enable = 1'b0;

// AXI transfer
cfg_axi_rd_xfer_beats = 8'd8;            // Small bursts
cfg_axi_wr_xfer_beats = 8'd8;

// Performance profiler
cfg_perf_enable = 1'b0;                  // Disabled
```

28.8.27.2 Balanced Configuration (Typical FPGA)

Use Case: General-purpose DMA, moderate channels, balanced performance/area

```
// Channel control
cfg_channel_enable = 8'hFF;              // All 8 channels

// Scheduler
cfg_sched_enable = 1'b1;
cfg_sched_timeout_cycles = 16'd5000;     // DDR4 timeout
```

```
cfg_sched_timeout_enable = 1'b1;
cfg_sched_err_enable = 1'b1;
cfg_sched_compl_enable = 1'b0;          // Errors only
cfg_sched_perf_enable = 1'b0;

// Descriptor engine
cfg_desceng_enable = 1'b1;
cfg_desceng_prefetch = 1'b1;            // Enable prefetch
cfg_desceng_fifo_thresh = 4'h8;         // Balanced FIFO

// Descriptor monitor (errors only)
cfg_desc_mon_enable = 1'b1;
cfg_desc_mon_err_enable = 1'b1;
cfg_desc_mon_perf_enable = 1'b0;
cfg_desc_mon_timeout_enable = 1'b1;
cfg_desc_mon_timeout_cycles = 32'd10000;
cfg_desc_mon_pkt_mask = 16'h0060;       // Errors + timeouts

// Read/write monitors (errors only)
cfg_rdeng_mon_enable = 1'b1;
cfg_rdeng_mon_err_enable = 1'b1;
cfg_rdeng_mon_perf_enable = 1'b0;

cfg_wreng_mon_enable = 1'b1;
cfg_wreng_mon_err_enable = 1'b1;
cfg_wreng_mon_perf_enable = 1'b0;

// AXI transfer
cfg_axi_rd_xfer_beats = 8'd32;          // Moderate bursts
cfg_axi_wr_xfer_beats = 8'd32;

// Performance profiler
cfg_perf_enable = 1'b1;                 // Enable profiling
cfg_perf_mode = 1'b1;                   // Elapsed time mode
```

28.8.3 7.3 High-Performance Configuration (ASIC/Datacenter)

Use Case: Maximum throughput, all channels active, full monitoring

```
// Channel control
cfg_channel_enable = 8'hFF;             // All channels

// Scheduler
cfg_sched_enable = 1'b1;
cfg_sched_timeout_cycles = 16'd20000;   // High latency tolerance
```

```
cfg_sched_timeout_enable = 1'b1;
cfg_sched_err_enable = 1'b1;
cfg_sched_compl_enable = 1'b1;          // Full monitoring
cfg_sched_perf_enable = 1'b1;

// Descriptor engine
cfg_desceng_enable = 1'b1;
cfg_desceng_prefetch = 1'b1;
cfg_desceng_fifo_thresh = 4'hF;         // Max FIFO depth

// All monitors ENABLED with full profiling
cfg_desc_mon_enable = 1'b1;
cfg_desc_mon_err_enable = 1'b1;
cfg_desc_mon_perf_enable = 1'b1;        // Performance monitoring
cfg_desc_mon_timeout_enable = 1'b1;
cfg_desc_mon_timeout_cycles = 32'd20000;
cfg_desc_mon_pkt_mask = 16'hFFFF;       // All packet types

// Same for read/write monitors
cfg_rdeng_mon_enable = 1'b1;
cfg_rdeng_mon_err_enable = 1'b1;
cfg_rdeng_mon_perf_enable = 1'b1;

cfg_wreng_mon_enable = 1'b1;
cfg_wreng_mon_err_enable = 1'b1;
cfg_wreng_mon_perf_enable = 1'b1;

// AXI transfer
cfg_axi_rd_xfer_beats = 8'd128;         // Large bursts
cfg_axi_wr_xfer_beats = 8'd128;

// Performance profiler
cfg_perf_enable = 1'b1;
cfg_perf_mode = 1'b0;                   // Timestamp mode
(correlation)
```

28.8.4 7.4 Debug Configuration (Verbose Monitoring)

Use Case: Debugging integration issues, detailed trace analysis

```
// Enable specific channel for debug
cfg_channel_enable = 8'b00000001;       // Channel 0 only

// Scheduler
cfg_sched_enable = 1'b1;
```

```
cfg_sched_timeout_cycles = 16'd65535;    // Long timeout for debug
cfg_sched_timeout_enable = 1'b1;
cfg_sched_err_enable = 1'b1;
cfg_sched_compl_enable = 1'b1;           // All events
cfg_sched_perf_enable = 1'b1;

// Descriptor engine
cfg_desceng_enable = 1'b1;
cfg_desceng_prefetch = 1'b0;             // Simpler for debug

// All monitors ENABLED with verbose trace
cfg_desc_mon_enable = 1'b1;
cfg_desc_mon_err_enable = 1'b1;
cfg_desc_mon_perf_enable = 1'b1;
cfg_desc_mon_timeout_enable = 1'b1;
cfg_desc_mon_pkt_mask = 16'hFFFF;        // ALL packets
cfg_desc_mon_compl_mask = 8'h01;         // Channel 0 completions
cfg_desc_mon_debug_mask = 8'h01;         // Channel 0 debug packets

// Same for read/write monitors
cfg_rdeng_mon_enable = 1'b1;
cfg_rdeng_mon_pkt_mask = 16'hFFFF;       // Verbose
cfg_rdeng_mon_compl_mask = 8'h01;

cfg_wreng_mon_enable = 1'b1;
cfg_wreng_mon_pkt_mask = 16'hFFFF;
cfg_wreng_mon_compl_mask = 8'h01;

// AXI transfer (small for debug)
cfg_axi_rd_xfer_beats = 8'd4;
cfg_axi_wr_xfer_beats = 8'd4;

// Performance profiler
cfg_perf_enable = 1'b1;
cfg_perf_mode = 1'b0;                    // Timestamps
```

28.9 8. Configuration Best Practices

28.9.1 8.1 Monitor Configuration Guidelines

General Rules:

1. **Start with errors only:** Enable only `cfg_*_mon_err_enable` initially. Add other packets as needed.
2. **Completion packets are expensive:** Only enable `cfg_*_mon_compl_enable` for specific channels during debug.
3. **Performance packets flood MonBus:** Enable `cfg_*_mon_perf_enable` sparingly (1-2 channels maximum).
4. **Use masks aggressively:** Set channel masks to enable monitoring only on channels of interest.

28.9.2 8.2 Timeout Configuration

Calculation Method:

```
Recommended timeout = (Expected latency × Safety factor) + Margin

Safety factor:
- SRAM: 2-5x
- DDR3/DDR4: 5-10x
- High-latency: 10-20x

Margin: +100 cycles minimum
```

Examples:

```
SRAM @ 200 MHz:
- Expected: 20 cycles (100 ns)
- Safety: 5x
- Timeout: 20 × 5 + 100 = 200 cycles

DDR4 @ 200 MHz:
- Expected: 100 cycles (500 ns)
- Safety: 10x
- Timeout: 100 × 10 + 100 = 1100 cycles
```

28.9.3 8.3 Prefetch Configuration

Enable prefetch when: - Descriptor chains > 2 descriptors - Memory latency > 50 cycles - Throughput is priority

Disable prefetch when: - Area is constrained - Single descriptors only - Simplicity is priority

---

28.9.4 8.4 Burst Size Selection

Read Burst Size:

```
Optimal burst size = min(
    Memory controller page size,
    4KB (AXI limit),
    SRAM FIFO depth / 2
)

Example for DDR4 (8KB page), FIFO depth 512 entries:
- Page size: 8192 bytes = 128 beats (512-bit)
- AXI limit: 4096 bytes = 64 beats
- FIFO limit: 512/2 = 256 beats
- Optimal: min(128, 64, 256) = 64 beats
```

Write Burst Size: - Usually same as read burst size - May be smaller if write FIFO depth is limited

---

28.10 9. Configuration Register Map Summary

9. Configuration Register Map Summary

Address	Register Name	Fields	Section
0x100	GLOBAL_CTRL	global_en, global_rst	-
0x120	CHANNEL_ENABLE	ch_en[7:0]	1
0x124	CHANNEL_RESET	ch_rst[7:0]	1
0x200	SCHED_TIMEOUT_CYCLES	timeout_cycles[15:0]	2
0x204	SCHED_CONFIG	enable, timeout_en, err_en, compl_en, perf_en	2
0x220	DESCENG_CONFIG	DESCENG_EN, PREFETCH_EN, FIFO_THRESH	3
0x224	DESCENG_ADDR0_BASE	addr0_base[31:0]	3
0x228	DESCENG_ADDR0_LIMIT	addr0_limit[31:0]	3
0x22C	DESCENG_ADDR1_BASE	addr1_base[31:0]	3
0x230	DESCENG_ADDR1_LIMIT	addr1_limit[31:0]	3
0x10C0-0x10DF	DAXMON_*	Descriptor monitor (15 signals)	4.1
0x10E0-0x10FF	RDMON_*	Read engine monitor (15 signals)	4.2
0x1100-0x111F	WRMON_*	Write engine	4.3

Address	Register Name	Fields	Section
		monitor (15 signals)	
0x2A0	AXI_XFER_CONFIG	RD_XFER_BEATS, WR_XFER_BEATS	5
0x2B0	PERF_CONFIG	PERF_EN, PERF_MODE, PERF_CLEAR	6

Total Address Space: 0x000-0x3FF (1KB)

28.11 10. Software Configuration Examples

28.11.1 10.1 C/C++ Initialization (Minimal)

```
// Minimal configuration for single-channel operation
void stream_init_minimal(volatile uint32_t *base_addr) {
    // Global enable
    base_addr[0x100/4] = 0x00000001;  // GLOBAL_CTRL.global_en

    // Enable channel 0 only
    base_addr[0x120/4] = 0x00000001;  // CHANNEL_ENABLE.ch_en

    // Scheduler config
    base_addr[0x200/4] = 500;         // SCHED_TIMEOUT_CYCLES (SRAM)
    base_addr[0x204/4] = 0x00000007;  // SCHED_CONFIG: enable |
timeout_en | err_en

    // Descriptor engine
    base_addr[0x220/4] = 0x00000041;  // enable | fifo_thresh=4

    // Disable all monitors (minimal)
    base_addr[0x10C0/4] = 0x00000000;  // DAXMON_ENABLE
    base_addr[0x10E0/4] = 0x00000000;  // RDMON_ENABLE
    base_addr[0x1100/4] = 0x00000000;  // WRMON_ENABLE

    // AXI transfer config
    base_addr[0x2A0/4] = 0x00000808;  // 8 beats read + write
}
```

28.11.2 10.2 C/C++ Initialization (Balanced)

```
// Balanced configuration for typical FPGA
void stream_init_balanced(volatile uint32_t *base_addr) {
    // Global enable
    base_addr[0x100/4] = 0x00000001;

    // Enable all 8 channels
    base_addr[0x120/4] = 0x000000FF;

    // Scheduler config
    base_addr[0x200/4] = 5000;        // SCHED_TIMEOUT_CYCLES (DDR4)
    base_addr[0x204/4] = 0x00000007;  // SCHED_CONFIG: enable |
timeout_en | err_en

    // Descriptor engine
```

```
    base_addr[0x220/4] = 0x00000083;  // enable | prefetch |
fifo_thresh=8

    // Descriptor AXI monitor (errors only)
    base_addr[0x10C0/4] = 0x0000000B;  // enable | err_en | timeout_en
    base_addr[0x10C4/4] = 10000;       // timeout_cycles
    base_addr[0x10CC/4] = 0x00000060;  // pkt_mask: errors + timeouts

    // Read/write monitors (errors only)
    base_addr[0x10E0/4] = 0x0000000B;  // RDMON: enable | err_en |
timeout_en
    base_addr[0x1100/4] = 0x0000000B;  // WRMON: enable | err_en |
timeout_en

    // AXI transfer config
    base_addr[0x2A0/4] = 0x00002020;  // 32 beats read + write

    // Enable performance profiler
    base_addr[0x2B0/4] = 0x00000003;  // enable | elapsed mode
}
```

28.12 11. Troubleshooting Configuration Issues

28.12.1 Problem: No transfers occurring

Check: 1. `cfg_channel_enable[n]` set for channel n? 2. `cfg_sched_enable = 1`? 3. `cfg_desceng_enable = 1`? 4. Global enable set?

28.12.2 Problem: Timeout errors

Check: 1. `cfg_sched_timeout_cycles` too small? 2. Memory latency higher than expected? 3. AXI backpressure not handled?

Solution: - Increase timeout: `cfg_sched_timeout_cycles = 20000` - Disable temporarily: `cfg_sched_timeout_enable = 0`

28.12.3 Problem: MonBus overflow

Check: 1. Too many completion packets enabled? 2. Performance packets enabled on all channels? 3. Debug packets enabled?

Solution: - Disable completion: `cfg_*_mon_compl_enable = 0` - Use channel masks: `cfg_*_mon_compl_mask = 8'h01` (channel 0 only) - Reduce packet types: `cfg_*_mon_pkt_mask = 16'h0060` (errors + timeouts)

28.12.4 Problem: Low throughput

Check: 1. Prefetch disabled? 2. Burst size too small? 3. FIFO threshold too conservative?

Solution: - Enable prefetch: `cfg_desceng_prefetch = 1` - Increase bursts: `cfg_axi_rd_xfer_beats = 64` - Increase FIFO: `cfg_desceng_fifo_thresh = 12`

28.13 Related Documentation

- **Register Map** - Complete APB register specification
- **Clocks and Reset** - Timing requirements
- **Programming Guide** - Software API examples

Last Updated: 2025-12-01 **Maintained By:** STREAM Architecture Team

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29 Product Requirements Document (PRD)

29.1 STREAM - Scatter-gather Transfer Rapid Engine for AXI Memory

Version: 1.0 **Date:** 2025-10-17 **Status:** Nearly Complete - Final Integration Pending **Owner:** RTL Design Sherpa Project **Parent Document:** `/PRD.md`

29.2 1. Executive Summary

The **STREAM** (Scatter-gather Transfer Rapid Engine for AXI Memory) is a simplified DMA-style accelerator designed for efficient memory-to-memory data movement with descriptor-based scatter-gather support. STREAM serves as a beginner-friendly tutorial demonstrating descriptor-based DMA engine design patterns while maintaining production-quality RTL practices.

29.2.1 1.1 Quick Stats

- **Modules:** ~8-10 SystemVerilog files (estimated)
- **Channels:** Maximum 8 independent channels
- **Interfaces:** APB (config), AXI4 (descriptor fetch + data read/write), MonBus (monitoring)
- **Architecture:** Simplified from RAPIDS - pure memory-to-memory
- **Tutorial Focus:** Aligned addresses only, straightforward data flow
- **Status:** Nearly complete - config interface and top-level wrapper pending

29.2.2 1.2 Project Goals

- **Primary:** Educational DMA engine demonstrating scatter-gather descriptor chains
- **Secondary:** Production-quality RTL suitable for FPGA/ASIC implementation
- **Tertiary:** Foundation for understanding more complex DMA architectures (e.g., RAPIDS)

29.3 2. Key Design Principles

29.3.1 2.1 Simplifications from RAPIDS

STREAM is intentionally simplified for tutorial purposes:

Feature	RAPIDS	STREAM
Network Interfaces	Yes (Network master/slave)	✗ No (pure memory-to-memory)
Address Alignment	Complex fixup logic	√ **Aligned addresses only**
Credit Management	Exponential encoding	√ Simple transaction limits
Control Engines	Control read/write engines	✗ No (direct APB config)
Descriptor Length	Chunks (4-byte)	√ **Beats** (data width)
Program Engine	Complex alignment FSM	√ Simplified coordination

29.3.2 2.2 Tutorial-Friendly Features

- **Aligned Addresses:** Source and destination addresses must be aligned to data width
- **Length in Beats:** Descriptor length specified in data beats (not bytes/chunks)
- **Single APB Write:** One APB register write kicks off entire descriptor chain
- **No Circular Buffers:** Chained descriptors with explicit termination
- **Parameterized Engines:** Multiple AXI engine versions (compile-time selection)

29.4 3. Architecture Overview

29.4.1 3.1 Top-Level Block Diagram

```
STREAM (Scatter-gather Transfer Rapid Engine for AXI Memory)
├── APB Config Slave
│   └── Channel registers (8 channels, kick-off via write)
│
├── Descriptor Fetch
│   └── AXI Master (Read) - Fetch descriptors from memory
│
├── Scheduler Group (shared across all channels)
│   ├── Descriptor Engine   (FIFO, parsing - FROM RAPIDS)
│   ├── Scheduler           (Simplified FSM coordination)
│   └── Channel Arbitration (8 independent channels)
│
├── Data Path (shared resources)
│   ├── AXI Master (Read)  - Parameterized data width
│   ├── Simple SRAM        - Dual-port buffer (FROM RAPIDS)
│   └── AXI Master (Write) - Parameterized data width
│
└── MonBus Reporter
    └── MonBus Master - 64-bit monitoring packets
```

29.4.2 3.2 Data Flow

Descriptor-Based Transfer Sequence:

```
1. Software writes to APB channel register
   ↓
2. Descriptor fetch via AXI descriptor master
   ↓
3. Descriptor Engine parses descriptor fields
   ↓
4. Scheduler coordinates data transfer:
   a. AXI Read Engine fetches source data → SRAM buffer
   b. AXI Write Engine writes SRAM → destination
   ↓
5. Check for chained descriptor (next_descriptor_ptr != 0)
   ↓ (if chained)
6. Fetch next descriptor, repeat from step 3
   ↓ (if last)
7. Generate MonBus completion packet
```

Channel Independence: - 8 channels operate independently - All channels share: SRAM, AXI data masters, descriptor fetch master - Arbitration required for shared resources

---

29.5 4. Interfaces

29.5.1 4.1 External Interfaces

Interface	Type	Width	Purpose	Notes
APB Slave	Slave	32-bit	Configuration, channel kick-off	Write to channel register starts transfer
AXI Master (Descriptor)	Master	256-bit	Fetch descriptors from memory	Dedicated descriptor fetch path
AXI Master (Data Read)	Master	Parameterizable	Read source data	Multiple engine versions (compile-time)
AXI Master (Data Write)	Master	Parameterizable	Write destination data	Multiple engine versions (compile-time)
MonBus Master	Master	64-bit	Monitor packet output	Standard AMBA format

29.5.2 4.2 Descriptor Format

256-bit Descriptor Structure:

Bits	Field	Description
[63:0]	`src_addr`	Source address (64-bit, must be aligned to data width)
[127:64]	`dst_addr`	Destination address (64-bit, must be aligned to data width)
[159:128]	`length`	Transfer length in **BEATS** (not bytes!)
[191:160]	`next_descriptor_p`	Address of next descriptor (0 = last in

Bits	Field	Description
	tr	chain)
[192]	valid	Descriptor is valid
[193]	interrupt	Generate interrupt on completion
[194]	last	Last descriptor in chain (explicit flag)
[195]	error	Error status (used for reporting)
[199:196]	channel_id	Channel ID (0-7)
[207:200]	priority	Transfer priority (for arbitration)
[255:208]	reserved	Reserved for future use

Key Descriptor Features: - ✓ **Chained descriptors:** `next_descriptor_ptr` links to next descriptor - ✗ **No circular buffers:** Explicit termination (last flag or ptr=0) - ✓ **Length in beats:** Simplified for tutorial (no byte/chunk conversion) - ✓ **Aligned addresses:** Tutorial constraint (performance hidden for now)

29.6 5. Key Components

29.6.1 5.1 Descriptor Engine (APB-Only for STREAM)

Source: Adapted from RAPIDS `descriptor_engine.sv`

Purpose: - Autonomous descriptor fetch and chaining - APB interface for initial descriptor address - AXI read interface for descriptor memory fetches - Descriptor FIFO storage and distribution

Key Features: - ✓ **Autonomous chaining:** Automatically fetches next descriptor if `next_descriptor_ptr != 0` AND `last == 0` - ✓ **Address validation:** Validates next descriptor addresses against `cfg_addr0/1_base/limit` - ✓ **APB blocking:** APB blocked until `channel_idle == 1` (channel fully idle) - ✓ **Error handling:** AXI errors stop chaining, set `descriptor_error`, block `descriptor_valid`

Adaptations from RAPIDS: - ✓ **APB-only:** Single APB write kicks off entire descriptor chain - ✓ **Descriptor Read Address FIFO:** 2-deep FIFO stores addresses for AXI fetch (APB + chaining) - ✓ **Chaining logic:** Descriptor engine autonomously manages `next_descriptor_ptr` chaining

Idle Signal: - `descriptor_engine_idle` asserted when: - FSM in `RD_IDLE` state - No pending descriptor fetches (address FIFO empty) - No active AXI transactions

29.6.2 5.2 Scheduler Group (Integration Wrapper)

Purpose: Wraps descriptor engine and scheduler into a single channel processing unit

Architecture:

```
scheduler_group (
    // APB interface (from APB config slave)
    .apb_valid          (apb_valid),
    .apb_ready          (apb_ready),   // Blocked when channel not idle
    .apb_addr           (descriptor_addr),

    // Channel idle signal composition (CRITICAL!)
    .channel_idle       (channel_idle),

    // Descriptor → Scheduler flow
    .descriptor_valid   (desc_valid),
    .descriptor_ready   (desc_ready),
    .descriptor_packet  (desc_packet),

    // Data engine interfaces
    .datard_*           (datard_*),    // Read engine
    .datawr_*           (datawr_*),    // Write engine

    // Status
```

```
    .scheduler_idle    (sched_idle),
    .descriptor_idle   (desc_idle)
);
```

Channel Idle Signal Composition:

```
// Channel is idle ONLY when BOTH sub-blocks are idle
assign channel_idle = scheduler_idle && descriptor_engine_idle;
```

Why Both Signals Matter:

Signal	Indicates	Used For
`scheduler_idle`	No active data transfers, all descriptors processed	Prevents new APB request during active transfer
`descriptor_engine_idle`	No pending descriptor fetches (FIFO empty)	Prevents new APB request during chaining
`channel_idle` (AND of both)	Channel fully quiescent	**Gates APB interface**

APB Blocking Logic:

```
// Descriptor engine blocks APB when channel not idle
assign apb_ready = apb_skid_ready_in &&
                   !r_channel_reset_active &&
                   w_desc_addr_fifo_empty &&    // No pending fetches
                   channel_idle;                 // Scheduler +
descriptor idle
```

Example Scenario:

```
1. Software writes APB → descriptor_addr = 0x1000
   - channel_idle = 1 (both idle)
   - APB accepted

2. Descriptor engine fetches descriptor @ 0x1000
   - descriptor_engine_idle = 0 (fetch in progress)
   - channel_idle = 0
   - APB BLOCKED

3. Descriptor pushed to scheduler
   - descriptor_engine_idle = 1 (fetch complete)
   - scheduler_idle = 0 (transfer starting)
   - channel_idle = 0
   - APB BLOCKED

4. Scheduler completes data transfer
   - Descriptor has next_descriptor_ptr = 0x1100 (chained!)
```

```
   - Descriptor engine autonomously fetches @ 0x1100
   - descriptor_engine_idle = 0 (autonomous fetch)
   - channel_idle = 0
   - APB BLOCKED

5. Final descriptor completes (last = 1 OR next_ptr = 0)
   - scheduler_idle = 1 (transfer done)
   - descriptor_engine_idle = 1 (no more fetches)
   - channel_idle = 1
   - APB UNBLOCKED (ready for next transfer!)
```

Key Insight: The AND gate ensures software cannot interrupt a descriptor chain in progress!

29.6.3 5.3 Scheduler (Simplified from RAPIDS)

Purpose: - Coordinate descriptor-to-data-transfer flow - Manage 8 independent channels - Arbitrate shared resources (SRAM, AXI masters)

FSM States:

```
typedef enum logic [7:0] {
    SCHED_IDLE              = 8'b00000001,  // Idle, waiting for
channel activation
    SCHED_FETCH_DESCRIPTOR  = 8'b00000010,  // Fetch descriptor via
AXI master
    SCHED_PARSE_DESCRIPTOR  = 8'b00000100,  // Parse descriptor fields
    SCHED_READ_PHASE        = 8'b00001000,  // Coordinate read engine
    SCHED_WRITE_PHASE       = 8'b00010000,  // Coordinate write engine
    SCHED_CHAIN_CHECK       = 8'b00100000,  // Check for next
descriptor
    SCHED_COMPLETE          = 8'b01000000,  // Transfer complete,
report status
    SCHED_ERROR             = 8'b10000000   // Error state
} scheduler_state_t;
```

Key Differences from RAPIDS: - ✗ No credit management (just simple transaction limits) - ✗ No program engine coordination (no alignment fixup) - √ Simplified FSM (no control read/write phases)

29.6.4 5.3 AXI Read Engine (Streaming Pipeline - NO FSM)

Purpose: High-performance streaming reads from memory to SRAM buffer

Architecture: Pipelined streaming design (NO FSM for performance)

Key Insight: FSMs are horrible for performance! Instead, use: - **Arbiter** selects which channel's `datard_*` interface gets access - **Streaming pipeline** continuously moves data when granted - **Data interface** (`datard_valid`, `datard_ready`, `datard_beats_remaining`) controls flow

Data Read Interface (per channel):

```
// Channel requests read access
input  logic        datard_valid;          // Channel has read request
output logic        datard_ready;          // Engine ready for request
input  logic [63:0] datard_addr;           // Source address (aligned)
input  logic [31:0] datard_beats_remaining; // Beats left to read
input  logic [7:0]  datard_burst_len;      // Preferred burst length
input  logic [3:0]  datard_channel_id;     // Channel ID for tracking
```

Multiple Versions (Compile-Time Selection): 1. **Version 1 - Basic:** Single outstanding read, fixed burst length 2. **Version 2 - Pipelined:** Multiple outstanding reads, configurable bursts 3. **Version 3 - Adaptive:** Dynamic burst sizing based on remaining beats

Operation:

```
1. Arbiter grants channel access based on priority
2. Engine accepts datard_* request (continuous streaming)
3. AXI AR channel issues read burst
4. AXI R channel streams data → SRAM (no FSM stalls!)
5. Engine updates beats_remaining, accepts next request
6. Different channels can have different burst lengths (e.g., CH0: 8
beats, CH1: 16 beats)
```

Example: Different Burst Lengths per Channel

```
// Channel 0 prefers 8-beat bursts
datard_burst_len[0] = 8'd8;

// Channel 1 prefers 16-beat bursts
datard_burst_len[1] = 8'd16;

// Engine adapts to requested burst length (within MAX_BURST_LEN)
```

29.6.5 5.4 AXI Write Engine (Streaming Pipeline - NO FSM)

Purpose: High-performance streaming writes from SRAM buffer to memory

Architecture: Pipelined streaming design (NO FSM for performance)

Key Insight: Same as read engine - no FSMs! Use streaming pipeline with arbiter.

Data Write Interface (per channel):

```
// Channel requests write access
input  logic        datawr_valid;          // Channel has write
request
output logic        datawr_ready;          // Engine ready for
request
input  logic [63:0] datawr_addr;           // Destination address
(aligned)
input  logic [31:0] datawr_beats_remaining; // Beats left to write
```

```
input  logic [7:0]  datawr_burst_len;       // Preferred burst length
input  logic [3:0]  datawr_channel_id;      // Channel ID for tracking
```

Multiple Versions (Compile-Time Selection): 1. **Version 1 - Basic:** Single outstanding write, fixed burst length 2. **Version 2 - Pipelined:** Multiple outstanding writes, configurable bursts 3. **Version 3 - Adaptive:** Dynamic burst sizing based on remaining beats

Operation:

```
1. Arbiter grants channel access based on priority
2. Engine accepts datawr_* request (continuous streaming)
3. Engine reads data from SRAM
4. AXI AW channel issues write address
5. AXI W channel streams data (no FSM stalls!)
6. AXI B channel receives response, updates beats_remaining
7. Different channels can have different burst lengths (e.g., CH0: 8
beats, CH2: 32 beats)
```

Read/Write Asymmetry Example:

```
// Channel can use different burst lengths for read vs write
// Example: Read in small bursts, write in large bursts
datard_burst_len[0] = 8'd8;   // Read: 8 beats
datawr_burst_len[0] = 8'd16;  // Write: 16 beats

// Engine handles asymmetry via SRAM buffering
```

29.6.6 5.5 SRAM Buffering

Source: Originally a direct copy of RAPIDS `simple_sram.sv`; now implemented by `rtl/fub/sram_controller.sv` + `sram_controller_unit.sv` (wrapping the shared `gaxi_fifo_sync` primitive)

Purpose: - Dual-port SRAM buffer - Decouples read and write engines - Shared across all channels (arbitration required)

Why Reuse: - Standard dual-port SRAM design - Proven in RAPIDS integration tests - Parameterizable depth and width

---

29.7 6. Configuration and Control

29.7.1 6.1 APB Register Map

Offset	Register	Access	Description
0x0000	GLOBAL_CTRL	RW	Global enable, reset
0x0004	GLOBAL_STATUS	RO	Global status, error flags
0x0100	CH0_CTRL	WO	**Channel 0 kick-off** (write descriptor address)
0x0104	CH0_STATUS	RO	Channel 0 status
0x0108	CH0_DESC_ADDR	RO	Channel 0 current descriptor address
0x010C	CH0_BYTES_XFER	RO	Channel 0 bytes transferred
...	...	...	... (repeat for channels 1-7)
0x0200	CH1_CTRL	WO	Channel 1 kick-off
...			
0x0700	CH7_CTRL	WO	Channel 7 kick-off

Kick-Off Sequence: 1. Software writes descriptor address to CHx_CTRL register 2. STREAM fetches descriptor from memory 3. Transfer begins automatically 4. If chained, STREAM follows next_descriptor_ptr automatically 5. Completion reported via MonBus packet

29.7.2 6.2 Channel Configuration

Each channel independently configurable: - **Descriptor start address:** Written to CHx_CTRL - **Priority:** Encoded in descriptor (arbitration) - **Interrupt enable:** Per-descriptor flag - **Status monitoring:** Read CHx_STATUS

29.8 7. Resource Sharing and Arbitration

29.8.1 7.1 Shared Resources

All channels share: 1. **Descriptor Fetch AXI Master** - Fetches descriptors for all channels 2. **Data Read AXI Master** - Reads source data for all channels 3. **Data Write AXI Master** - Writes destination data for all channels 4. **SRAM Buffer** - Shared buffer (dual-port, but still arbitrated) 5. **MonBus Reporter** - Single monitor output

29.8.2 7.2 Arbitration Strategy

Priority-Based Round-Robin: - Channels have priority field in descriptor - Higher priority = serviced first - Within same priority: round-robin - Prevents starvation with timeout

Example Arbitration:

```
Channel 0: Priority 7 (highest)
Channel 1: Priority 5
Channel 2: Priority 5
Channel 3: Priority 3
```

Service order: CH0 → CH1 → CH2 (round-robin) → CH0 → CH1 → CH2 → CH3 ...

29.9 8. Error Detection and Recovery

29.9.1 8.1 Error Types

Error Type	Detection	Response
Invalid descriptor	Valid bit = 0	Skip, move to next
Alignment error	Address not aligned	Set error flag, halt channel
AXI SLVERR	AXI response	Set error flag, halt channel
AXI DECERR	AXI response	Set error flag, halt channel
Timeout	Transaction timeout	Set error flag, halt channel
SRAM overflow	Buffer full	Backpressure, wait

29.9.2 8.2 Error Recovery

Per-Channel Error Handling: - Error sets channel to `CH_ERROR` state - Channel halts, does not affect other channels - Software must: 1. Read `CHx_STATUS` to identify error 2. Clear error condition 3. Re-kick channel with new descriptor

No Automatic Retry: - Tutorial design keeps error handling simple - Software responsible for retry logic

---

29.10 9. MonBus Integration

29.10.1 9.1 Standard MonBus Format

Uses standard 64-bit MonBus packet format: - [63:60] Packet Type (0=ERROR, 1=COMPL, etc.) - [59:57] Protocol (custom STREAM protocol) - [56:53] Event Code (STREAM-specific events) - [52:47] Channel ID (0-7) - [46:43] Unit ID (unused for STREAM) - [42:35] Agent ID (unused for STREAM) - [34:0] Event Data (address, byte count, etc.)

29.10.2 9.2 STREAM Event Codes

Code	Event	Description
0x0	`DESC_START`	Descriptor started
0x1	`DESC_COMPLETE`	Descriptor completed
0x2	`READ_START`	Read phase started
0x3	`READ_COMPLETE`	Read phase completed
0x4	`WRITE_START`	Write phase started
0x5	`WRITE_COMPLETE`	Write phase completed
0x6	`CHAIN_FETCH`	Chained descriptor fetch
0xF	`ERROR`	Error occurred

29.10.3 9.3 Default Configuration

Tutorial-Friendly Defaults: - **Errors only:** `cfg_error_enable = 1`, all others = 0 - **Interrupts:** Descriptor flag controls per-transfer interrupt - **Reduces MonBus traffic** for beginner understanding

---

29.11 10. Design Constraints

29.11.1 10.1 Tutorial Constraints (Intentional Simplifications)

Constraint	Rationale
Aligned addresses only	Simplify data path, hide alignment complexity
Length in beats	Avoid byte/chunk conversion math
No circular buffers	Explicit termination easier to understand
Single APB kick-off	Simple software model
No credit management	Avoid exponential encoding complexity

29.11.2 10.2 Implementation Constraints

Parameter	Value	Notes
Max channels	8	Compile-time parameter
Max burst length	256	AXI4 spec limit
Descriptor size	256 bits	4 × 64-bit words
SRAM depth	Parameterizable	Typical: 1024-4096 entries
Data width	Parameterizable	Typical: 512-bit

29.12 11. Verification Strategy

29.12.1 11.1 Test Organization

```
projects/components/dma-ip/stream/dv/tests/
├── fub/                        # Functional Unit Block tests
│   ├── test_descriptor_engine.py
│   ├── test_scheduler.py
│   ├── test_sram_controller.py (+ test_sram_controller_alloc.py)
│   ├── test_perf_profiler.py
│   └── test_stream_latency_bridge.py
├── macro/                      # Multi-block scenarios (stream_core,
datapaths)
└── top/                        # Top-level (stream_top_ch8) tests
```

29.12.2 11.2 Test Levels

Basic (Quick Smoke Tests): - Single descriptor, single channel - Aligned addresses, simple transfers - ~30 seconds runtime

Medium (Typical Scenarios): - Multiple descriptors, 2-4 channels - Chained descriptors (2-3 deep) - ~90 seconds runtime

Full (Comprehensive Validation): - All 8 channels concurrent - Long descriptor chains (10+ deep) - Error injection, stress testing - ~180 seconds runtime

---

29.13 12. Performance Characteristics

29.13.1 12.1 Throughput by Engine Version

V1 (Low Performance - Tutorial Mode): - **Throughput:** 0.14 beats/cycle (DDR4), 0.40 beats/cycle (SRAM) - **Architecture:** Single outstanding transaction, blocks on completion - **Use Case:** Tutorial examples, embedded systems, low-latency SRAM

V2 (Medium Performance - Command Pipelined): - **Throughput:** 0.94 beats/cycle (DDR4), 0.85 beats/cycle (SRAM) - **Improvement:** 6.7x over V1 (DDR4), 2.1x over V1 (SRAM) - **Architecture:** Command queue (4-8 deep), hides memory latency - **Use Case:** General-purpose FPGA, DDR3/DDR4, best area efficiency

V3 (High Performance - Out-of-Order): - **Throughput:** 0.98 beats/cycle (DDR4), 0.92 beats/cycle (SRAM) - **Improvement:** 7.0x over V1 (DDR4), 2.3x over V1 (SRAM) - **Architecture:** OOO command selection, maximizes memory controller efficiency - **Use Case:** Datacenter, ASIC, HBM2, high-performance memory

Key Insight: Benefit scales with memory latency. V1 throughput degrades from 40% (SRAM) to 14% (DDR4), while V2/V3 maintain 94-98% throughput regardless of latency.

Configuration Parameters: - `ENABLE_CMD_PIPELINE = 0`: V1 (default, tutorial mode) - `ENABLE_CMD_PIPELINE = 1`: V2 (command pipelined) - `ENABLE_CMD_PIPELINE = 1`, `ENABLE_OOO_DRAIN = 1` (write) or `ENABLE_OOO_READ = 1` (read): V3

Factors Affecting Throughput: - Memory latency (V2/V3 hide latency via pipelining) - SRAM buffer size (limits burst pipelining) - Channel arbitration overhead - Descriptor fetch latency - Engine version (V1/V2/V3 configuration)

29.13.2 12.2 Latency

Transfer Latency Breakdown: - Descriptor fetch: ~10-50 cycles (depends on memory) - Read phase: `(length / burst_len) × burst_latency` - Write phase: `(length / burst_len) × burst_latency` - End-to-end: Typically <200 cycles for small transfers (V1), <100 cycles (V2/V3 pipelined)

V2/V3 Latency Hiding: - Command pipelining overlaps descriptor fetch, read, write operations - Multiple outstanding transactions hide memory latency - OOO completion (V3) reduces head-of-line blocking

29.13.3 12.3 Resource Utilization by Engine Version

V1 Configuration (Tutorial - Minimum Area): - Total: ~9,500 LUTs + 64 KB SRAM - Descriptor Engine (8×): ~2,400 LUTs - Scheduler (8×): ~3,200 LUTs - AXI Read Engine: ~1,250 LUTs - AXI Write Engine: ~1,250 LUTs - SRAM Controller: ~1,600 LUTs - APB Config: ~350 LUTs - MonBus AXIL Group: ~1,000 LUTs

V2 Configuration (Balanced - Best Area Efficiency): - Total: ~11,000 LUTs + 64 KB SRAM (1.16x area) - AXI Read Engine: ~2,000 LUTs (1.6x increase, 6.7x throughput) - AXI Write Engine: ~2,500 LUTs (2.0x increase, 6.7x throughput) - Other blocks: Same as V1

V3 Configuration (Maximum Performance): - Total: ~14,000 LUTs + 64 KB SRAM (1.47x area) - AXI Read Engine: ~3,500 LUTs (2.8x increase, 7.0x throughput) - AXI Write Engine: ~4,000 LUTs (3.2x increase, 7.0x throughput) - Other blocks: Same as V1

Area Efficiency Comparison: - V1: 1.00 throughput / 1.00 area = 1.00 - V2: 6.70 throughput / 1.16 area = 5.78 (best efficiency) - V3: 7.00 throughput / 1.47 area = 4.76

Recommendation: V2 provides best area efficiency for most use cases. V3 justified only for high-performance memory controllers that support OOO responses.

29.14 13. Development Roadmap

29.14.1 13.1 Phase 1: Foundation (Current)

- ✓ Directory structure
- ✓ Package definitions (`stream_pkg.sv`)
- ✓ Imports header (`stream_imports.svh`)
- ⏳ Documentation (this PRD)

29.14.2 13.2 Phase 2: Core Blocks

- Adapt descriptor engine from RAPIDS
- Design simplified scheduler
- Create APB config interface
- Copy simple SRAM from RAPIDS

29.14.3 13.3 Phase 3: Data Path

- AXI read engine (version 1 - basic)
- AXI write engine (version 1 - basic)
- SRAM integration
- Channel arbitration

29.14.4 13.4 Phase 4: Integration

- Top-level module
- MonBus reporter
- Integration testbench
- Single-channel validation

29.14.5 13.5 Phase 5: Multi-Channel

- 8-channel support
- Arbiter implementation
- Multi-channel tests
- Performance tuning

29.14.6 13.6 Phase 6: Advanced Engines (Future - V2/V3)

Goal: Add parameterized high-performance engine variants

V2 - Command Pipelined (Medium Performance): - Command queue implementation (4-8 deep) - W drain FSM for write engine - B response scoreboard (write) or in-order R reception (read) -

Per-command SRAM pointer tracking - Parameter: `ENABLE_CMD_PIPELINE = 1` - Expected: 6.7x throughput improvement over V1

V3 - Out-of-Order Completion (High Performance): - OOO command selection logic - Transaction ID matching (AXI ID to queue entry) - SRAM data availability checking (write engine) - R beat matching to queue entry (read engine) - Parameters: `ENABLE_CMD_PIPELINE = 1`, `ENABLE_OOO_DRAIN = 1` (write) or `ENABLE_OOO_READ = 1` (read) - Expected: 7.0x throughput improvement over V1

Deliverables: - Updated RTL with parameterization - Performance comparison tests (V1 vs V2 vs V3) - Tutorial documentation explaining trade-offs - Area/throughput measurements on target FPGA

29.15 14. Educational Value

29.15.1 14.1 Learning Objectives

STREAM teaches: 1. **Descriptor-based DMA design patterns** - Descriptor structure and parsing - Chained descriptors (scatter-gather) - Descriptor fetch via AXI

2. **AXI4 Memory Interface Usage**
   - Burst transactions
   - Address/data/response channels
   - Outstanding transactions
3. **Resource Sharing and Arbitration**
   - Multiple channels sharing AXI masters
   - Priority-based arbitration
   - Conflict resolution
4. **FSM Design and Coordination**
   - Multiple interconnected FSMs
   - State machine composition
   - Error handling
5. **Buffer Management**
   - SRAM-based buffering
   - Rate matching
   - Flow control

29.15.2 14.2 Progression Path

Learning Sequence: 1. **STREAM (this project):** Memory-to-memory DMA, aligned addresses 2. **STREAM Extended:** Add alignment fixup, more complex scenarios 3. **RAPIDS:** Add network interfaces, credit management, full complexity

---

29.16 15. Success Criteria

29.16.1 15.1 Functional

- √ Single descriptor transfer working
- √ Chained descriptors (2-3 deep)
- √ Multi-channel operation (8 channels concurrent)
- √ Error detection and reporting
- √ MonBus packet generation
- √ >90% functional test coverage

29.16.2 15.2 Quality

- √ Verilator compiles with 0 warnings
- √ All tests passing (100% success rate)
- √ Comprehensive documentation
- √ Tutorial-quality code comments

29.16.3 15.3 Performance

- √ Achieves >80% of theoretical AXI bandwidth
- √ <5K LUT utilization (excluding SRAM)
- √ <200 cycle end-to-end latency for small transfers

29.17 16. Open Questions (For Review)

29.17.1 16.1 Descriptor Engine Adaptation

- **Q:** Should descriptor engine use APB-only, or support additional modes?
- **A (pending):** TBD - depends on software use case preference

29.17.2 16.2 AXI Descriptor Master

- **Q:** Fixed 256-bit width, or parameterizable?
- **A (pending):** Propose fixed 256-bit for simplicity

29.17.3 16.3 Channel Arbitration

- **Q:** Fixed priority, or dynamic priority based on age/fairness?
- **A (pending):** Propose fixed priority with round-robin for tutorial simplicity

29.17.4 16.4 SRAM Partitioning

- **Q:** Single shared SRAM, or per-channel SRAMs?
- **A (pending):** Propose single shared SRAM with arbitration (matches RAPIDS pattern)

29.18 16.2 PDF Generation Location

IMPORTANT: PDF files should be generated in the docs directory:

```
/mnt/data/github/rtldesignsherpa/projects/components/dma-ip/stream/
docs/
```

Quick Command: Use the provided shell scripts:

```
cd projects/components/dma-ip/stream/docs
./generate_has_pdf.sh   # STREAM HAS
./generate_mas_pdf.sh   # STREAM MAS
```

The shell scripts will automatically: 1. Use the md_to_docx.py tool from bin/ 2. Process the stream_has/ and stream_mas/ index files 3. Generate both DOCX and PDF files in the docs/ directory 4. Create table of contents and title page

📖 **See:** `bin/md_to_docx.py` for complete implementation details

29.19 17. References

29.19.1 17.1 Internal Documentation

- **RAPIDS PRD:** `projects/components/dma-ip/rapids/PRD.md` - Parent architecture
- **RAPIDS Descriptor Engine:** `projects/components/dma-ip/rapids/rtl/fub_beats/descriptor_engine_beats.sv` (beats rearchitecture of the original)
- **Shared SRAM/FIFO primitives:** `rtl/amba/gaxi/gaxi_fifo_sync.sv`, `rtl/amba/shared/sdpram_core.sv` (the old `simple_sram.sv` was removed)
- **AMBA PRD:** `docs/markdown/rtl-amba/index.md` - MonBus integration
- **Repository Guide:** `/CLAUDE.md` - Design patterns and conventions

29.19.2 17.2 External References

- **AXI4 Specification:** ARM IHI0022E
- **APB Specification:** ARM IHI0024C
- CocoTB Documentation: https://docs.cocotb.org/
- Verilator Manual: https://verilator.org/guide/latest/

Document Version: 1.0 **Last Updated:** 2025-10-17 **Review Cycle:** Weekly during initial design **Next Review:** TBD (after user feedback) **Owner:** RTL Design Sherpa Project

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29.20 Navigation

- **← Back to Root:** `/PRD.md`
- **Parent Architecture:** `projects/components/dma-ip/rapids/PRD.md`
- **AI Guidance:** `CLAUDE.md` (to be created)
- **Quick Start:** `README.md` (to be created)

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30 Claude Code Guide: STREAM Subsystem

Version: 1.0 **Last Updated:** 2025-10-17 **Purpose:** AI-specific guidance for working with STREAM subsystem

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30.1 Quick Context

What: STREAM - Scatter-gather Transfer Rapid Engine for AXI Memory **Status:** √ Complete - tutorial-focused DMA engine (fub/macro/top all passing; on-board characterization flows exist) **Your Role:** Help users build a beginner-friendly descriptor-based DMA engine

📖 **Complete Specification:** `projects/components/dma-ip/stream/PRD.md` ← **Always reference this for technical details**

30.2 📖 Global Requirements Reference

IMPORTANT: Review `/GLOBAL_REQUIREMENTS.md` for all mandatory requirements

All mandatory requirements are consolidated in the global requirements document: - **See:** `/GLOBAL_REQUIREMENTS.md` - Repository-wide mandatory requirements - **STREAM-Specific:** Attribution format, tutorial focus (intentional simplifications) - **Universal:** TB location, three methods, TBBase inheritance, 100% success

This CLAUDE.md provides STREAM-specific guidance. Also review: - Root `/CLAUDE.md` - Repository-wide patterns - `projects/components/CLAUDE.md` - Project area standards (reset macros, FPGA attributes) - `bin/TBClasses/` - Shared TB framework (full CocoTBFramework in the RTLDesignSherpa-DV repo)

30.3 Critical Rules for This Subsystem

30.3.1 Rule #0.1: TUTORIAL FOCUS - Intentional Simplifications

⚠️ STREAM is INTENTIONALLY SIMPLIFIED for educational purposes ⚠️

Key Simplifications (DO NOT "fix" these): 1. ✓ **Aligned addresses only** - No alignment fixup logic (kept for RAPIDS) 2. ✓ **Length in beats** - Not bytes or chunks (simplifies math) 3. ✓ **No circular buffers** - Explicit chain termination only 4. ✓ **No credit management** - Simple transaction limits 5. ✓ **Pure memory-to-memory** - No network interfaces

When users ask "Can we add alignment fixup?": - ✓ **Correct answer:** "STREAM intentionally keeps addresses aligned for tutorial simplicity. For complex alignment, see RAPIDS." - **✗ Wrong answer:** "Sure, let me add alignment logic..." (defeats tutorial purpose!)

30.3.2 Rule #0.1: Testbench Location and Test Structure (MANDATORY)

📖 **See:** `/GLOBAL_REQUIREMENTS.md` Section 2.1 for complete requirement

STREAM-Specific Directory Structure:

```
projects/components/dma-ip/stream/dv/
├── tbclasses/                    # ⭐ STREAM TB classes (project
area!)
│   ├── scheduler_tb.py           # Scheduler testbench
│   ├── descriptor_engine_tb.py   # Descriptor engine testbench
│   ├── sram_controller_tb.py     # SRAM controller testbench
│   ├── stream_core_tb.py         # Macro-level core testbench
│   └── stream_top_tb.py          # Top-level testbench
└── tests/                        # Test runners import from
tbclasses/
    ├── fub/                      # Per-block tests
    ├── macro/                    # Integration tests
    └── top/                      # Top-level tests
```

STREAM Import Pattern:

```python
# Import STREAM TB from project area
from projects.components.dma_ip.stream.dv.tbclasses.scheduler_tb
import SchedulerTB

# Shared utilities from framework
from TBClasses.shared.tbbase import TBBase
```

📖 **Complete Pattern:** `projects/components/dma-ip/rapids/CLAUDE.md` Rule #0.1

30.3.3 Rule #0.2: Three Mandatory TB Methods (MANDATORY)

📖 **See:** `/GLOBAL_REQUIREMENTS.md` Section 2.2 for complete requirement

STREAM-Specific Context:

STREAM has simpler config requirements than RAPIDS - most config can be set after reset.

Standard STREAM Pattern:

```
class SchedulerTB(TBBase):
    async def setup_clocks_and_reset(self):
        """Standard STREAM initialization"""
        await self.start_clock('clk', freq=10, units='ns')
        await self.assert_reset()
        await self.wait_clocks('clk', 10)
        await self.deassert_reset()
        await self.wait_clocks('clk', 5)

    async def assert_reset(self):
        self.dut.rst_n.value = 0

    async def deassert_reset(self):
        self.dut.rst_n.value = 1
```

Note: Unlike RAPIDS, STREAM typically doesn't need config set before reset.

30.3.4 Rule #1: REUSE from RAPIDS Where Appropriate

Direct Reuse (No Modification): - ✓ `descriptor_engine.sv` - Works with STREAM descriptors - ✓ Shared FIFO/SRAM primitives (`gaxi_fifo_sync`, used inside `sram_controller_unit.sv`; the old `simple_sram.sv` was removed) - ✓ Descriptor engine tests - Adapt imports only

Adapt from RAPIDS: - ⚠️ `scheduler.sv` - **Simplify FSM** (no credit management, no control engines) - ⚠️ AXI engines - **Create simplified versions** (no alignment fixup)

Create New for STREAM: - 📝 APB config interface - PeakRDL-generated (like HPET), 8 channels, kick-off registers - 📝 Top-level integration - Different interface set

Always Ask Yourself: "Can I reuse from RAPIDS instead of creating new?"

30.3.5 Rule #2: Descriptor Format is DIFFERENT from RAPIDS

STREAM Descriptor (256-bit): - `src_addr` (64-bit), `dst_addr` (64-bit), `length` (**beats**), `next_descriptor_ptr` (32-bit) - **Length is in BEATS, not chunks!**

RAPIDS Descriptor: - Uses chunks (4-byte units) - Has alignment metadata

When comparing/referencing RAPIDS: - ✓ "RAPIDS uses chunks, STREAM uses beats for tutorial simplicity" - ✗ Don't assume RAPIDS descriptor format applies to STREAM

30.3.6 Rule #3: Know the Shared Resources

All channels (8 or 16, by top) share: 1. Descriptor fetch AXI master – and, by default (`SHARED_ENGINES=1`), ONE `descriptor_engine_shared` that fetches for every channel (ARID = channel). `SHARED_ENGINES=0` restores one `descriptor_engine` per channel behind an AR arbiter. 2. Data read AXI master 3. Data write AXI master 4. SRAM buffer 5. MonBus reporter

Arbitration is required! - Never assume exclusive access - All shared resources need arbiter logic - Priority-based round-robin (descriptor priority field)

30.4 Architecture Quick Reference

30.4.1 Block Organization

```
STREAM Architecture (as implemented)
├── rtl/fub/                          # Functional unit blocks
│   ├── descriptor_engine.sv          # Descriptor fetch/parse, one per
channel (SHARED_ENGINES=0)
│   ├── descriptor_engine_shared.sv # One fetch engine for all
channels (default; ported from RAPIDS)
│   ├── scheduler.sv                  # Simplified per-channel scheduler
│   ├── axi_read_engine.sv            # Data read engine
│   ├── axi_write_engine.sv           # Data write engine
│   ├── sram_controller.sv            # SRAM buffering (+
sram_controller_unit.sv per channel)
│   ├── stream_alloc_ctrl.sv          # Space allocation tracking
│   ├── stream_drain_ctrl.sv          # Drain control
│   ├── stream_latency_bridge.sv      # Latency decoupling
│   ├── (moved) stream_run_addr_gen.sv -> projects/components/utility-
ip/misc/rtl/  # shared with RAPIDS
│   └── perf_profiler.sv              # Performance profiling
│
├── rtl/macro/                        # Integration blocks
│   ├── scheduler_group.sv            # Per-channel scheduler group
│   ├── scheduler_group_array.sv      # NUM_CHANNELS array + descriptor
fetch (SHARED_ENGINES)
│   ├── stream_core.sv                # Core integration
│   ├── stream_regs.rdl               # PeakRDL register source (8-
channel, hand-written)
│   └── generated_16ch/               # 16-channel RDL, DERIVED by
bin/gen_stream_regs_16ch.py
│
└── rtl/top/                          # Top level
    ├── stream_top_ch8.sv             # 8-channel top-level wrapper
(hand-written)
    ├── stream_top_ch16.sv            # 16-channel top, GENERATED from
stream_top_ch8.sv
    ├── generated/regsel_{8,16}ch/  # regblock/hwif/kick glue
includes; the filelist picks one
    └── stream_config_block.sv        # Configuration block
```

16 channels: edit the 8-channel sources (`stream_regs.rdl`, `stream_mon_regs.rdl`, `stream_top_ch8.sv`), then run `python3 bin/gen_stream_regs_16ch.py` and regenerate BOTH regblocks with `bin/peakrdl_generate.py`; `--check` and `bin/check_rdl_regen.py` catch a stale 16-channel copy. Software and tests go by register name (`rtl/stream_regmap.py` / `rtl/stream_regmap_16ch.py`): the 16-channel map moves CH_STATE to 0x180 and OBS_CTRL.CAT_SEL to [5:4].

Monitoring uses the shared monbus group modules from `rtl/amba/ocelus/` (e.g. `monbus_axil4_axil4_group.sv`) rather than a STREAM-local reporter.

30.5 Common User Questions and Responses

30.5.1 Q: "How is STREAM different from RAPIDS?"

A: STREAM is intentionally simplified for tutorial purposes:

Simplifications: | Feature | RAPIDS | STREAM | |———|———|———| | **Interfaces** | APB + AXI + Network | APB + AXI only | | **Address Alignment** | Complex fixup | Aligned only | | **Credit Management** | Exponential encoding | Simple limits | | **Descriptor Length** | Chunks (4-byte) | Beats (data width) | | **Control Engines** | Control read/write | None (direct APB) |

Learning Path: 1. **STREAM** - Basic DMA with scatter-gather 2. **STREAM Extended** - Add alignment fixup 3. **RAPIDS** - Full complexity with network + credit management

📖 **See:** PRD.md Section 2.1 for complete comparison table

30.5.2 Q: "How do I kick off a transfer?"

A: Single APB write starts descriptor chain:

```
// Software writes descriptor address to channel register
write_apb(ADDR_CH0_CTRL, 0x1000_0000);  // Start address of descriptor

// STREAM automatically:
// 1. Fetches descriptor from 0x1000_0000
// 2. Parses src_addr, dst_addr, length
// 3. Reads source data → SRAM
// 4. Writes SRAM → destination
// 5. Checks next_descriptor_ptr
//    - If != 0: Fetch next descriptor, repeat
//    - If == 0 or last flag set: Complete
```

Chained Descriptors:

```
Descriptor 0 @ 0x1000_0000:
  src_addr = 0x2000_0000
  dst_addr = 0x3000_0000
  length = 64 beats
  next_descriptor_ptr = 0x1000_0100  ← Points to next descriptor

Descriptor 1 @ 0x1000_0100:
  src_addr = 0x2000_1000
  dst_addr = 0x3000_1000
  length = 32 beats
  next_descriptor_ptr = 0x0000_0000  ← Last descriptor (0 = stop)
```

📖 **See:** PRD.md Section 3.2 for complete data flow

30.5.3 Q: "How many channels can I use?"

A: 8 or 16 independent channels, by top (`stream_top_ch8` / `stream_top_ch16`):

Channel Operation: - Each channel has own FSM state - Each channel can have separate descriptor chain - All channels share: AXI masters, SRAM, MonBus

Arbitration: - Priority-based (descriptor priority field) - Round-robin within same priority - Prevents starvation with timeout

Example:

```
// Kick off 3 channels concurrently
write_apb(ADDR_CH0_CTRL, desc0_addr);  // Channel 0: Priority 7
write_apb(ADDR_CH1_CTRL, desc1_addr);  // Channel 1: Priority 5
write_apb(ADDR_CH2_CTRL, desc2_addr);  // Channel 2: Priority 5

// Service order: CH0 → CH1 → CH2 → CH0 → CH1 → CH2 ...
```

📖 **See:** `PRD.md` Section 7 for arbitration details

30.5.4 Q: "What's the descriptor format?"

A: 256-bit descriptor (4 × 64-bit words):

```
typedef struct packed {
    logic [63:0]  reserved;             // [255:192] Reserved
    logic [7:0]   priority;             // [207:200] Priority
    logic [3:0]   channel_id;           // [199:196] Channel ID
    logic         error;                // [195] Error flag
    logic         last;                 // [194] Last in chain
    logic         interrupt;            // [193] Generate interrupt
    logic         valid;                // [192] Valid descriptor
    logic [31:0]  next_descriptor_ptr;  // [191:160] Next descriptor
address
    logic [31:0]  length;               // [159:128] Length in BEATS
★
    logic [63:0]  dst_addr;             // [127:64] Destination
address
    logic [63:0]  src_addr;             // [63:0] Source address
} descriptor_t;
```

★ **CRITICAL:** `length` is in **BEATS**, not bytes or chunks!

Example:

```
Data width = 512 bits = 64 bytes
Length = 16 beats
Total transfer = 16 × 64 = 1024 bytes
```

📖 **See:** PRD.md Section 4.2 for complete descriptor specification

30.5.5 Q: "How do I run STREAM tests?"

A: Multi-layered test approach (same as RAPIDS):

```
# 1. FUB (Functional Unit Block) Tests - Individual blocks
pytest
projects/components/dma-ip/stream/dv/tests/fub/test_scheduler.py -v
pytest
projects/components/dma-ip/stream/dv/tests/fub/test_descriptor_engine.
py -v
pytest projects/components/dma-ip/stream/dv/tests/fub/ -v  # All FUB
tests

# 2. Macro / Top Tests - Multi-block scenarios
pytest projects/components/dma-ip/stream/dv/tests/macro/ -v
pytest projects/components/dma-ip/stream/dv/tests/top/ -v

# Run with waveforms for debugging
WAVES=1 pytest
projects/components/dma-ip/stream/dv/tests/fub/test_scheduler.py -v
```

Test Organization: - **FUB tests:** Focus on individual block functionality (`dv/tests/fub/`) - **Macro/top tests:** Verify block-to-block interfaces and complete data flows (`dv/tests/macro/`, `dv/tests/top/`)

30.6 Integration Patterns

30.6.1 Pattern 1: Basic STREAM Instantiation

```
stream_top #(
    .NUM_CHANNELS(8),
    .DATA_WIDTH(512),
    .ADDR_WIDTH(64),
    .SRAM_DEPTH(4096)
) u_stream (
    // Clock and Reset
    .aclk               (system_clk),
    .aresetn            (system_rst_n),

    // APB Configuration Interface
    .s_apb_paddr        (apb_paddr),
    .s_apb_psel         (apb_psel),
    .s_apb_penable      (apb_penable),
    .s_apb_pwrite       (apb_pwrite),
    .s_apb_pwdata       (apb_pwdata),
    .s_apb_pready       (apb_pready),
    .s_apb_prdata       (apb_prdata),
    .s_apb_pslverr      (apb_pslverr),

    // AXI Master - Descriptor Fetch (256-bit)
    .m_axi_desc_araddr  (desc_araddr),
    .m_axi_desc_arlen   (desc_arlen),
    .m_axi_desc_arsize  (desc_arsize),
    .m_axi_desc_arvalid (desc_arvalid),
    .m_axi_desc_arready (desc_arready),
    .m_axi_desc_rdata   (desc_rdata),
    .m_axi_desc_rresp   (desc_rresp),
    .m_axi_desc_rlast   (desc_rlast),
    .m_axi_desc_rvalid  (desc_rvalid),
    .m_axi_desc_rready  (desc_rready),

    // AXI Master - Data Read (parameterizable width)
    .m_axi_rd_araddr    (rd_araddr),
    // ... (full AXI4 AR + R channels)

    // AXI Master - Data Write (parameterizable width)
    .m_axi_wr_awaddr    (wr_awaddr),
    // ... (full AXI4 AW + W + B channels)

    // MonBus Output
    .monbus_pkt_valid   (stream_mon_valid),
    .monbus_pkt_ready   (stream_mon_ready),
```

```
    .monbus_pkt_data    (stream_mon_data)
);
```

30.6.2 Pattern 2: Descriptor Creation (Software Model)

```
// C/C++ software model for descriptor creation
typedef struct {
    uint64_t src_addr;          // Source address (must be aligned!)
    uint64_t dst_addr;          // Destination address (must be
aligned!)
    uint32_t length;            // Transfer length in BEATS
    uint32_t next_descriptor_ptr; // Next descriptor address (0 =
last)
    uint8_t  control;           // valid | interrupt | last | error |
channel_id | priority
} stream_descriptor_t;

// Create descriptor chain
stream_descriptor_t desc[2];

// Descriptor 0
desc[0].src_addr = 0x80000000ULL;  // Aligned to 64B (512-bit data)
desc[0].dst_addr = 0x90000000ULL;
desc[0].length = 64;  // 64 beats × 64 bytes = 4KB transfer
desc[0].next_descriptor_ptr = (uint32_t)&desc[1];  // Chain to next
desc[0].control = 0x01;  // valid = 1

// Descriptor 1 (last)
desc[1].src_addr = 0x80001000ULL;
desc[1].dst_addr = 0x90001000ULL;
desc[1].length = 32;  // 32 beats × 64 bytes = 2KB transfer
desc[1].next_descriptor_ptr = 0;  // Last descriptor
desc[1].control = 0x45;  // valid | last | interrupt

// Kick off transfer
write_apb_reg(CH0_CTRL, (uint32_t)&desc[0]);
```

30.6.3 Pattern 3: MonBus Integration

```
// Always add downstream FIFO for MonBus
gaxi_fifo_sync #(
    .DATA_WIDTH(64),
    .DEPTH(256)
) u_stream_mon_fifo (
    .axi_aclk   (aclk),
    .axi_aresetn(aresetn),
    // producer side: wr_ready is "not full", so it is what
backpressures MonBus
    .wr_valid   (monbus_pkt_valid),
```

```
    .wr_data    (monbus_pkt_data),
    .wr_ready   (monbus_pkt_ready),
    // consumer side
    .rd_valid   (fifo_mon_valid),
    .rd_data    (fifo_mon_data),
    .rd_ready   (consumer_ready)
);
```

30.7 Anti-Patterns to Catch

30.7.1 ✗ Anti-Pattern 1: Adding Alignment Fixup

✗ WRONG:
```
"Let me add alignment fixup logic to handle unaligned addresses..."
```

✓ CORRECTED:
```
"STREAM intentionally requires aligned addresses for tutorial
simplicity.
If you need unaligned transfers, that's a great learning exercise for
extending STREAM, or use RAPIDS which has full alignment support."
```

30.7.2 ✗ Anti-Pattern 2: Using Length in Bytes

✗ WRONG:
```
descriptor.length = 4096;  // Thinking this is 4096 bytes
```

✓ CORRECTED:
```
// Length is in BEATS, not bytes!
// For 512-bit data width (64 bytes per beat):
descriptor.length = 4096 / 64;  // = 64 beats for 4096 bytes
```

30.7.3 ✗ Anti-Pattern 3: Circular Buffer Descriptors

✗ WRONG:
```
// Descriptor chain that loops back
desc[9].next_descriptor_ptr = &desc[0];  // Circular!
```

✓ CORRECTED:
```
"STREAM does not support circular buffers (no stop condition).
Always terminate chains explicitly:
  desc[last].next_descriptor_ptr = 0;  // Stop
  desc[last].last = 1;  // Explicit last flag
```

30.7.4 ✗ Anti-Pattern 4: Assuming Exclusive Channel Access

✗ WRONG:
```
// Assume channel has exclusive AXI master access
assign m_axi_arvalid = channel_request;  // No arbitration!
```

✓ CORRECTED:
```
// All channels share AXI masters - arbitration required
channel_arbiter u_arbiter (
    .ch_requests (channel_requests[7:0]),
    .ch_grant    (channel_grant[7:0]),
    .axi_master_available (axi_master_idle)
);
```

30.8 Debugging Workflow

30.8.1 Issue: Descriptor Not Fetched

Check in order: 1. ✓ APB write to CHx_CTRL register successful? 2. ✓ Descriptor address valid? 3. ✓ AXI descriptor master not stalled? 4. ✓ Descriptor memory accessible? 5. ✓ Arbiter granting descriptor fetch?

Debug commands:

```
pytest
projects/components/dma-ip/stream/dv/tests/fub/test_scheduler.py -v -s
# Verbose test
WAVES=1 pytest
projects/components/dma-ip/stream/dv/tests/fub/test_scheduler.py -v
gtkwave <logs dir>/*.fst  # Inspect FSM state transitions
```

30.8.2 Issue: Data Transfer Stalls

Check in order: 1. ✓ SRAM buffer depth sufficient? 2. ✓ Source/destination addresses aligned? 3. ✓ AXI read/write engines getting grants? 4. ✓ Downstream AXI ready signals asserted? 5. ✓ Channel not in error state?

Waveform Analysis: - Check SRAM read/write pointers - Verify AXI AR/AW/R/W/B handshakes - Inspect scheduler FSM state - Check arbiter grant signals

30.8.3 Issue: Chained Descriptors Not Following

Check in order: 1. ✓ `next_descriptor_ptr` != 0? 2. ✓ `last` flag not set prematurely? 3. ✓ Descriptor fetch completing successfully? 4. ✓ Scheduler transitioning to CHAIN_CHECK state? 5. ✓ MonBus showing CHAIN_FETCH event?

30.9 Testing Guidance

30.9.1 Test Organization

```
projects/components/dma-ip/stream/dv/tests/
├── fub/                        # Individual block tests
│   ├── test_descriptor_engine.py
│   ├── test_scheduler.py
│   ├── test_sram_controller.py (+ test_sram_controller_alloc.py)
│   ├── test_perf_profiler.py
│   └── test_stream_latency_bridge.py
├── macro/                      # Multi-block scenarios (stream_core,
scheduler_group, datapaths)
└── top/                        # Top-level (stream_top_ch8) tests
```

30.9.2 Test Levels

Basic (Quick Smoke Tests): - Single descriptor, single channel - Aligned addresses, simple transfers - ~30 seconds runtime

Medium (Typical Scenarios): - Multiple descriptors, 2-4 channels - Chained descriptors (2-3 deep) - ~90 seconds runtime

Full (Comprehensive Validation): - All 8 channels concurrent - Long descriptor chains (10+ deep) - Error injection, stress testing - ~180 seconds runtime

---

30.10 Key Documentation Links

30.10.1 Always Reference These

This Subsystem: - `projects/components/dma-ip/stream/PRD.md` - **Complete specification** - `projects/components/dma-ip/stream/README.md` - Quick start guide - `projects/components/dma-ip/stream/known_issues/` - Bug tracking - `projects/components/dma-ip/stream/docs/stream_has/`, `docs/stream_mas/` - HAS/MAS spec trees

Related: - `projects/components/dma-ip/rapids/PRD.md` - Parent architecture (for comparison) - `docs/markdown/rtl-amba/index.md` - MonBus integration - `/PRD.md` - Master requirements - `/CLAUDE.md` - Repository guide

---

30.11 Quick Commands

```
# View complete specification
cat projects/components/dma-ip/stream/PRD.md

# Check package definition
cat projects/components/dma-ip/stream/rtl/includes/stream_pkg.sv

# Run tests
pytest projects/components/dma-ip/stream/dv/tests/fub/ -v
pytest projects/components/dma-ip/stream/dv/tests/macro/ -v
pytest projects/components/dma-ip/stream/dv/tests/top/ -v

# Lint
verilator --lint-only
projects/components/dma-ip/stream/rtl/fub/scheduler.sv

# Search for modules
find projects/components/dma-ip/stream/rtl/ -name "*.sv" -exec grep -H
"^module" {} \;
```

30.12 Remember

1. 📖 **Tutorial focus** - Intentional simplifications for learning
2. 🔄 **Reuse from RAPIDS** - Descriptor engine, SRAM, patterns
3. 📏 **Length in beats** - Not bytes or chunks!
4. 🔗 **Aligned addresses** - No fixup logic
5. ⛓️ **Chained descriptors** - No circular buffers
6. 🎯 **8 or 16 channels** - Shared resources, arbitration required
7. 🔍 **MonBus standard** - Same format as AMBA/RAPIDS
8. 🏗️ **Testbench reuse** - Always create TB classes in `projects/components/dma-ip/stream/dv/tbclasses/` (project area, not the framework)

30.13 PDF Generation Location

IMPORTANT: PDF files should be generated in the docs directory:

`/mnt/data/github/rtldesignsherpa/projects/components/dma-ip/stream/docs/`

Quick Command: Use the provided shell scripts:

```
cd projects/components/dma-ip/stream/docs
./generate_has_pdf.sh   # STREAM HAS
./generate_mas_pdf.sh   # STREAM MAS
```

The shell scripts will automatically: 1. Use the md_to_docx.py tool from bin/ 2. Process the stream_has/ and stream_mas/ index files 3. Generate both DOCX and PDF files in the docs/ directory 4. Create table of contents and title page

📖 **See:** `bin/md_to_docx.py` for complete implementation details

Version: 1.0 **Last Updated:** 2025-10-17 **Maintained By:** RTL Design Sherpa Project

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31 STREAM Register Definitions

Purpose: the PeakRDL-generated register block STREAM's config port exposes, and the one hand-written file that has to sit beside it. This page is a link page: the register map itself is the generated documentation, the fields are the RDL, and the wiring is the RTL that instantiates the block.

31.1 What is here

Path	What it is	Edit?
`generated/rtl/ stream_regs.sv, stream_regs_pkg.sv`	the regblock (`s_cpuif_*` passthrough cpuif in, `hwif_in`/`hwif_out` structs out)	never – regenerate
`generated/docs/ stream_regs.md, .html`	the register map, field by field	never – regenerate
`generated/ stream_regs_regmap.py`	the Python map the DV and host code read registers BY NAME from	never – regenerate
`stream_regs.vlt`	Verilator waiver: PeakRDL's per-field `always_comb` blocks trip a strict MULTIDRIVEN reading; waived for the generated file only. Lives outside `generated/` so a regen cannot erase it	yes

The source is NOT in this directory: `../rtl/macro/stream_regs.rdl`, which includes `../rtl/macro/stream_mon_regs.rdl` for the per-monitor block.

31.2 Who uses it

`../rtl/top/stream_config_block.sv` wraps the regblock behind the APB config port; `../rtl/macro/stream_core.sv` and `../rtl/top/stream_top_ch8.sv` instantiate it. Look there for the port wiring rather than at an example here – an example copied into a README is the copy nobody regenerates, and this page carried one for months that named ports the block never had.

31.3 Regenerating

Any `.rdl` edit regenerates ALL of `generated/` (root `CLAUDE.md`, Critical Rule #0), through the repo wrapper and never raw `peakrdl` – the wrapper is what also emits the docs and the regmap:

```
python3 bin/peakrdl_generate.py
projects/components/dma-ip/stream/rtl/macro/stream_regs.rdl \
    -o projects/components/dma-ip/stream/regs/generated --no-html
```

then `cd dv/tests && make clean-all && make run-all`. Method: `vault/handbook/dv/registers-by-name.md` (registers by name, never by offset) and `vault/handbook/design/generated-rtl-discipline.md`.